Psychological Statistics

Stats for Kids Who Don't Read Good

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Preface

I'm dyslexic, I don't like reading, and I wrote a stats textbook. Make it make sense.

Here's the deal: I learn by *doing* things and *seeing* things. Dense paragraphs of text make my brain go on vacation. Pretty pictures, clean code, and silly examples! So that's what this book is. Minimal words. Maximum visuals. Actual R code you can run yourself. The early chapters are more wordy, but as you get the basic concepts down, I start moving into shorthand and stop re-explaining everything because that just makes more words.

This book teaches **statistics and R at the same time** — because honestly, you learn both better together. Stats without code is just memorizing formulas. Code without stats is just typing. Together they're actually useful.

A few things you should know going in:

I hate ANOVA. It's old. It's clunky. Psychologists have been running ANOVAs since the Eisenhower administration and it needs to stop. Everything you can do with an ANOVA you can do with regression—and regression scales up to the real world in ways ANOVA never will. So we're going full regression in this book. ANOVA gets a brief appearance so you know what it is when someone inflicts it on you. That's it. The undergraduate chapters assume you have zero statistics background and approximately zero R experience. That's fine. We start from scratch.

Most of these chapters are deeply misunderstood summaries of *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences* by Cohen, Cohen, West, and Aiken (Cohen et al. 2003). That book is the holy bible of regression. This book is not trying to replace it. The Cohen book provides more theory, more detail, more qualifications, and substantially more words. If you need the full argument or plan to conduct advanced work, use it as a reference and read the relevant chapters.

This book has a smaller and cheaper job. It is a free knockoff assembled from roughly two decades of me ranting about statistics, including statistics lectures, class materials, examples, mistakes, revised examples, and explanations that finally worked after the first three explanations failed. Its goal is to get undergraduate and graduate students up to speed on the statistical theory they must know to survive, do research responsibly, and recognize when an analysis has gone feral.

> **About the AI Editor Standing Behind the Curtain**
>
> AI was the editor for this book. Claude and ChatGPT have helped clean up my grammar and spelling, merge material from different courses, rework examples because I am old and I bet people do not even know that the title of this book is a reference to *Zoolander*, fix the formatting, fix my stupid R code errors, check consistency across chapters, and suggest reworded explanations so they make sense to people who are not dyslexic and cannot automatically translate Alex into English.
>
> Many people have negative feelings about AI. I do not. I believe this technology can open doors for people like me who would never otherwise dream of turning decades of lectures and course materials into one coherent, free book. A traditional textbook publisher would have been helpful, but the resulting cost to students could have been astronomical. That is not my goal. My goal is to make statistics accessible, and I will use every reasonable tool at my disposal to do it.
>
> AI can also produce confident nonsense. I have tried to catch its hallucinations, check the code, preserve my own voice, and remain responsible for what appears here. As I already warned you, however, I do not like to read. If you find something wrong, please tell me before it reproduces.

None of this could have come together without the many students who have suffered in my classes, particularly my Spring 2026 class, who gave me feedback on drafts of several chapters. I also owe thanks to my graduate-student TAs and my many colleagues. I should apologize in advance if I borrowed material from my super-smart and awesome statistics colleagues Ryne Estabrook and Linda Skitka, or saw something on the web, thought, *Wow, that looks cool for my lecture*, and promptly forgot where I found it. I am 1000% sure Ryne contributed to many places in this book because we have edited each other's R Markdown lectures. Thanks, Ryne. I will make sure to share all the profits with you. Oh wait, it is free online!

Anyway, I have the memory of a goldfish. If I forgot to acknowledge you, tell me and I will happily fix it. Thank you as well to everyone who believes in sharing teaching materials on the web. Feel free to borrow whatever you want from me. If you remember to cite me, that is great; I can show it to my department to remind them not to get rid of me, since I am often more trouble than I am worth.

The goal is that by the end, you can look at a dataset, run an appropriate model in R, make a plot that actually communicates something, and write up the results without crying. That's it. That's the whole thing.

Happy coding!

Part I

Regression Crash Course

1 Why You Have to Learn Statistics (Sorry not sorry)

1.1 Why Should You Care?

You are taking a statistics class. I assume this was always your dream.

Maybe you woke up as a child, looked at the ceiling, and thought, *One day I hope to calculate a standard error.* More likely, you have to read this book because someone has blocked your graduation until you pass a statistics class. Either way, statistics matters because psychology runs on data, and data are messy!

Before we go any farther, here are three phrases researchers use constantly and explain approximately never:

- **Statistically significant** means something like, "This result probably, maybe did not happen just because chance was being weird." That definition is deliberately incomplete. We will give it a real definition after you learn some probability theory and are emotionally prepared for it.
- An **asterisk** is the little star researchers sometimes put after a number to show that it was statistically significant: for example, `r=.32*`. More stars may indicate smaller p-values, depending on the researchers' chosen rules, but more does not mean better or "more better," as people often hear me say. Asterisks do not mean the effect is large, important, true, or blessed by the Statistics Gods.
- An **effect size** asks how strong or large an effect is. Think of a small effect as being hit over the head with a coffee stirrer, a medium effect as a Nerf bat, a large effect as a wooden bat, and a huge effect as a metal bat. A **gargantuan effect**—a made-up but extremely technical term—is your mother hitting you over the head with her slipper or the hanger she happens to be holding. The effect is so devastating that you would have preferred to have taken the metal bat.

Statistics does four jobs for us:

The Job	Psychological Example	Why You Care
Analyze and interpret data	Two thousand students answer a mental-health survey.	Statistics turns 2,000 separate cries for help into a pattern we can understand.
Make decisions	A therapy group improves more than a control group.	We need to decide whether the difference looks like a real effect or the kind of nonsense random samples produce.
Design research	We want to test whether sleep improves memory.	Statistics helps us choose the sample size, variables, comparisons, and analysis *before* we start throwing undergraduates at the problem.
Communicate findings	We found a relationship between stress and exam performance.	A graph, effect size, and confidence interval tell people how large and uncertain the result is. "Trust me, bro, it looked huge in Excel" does not cut it.

These are not four separate activities. Your design determines the data you get. Your measurement determines what the numbers mean. Your analysis determines what claims are defensible. Then your graph determines whether anyone understands you before they wander away.

Statistics is therefore not the decorative math placed at the end of a study. It is part of the logic of the entire study.

Abelson called statistics a form of **principled argument** (Abelson 1995). The software will calculate whatever you ask it to calculate, including complete garbage. Your job is to build an argument in which the design, measurement, analysis, and conclusion are all talking about the same thing.

That argument usually asks whether **chance is enough** to explain what happened or whether we need some systematic explanation on top of it. Unfortunately, chance is fickle. Random data do not arrive evenly spaced, alphabetized, and wearing little name tags that say, "Hello, I am meaningless noise." They form streaks, clusters, suspicious-looking differences, and occasionally an image of a saint on a piece of toast. Statistics helps us judge how much of that weirdness a reasonable chance process could produce.

Abelson argued that doing this well requires the combined skills of an honest lawyer, a detective, and a storyteller (Abelson 1995). The lawyer builds a defensible case. The detective looks for alternative explanations and evidence that does not fit. The storyteller explains why the comparison matters. The word **honest** is doing considerable work here. Statistics does not let us torture the data until they confess to whatever we hoped was true.

1.2 The Replication Crisis: Our Extremely Rude Wake-Up Call

A **replication** repeats a study to see whether the result appears again. If a finding is "real," we should be able to find it more than once. Not every single time—chance is fickle—but often enough that the result becomes more trustworthy.

In 2015, the Open Science Collaboration repeated 100 published psychology studies. Ninety-seven of the original studies reported statistically significant results. Only 36 replications did, and the replication effects were, on average, about half the original size (Open Science Collaboration 2015). Psychology—well, mostly social psychology—responded with the scientific equivalent of waking up to find the house on fire. *They are still sort of overcorrecting and have totally lost the forest for the trees, but we will love them for caring.*

This was not proof that 64% of psychology is fake. Replication is not a divine courtroom where one failed study gets to bang a tiny gavel from the Statistics Gods and declare a theory dead. The projects differed in methods, samples, measures, and statistical power. Still, the result was far worse than psychologists expected.

1.2.1 Why Results Fail to Replicate

The problem is partly misunderstanding statistics:

- Treating the **null hypothesis** as if it must be exactly true, even though tiny relationships are almost everywhere (Meehl 1978).
- Treating a **p-value** as the probability that the theory is wrong. It is not (Cohen 1994).
- Using small samples that can only detect cartoonishly large effects (Cohen 1992; Button et al. 2013).
- Ignoring effect sizes and power, then acting shocked when a weak study produces unstable results.
- Calculating **post hoc power** from the result we already observed, which mostly turns the p-value into a new number wearing a fake mustache (Hoenig and Heisey 2001).
- Publishing exciting positive results while null results live forever in someone's forgotten `Final_Final_REAL_Results.docx` file (Ioannidis 2005).

The problem is also systemic. Journals want novelty. Universities want publications. Funders want breakthroughs. Nobody gets a parade for carefully discovering that the exciting effect is probably 0.08 instead of 0.80. Those incentives create publication bias and reward researchers for finding asterisks rather than finding the truth (Ioannidis 2012). You cannot personally repair every journal, university, and funding agency this semester. I checked, and maybe wrote, your syllabus. We do not have time.

But you *can* stop contributing avoidable statistical nonsense.

1.2.2 This Still Affects Your Life in a Serious Way

The crisis did not end in 2015, and it is not confined to social-psychology trivia.

- In preclinical cancer biology, researchers repeated 50 experiments from 23 influential papers. Across several definitions of success, the replications produced weaker evidence; replicated positive effects were typically much smaller, with a median effect size 85% below the originals (Errington et al. 2021). Those studies help determine which treatments deserve more research. This is not an argument about whether people unconsciously prefer the color blue on Tuesdays.
- A 2022 analysis of brain-wide association studies found that common samples below 500 people produced replication rates around 5% for links between brain features and behavior. Stable brain-behavior associations often required **thousands** of people (Marek et al. 2022). So when a headline announces that scientists found "the brain region for procrastination" after scanning 23 sophomores, perhaps continue procrastinating before sharing it.
- In a large project examining urgent COVID-era social and behavioral claims, 19 of 29 sufficiently powered replications succeeded, or about 65% (Marcoci et al. 2025). That is better than the 2015 wake-up call, but these were different claims selected and tested in different ways. It is not a national replication scoreboard.

There has also been real progress. Researchers preregister hypotheses and analysis plans, share data and code, conduct large multi-lab studies, and use **Registered Reports**, in which journals judge the question and method before knowing whether the result is exciting. More than 300 journals had adopted Registered Reports by 2024 (Nature Neuroscience 2024). A large preregistered national study of growth-mindset training, for example, found a small benefit for lower-achieving students and showed that the effect depended on the school context (Yeager et al. 2019). That is a much more useful conclusion than either "growth mindset fixes education" or "growth mindset is fake."

The lesson is not that science is broken beyond repair. The lesson is that science is a repair process.

Suppose you put 20 people in therapy. The Probability Gods might hand you 20 unusually cooperative people who respond beautifully. Or they might give you 20 people who hate your face, refuse to listen to anything you say, and make the treatment look useless. With 200 people, it becomes much harder for the Probability Gods to find enough people who all hate your face. The oddballs begin to balance one another out, so the estimated effects tend to cluster closer to the true effect. Probability remains fickle; it just has less room to wreck the study.

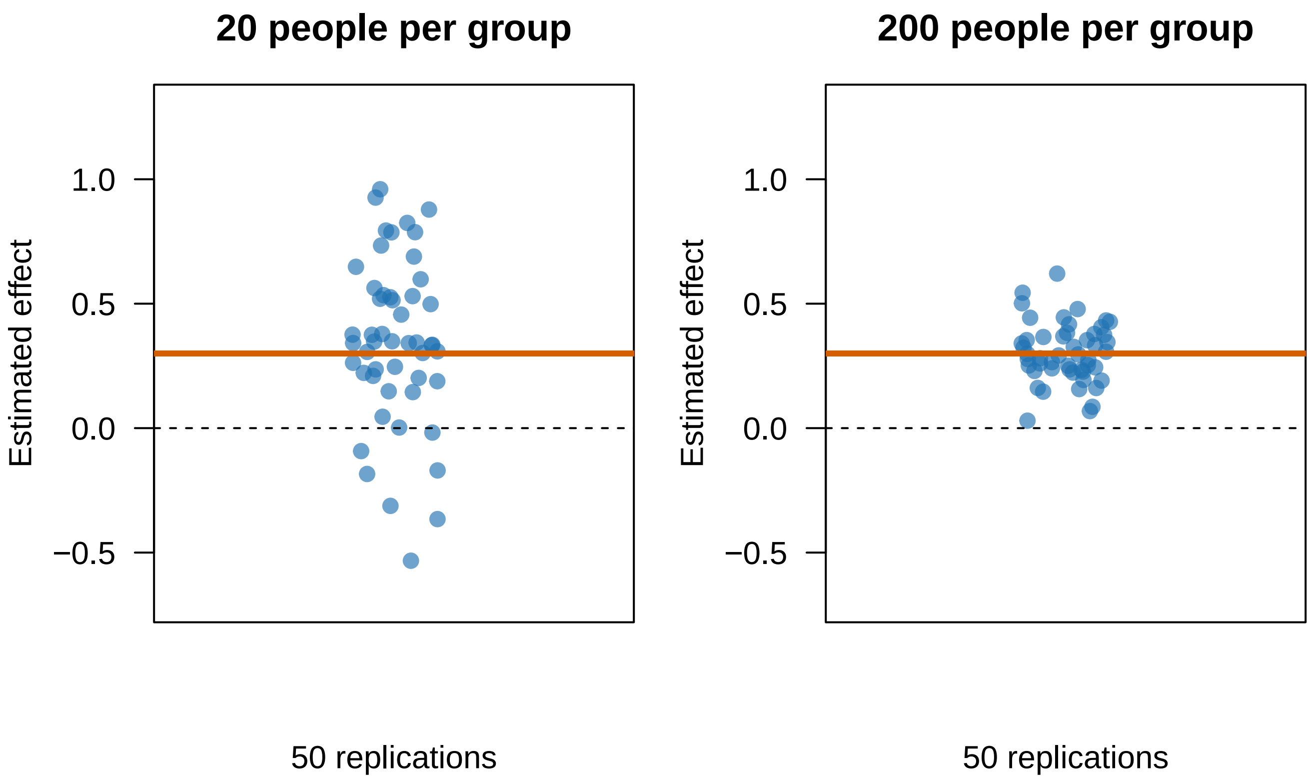


Figure 1.1: A real effect can produce wildly different estimates in small studies. Each point is one simulated replication; the orange line is the true effect.

The point of this graph is that collecting more people makes our estimate more stable. Small samples can produce wildly inaccurate and unstable guesses because probability is fickle. What does *probability is fickle* mean? It means the Gods who govern probability decide whether to do helpful or unhelpful things because they feel like it.

1.3 Do Not Just Apply Statistics

You could memorize a decision tree:

> Two groups? Push the t-test button. Three groups? Push the ANOVA button. More variables? Panic, then push buttons until Reviewer 2 stops emailing.

That approach can generate output. It cannot tell you whether the output answers your question.

To use statistics responsibly, you need to understand **why** a method works, what assumptions it makes, what uncertainty it leaves behind, and what your design allows you to claim. A perfectly executed analysis of a badly measured variable is still a beautifully formatted mistake.

1.4 Paradigms and the Stories We Tell About Causes

Thomas Kuhn used **paradigm** to describe the shared assumptions that tell scientists what counts as a sensible question, acceptable evidence, and a reasonable explanation (Kuhn 1962). Most of the time, scientists work inside a paradigm without discussing it. Fish probably spend very little time defining water. Also, did you know there are no such things as fish? (Miller 2020)

Psychology usually relies on **linear causal logic**:

$$X \longrightarrow Y$$

Sleep affects memory. Stress affects grades. A treatment affects symptoms. This does **not** mean every relationship must be a straight line. It means our explanation has a direction: X does something to Y.

But human behavior also contains feedback. Stress disrupts sleep, poor sleep increases stress, both encourage caffeine, and caffeine returns at 2:00 a.m. to finish the job. In a dynamical-systems perspective, causes can be circular, reciprocal, and dependent on what the system was doing a moment ago. Asking "Which came first?" can be a stupid question when each variable keeps causing the other one.

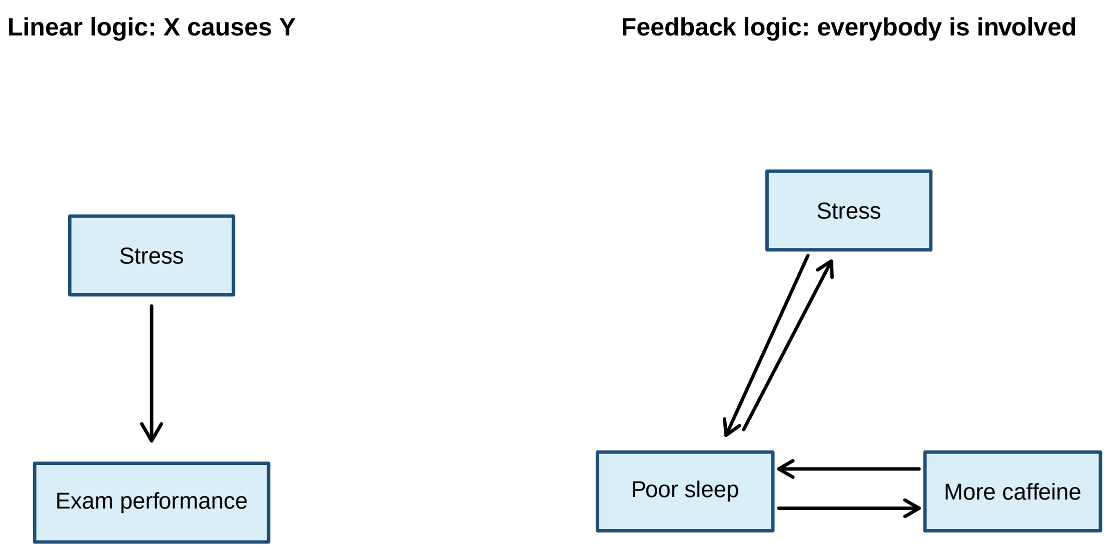


Figure 1.2: Linear causal logic gives us a direction. Feedback logic gives us a small committee of variables making one another worse.

I study systems like the second picture. Most statistics in this book will use the cleaner X-predicts-Y structure because that is how classical statistical models and most psychological research questions are organized. The nonlinear perspective does not make those statistics useless. It changes the questions I ask, the boundaries I place around an answer, and how suspicious I become when someone claims to have found the one true cause of a human behavior.

1.5 Measurement: Numbers Are Not Magic

Before analyzing a variable, we must decide what our numbers mean.

Stevens defined measurement broadly as assigning numbers to objects or events according to rules. He organized measurements into four scales (Stevens 1946):

Scale	What the numbers tell us	Psychology example	A better example
Nominal	Different categories; no order	Therapy condition: CBT, exposure, or waitlist	Volcano A, B, or C into which an undergraduate is dropped
Ordinal	Categories have an order; gaps between categories may be unequal	Mild, moderate, or severe symptoms	How much the student regrets volunteering for "easy" extra credit
Interval	Equal differences are meaningful; zero is arbitrary	Temperature in the overheated laboratory	Years on a calendar counting down to when Reviewer 2 retires
Ratio	Equal differences plus a meaningful zero	Heart rate after a student realizes the exam is today and they forgot to study	Number of chainsaw-carrying research assistants chasing the participant: ideally zero

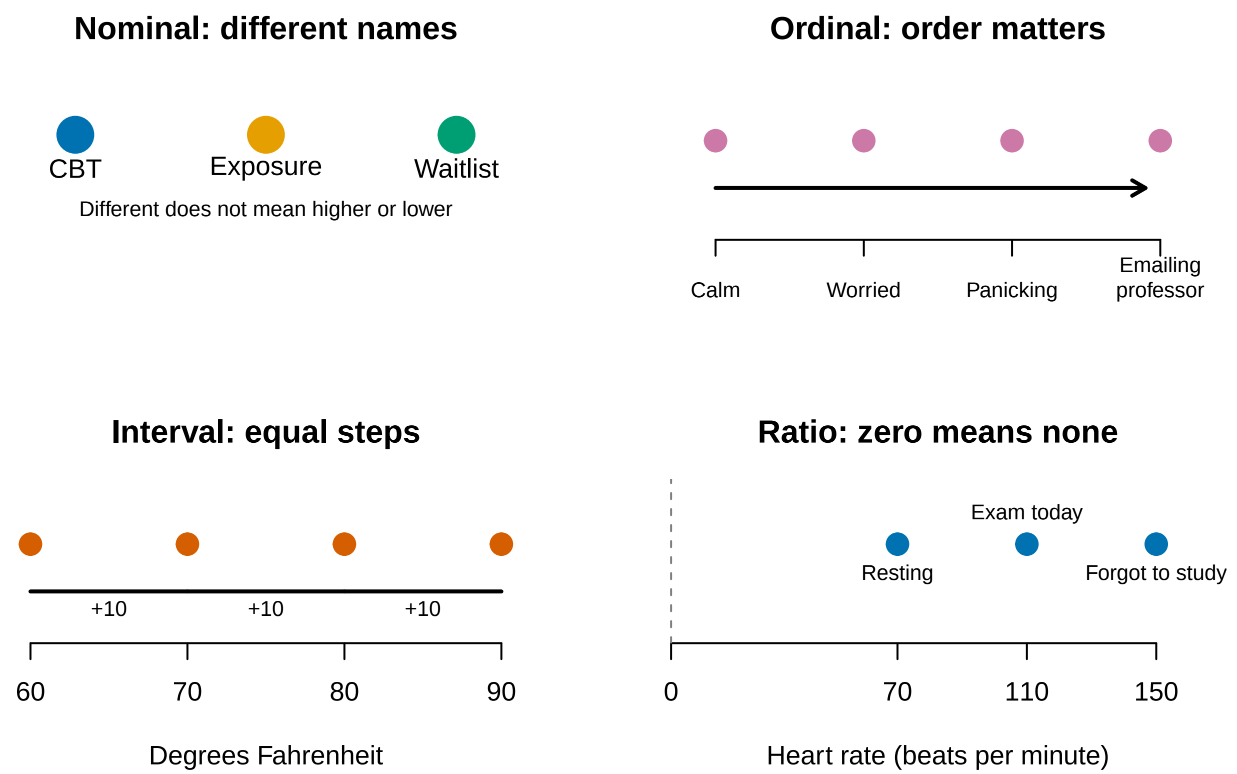


Figure 1.3: Four scales of measurement. The pictures are simple; the assumptions hiding underneath them are not.

That classification is useful, but Joel Michell points out a rather large problem: assigning a number does not prove that the attribute itself is quantitative (Michell 1997). I can assign you a student ID number. That does not mean a student with ID 800000 is twice as much student as someone with ID 400000.

Michell distinguishes two jobs. The **instrumental task** creates a procedure that produces numbers. The **scientific task** asks whether the attribute actually has the structure those numbers assume. Psychologists are often excellent at the first job and quietly assume the second one has been handled by someone else. Possibly an undergrad. Probably for extra credit.

This matters because psychological constructs are abstractions. Anxiety is not sitting inside the skull like a kidney waiting to be weighed. We infer it from behavior, self-report, physiology, and context. Those measurements may be useful without being perfect, but their meaning depends on the theory and the measurement process. Numbers do not rescue a vague construct. Sometimes they merely give the vagueness decimal places.

> ⚠ The Institutional Review Board has entered the chat
>
> Undergraduates may volunteer for ordinary research or earn reasonable extra credit. We cannot dissect their brains, drop them into volcanoes to appease Dorkaos (Ancient Greek God of Statistics), or chase them with chainsaws to increase the relationship between fear and exercise. "The effect size would be enormous" is not an ethical argument, or so I am told.

1.6 Research Design: The Extremely Short Version

Here are the terms you need before the rest of the book starts moving:

Term	Meaning
Theory	An organized explanation of how or why variables are related.
Hypothesis	A specific, testable prediction derived from a theory.
Operational definition	The exact procedure used to manipulate or measure a construct.
Independent variable (IV)	The proposed cause or predictor. In an experiment, the researcher manipulates it.
Dependent variable (DV)	The measured outcome we are trying to explain or predict.
Population	Everyone or everything we want our conclusion to describe.
Sample	The smaller group we actually observe because surveying all humans is expensive and unpleasant.
Parameter	A numerical description of the population, usually unknown.
Statistic	A numerical description calculated from the sample, which we use to guess the parameter.
Random sampling	Selecting people so members of the population have a known chance of entering the sample. This helps generalization.
Random assignment	Assigning sampled participants to conditions by chance. This helps causal inference. It is not the same as random sampling.
Experimental design	The researcher manipulates an IV and uses control plus random assignment to test a causal claim.
Correlational design	Variables are observed as they naturally occur. We can estimate relationships, but causal direction remains unresolved.
Confound	An uncontrolled alternative explanation that changes systematically with the IV.
Internal validity	How confidently we can say the IV, rather than a confound, caused the DV.

Term	Meaning
External validity	How confidently the result can generalize beyond this study to other people, settings, situations, or times.

The basic research sequence is:

1. Start with a problem or question.
2. Use **strong**, not soft, theory to make a hypothesis.
3. Define and measure the variables.
4. Choose a design and a sample.
5. Collect the data.
6. Analyze the data and quantify uncertainty.
7. Decide what the result does, and does not, support.
8. Communicate it clearly.
9. Replicate, revise, and try again.

If the hypothesis fails, the theory may be wrong. The operational definition may be bad. The manipulation may have failed. The sample may be weird. Or the data may simply have had a day. Statistics cannot automatically tell us which explanation is correct. It gives us disciplined ways to narrow the possibilities.

> ⚠ A Warning about Soft Theories
>
> Meehl argued that theories in "soft" psychology often do not win, lose, or improve. They become fashionable, collect a mixture of significant and nonsignificant results, acquire emergency auxiliary explanations, and eventually fade away when everyone gets bored (Meehl 1978). The theory is not killed. It is placed in an attic with several dissertations and a box of unidentified power cords.
> His deeper complaint was that a significant result is usually a weak test of a theory. Predicting "the groups will differ" is easy because groups almost always differ a little. A stronger theory predicts **how much**, **in which direction**, **under what conditions**, and **what result would make us abandon or revise it**. Good science takes theoretical risks. Collecting asterisks is not the same as understanding behavior.

That is why this book will not only show you **how** to run statistics. You will learn what the statistics are doing, why they behave that way, and how to notice when your analysis is answering a question no sensible person asked.

Statistics does not manufacture truth. It helps us make increasingly defensible guesses while being honest about how wrong we could be.

That is less comforting than a magic truth machine, but considerably more useful.

1.7 How to Read This Book Without Reading All of It

This book serves two groups that have very different reasons for being tired.

The main chapters are written for PSCH 242/343 undergraduates with little or no statistics background. Read those chapters in order. Each chapter assumes you survived the chapters before it, although it may occasionally provide a short recap because forgetting is one of the brain's most dependable skills.

The book also contains material labeled **Advanced**. Those sections and chapters are intended for:

- Graduate students who need a compact statistics and regression reference.
- Advanced undergraduates who want to understand what is hiding behind the friendly introductory explanation.

- Researchers who remember learning the topic but cannot remember where, when, or why someone decided there should be three different kinds of residuals.

You may skip advanced material during the undergraduate course unless your instructor assigns it or curiosity has defeated your instinct for self-preservation. Skipping an advanced chapter will not prevent you from understanding the next undergraduate chapter. Graduate students should treat the advanced material as a rapid overview, not a replacement for a full course or a detailed reference such as Cohen, Cohen, West, and Aiken (Cohen et al. 2003).

> **A Sensible Way to Use Each Chapter**
>
> 1. Read the explanation and inspect the figures before running code.
> 2. Run the R code yourself. Looking at code is not the same as discovering that you replaced a comma with a semicolon and angered R.
> 3. Change one value and predict what will happen before rerunning it.
> 4. Use the callout boxes to find warnings, common mistakes, and the short version of an argument.
> 5. Return to the mathematics after the example makes sense. Symbols are much friendlier once you know what they are trying to describe.

> **Advanced Does Not Mean Optional Assumptions**
>
> An advanced model does not rescue a weak design, a bad measure, or a causal claim assembled from wishful thinking. More complicated software merely allows you to be wrong with additional parameters.

2 R: The Extremely Short Survival Guide

2.1 Why R?

R is a free, open-source programming language built for statistics, data analysis, and graphics. It was developed by people who enjoy statistics enough to create an entire language for it. We should probably be grateful and slightly concerned.

Why are we using it?

- **It is free.** You do not lose access when you graduate, change jobs, or anger the university licensing office.
- **It is powerful.** R can run the analyses in this book, make publication-quality figures, simulate entire populations, and automate repetitive work.
- **It is reproducible.** Your script records what you did. Six months later, you can rerun the analysis instead of trying to remember which mysterious buttons you clicked.
- **It is extensible.** Researchers build packages that add new methods. This is wonderful, although packages sometimes change, break, or disappear into the forest.
- **It is used by actual researchers.** Learning R gives you a skill that travels beyond your class.

R changes more frequently than a pirate changes their underwear. Packages change too. This is the price of living in a flexible, community-driven ecosystem. When old code breaks, it does not mean you are stupid. It means you have joined the rest of us.

2.2 R Is Not RStudio

R is the language and the statistical engine. **RStudio** is an integrated development environment, or IDE, that gives R a useful place to live. R does the calculations; RStudio provides the script editor, console, plots, files, help, and many little buttons you will ignore until the day they save you.

Think of R as a temperamental but extremely capable chef. RStudio is the kitchen. Installing only RStudio is like building a lovely kitchen and forgetting the chef.

2.3 Install Them in This Order

1. Download and install the current version of **R** from CRAN.
2. Download and install the free **RStudio Desktop** from Posit.
3. Open RStudio. In the Console, run `R.version.string`. If R replies with a version, the creature is alive.

Windows users can usually keep multiple versions of R. On any operating system, install a version that matches your computer. For example, Apple Silicon Macs use the ARM64 build. The download pages provide the current choices, which is safer than this textbook pretending the year will remain 2026 forever.

Give yourself time to install software. Starting 30 minutes before class is not “giving yourself time.” That is creating a small emergency and assigning it to Future You.

Imprecations to the Statistics Gods

If RStudio behaves strangely after an update, close it completely and open it again. This solves an embarrassing number of problems. Sometimes it fails for you, you bring the computer to me, and it works in my mere presence. The explanation is simple: I am a very sophisticated android. If you do not have access to me, find someone you suspect is also a sophisticated android.

2.4 The Console Is for Experiments; Scripts Are for Work

You can type directly into the **Console**, but console commands are easy to lose. Put work you care about in an R script: a plain-text file ending in `.R`.

In RStudio, choose **File** > **New File** > **R Script**. Type the following into the script:

```
# This is a comment. R ignores it. Future You should not.
2 + 2
```

```
[1] 4
```

Run a line with **Ctrl** + **Enter** on Windows or **Command** + **Enter** on a Mac. You may also highlight several lines and run them together.

Comments begin with `#`. Write comments that explain *why* the code exists. `# calculate the mean` above `mean(scores)` is not a revelation. `# exclude the practice trial before calculating each person's mean` may prevent a future disaster.

For this book, make an **RStudio Project** for your work. Projects keep scripts, data, and output together and reduce the need for `setwd()`, which is the command people use to make code work only on their own computer.

2.5 PEMDAS Has Returned

I regret to report that middle-school mathematics has returned!

R follows the usual order of operations:

1. **P**arentheses: `()`
2. **E**xponents: `^`
3. **M**ultiplication and **D**ivision: `*` and `/`
4. **A**ddition and **S**ubtraction: `+` and `-`

Operations at the same level are generally handled in their defined order. The humane solution is to use parentheses whenever a tired person might have to stare at the expression.

```
2 + 3 * 4
```

```
[1] 14
```

```r
(2 + 3) * 4
```

```
[1] 20
```

```r
10 / 2 * 5
```

```
[1] 25
```

```r
10 / (2 * 5)
```

```
[1] 1
```

```r
(5 + 3)^2 / 4
```

```
[1] 16
```

Those expressions are not equivalent. Parentheses are cheap.

Suppose a stress score is 10, and our terrible theory predicts performance using

$$\widehat{Y} = 80 - 2(\text{Stress}).$$

In R:

```r
stress <- 10
predicted_performance <- 80 - 2 * stress
predicted_performance
```

```
[1] 60
```

R does not understand implied multiplication. Write `2 * stress`, not `2stress`. R is powerful, not psychic.

2.6 Objects: Put a Name on the Thing

The assignment arrow `<-` stores a value inside an **object**.

```r
coffee_cups <- 4
hours_of_sleep <- 3
regret <- coffee_cups^2 / hours_of_sleep
regret
```

```
[1] 5.333333
```

Read `<-` as "gets." `coffee_cups <- 4` means *coffee_ cups gets 4*. You may see `=` used for assignment. It often works, but this book will generally use `<-` because it makes assignment easier to recognize.

Object names cannot contain spaces. Use names such as `exam_score`, `ExamScore`, or `exam.score`. Pick a system and try not to invent a new one every seven minutes.

2.7 Vectors and Functions

A **vector** stores several values of the same basic type. The function `c()` combines values into a vector:

```
anxiety <- c(4, 7, 6, 9, 3)
anxiety
```

```
[1] 4 7 6 9 3
```

```
mean(anxiety)
```

```
[1] 5.8
```

```
sd(anxiety)
```

```
[1] 2.387467
```

```
length(anxiety)
```

```
[1] 5
```

A **function** performs a job. Its arguments go inside parentheses. In `mean(anxiety)`, the function is `mean()` and the object we hand it is `anxiety`.

Arguments can change what a function does:

```
anxiety_with_missing <- c(4, 7, NA, 9, 3)

mean(anxiety_with_missing)
```

```
[1] NA
```

```
mean(anxiety_with_missing, na.rm = TRUE)
```

```
[1] 5.75
```

`NA` means a value is missing. R refuses to quietly pretend it is not there. The argument `na.rm = TRUE` tells `mean()` to remove missing values for this calculation.

2.8 Data Frames: Where the Undergraduates Are Stored

A **data frame** is a rectangular data set. Rows usually represent cases or observations; columns represent variables.

```r
study <- data.frame(
  student = c("A", "B", "C", "D"),
  therapy = c("CBT", "CBT", "Control", "Control"),
  anxiety = c(4, 7, 8, 6)
)

study
```

```
  student therapy anxiety
1       A     CBT       4
2       B     CBT       7
3       C Control       8
4       D Control       6
```

```r
summary(study)
```

```
    student            therapy            anxiety    
 Length   :4   Length   :4   Min.   :4.00  
 N.unique :4   N.unique :2   1st Qu.:5.50  
 N.blank  :0   N.blank  :0   Median :6.50  
 Min.nchar:1   Min.nchar:3   Mean   :6.25  
 Max.nchar:1   Max.nchar:7   3rd Qu.:7.25  
                             Max.   :8.00  
```

In a real ethical study, those rows represent participants who consented to research. They are not a convenient supply of organs, volcano offerings, or targets for a chainsaw-based manipulation of fear. The Institutional Review Board remains stubborn about this.

2.9 Packages: Install Once, Load Each Session

Packages add functions to R. Install a package once:

```r
install.packages("tidyverse")
```

Load it each time you start a new R session and need it:

```r
library(tidyverse)
```

The quotation marks belong in `install.packages("tidyverse")`, but not in `library(tidyverse)`. This distinction exists because the Probability Gods were briefly allowed to help design software.

2.10 Errors Are Information, Delivered Rudely

An **error** means R stopped. A **warning** means R finished but wants you to know something suspicious happened. A **message** is usually informational. Read the first error carefully, because the next 47 errors may merely be its panicked descendants.

When code fails:

1. Read the error.
2. Check spelling, capitalization, commas, quotation marks, and parentheses.
3. Run the code in smaller pieces.
4. Confirm that the objects and packages it needs exist.
5. Restart R and try again.
6. Ask for help and show the code **plus the complete error**. "It hates me" is emotionally valid but diagnostically weak.

That is enough R to begin. You do not need to memorize the language before doing statistics. You need to keep your work in scripts, run small pieces, inspect the result, and understand what you asked R to do. R will calculate nonsense with breathtaking speed. The thinking remains your job.

3 Probability: The Gods Are Fickle

3.1 A Car, Two Goats, and a Terrible Decision

You are on a game show. There are three doors. Behind one is a car. Behind the other two are goats. *My guess is you want the car, but I don't know why. Goats have several advantages: a) they make funny sounds when they scream, b) some faint when frightened, and c) you can use one to cast a Greek curse on the Chicago Cubs so they never win another game!*

You choose Door 1. The host, Monty, knows where the car is. He opens one of the other doors and reveals a goat. He then offers you a choice: **stay** with Door 1 or **switch** to the remaining closed door.

Should you **stay**, **switch**, or **does it make no difference** what you do? **I think you should declare that goats are better than cars and ask for a goat!**

So what should you do?

> **What Most People Do**
>
> Most people stay. Some believe a deity secretly loves them above all others and guided them to the correct door. Switching would insult the deity. Others know they would be furious if they chose the correct door and then switched away from it. Finally, some people use their giant frontal cortex to conclude that two remaining doors must mean a 50/50 chance.
> All three groups stay. All three groups are wrong.

Well, people are really bad at probability, and it is not 50/50. *You should switch.* Staying with your first door wins about 1/3 of the time; switching wins about 2/3 of the time.

Why? Your original choice had a 1/3 chance of being correct and a 2/3 chance of being wrong. Monty did not open a random door. He used his knowledge to remove a goat from the two doors you did not choose. If your first choice was wrong—which happens 2/3 of the time—switching fixes it.

This answer depends on the rules: Monty knows where the car is, always opens a goat door, and always offers the switch. If the host sometimes opens the car, chooses doors randomly, or offers a switch only when he is bored, the probability changes. A probability depends on the rules that produce the outcomes. Change the rules, and you may change the probability. Keeping the old probability after quietly changing the rules is how nonsense enters wearing a lab coat.

The Monty Hall problem feels wrong because people often treat all visible options as equally likely. They are not. Two possible outcomes do not automatically mean 50/50 (Ang and Shahrill 2014). A hypothesis test can also end with two possible decisions. That does not mean each decision had a 50% probability.

3.2 What Probability Means

A probability is a number from 0 to 1 describing how likely an event is under stated assumptions.

- An impossible event has probability 0.

- A certain event has probability 1.
- $P(A) = .50$ means that across many comparable opportunities, event A should occur about half the time.

That last sentence is important. Probability does not promise that exactly 10 of your next 20 coin flips will be heads. It describes what happens over many repetitions. In a small sample, chance can behave like a raccoon inside a convenience store: energetic, destructive, and technically following the laws of nature.

The **sample space** is the set of possible outcomes. For one ordinary six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

If the die is fair, the probability of rolling a 2 is

$$P(2) = \frac{1}{6} = .167.$$

The word **fair** is an assumption. Statistics is full of assumptions. We will test some and defend others. Occasionally, we will discover that an assumption came from someone in 1974 who "felt" it was right. Everyone else cited them and put their fingers in their ears.

3.3 Three Rules That Do Most of the Work

3.3.1 The complement rule: "not A"

The probability that an event does not occur is

$$P(\text{not } A) = 1 - P(A).$$

Let event A mean that a participant notices clues that reveal what the researcher expects them—the participant—to do or feel. Suppose:

$$P(A) = .20$$

The probability that the participant does **not** notice them is:

$$P(\text{not } A) = 1 - .20 = .80$$

This example does not include a research assistant revving a chainsaw to increase the participant's heart rate. Everyone would notice that, so $P(\text{not } A) = 0$. More importantly, the ethics board already rejected it.

I do not know why they keep telling me I need help. I think my manipulations are very ethical!

3.3.2 The addition rule: "A or B"

For events that cannot occur together,

$$P(A \text{ or } B) = P(A) + P(B).$$

On one die roll, the probability of a 1 or a 2 is $1/6 + 1/6 = 2/6$.

If events can overlap, subtract the overlap so it is not counted twice:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

3.3.3 The multiplication rule: "A and B"

If two events are **independent**, knowing one occurred tells us nothing about the probability of the other:

$$P(A \text{ and } B) = P(A)P(B).$$

The probability of getting heads twice with a fair coin is $.5 \times .5 = .25$.

Independent does not mean "different." It means one event does not change the probability of the other. Drawing two hearts **with replacement** gives

$$\frac{13}{52} \times \frac{13}{52} = .0625.$$

Without replacement, the first card changes what remains:

$$\frac{13}{52} \times \frac{12}{51} = .0588.$$

The second probability is **conditional** on what happened first.

> ⚠ Mutually Exclusive Is Not the Same as Independent
>
> **Mutually exclusive** events cannot happen together. If one occurs, the other becomes impossible.
> **Independent** events can happen together, but one does not change the probability of the other.
> For one die roll, rolling a 2 and rolling a 5 are mutually exclusive. They are not independent. Once you roll a 2, the probability that the same roll was a 5 collapses to zero.
> These terms sound vaguely similar and are waiting for an opportunity to ruin your exam.

3.4 Conditional Probability: The Vertical Bar Is Not Decoration

$P(B \mid A)$ means "the probability of B given that A occurred."

The general multiplication rule is

$$P(A \text{ and } B) = P(A)P(B \mid A).$$

Conditional probabilities answer questions about a restricted group. For example:

- $P(\text{passes})$ asks about everyone.
- $P(\text{passes} \mid \text{attended review})$ asks only about students who attended the review.
- $P(\text{attended review} \mid \text{passes})$ asks something different.

Reversing the condition reverses the question. The probability that a person is tall given that they play basketball is not the probability that they play basketball given that they are tall. Likewise, a p-value will eventually describe how probable data this extreme—or more extreme—would be if H_0 and the model assumptions were true. It does not describe the probability that H_0 is true. Swapping those is one of psychology's favorite avoidable catastrophes.

! The Direction of the Bar Matters

$$P(B \mid A) \neq P(A \mid B)$$

The probability of playing basketball **given that someone is tall** is not the probability of being tall **given that someone plays basketball**.
Read everything after the vertical bar first. It identifies the group we are standing inside. Reversing the bar changes the group and usually changes the answer.

3.5 Chance Is Fickle, Not Obligated to Look Random

Imagine 20 coin flips. Humans usually invent sequences with roughly equal heads and tails, few long streaks, and tasteful amounts of alternation. Real coins have no taste. They produce clusters and streaks because **streaks are part of randomness**.

The chance of getting exactly 10 heads in 20 fair flips is only about 17.6%. Getting between 8 and 12 heads happens about 73.7% of the time. A result outside that range is not impossible. The Probability Gods did not violate a contract; you invented a contract they never signed.

Larger samples do not force the estimate to be correct. They make wildly inaccurate estimates less common. With 20 therapy clients, the Probability Gods might hand you 20 people who adore the treatment, or 20 people who hate your face. With 200 clients, finding enough unusually delighted or face-hating people to hijack the entire result becomes much harder.

This is the **law of large numbers**: as the number of independent observations increases, a sample average or proportion tends to settle near its expected value. It is not the law of small numbers. Twenty people do not become a population because the grant ran out.

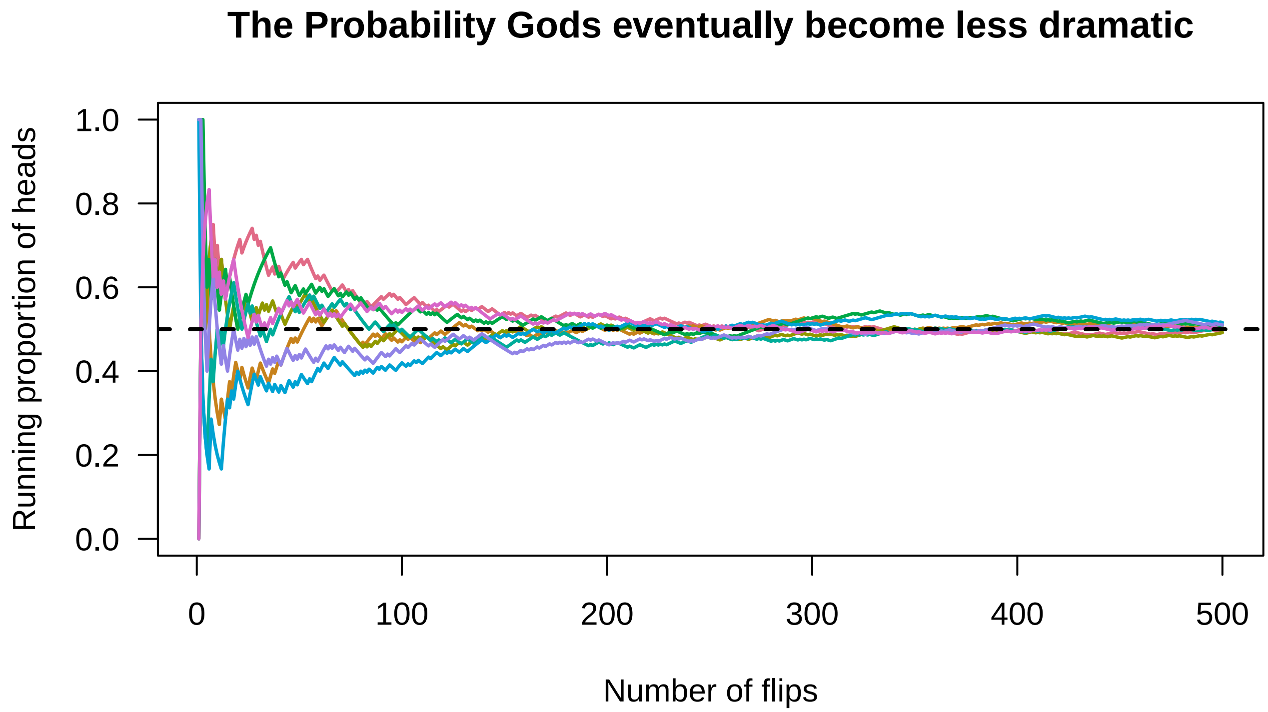


Figure 3.1: The running proportion of heads in eight fair-coin experiments. Early estimates stagger around; with more flips, they become more stable around .50.

3.6 Expected Value Is a Long-Run Average, Not a Prophecy

The **expected value** is the probability-weighted average outcome over many repetitions:

$$E(X) = \sum xP(x).$$

Suppose a deeply legitimate carnival game charges \$2. You win \$10 with probability .10 and \$0 otherwise. Your expected winnings are

$$E(X) = \$10(.10) + \$0(.90) = \$1.$$

After paying \$2, your expected net result is $-\$1$ per game. You could still win \$8 on the first play. Expected value does not predict one event; it describes the long-run average. The carnival can stay in business because it has patience and your money.

> **i Expected Does Not Mean Most Likely**
>
> An expected value does not have to be an outcome that can actually occur.
> The expected number of children in a family might be 1.7. Nobody has to produce seven-tenths of a child. The value describes the long-run average across many families.
> Expected value is an average, not a prophecy or a recommendation from the Probability Gods.

3.7 Advanced Section: Bayes, Base Rates, and a Scary Positive Test

You test positive for a rare condition that will turn you into a goat. The condition affects 1% of people. The test correctly detects 95% of cases and has a 5% false-positive rate. What is the probability that you actually have the condition and turn into a goat?

The tempting answer is 95%. That answer has ignored almost everyone who does not have the condition.

Imagine testing 10,000 people. Assume:

- 1% actually have the condition.
- The test correctly detects 95% of people who have it.
- The test incorrectly gives a positive result to 5% of people who do not have it.

3.7.1 Step 1: Who Has the Condition?

$$10{,}000 \times .01 = 100$$

So 100 people have the condition. The other 9,900 do not.

3.7.2 Step 2: Test the 100 People Who Have It

The test correctly identifies 95% of them:

$$100 \times .95 = 95$$

Therefore:

- 95 receive a **true positive**.
- 5 receive a **false negative**.

3.7.3 Step 3: Test the 9,900 People Who Do Not Have It

The test incorrectly gives a positive result to 5% of them:

$$9{,}900 \times .05 = 495$$

Therefore:

- 495 receive a **false positive**.
- 9,405 receive a **true negative**.

3.7.4 Step 4: Count Everyone Who Tested Positive

The positive results include 95 true positives and 495 false positives:

$$95 + 495 = 590$$

A total of 590 people test positive. However, only 95 of them actually have the condition.

3.7.5 Step 5: Find the Probability of Having the Condition After Testing Positive

$$P(\text{Condition} \mid \text{Positive}) = \frac{95}{590} = .161$$

Therefore, a person who tests positive has about a **16.1% probability** of actually having the condition.

The test is fairly accurate, but the condition is rare. That leaves the Probability Gods plenty of healthy people to terrify with false predictions that they will turn into goats tomorrow.

> ⚠ A 95% Detection Rate Is Not a 95% Diagnosis
>
> The test detects 95% of people who have the condition:
>
> $$P(\text{Positive} \mid \text{Condition}) = .95$$
>
> Our actual question reverses the condition:
>
> $$P(\text{Condition} \mid \text{Positive}) = .161$$
>
> Those are different probabilities. Test performance, false-positive rates, and the base rate must all be considered before the test can accuse you of becoming a goat.

The **base rate** describes how common the condition is before we consider the new evidence. When the condition is rare, even a fairly accurate test can produce many false alarms. This is why probability questions are easier when we turn percentages into counts of actual imaginary people. The imaginary people are easier to organize and much less likely to contact my department chair to complain about being turned into goats.

Bayes' theorem formalizes the update:

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{P(E \mid H)P(H) + P(E \mid \text{not } H)P(\text{not } H)}.$$

$P(H)$ is the **prior probability** or base rate. $P(E \mid H)$ tells us how expected the evidence is if the hypothesis is true. $P(H \mid E)$ is the **posterior probability** after combining the prior and evidence.

You do not need to become a Bayesian statistician today. You do need to remember that evidence does not arrive in an empty universe. Base rates matter, and $P(E \mid H)$ is not interchangeable with $P(H \mid E)$.

3.8 What This Has to Do with Statistics

Every study will produce a pattern. One group will have a higher mean. One correlation will be farther from zero. One brain image will contain a bright blob, possibly inside a dead salmon if you run enough tests.

Statistics asks whether the pattern is larger than we would reasonably expect from a specified chance process. To answer that question, we need to know:

- what outcomes were possible;
- what assumptions define the probability model;
- whether events are independent;
- what information we conditioned on;
- how much data we collected; and
- how often our entire analysis gave chance an opportunity to create something exciting.

Probability is not a promise about what must happen next. It is a language for uncertainty. The Probability Gods remain fickle, but they are not completely lawless. Our job is to learn the laws well enough that we stop blaming divine intervention for bad design.

4 Distributions and their Moments

4.1 What is a distribution?

A **distribution** is a picture of how values spread out in a population:

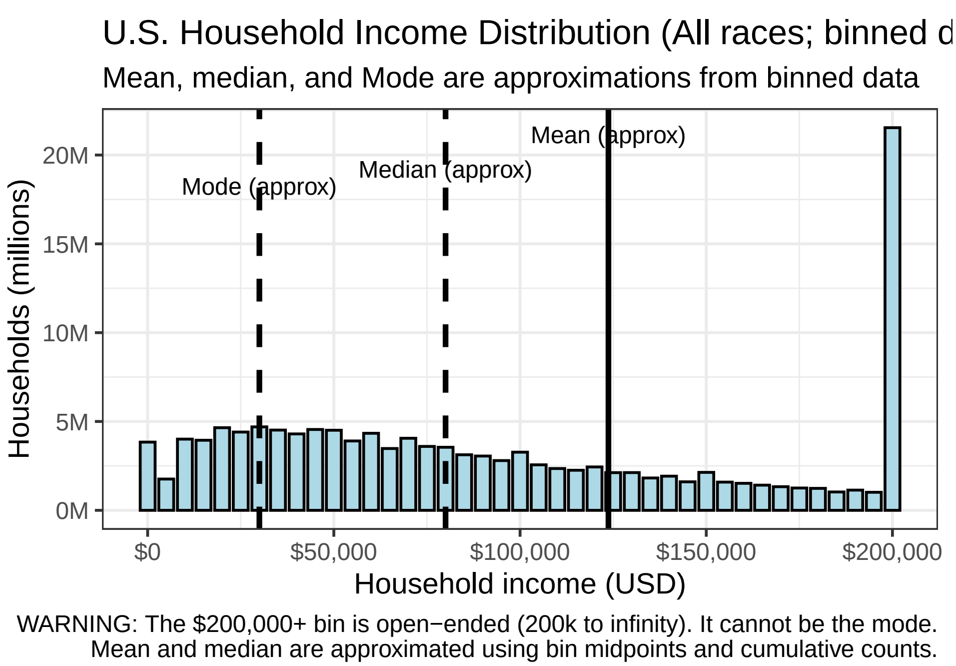
- Where do most observations land?
- How much do they vary?
- Is it symmetric, or does it have a long tail?

The average alone cannot answer these questions. Two groups can have the same mean while one forms a tidy mound and the other consists of people at both extremes glaring across an empty middle. A mean without its distribution is a witness who has omitted several important details.

4.1.1 A real distribution

Income in the US is a classic **positively-skewed** distribution: most households cluster around moderate incomes, with a few earning *very* large amounts.

This plot uses binned Census data (U.S. Census Bureau 2024), so the mean and median are approximations. Also: the \$200k+ bin is *everything* from \$200k to infinity — so if it's the tallest bar, that doesn't mean \$200k is the most common income. It just means rich people got lumped together.



Think about it:

1. Why is there a positive skew (the tail points right on the x-axis, toward the larger numbers)?
2. Which number (mean = 123K, median = 80K, or mode = 32.5K) best represents the "average" American? Why?

4.1.2 Historical sidenote

In the 1950s, the top marginal tax rate in the US was ~91%. Imagine that money went back into schools, housing, and infrastructure.

Predict: how would the income distribution shift? What would happen to the mean, median, and mode? Be explicit about why each one would (or wouldn't) change.

4.1.3 Distributions have meaning

Governments tend to use median income to decide who qualifies for financial aid, housing assistance, school lunch programs, tax credits, and healthcare subsidies.

When a distribution is highly skewed, these numbers tell very different stories. A rising *mean* can make a country look richer even when **most** households see no change — the median may stay flat. Understanding the shape helps us see who's actually benefiting from policy decisions.

> ❗ The Mean Can Be Correct and Still Mislead You
>
> In a strongly skewed distribution, a small number of enormous values can drag the mean upward.
> That does not make the mean mathematically wrong. It may simply describe the money rather than the experience of a typical person.
> Always inspect the distribution before allowing the mean to speak for everyone.

4.2 Gaussian (Normal) Distribution

Imagine a broad cognitive score influenced by many small factors: sleep, attention, schooling, mood, caffeine, random neural noise, and whether a chainsaw-wielding researcher was standing behind the participant. When many small influences add together, scores can form a smooth, symmetric hill. That hill is the **normal distribution**.

The Gaussian distribution (named after Carl Friedrich Gauss, 1777–1855) is the foundation of classical parametric statistics because it describes a huge range of psychological phenomena.

Two parameters fully define it:

- μ (mean) — where the center is
- σ (standard deviation) — how spread out it is

A normal distribution has skewness of zero and excess kurtosis of zero. However, those two values alone do **not** prove that a distribution is normal; very strange distributions can share the same moments. The mean, variance, skewness, and kurtosis are commonly described through the distribution's **moments**. They summarize important features, but they are not a magical distribution-identification scanner.

> **Normality Is a Model, Not a Moral Achievement**
>
> "Normal" describes a mathematical distribution. It does not mean healthy, desirable, typical, or free from chainsaw-wielding researchers.
> A normal distribution has zero skewness and zero excess kurtosis, but matching those two values does not prove that your data are normal. Plot the distribution. Numbers are capable of lying by omission.

4.2.1 Simulating a Population in R

We'll use the normal distribution as our population. R makes this easy with `rnorm()`.

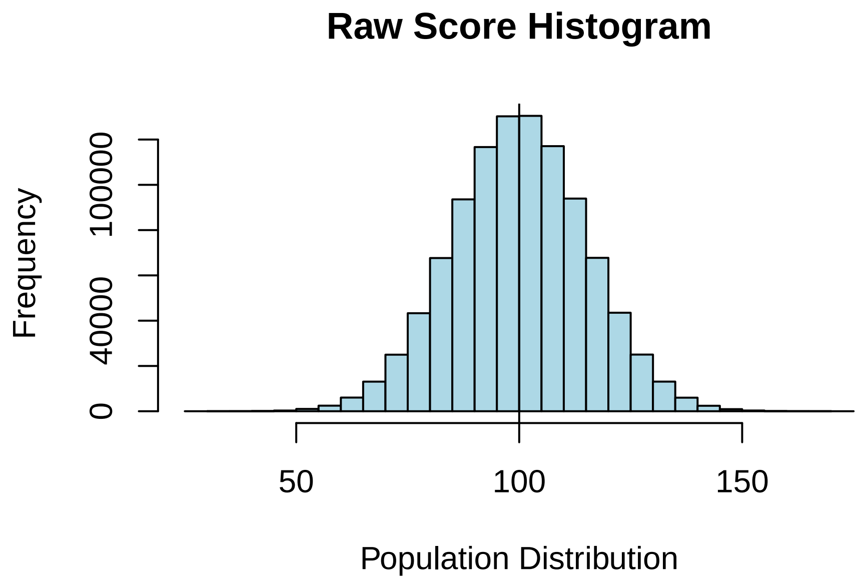
Let's create a population with:

- $\mu = 100$ points (cognitive score)
- $\sigma = 15$ points
- $N = 1,000,000$

```r
set.seed(42) # the answer to the universe
n  <- 1e6   # 1 million people
Mu <- 100   # population mean
S  <- 15    # population SD
Normal.Sample.1 <- rnorm(n, mean = Mu, sd = S)
```

4.2.2 Histogram (Base R)

```r
hist(Normal.Sample.1,
  breaks = 30,
  col    = "lightblue",
  border = "black",
  main   = "Raw Score Histogram",
  xlab   = "Population Distribution",
  ylab   = "Frequency")
abline(v = Mu, col = "black")
```



Looks normal. Now let's describe its shape using the 4 moments.

4.3 Moments

Four numbers capture the important features of a distribution: center, spread, symmetry, and peakiness.

4.3.1 Moment 1: Central Tendency

The center of a distribution can be described three ways:

Mean (the average):

- Standard measure of central tendency
- Heavily influenced by outliers
- Most stable estimate from sample to sample

Median (the middle number):

- Not sensitive to outliers
- Best measure when the distribution is skewed

Mode (most common number):

- Least stable from sample to sample

4.3.2 Mean

$$M = \frac{\Sigma X}{N}$$

```
SumX <- sum(Normal.Sample.1)
N    <- length(Normal.Sample.1)
M    <- SumX / N
M
```

```
[1] 100.0086
```

```
# Or all at once:
M <- sum(Normal.Sample.1) / length(Normal.Sample.1)
```

The mean was 100.01 — close to our target μ of 100. But just use the built-in:

```
mean(Normal.Sample.1)
```

```
[1] 100.0086
```

4.3.3 Median

```
median(Normal.Sample.1)
```

```
[1] 100.0208
```

Want to do it by hand? Here's a custom function (mostly to show you why built-ins exist):

```
calculate_median <- function(x) {
  sorted_x <- sort(x)
  n <- length(sorted_x)
  if (n %% 2 == 1) {
    sorted_x[(n + 1) / 2]
  } else {
    (sorted_x[n / 2] + sorted_x[(n / 2) + 1]) / 2
  }
}
calculate_median(Normal.Sample.1)
```

```
[1] 100.0208
```

Same answer. Lots of code. This is why R has built-in functions!

4.3.4 Mode

No built-in mode function in R for continuous data — and it's almost never used for normal distributions. Note: `mode()` in R tells you the *type* of an object (numeric, factor, etc.), not the statistical mode.

4.3.5 Moment 2: Spread

Variance describes how spread out a distribution is. In psychology we mostly use **variance** and **standard deviation** because they work well with normal distributions and parametric statistics.

Variability comes from:

- Individual differences (genetics, mood, experience)
- Research/environment factors (testing conditions, context)
- Measurement effects

When estimating variance from a sample, we divide by $N-1$ instead of N to get an unbiased estimate. This is called **degrees of freedom**.

4.3.6 Variance and SD

Population variance (when we have the entire population and know μ):

$$\sigma^2 = \frac{\Sigma(X-\mu)^2}{N}$$

Sample variance, using $N-1$ to estimate population variance:

$$S^2 = \frac{\Sigma(X-M)^2}{N-1}$$

R vectorizes this for you, but watch your parentheses. The built-in `var()` uses M and $N-1$:

```
Variance.M <- var(Normal.Sample.1)
```

Sample variance $= 225.47$

$$S = \sqrt{S^2}$$

```
SD.calc <- sd(Normal.Sample.1)
```

Sample SD $= 15.02$

> **Variance Has Weird Units; SD Comes Back Home**
>
> Variance is measured in **squared units**.
>
> - If reaction time is measured in seconds, variance is measured in seconds squared.
> - If exam scores are measured in points, variance is measured in points squared.
> - If cookie-dough devotion is measured in questionable rating units, variance uses questionable rating units squared.
>
> Standard deviation takes the square root of variance, returning to the original units. That is one reason SD is easier to interpret.

4.3.7 A Dot Plot: Where Does One Standard Deviation Live?

The standard deviation is a measure of the **typical distance of scores from the mean**. It is calculated from every score's distance from the mean. It is not defined by capturing a particular percentage of people.

However, when a distribution is approximately normal, standard deviations divide the distribution in a very useful and predictable way:

- About 34% of scores fall between the mean and one SD **below** the mean.
- About 34% fall between the mean and one SD **above** the mean.
- Put those pieces together, and about 68% fall within one SD of the mean.
- About 95% fall within two SDs of the mean.

Let us make that visible with a deeply important psychological construct: love of cookie-dough ice cream.

We ask 240 people:

> **How much do you like cookie-dough ice cream?**

They answer on a 1–10 unipolar rating scale:

- **1 = Not at all**
- **10 = I would trade my mother for cookie-dough ice cream**

The scale is **unipolar** because it begins with none of the feeling and moves in one direction toward more of it. The opposite endpoint is not "I hate cookie-dough ice cream with the fire of a thousand suns." That would be a bipolar scale and possibly a cry for help.

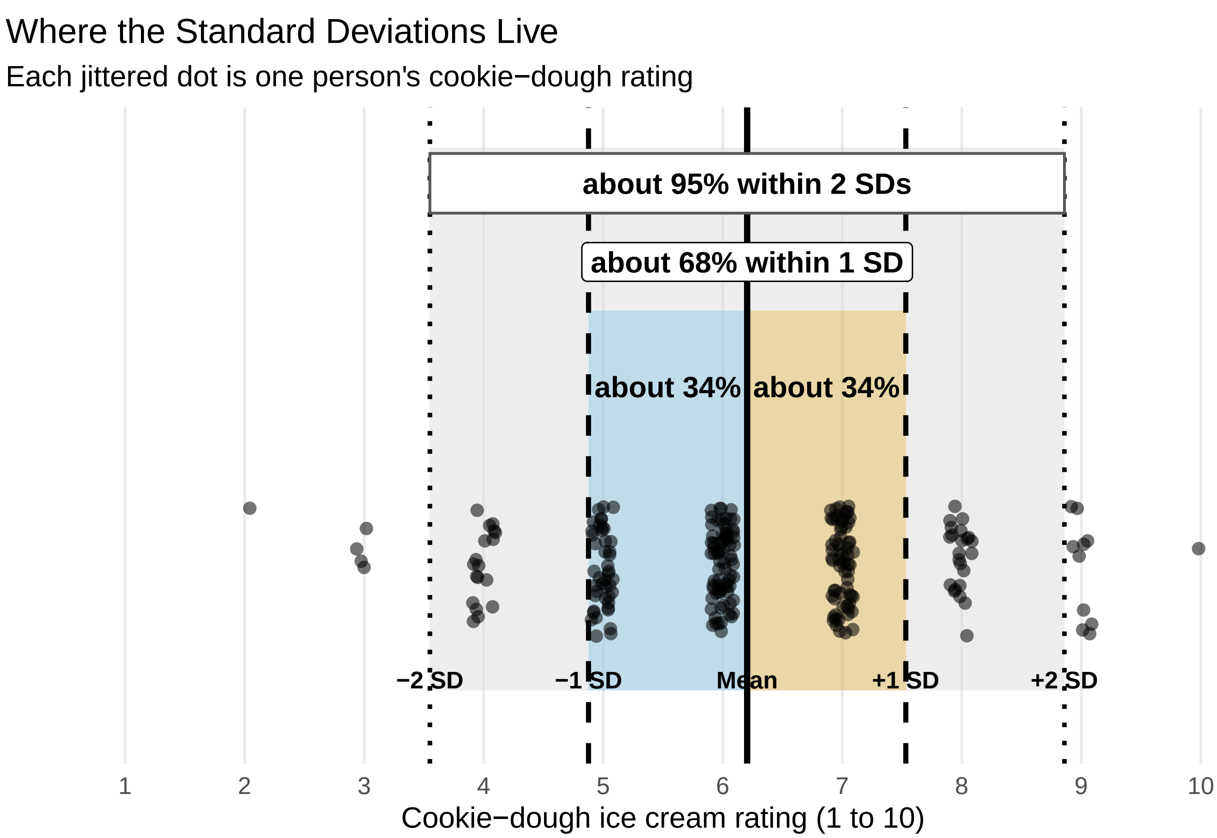


Figure 4.1: Simulated cookie-dough ratings. The dots are people; the lines mark the mean and standard-deviation boundaries. The 34%, 68%, and 95% values describe an approximately normal distribution.

In this particular simulated sample, 76.2% of ratings fall within one SD of the mean and 93.3% fall within two SDs. Those values will not equal exactly 68% and 95%. These are discrete ratings from one sample, not obedient dots hired to impersonate a perfect normal curve. Chance is fickle, and a 1–10 scale also has walls at both ends.

The important visual intuition is this: **one SD takes us from the mean through the thick central part of a normal distribution. Two SDs cover nearly everybody.** A score farther than two SDs from the mean is unusual, although "unusual" does not automatically mean impossible, important, fraudulent, or possessed.

4.3.8 Relationship between M and S

The standard deviation is expressed in the same units as the scores, but it does not automatically grow just because the mean grows. For positive **ratio-scale** variables, the **coefficient of variation** can compare spread relative to the mean:

$$CV = \frac{S}{|M|}$$

```r
CV <- SD.calc / abs(M)
```

$CV = 0.15$.

> **The CV Needs a Mean Worth Dividing By**
>
> A problem occurs when the mean is close to zero. Dividing by a tiny mean can produce an enormous and unstable CV. If the mean is zero, the CV is undefined.
> Therefore, interpret the CV most confidently when zero has a meaningful interpretation, ratios are sensible, and the mean is not hovering near zero while laughing at you.

4.3.9 Moment 3: Skewness

Skewness is how asymmetrical the hill is. Generally, the farther the value is from zero, the more skewed the distribution. Human data will almost always be somewhat asymmetrical, like most of our faces. When people's ears or eyes are slightly asymmetrical, we generally choose to ignore the problem for reasons—assumptions, really—that we hope are true. More on that later.

The skewness formula is below for completeness. Almost nobody reports the formula itself.

$$Skewness = \frac{\Sigma(X - M)^3/N}{S^3}$$

```r
library(moments)
SK.Calc <- skewness(Normal.Sample.1)
```

Skewness $= 0$

4.3.10 Moment 4: Kurtosis

This is peakiness! You can have flattish, longish hills (**platykurtic**) or skinny, tall hills (**leptokurtic**). I would be remiss to my mother if I did not point out that these are Greek words. I would also like to point out that nobody has ever confused my body shape with leptokurtotic.

The kurtosis formula is below for completeness. Not only does almost nobody report it, I do not think many people know how to make sense of the numbers it spews out.

$$Kurtosis = \frac{\Sigma(X - M)^4/N}{S^4} - 3$$

```
K.Calc <- kurtosis(Normal.Sample.1) - 3
```

Kurtosis $= 0$

4.3.11 All Four Moments via Tidyverse

The tidyverse is a collection of R packages that makes data work readable and consistent. Think of it as data operations in near-plain English: take this data, group it, summarize it, done.

We need three verbs for now:

- `group_by()` — split data into groups
- `summarise()` — compute statistics per group
- `filter()` — keep only rows we want

First, put our vector into a **data frame** (R's version of a spreadsheet — rows are observations, columns are variables):

```
DF <- data.frame(X = Normal.Sample.1)
```

Now compute all four moments at once:

```
library(tidyverse)
library(moments)
Desc.table <-
  DF %>%
  summarise(N    = n(),
            M    = mean(X),
            S    = sd(X),
            CV   = S / abs(M),
            Skew = skewness(X),
            Kurt = kurtosis(X) - 3)
round(Desc.table, 2)
```

```
      N      M     S   CV Skew Kurt
1 1e+06 100.01 15.02 0.15    0    0
```

4.4 Distributions and Probability

A distribution isn't just a picture of data — it lets us make **probability statements**.

The height of the curve at any point tells us how likely values near that point are. This is the **Probability Density Function** (PDF).

4.4.1 Probability Density Function (PDF)

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Looks scary. The concept is simple:

- Values near the mean → highest density
- Values far from the mean → lower density
- Shape is fully determined by μ and σ

In practice, we convert to **z-scores** so that mean = 0 and SD = 1, which makes everything easier.

4.4.2 Z-scores

A **z-score** tells you how far a value is from the mean in standard deviation units:

- z = 0 → exactly at the mean
- z = 1 → one SD above the mean
- z = −1 → one SD below the mean

Converting to z-scores does **not change the shape** of the distribution — it just re-labels the axis.

$$Z = \frac{X - M}{S}$$

> **! A z-Score Is Not a Probability**
>
> A z-score tells us how far a score is from the mean in standard-deviation units.
> It does **not** directly tell us the percentage of people below that score. To obtain a percentile, we must use the appropriate distribution—usually with `pnorm()` when normality is a reasonable model.
> A z-score of 1 means one SD above the mean. It does not mean 1%, 10%, or that the participant has won statistics.

4.4.3 Making Z-scores in R

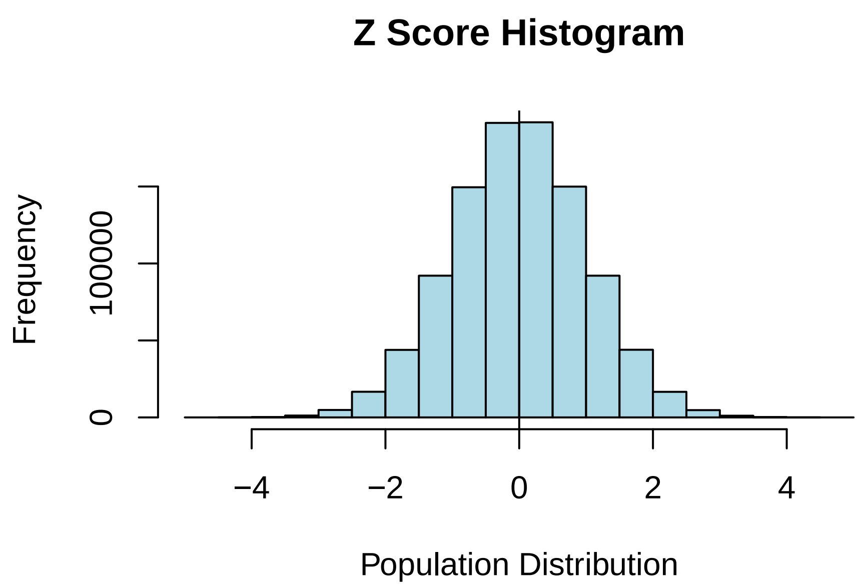
```r
Normal.Sample.Z <- scale(Normal.Sample.1, center = TRUE, scale = TRUE)
```

```r
DF.Z <- data.frame(X = Normal.Sample.Z)
Desc.table.Z <-
  DF.Z %>%
  summarise(N = n(), M = mean(X), S = sd(X))
round(Desc.table.Z, 2)
```

```
       N M S
1 1e+06 0 1
```

The shape doesn't change:

```
hist(Normal.Sample.Z,
  breaks = 30,
  col    = "lightblue",
  border = "black",
  main   = "Z Score Histogram",
  xlab   = "Population Distribution",
  ylab   = "Frequency")
abline(v = 0, col = "black")
```



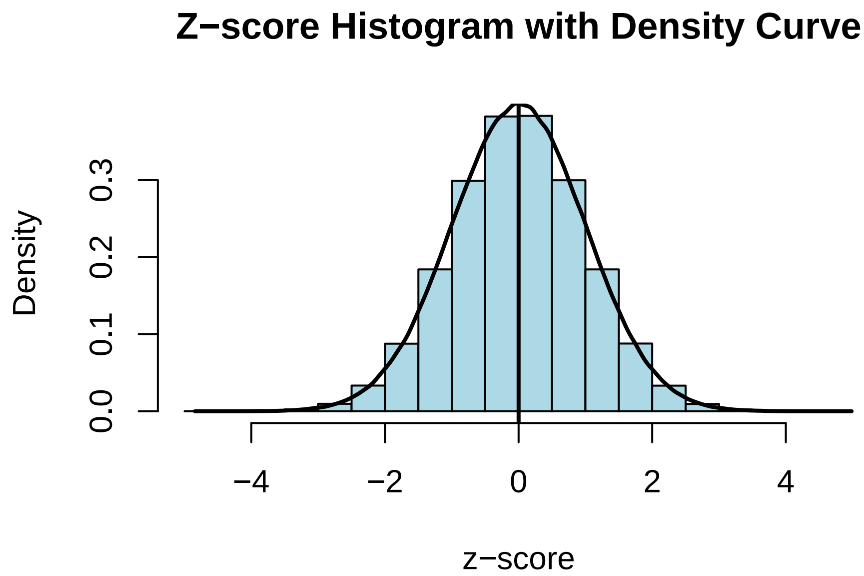
4.4.4 Finding Probability from Z-scores

Two functions to know:

- `dnorm()` → height of the curve (density) at a value
- `pnorm()` → area under the curve to the left of a value (probability)

The PDF `dnorm()` implements:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



4.4.5 How much of the population is below the mean?

Exactly 50% — the normal distribution is symmetric. Confirm it:

```
pnorm(0)
```

```
[1] 0.5
```

What proportion falls between 1 and 2 SDs above the mean?

```
pnorm(2) - pnorm(1)
```

```
[1] 0.1359051
```

No calculus. Just area under the curve.

4.5 Real-World Example: Mental Rotation Test

We now know how to:

- Describe distributions
- Standardize scores with z-scores
- Convert z-scores to probabilities

Let's apply all of it to a real psychological measure.

The **Mental Rotations Test (MRT)** measures spatial visualization ability. Peters et al. developed a redrawn version and examined factors affecting performance (Peters et al. 1995). Here's a summary by field and gender:

	Sciences			Social Sciences & Humanities		
	n	M	SD	n	M	SD
M	817	15.23	4.42	517	12.74	4.87
F	392	11.18	4.37	1358	8.75	4.18

> Advanced Code Has Arrived Early
>
> The code below is more advanced; you do not need to understand it yet. Why am I showing you crazy code now? To make you feel bad about yourself and the terrible life choices you made when you took a statistics class! Actually, I want you to see what R can do. I can go from a table to simulation, graphs, and analysis without a lot of code. Simulating data is my favorite way to make graduate students cry when they propose dissertations because it is the fastest way to discover whether the theory and hypotheses they proposed make any sense. When they do not, the students cry and Dorkaos, our God of Statistics, gains more strength.

4.5.1 Step 1: Simulate the population distributions

We only have summary stats, but because MRT scores are approximately normal we can **simulate each group** using `rnorm()`.

```
library(tidyverse)
set.seed(123)

params <- tribble(
  ~Field,             ~Group, ~n,    ~mean,  ~sd,
  "Sciences",         "M",    817,   15.23,  4.42,
  "Sciences",         "F",    392,   11.18,  4.37,
  "Social Sciences", "M",    517,   12.74,  4.87,
  "Social Sciences", "F",    1358,   8.75,  4.18
)

MRT.2 <- params %>%
  mutate(data = map2(n, mean, ~rnorm(.x, mean = .y, sd = params$sd[params$n == .x]))) %>%
  unnest(data) %>%
  select(Field, Group, Value = data)
```

Check that our simulation matches published values:

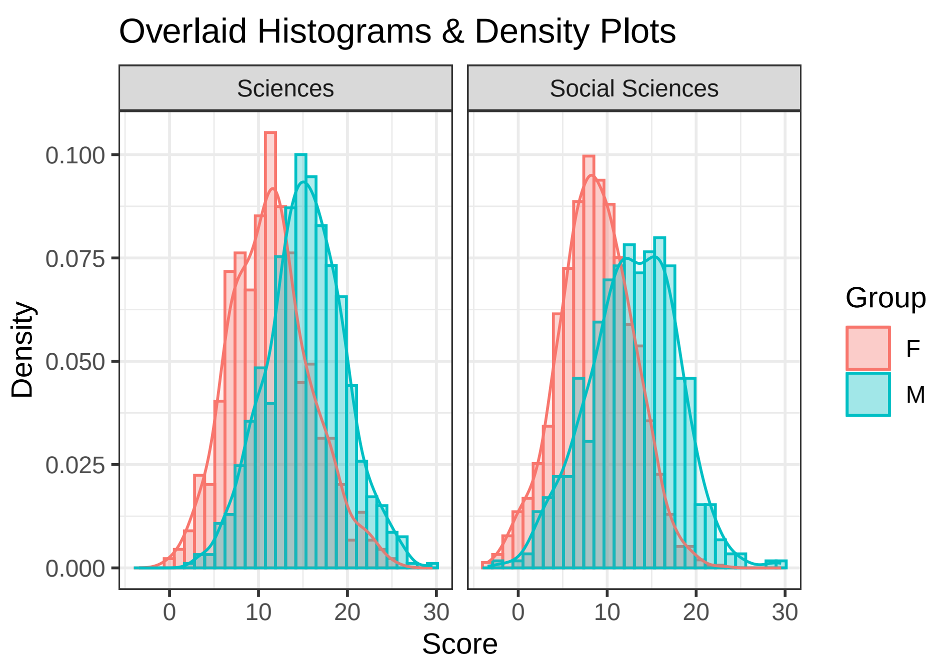
```
MRT.table <-
  MRT.2 %>%
  group_by(Field, Group) %>%
  summarise(N = n(), M = mean(Value), S = sd(Value))
MRT.table
```

```
# A tibble: 4 x 5
# Groups:   Field [2]
  Field           Group     N     M     S
  <chr>           <chr> <int> <dbl> <dbl>
```

```
1 Sciences        F       392 11.4   4.47
2 Sciences        M       817 15.3   4.34
3 Social Sciences F      1358  8.74  4.16
4 Social Sciences M       517 12.9   4.88
```

4.5.2 Step 2: Visualize the distributions

```
ggplot(MRT.2, aes(x = Value, fill = Group, color = Group)) +
  geom_histogram(aes(y = ..density..), position = "identity", alpha = 0.3, bins = 30) +
  geom_density(alpha = 0.1) +
  facet_wrap(~ Field) +
  labs(title = "Overlaid Histograms & Density Plots", x = "Score", y = "Density") +
  theme_bw()
```



4.5.3 Step 3: Convert to z-scores (within groups)

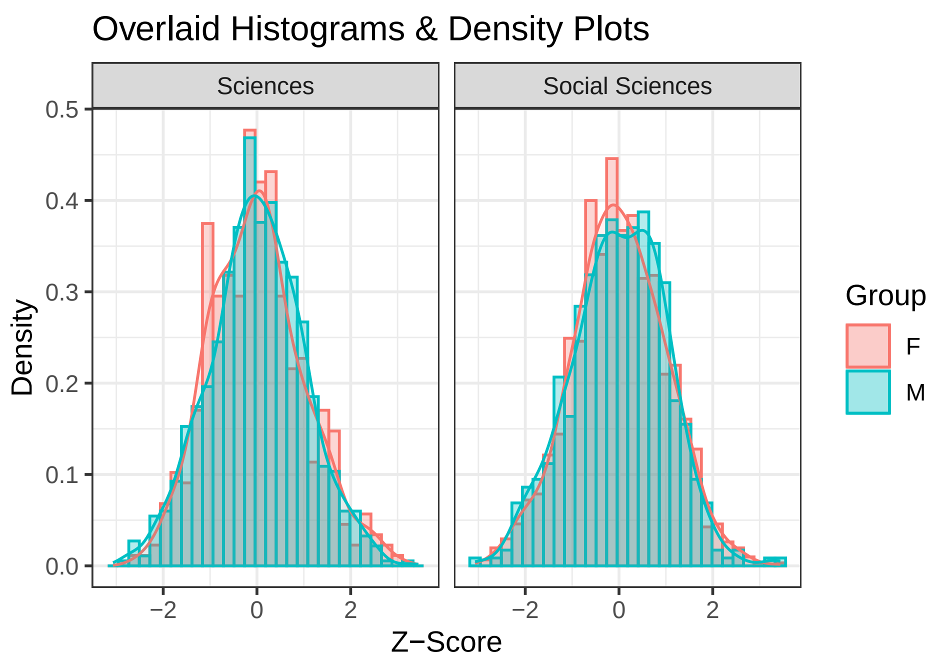
Standardize within each group — critical because each group has a different mean and SD.

```
MRT.2 <- MRT.2 %>%
  group_by(Field, Group) %>%
  mutate(Z = scale(Value))
MRT.2 %>% head()
```

```
# A tibble: 6 x 4
# Groups:   Field, Group [1]
  Field    Group Value   Z[,1]
  <chr>    <chr> <dbl>   <dbl>
1 Sciences M      12.8 -0.583
```

```
2 Sciences M      14.2 -0.247
3 Sciences M      22.1  1.58
4 Sciences M      15.5  0.0594
5 Sciences M      15.8  0.119
6 Sciences M      22.8  1.73
```

```r
ggplot(MRT.2, aes(x = Z, fill = Group, color = Group)) +
  geom_histogram(aes(y = ..density..), position = "identity", alpha = 0.3, bins = 30) +
  geom_density(alpha = 0.1) +
  facet_wrap(~ Field) +
  labs(title = "Overlaid Histograms & Density Plots", x = "Z-Score", y = "Density") +
  theme_bw()
```



Why do the curves overlap now? Because standardizing compares *relative position within each group*, not raw scores.

4.5.4 Step 4: Z-score to probability

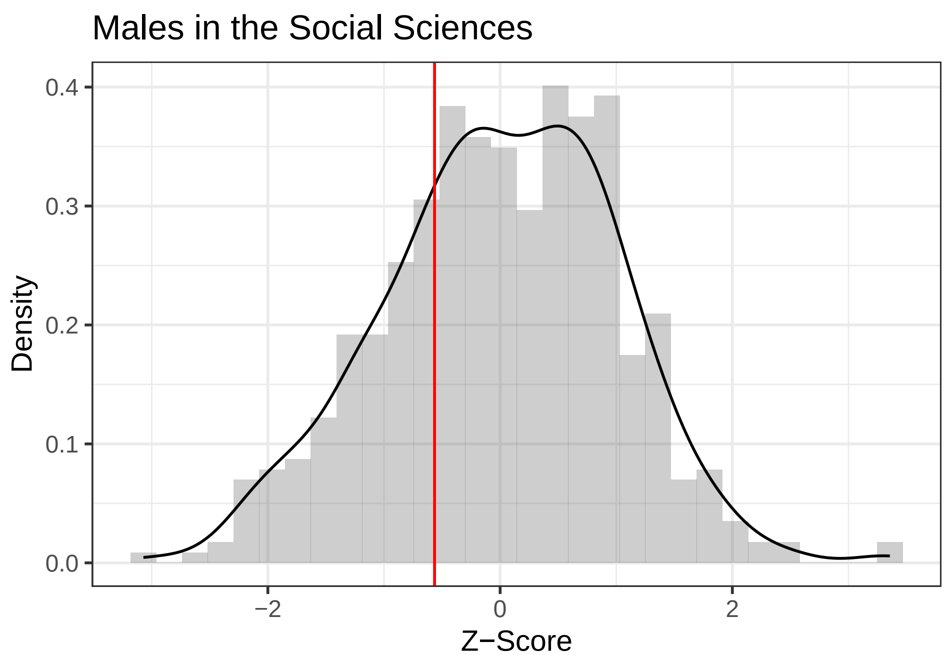
Alex (male, social sciences) got a score of 10. How did he do relative to his peers? (Males, Social Sciences: $M = 12.74$, $S = 4.87$)

```r
Z.Alex <- (10 - 12.75) / 4.87
Z.Alex
```

```
[1] -0.5646817
```

A little over half an SD below the mean. Visually:

```r
MRT.2.Males.Social.Sciences <- MRT.2 %>% filter(Group == "M" & Field == "Social Sciences")

ggplot(MRT.2.Males.Social.Sciences, aes(x = Z)) +
  geom_histogram(aes(y = ..density..), position = "identity", alpha = 0.3, bins = 30) +
  geom_density(alpha = 0.1) +
  labs(title = "Males in the Social Sciences", x = "Z-Score", y = "Density") +
  theme_bw() +
  geom_vline(xintercept = Z.Alex, color = 'red')
```




```r
pnorm(Z.Alex)
```

```
[1] 0.2861451
```

Only better than 28.6% of peers. That's also called a **percentile**.

4.5.5 Step 5: Same raw score, different meaning

Emma also scored a 10, but she's in a different group (Female, Social Sciences: $M = 8.75$, $S = 4.18$).

```r
Z.Emma <- (10 - 8.75) / 4.18
Z.Emma
```

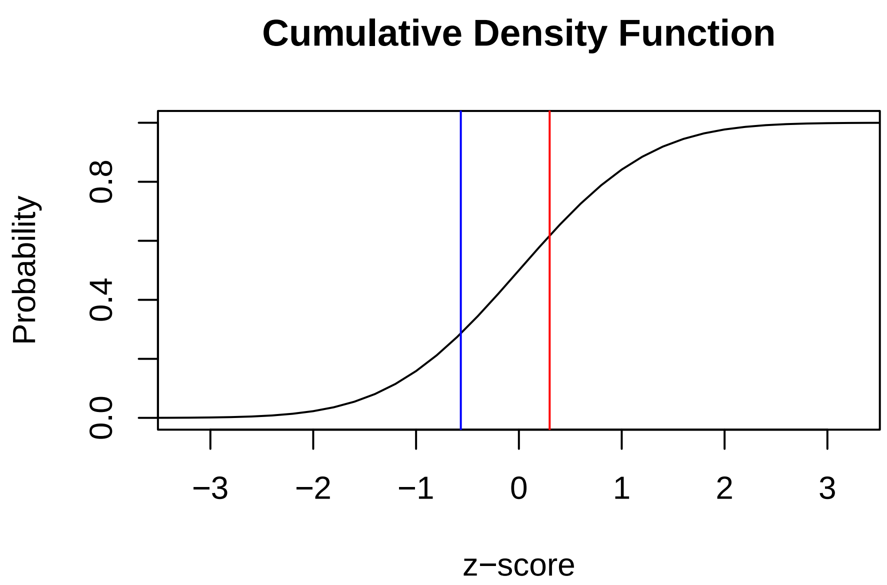
```
[1] 0.2990431
```

```r
pnorm(Z.Emma)
```

```
[1] 0.6175464
```

Emma scored better than 61.75% of her peers.

```
curve(pnorm, from = -10, to = 10,
  main = "Cumulative Density Function",
  xlab = "z-score", ylab = "Probability",
  xlim = c(-3.25, 3.25))
abline(v = Z.Emma, col = "red")
abline(v = Z.Alex, col = "blue")
```



Same raw score. Emma is above average for her group; Alex is below average for his. Raw scores mean nothing without context.

4.5.6 Big idea

- Raw scores mean nothing without context.
- Distributions give scores meaning.

4.6 The Short Story

- A distribution shows which values occur and how often.
- The mean locates the center; variance and standard deviation describe spread.
- Skewness describes asymmetry; kurtosis describes tail weight and peak shape relative to a reference distribution.
- Zero skewness and zero excess kurtosis do not prove normality. Strange distributions are capable of wearing fake mustaches.
- z-scores translate raw scores into standard-deviation units.
- The same raw score can mean very different things in different reference groups.
- Always inspect the distribution before asking one summary number to represent several hundred people without supervision.

5 Standard Error

5.1 Sampling Error and the Limits of Samples

Samples are guesses. Your sample mean will almost never equal the true population mean—and that is fine, as long as you know *how unstable the guess might be.*

Probability is fickle. With a tiny sample, the Probability Gods may hand you ten unusually delighted commuters or ten people whose train just vanished from the tracker. A larger random sample gives chance fewer opportunities to build your entire conclusion out of one weird subway car.

Suppose we randomly select 10 riders from a CTA subway car and ask:

How much do you typically like traveling on the CTA?

- 1 = Strongly dislike
- 2 = Dislike
- 3 = Neither dislike nor like
- 4 = Like
- 5 = Strongly like

Who we grab matters — daily commuters, tourists, and people who just missed their train all give different answers. The gap between our sample estimate and the true population value is called **sampling error**. It's not a mistake — it's just an unavoidable consequence of observing only part of the population.

The question is: *how bad is the error, and how can we estimate it from a single sample?*

5.1.1 Traveling on the CTA

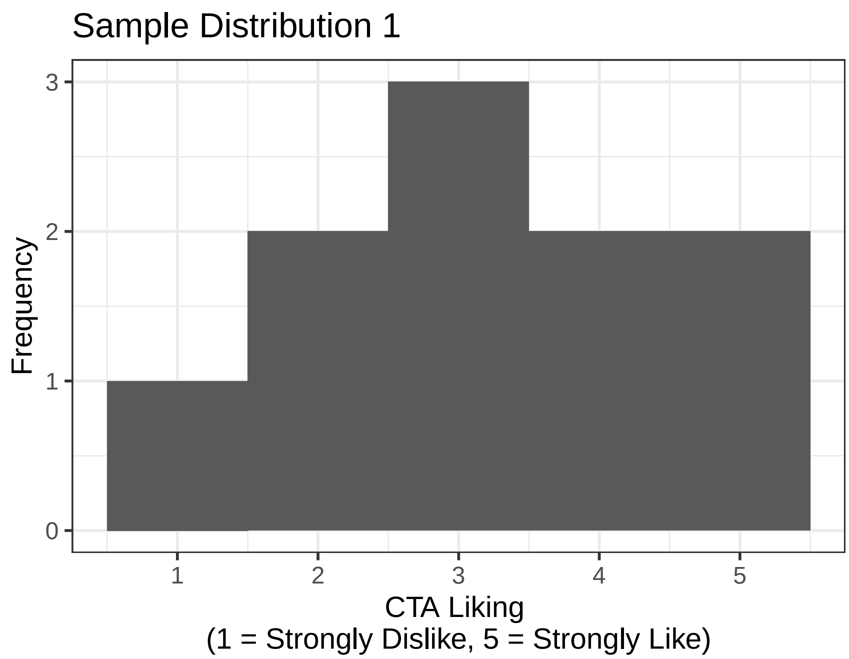
The CTA hires me to do a satisfaction survey on the subway. So I take my question, and I want to find out what the population attitude toward the CTA is.

Well, the CTA handles about 1 million riders a day, but (a) they did not pay me enough to hire all the people I would need to ask all those people, and (b) I do not want to talk to random strangers on the Chicago subway about the CTA. They might punch me in the face as an answer to my question.

So I am hoping that 10 people is enough to answer their question and also lets me run this study cheaply so I can pocket all the money. Have you seen how much groceries cost nowadays?

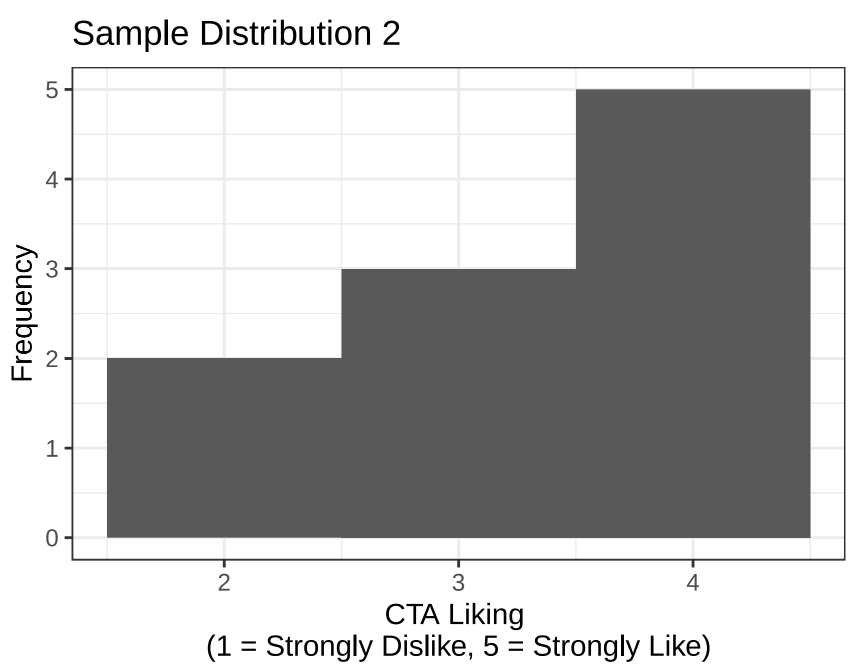
5.1.2 Randomly sample from the population

I enter a random subway car, at a random time, and randomly select 10 people and ask them to answer the question.



The mean of this sample is $M = 3.2$.

Okay, so is that true population mean? Who knows. Ugh, I should probably double check. So I head out again to a new random subway car, at a random time, and randomly select 10 new people and ask them to answer the question. Let's hope no one punches me...

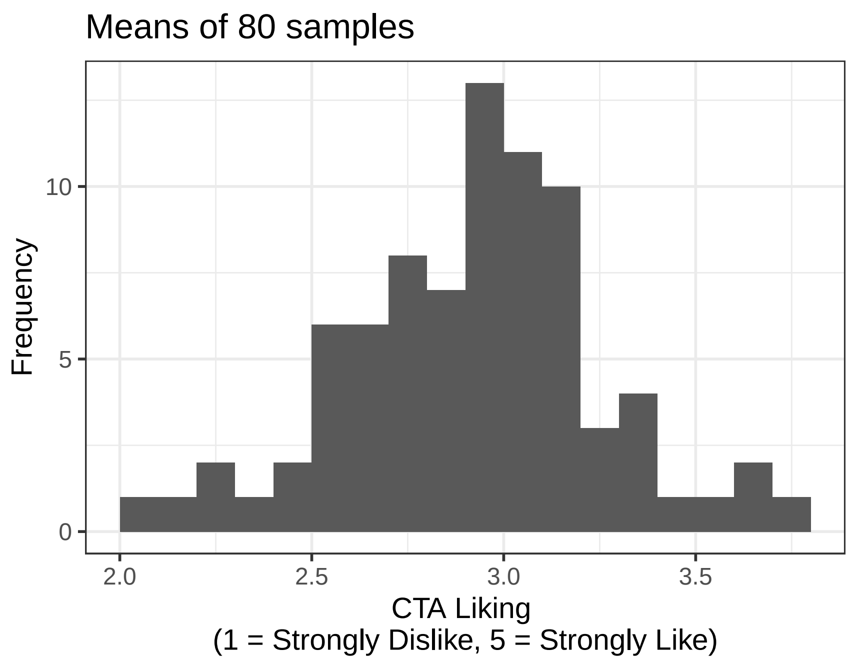


The new mean of this sample is $M = 3.3$.

The mean is higher now. Which sample (1 or 2) is correct?

5.1.3 Brute force solution?

Hmm, I am not sure. So to solve my problem, I will do what all **good** professors do: use coerced undergraduate labor! So I send all undergraduates in this class out to each randomly sample 10 people on different days, at different times, and on different subway lines, and collect the data. We will call it homework...yeah homework...



You will notice the distribution of means is normal-ish and centered around 3, and it was precisely: Mean of means = 2.97

5.1.4 SD of Distribution of Means

But you can also see that there is a wide range of sample means. So what does that mean?

Can I trust any individual sample to tell me the true population value?

Recall what a standard deviation (SD) represents: **the typical distance that observations fall from their mean.**

For a single sample, SD tells us how much individual observations vary around the sample mean.

But now we're doing something slightly different.

Just like we had a standard deviation (S) around a sample mean, we can also calculate the standard deviation of the sample means themselves, that is, how much the means vary from sample to sample.

For a single sample, SD is:

$$S = \sqrt{\frac{\Sigma(X - M)^2}{N - 1}}$$

For the distribution of means, we look at how each sample mean differs from the **grand mean**:

$$S_M = \sqrt{\frac{\Sigma(M - M_{Grand})^2}{N-1}}$$

This quantity is the standard error (SE). Conceptually, it tells us: **how much we expect a sample mean to vary simply due to random sampling**.

> **!** SD and SE Describe Different Crowds
>
> The **standard deviation** describes how individual scores vary around their sample mean.
> The **standard error** describes how sample means would vary across repeated samples.
>
> - SD asks: *How different are the people?*
> - SE asks: *How unstable is our estimate?*
>
> Same units. Different crowd. Confusing them will anger Dorkaos.

The grand mean $M_G = 2.97$

Standard error (spread of the sample means) $= 0.34$

Just like ± 2 SD captures about 95% of individual observations in a roughly normal distribution, ± 2 SE captures about 95% of the sample means in the sampling distribution.

In other words, if we repeatedly sample from this population, about 95% of the sample means should fall within two standard errors of the grand mean.

Thus, the population mean is probably between:2.3 and 3.64

(*Assuming random sampling and a roughly normal sampling distribution—both of which we're getting for free here.*)

> ⚠ Two Is a Shortcut, Not a Sacred Number
>
> Using $M \pm 2SE$ gives a rough 95% interval when the sampling distribution is approximately normal.
> For small samples, the proper confidence interval uses a value from the **t-distribution**:
>
> $$M \pm t^* SE$$
>
> For example, with $N = 10$, the 95% multiplier is about 2.26—not exactly 2. We will learn where that number comes from in the t-test chapter. For now, $\pm 2SE$ is a useful approximation.

5.1.5 The Standard Error of the Mean (SEM)

The Dean found out about my use of forced student labor. Apparently it's unethical or something and told me to solve my problem another way.

So... what if we just make some assumptions about the world and estimate our sampling error from one sample? Because, again, I am lazy and would like to keep all the money for myself.

The strange thing was that the distribution of the means from all those samples looked normal-ish. After a little digging in some old, dusty textbooks, I learned that what I was seeing was not a fluke. In fact, it's a well-known property of repeated sampling called the *Central Limit Theorem*.

Central Limit Theorem (CLT): *If you repeatedly take random samples from a population, the distribution of the sample means will be approximately normal, centered on the population mean, even if the original population is not normal, as long as the sample size is reasonably large.*

> The Raw Data Do Not Become Normal
>
> The Central Limit Theorem applies to the **distribution of sample means**. It does not promise that the individual scores—or the population they came from—will become normal.
> The commuters remain weird. Their means become better behaved.

Because of this, we can assume that if I took repeated samples, the means would be normally distributed around the population mean, and that the amount of error depends on the sample size.

That makes intuitive sense. If I sampled 100 people per sample, there would be less random error. If I only sampled 10 people, there's a much better chance my estimate would be off.

According to this dusty old textbook, we can approximate the standard error of the mean using:

> Two Different Counts Are Hiding in This Example
>
> Our simulation contains:
>
> - **10 people per sample**, which determines how stable each sample mean is.
> - **80 repeated samples**, which lets us see the sampling distribution.
>
> In a real study, we usually collect only one sample. We cannot directly observe its entire sampling distribution, so we estimate the standard error with $S/\sqrt{N}$.

$$S_M = \frac{S}{\sqrt{N}}$$

So, if our first sample has a standard deviation of $S = 1.32$ then our estimated standard error of the mean is:

$S_M = 0.42$

This value is not too far off from the value I estimated earlier by sending out 80 undergraduates to repeatedly run the study: SE from the "actual" sampling distribution $= 0.34$.

In other words: the math mostly agrees with the coerced labor.

5.1.6 Effect of Sample Size

Okay, so what would have happened if I ran my coerced labor study with different sample sizes?

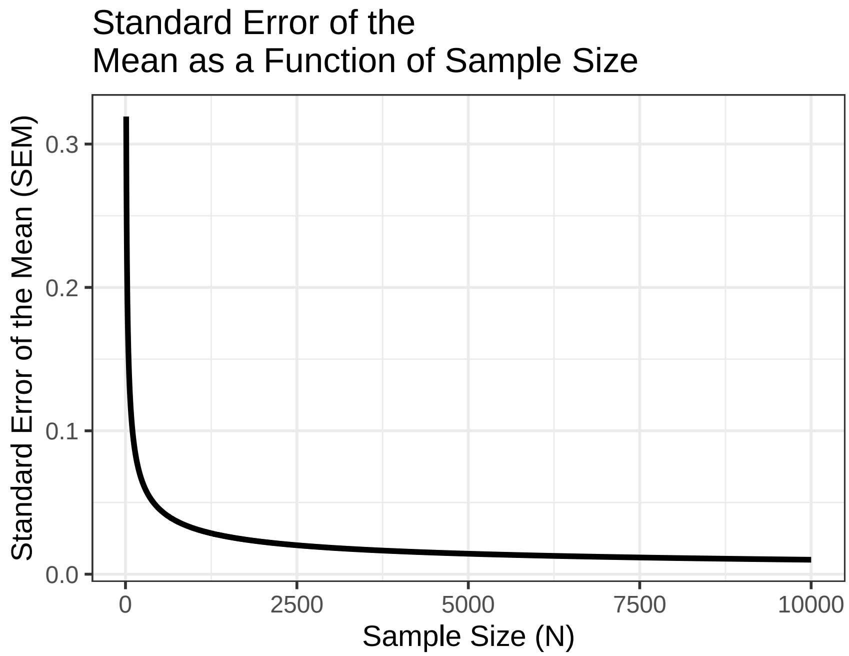
Instead of sending out more undergrads (sadly), I can just use math.

Recall:

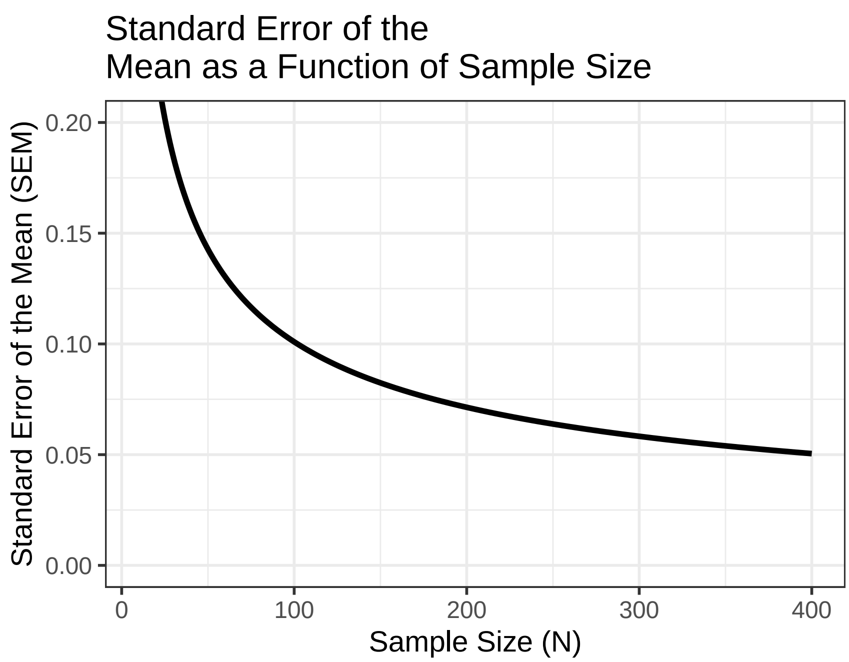
$$S_M = \frac{S}{\sqrt{N}}$$

So as N increases, the SEM must decrease and it does so nonlinearly.

Let's compute the SEM (S_M) for sample sizes from 10 to 10,000 people per sample and visualize it.



So look at that, my error drops really close to zero pretty quickly. So how many people would I need for this study?



In short, there are diminishing returns.

- When the sample size is small, the error is large and unstable.
- As the sample size gets very large, the error continues to shrink—but only a little at a time.

⚠ A Larger Sample Can Produce Very Precise Nonsense

Increasing N reduces random sampling error. It does **not** repair:

- biased sampling;
- bad measurements;
- confounds;
- dishonest responses; or
- collecting every participant from the same suspicious subway car.

A large biased sample can estimate the wrong value with extraordinary precision.

In short: A sample is just a guess about the population. How good that guess is depends on how variable the population is to begin with. If the distribution is tight, you can get away with a smaller sample. If it's spread out, you need a lot more people to pin it down.

You never actually know if your guess is right, which is why replication exists and why we all lose sleep. If you rerun the study and the first two moments (the mean and the variance) keep bouncing around, that's your data politely telling you: get more people.

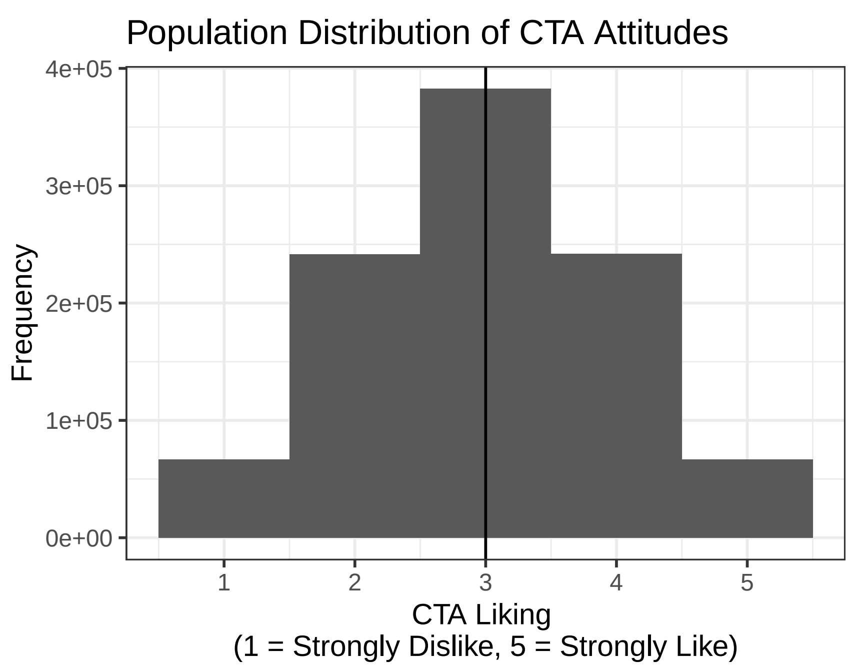
Statistics doesn't give you truth, but you can make increasingly expensive guesses.

5.1.7 What was the Truth?

So... you will never actually know the true population value unless you are either

- (a) an all-knowing God, or
- (b) good friends with a God.

Luckily, if you properly appease *Dorkaos*, the Ancient Greek God of Statistics, knower of all true distributions, with sufficient sacrifices, you may briefly glimpse the truth. Below, all is revealed:



The true mean of this population distribution is $\mu = 3$.

Note: Most people are not on speaking terms with Dorkaos, so in real life you would never truly know this value.

So the real question is: **How close did we come with our original samples?**

From sample 1

- $N = 10$,
- $M_{Sample1} = 3.2$,
- $S_{Sample1} = 1.32$,

- $S_M = 0.42$,
- So our 95% confidence guess (i.e., ±2 SE) would be:
  - 2.37 and 4.03

From sample 2

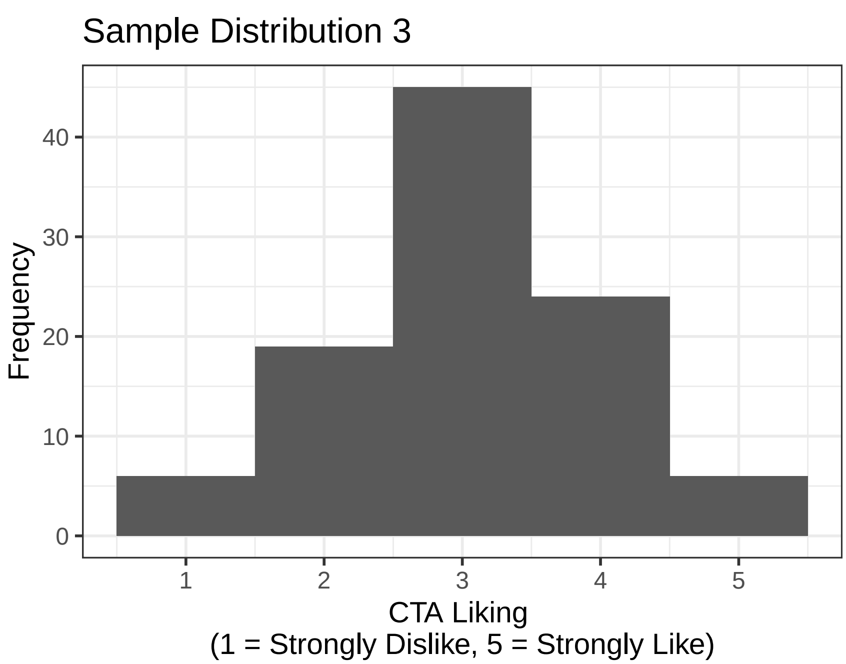
- $N = 10$,
- $M_{Sample2} = 3.3$,
- $S_{Sample2} = 0.82$,

- $S_M = 0.26$,
- So our 95% confidence guess (i.e., ±2 SE) would be:
  - 2.78 and 3.82

Importantly, both intervals include the true population mean. That doesn't mean we were brilliant; it means the intervals were wide. How do we shrink them?

5.1.8 What If We Sampled More People?

What if we ran a third sample with 100 people?



Remember: $S_M = \frac{S}{\sqrt{N}}$

Sample 3:

- $N = 100$,
- $M_{Sample3} = 3.05$,

- $S_{Sample3} = 0.96$,

- $S_M = \frac{.957}{\sqrt{100}} = 0.1$.

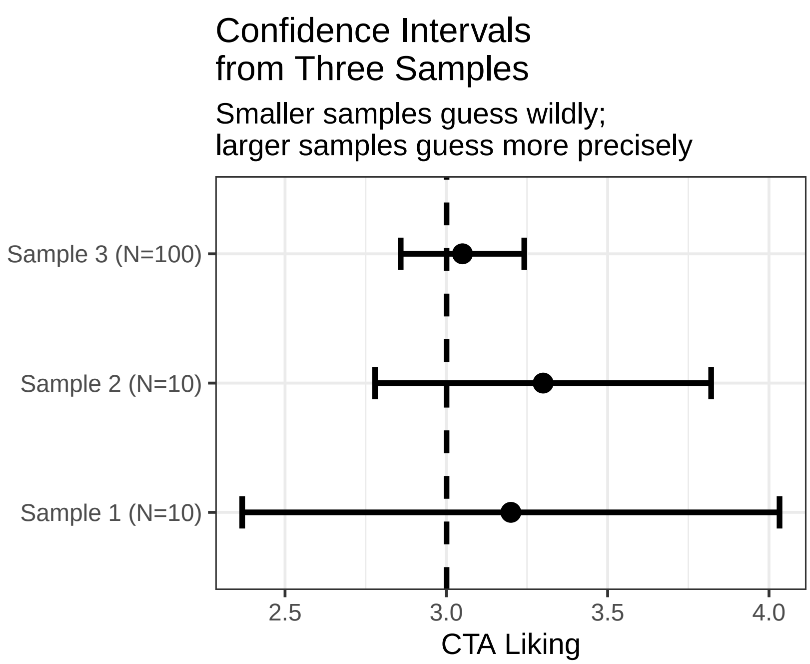
Our 95% confidence guess is now: 2.86 and 3.24

And now you can see the key difference: the interval is much narrower.

5.1.9 Final Report (to the CTA)

So, at last, I can report that the average Chicagoan is roughly neutral toward the CTA, with a true rating that likely falls somewhere between: 2.86 and 3.24.

Still just a guess, but now a much better one as you can see below:



5.1.10 What do we do with this?

Because samples are noisy guesses, and we still need to make decisions.

All of this stuff (SDs, SEs, sampling distributions, confidence intervals) exists for one reason: *to let us draw inferences about a population using imperfect data.*

We rarely get to see the truth. Instead, we:

- take a sample,

- quantify how uncertain that sample is, and
- ask whether what we observed is plausibly just random noise or something more interesting.

> **!** What a 95% Confidence Interval Means
>
> Across many repeated samples, a correctly constructed 95% confidence-interval procedure captures the true population value about 95% of the time, assuming the model is appropriate.
> After we calculate one interval, we do not say there is a 95% probability that the fixed population value is inside it. The procedure has 95% long-run coverage. The parameter is not wandering around trying to escape.

But sometimes we want to ask a sharper question: *Is this sample consistent with a specific claim about the population?*

That question is exactly what t-tests are designed to answer.

5.1.11 The Short Story

- The standard deviation describes variation among observations.
- The standard error describes variation in an estimate across hypothetical samples.
- Increasing sample size usually makes the standard error smaller at a rate of $1/\sqrt{n}$.
- Larger samples make estimates more stable; they do not purify biased samples with statistical holy water.
- Confidence intervals show uncertainty in the estimate, not a force field around truth.

6 One Sample t-Test

6.1 Experimental Logic

In psychology, we test claims by making comparisons.

Sometimes we compare a group to a known value: *Do more frequent train rides increase CTA liking above the neutral baseline of 3?* → one-sample t-test.

Sometimes we compare two groups: *Does CTA liking differ between people in quiet vs. regular subway cars?* → independent samples t-test.

Either way, the logic is the same: **are the differences we observe larger than we'd expect from random sampling noise alone?**

6.1.1 Logic behind testing: The Blind Men and the Elephant

A group of blind men were traveling together when they encountered a strange object blocking their path. Each man touched a different part of the object and described what he thought it was, based entirely on his limited experience.

One man, feeling the elephant's trunk, exclaimed, *"This must be a thick branch of a tree!"* Another, grasping the elephant's leg, disagreed: *"No, it's clearly a pillar!"* A third, holding the elephant's ear, insisted, *"You're both wrong—it's obviously a fan!"*

Each man was convinced he was right, believing his own perspective represented the full truth. Their disagreement quickly escalated into an argument, with each trying to prove the others wrong. Eventually, the elephant, thoroughly unimpressed, grew irritated by their foolishness and tossed them all into a nearby river for disturbing its peace.

— **A paraphrased ancient Indian parable**

This story is a useful reminder when trying to understand the world, and especially when building theories from data. Our perspectives are almost always limited, shaped by where we stand, what we've experienced, and what we want to believe. Just as each blind man grasped only one part of the elephant, our samples and hypotheses capture only small pieces of a much larger reality.

Statistical testing exists precisely because of this limitation: *it gives us a disciplined way to ask whether our small, noisy glimpses are consistent with the bigger picture—or whether we might just be arguing over trunks, legs, and ears.*

6.1.2 Experimental Example

- *Step 1*: Begin with a theory.
- *Step 2*: Develop a hypothesis derived logically from your theory.
- *Step 3*: Design a method to test your hypothesis, including measurements, sample recruitment, and analytical tools.
- *Step 4*: Collect data and compare it with your prediction.

When Step 4 fails, meaning the data don't match the prediction, we face a classic scientific dilemma. It is often difficult to determine why things went wrong:

- Was the theory flawed?
- Was the hypothesis poorly specified?
- Or were the methods inadequate?

This problem is well known in psychology and has been discussed extensively (Meehl 1978).

6.1.3 Theory:

Reinforcing good behavior in children makes them happy because it signals that they have pleased the adults around them. However, rewarding children with something they perceive as a forbidden treat, such as cookies, may have unintended consequences. Children know that cookies are usually off-limits and therefore associate them with rule-breaking. Receiving such a reward may overexcite them, pushing their self-control beyond its limits and leading to hyperactive behavior as they struggle to regulate their impulses.

6.1.4 Hypothesis:

Based on this theory, we hypothesize that giving children cookies will cause them to lose control of their impulses and become hyperactive.

To test this hypothesis, we design a simple study in which children are rewarded with chocolate chip cookies and their behavior is observed.

6.1.5 Important Note:

Regardless of the study's outcome, it would not rule out other plausible explanations. For example, one alternative account is that sugar in cookies is rapidly metabolized, producing a temporary spike in blood glucose that leads to increased activity levels.

This is the core difficulty of experimental inference: a single result rarely tells us which explanation is correct—it only tells us which explanations are still plausible.

6.1.6 Logic of Hypothesis Test

If X is true, then Y should also be true.

- We assume X is true.
- Therefore, we *expect* to observe Y.

However:

- If we observe Y, this does not necessarily confirm that X is true, because Y could result from other factors.
- If we do not observe Y, especially under rigorous and well-controlled conditions, this suggests that our theory may be flawed or incomplete.

Example:

- Theory: Reinforcing good behavior in children with treats leads to increased hyperactivity because children perceive the treat as a forbidden reward, triggering an excited and poorly controlled response
- Hypothesis: Giving children chocolate chip cookies as a reward will result in increased hyperactivity compared to children who do not receive such rewards.
- Testing the Hypothesis:
  - If children who receive cookies exhibit increased hyperactivity (Y is observed), this outcome is consistent with the theory but does not conclusively prove it, since alternative explanations (e.g., sugar content) could also account for the behavior.
    - If children do not exhibit increased hyperactivity (Y is not observed), this challenges the theory, suggesting it may be incorrect or incomplete. At that point, we must re-examine the theory, the methods, and consider alternative explanations.

In hypothesis testing, **falsification** plays a central role. Supporting evidence increases our confidence in a theory, but evidence that contradicts our predictions undermines it. Scientific progress depends on continuously testing and attempting to falsify our theories, refining them through rigorous scrutiny and new evidence.

4 steps for hypothesis testing:

- State the hypothesis
- Set the Criteria Decision
- Collect Data and Compute Sample Statistics
- Make a Decision

6.1.7 State the Hypothesis

6.1.8 The Null Hypothesis (H_0)

- The null hypothesis (H_0) asserts that in the general population, there is no change, no difference, or no relationship.
  - In an experiment, H_0 predicts that the independent variable (e.g., treatment) will have no effect on the dependent variable for the population.
- H_0 is the hypothesis that is subject to a final decision: to either *reject* or *fail to reject*.

6.1.9 The Scientific or Alternative Hypothesis (H_1)

- The alternative hypothesis (H_1) suggests that there is a change or a relationship in the general population.
  - In an experiment, H_1 predicts that the independent variable (e.g., treatment) will influence the dependent variable for the population.

6.1.10 Sample Experiment

6.1.11 Directional Hypothesis

- H_0: Chocolate chip cookies do not make children hyper.
- H_1: Chocolate chip cookies do make children hyper.

6.1.12 Non-directional Hypothesis

- H_0: Chocolate chip cookies do not change children's level of hyperactivity.
- H_1: Chocolate chip cookies do change children's level of hyperactivity.

Note: The default in psychology is to choose the non-directional test, so we will only talk about that version for the rest of below

6.1.13 Set the Decision Criteria

- The decision criteria depend on the type of test, the number of tests conducted, the sample size you have, and whether the hypothesis is directional or non-directional.
- A key decision is determining how far sample means need to be from the population mean to reject the null hypothesis (H_0).
- The challenge is identifying what constitutes abnormal behavior in the population.
  - Should we consider deviations in both tails, or just one tail (either lower or higher)?
  - Where does the tail begin, and the body of the distribution end?
- To address this, we set a cut-off point, known as the Alpha level (α).
  - Common Alpha levels are: $\alpha = .05$, $\alpha = .01$, $\alpha = .005$, $\alpha = .001$, $\alpha = .0001$.
- The Alpha level represents the threshold for what we consider "abnormal"—specifically, the area under the curve beyond this cut-off.

6.1.14 Accounting for Sample Size: The Transition to the t-Distribution

We also have to account for the sample size of our study, since in small samples, we know we will not get the best estimate of the population numbers (as we saw above). So, we apply a correction by moving from the standard Normal (z) distribution to the Student's t-distribution.

6.1.15 Why the t-Distribution?

- Estimation of Variance: we will never know the population standard deviation (σ). so we must estimate it using the sample standard deviation (S).
- The Penalty for Small Samples: Small samples are more prone to extreme fluctuations. Because our estimate of the population variance is less reliable when n is small, the t-distribution is "heavier" in the tails than the normal distribution. Degrees of Freedom (df): The shape of the t-distribution is determined by the degrees of freedom, calculated as $df = n - 1$.
  - As n (and thus df) increases, the t-distribution approaches the shape of the standard normal distribution.
  - As n decreases, the tails become "fatter," requiring a larger critical value to reach the same alpha level (α).

6.1.16 Adjusting the Critical Region

Because the t-distribution has more area in its tails, the "cut-off" points for our Alpha level move further away from the mean. This means: - It is harder to reject the null hypothesis with a small sample. - We are being mathematically conservative to account for the increased risk of error when we don't have a large amount of data.

The t-distribution acts as a "safety buffer." It ensures that we only label a result as "statistically significant" if the effect is strong enough to overcome the inherent noise of a small sample size.

6.1.17 Determining the Cut-Off Points for Two-Tailed Tests

For a two-tailed test with Alpha = 0.05, the cut-offs depend on sample size. For our study with 36 kids that would look like this:

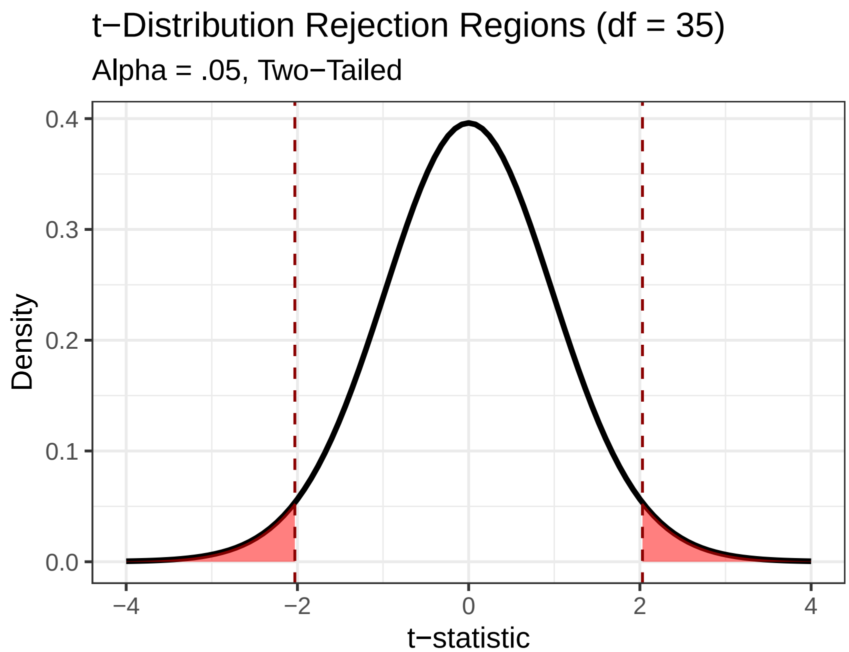


Figure 6.1: The t-distribution for df = 35. The shaded red areas represent the critical regions where we reject the null hypothesis.

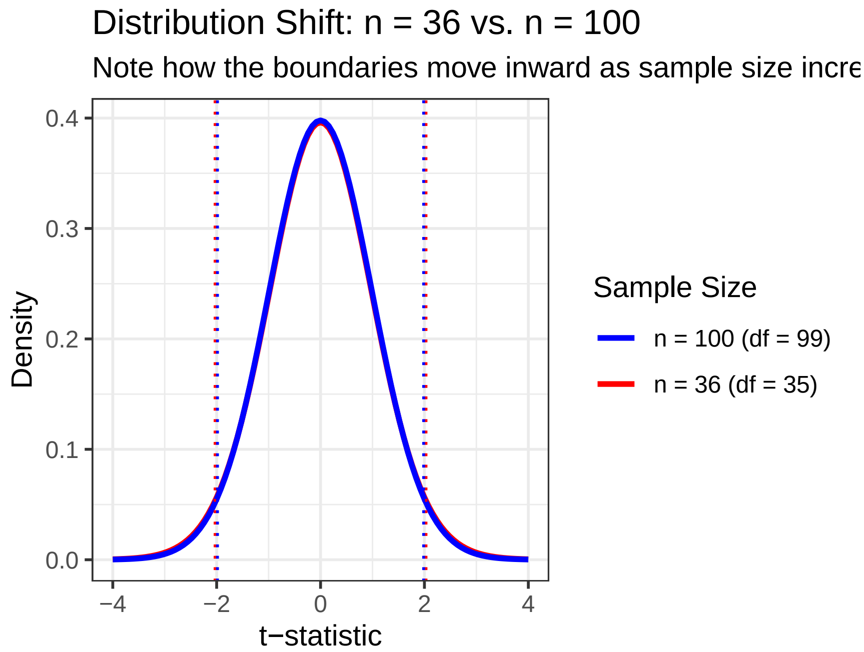
Our cut-off ($t_{critical}$) values for $df = 35$ would be:

- t_{lower} < -2.03
- t_{upper} > 2.03

6.1.18 The Impact of Sample Size on Critical Regions

If we were to increase our sample size from $n = 36$ to $n = 100$, our degrees of freedom would increase to $df = 99$.

As n increases, our confidence in our estimate of the population standard deviation also increases. Mathematically, the t-distribution begins to pull its “fat tails” inward, looking more and more like the standard Normal (z) distribution.



New Cut-Off ($t_{critical}$) Points ($n = 100$):

- $t_{lower} <$ -1.984
- $t_{upper} >$ 1.984

Why the boundaries are smaller: Reduced Uncertainty: With 100 children instead of 36, our sample mean is a much more reliable "anchor" for the population mean. We don't need as much of a "safety buffer.

Here is an example of our the boundaries shift with different sample sizes:

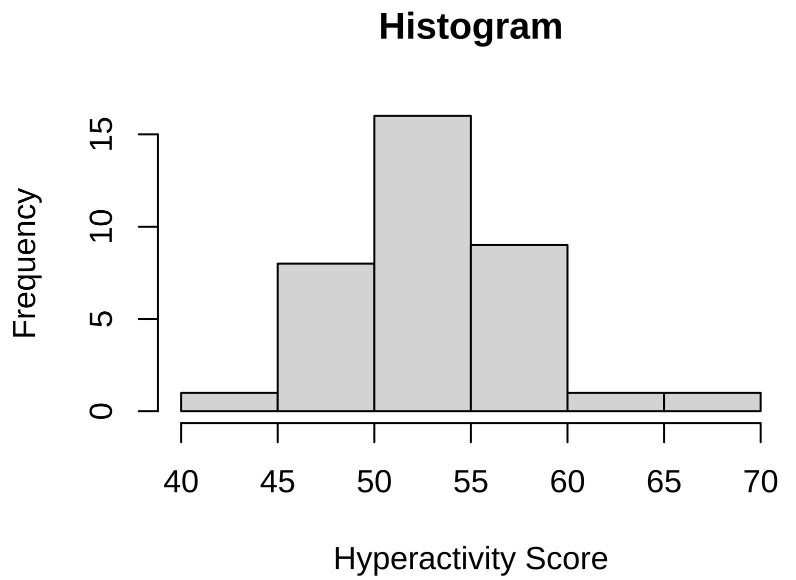
Table 6.1: How Sample Size Changes the Threshold for Significance

Sample Size (n)	Degrees of Freedom (df)	Absolute Critical Value
10	9	2.262
36	35	2.030
100	99	1.984
Infinity (z)	N/A	1.960

6.1.19 Collect Data and Compute Sample Statistics

- Get a sample of children, $n = 36$, and feed them a bunch of cookies! We collected that data and in R created their means in a object called `kids.1`
- **Dependent Variable (DV):** We will measure the children's hyperactivity, which we expect to have a random but generally positive effect on them.

Here is plot for kids.1:



6.1.20 Sample Results

The children who received cookies showed a mean of $M = 53.035$ with a standard deviation of $SD = 4.356$. Based on prior theory and past research, we hypothesize that the population mean is $\mu = 50$

We can then ask whether the data we observed are consistent with that claim, or whether the difference between the sample mean and the hypothesized population mean is larger than we would expect from random sampling error alone.

6.1.21 Compute Sample Statistics

The test statistic is computed using the following formula:

$$t_{test} = \frac{M - \mu}{S_M}$$

where the denominator is the standard error of the mean:

$$S_M = \frac{s}{\sqrt{n}}$$

We can do the math by hand, or use the t-test function in R. t-test by hand:

```
M = mean(kids.1)
s = sd(kids.1)
n = 36
Mu = 50 # given to you from the population
t.kids <- (M - Mu) / (s/sqrt(n))
t.kids
```

```
[1] 4.181321
```

one sample t-test in R:

```
Kids.t.test<-t.test(x = kids.1,
       alternative = c("two.sided"),
       mu = 50,
       var.equal = FALSE,
       conf.level = 0.95)
```

6.1.22 Test Statistic Result

We get a $t_{test} = 4.181$. Now, the question is: **does this value fall in the tail of the population distribution**?

Let's check our critical value.

Our $t_{critical}$ values for $df = 35$ would be:

- $t_{lower} <$ -2.03
- $t_{upper} >$ 2.03

This t-test value is greater than our upper value.

Further from the R t-test function, we can see our exact p-value which is well below $p < .05$.

We can also see 95 percent confidence interval from our R t-test function and that range does NOT include $\mu = 50$, which is another way of telling our results is greater than chance.

6.1.23 Make a Decision

There are two possible outcomes:

- **Reject the null hypothesis:** This decision is reached when the data fall within the critical region.
  - This indicates that the treatment has *likely* caused a difference between the population and the sample.
  - **We reject the null** because it is generally easier to demonstrate that something is false than to prove it true (See Bayesian Statistics).
- **Fail to reject the null hypothesis:** If the data do not provide strong evidence (i.e., they do not fall within the critical region), then the treatment is *assumed* to have no effect.

6.1.24 Decision Outcome

YES, our $t_{test} > t_{critical}$, so we REJECT the null hypothesis.

- We report: $t_{test} = 4.181$, $p < .001$ (since we know the exact p-value).
- In modern APA format, it is recommended to report the exact p-value.

6.1.25 One-sample t-test in APA format

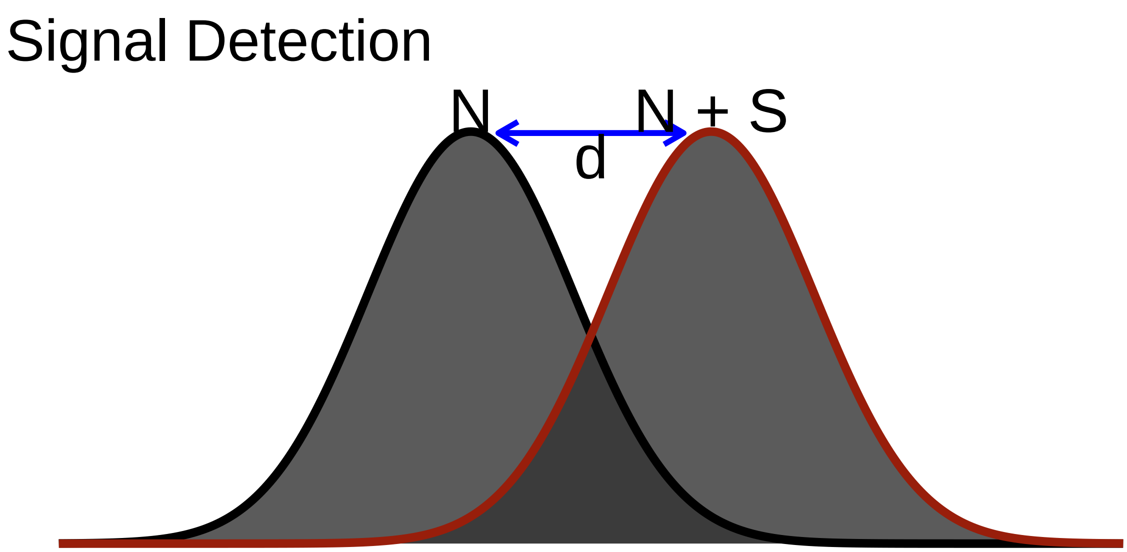
We can use the APA package to report it in APA format

```
library(apa)
#apa(Kids.t.test)
```

Wait what is the d thing at the end? The problem with significance testing is it does not tell you how "big" the effect is. It just tells you that is not probably not due the chance.

6.2 Effect Sizes

I need a way to see how "big" an effect might be. Cohen borrowed a useful idea at the time, signal detection theory, which was created during the invention of the radio and radar. The idea was that you want to know how strong a signal needed to be for you to detect it over the noise. The noise for us is the control group and the signal the effect of your treatment.



$$d = \frac{\mu_{(Noise+Signal)} - \mu_{(Noise)}}{\sigma_{(Noise)}}$$

Note this implicit assumption:

$$\sigma_{(Noise)} = \sigma_{(Noise+Signal)}$$

6.2.1 Cohen's d

Cohen surmised that we could apply the same logic to experiments to determine how big the effect would be (in other words how easy it is to distinguish the signal from the noise).

$$d = \frac{M_{H1} - \mu_{H0}}{S_{H1}}$$

So Cohen's d is in standard deviation units:

Size	d
Small	.2
Medium	.5
Large	.8
Huge	1.2

6.2.2 Small Visualized

6.2.3 Medium Visualized

6.2.4 Large Visualized

6.2.5 Huge Visualized

You notice that even in a huge effect there is a lot of overlap between the curves.

6.3 APA Style Report

Children who received cookies showed higher hyperactivity scores ($M = 53.04$, $SD = 4.36$) than the hypothesized population value ($\mu = 50$), $t(35) = 4.18$, $p < .001$, 95% CI [51.56, 54.51], $d = 0.7$. The standardized difference describes magnitude; whether the effect is practically meaningful requires context and better evidence than our imaginary cookie operation.

6.4 Uncertainty and Error in Hypothesis Testing

- Hypothesis testing is an inferential process because we do **not** know the actual population data, and our samples could be in error.
- When we infer, we can make at least two types of errors (though there are others).

6.4.1 Type I Error

- Also called a "false alarm" or "alpha error."
- This occurs when a researcher rejects a null hypothesis that is actually **true**. In other words, we reject the null even though the treatment does **nothing**.
- **Example:** Testing zinc or ColdEase on colds using self-report as a measure of improvement.
- Before observing the data, the α level sets the long-run Type I error rate for the testing procedure when the null hypothesis and its assumptions hold.
- Choosing $\alpha = .05$ does not mean there is a 5% probability that this particular rejected null hypothesis is true. The p-value cannot read the hidden state of the universe.

6.4.2 Type II Error

- Also called a "miss" or "beta error."
- A Type II error occurs when a researcher fails to reject the null hypothesis that is actually **false**.
- In other words, a Type II error occurs when the treatment has an effect, but it seems like the treatment has had no effect because the hypothesis test fails to detect it.

- This can occur when the treatment has an effect but the study has too little power to detect it—for example, because the effect is modest, the measurement is noisy, or the sample is small.
- **Example:** Taking kava tea to reduce stress.

Your decision	Treatment does not work (H_0=TRUE)	Treatment does work (H_0=FALSE)
Reject H_0	Type 1	Correct Decision
Fail to reject H_0	Correct Decision	Type II

6.4.3 Does our Alpha Set the Error/Miss Level?

- The answer is: it depends. We will revisit this issue when we discuss *statistical power*.

6.4.4 The Short Story

- A one-sample t-test compares a sample mean with a hypothesized population value.
- The t statistic is the observed difference divided by its standard error.
- A p-value measures compatibility with the null model; it is not the probability that the null is true.
- "Fail to reject" does not mean "prove no effect." Sometimes the study simply arrived carrying a tiny sample and no flashlight.
- Report the estimate, confidence interval, p-value, and effect size. The asterisk cannot do all the talking.

! The t Statistic Is Signal Divided by Noise

The numerator is how far the sample mean sits from the null value. The denominator is the standard error of that mean. A large t statistic can come from a large difference, a precise estimate, or both—so report the effect and its uncertainty, not merely the ceremonial p-value.

i Which t-Test Comes Next?

Use the one-sample t-test when one group is being compared with one number. If you want to ask whether **Group A differs from Group B**, continue to the independent-samples t-test chapter. For example, you might compare the running speed of one group of undergraduates chased by a goose with a completely different group chased by two geese. The ethics board will compare your application with a paper shredder.
If you measure the **same people twice**, continue to the paired-samples t-test chapter. For example, you might measure the same students' stress before and after telling them that the group presentation they forgot about has been moved to today. Same humans, two measurements, one sudden desire to transfer universities.

7 Independent Samples t-Test

7.1 The Research Question

What would happen if the CTA actually ran every 7 minutes?

Would people like using it more?

7.1.1 Design

Two independent groups:

- **Control** = Regular CTA timing ("scheduled" is a flexible concept)
- **Experimental** = Trains arrive every 7 minutes. Like clockwork. Almost suspiciously efficient.

7.1.2 Dependent Variable

A **1-7 bipolar Likert scale** measuring how much participants like using the CTA to get around Chicago:

- 1 = Absolutely hate using the CTA
- 4 = Neutral / It's fine
- 7 = Love it, I now defend the CTA on the internet

We expect:

- Control mean is approximately **3** (mild frustration, classic Chicago baseline)
- Experimental mean is approximately **5** ("Wait... this works?")

Now we walk through how to test this.

7.2 Step 1: State the Hypotheses

$$H_0 : \mu_1 = \mu_2$$

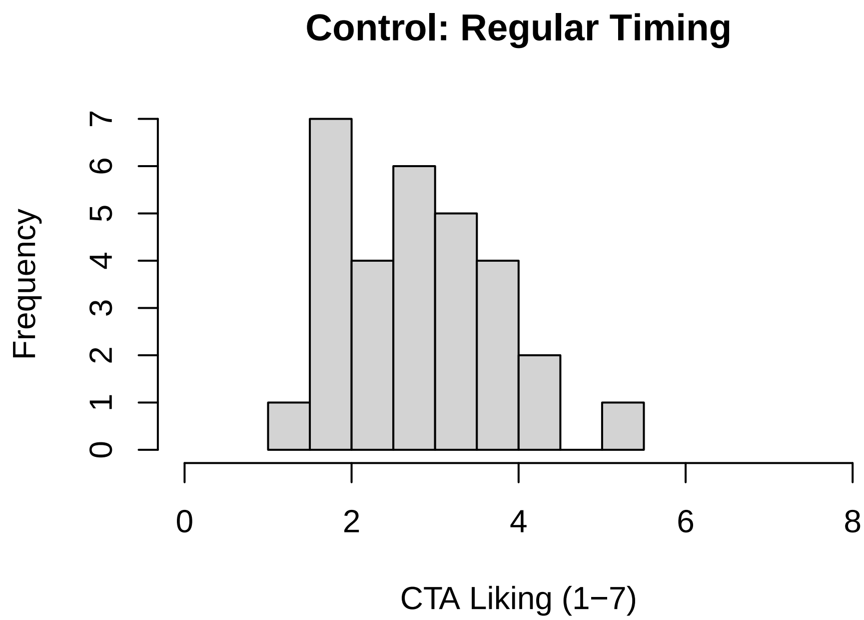
$$H_1 : \mu_1 \neq \mu_2$$

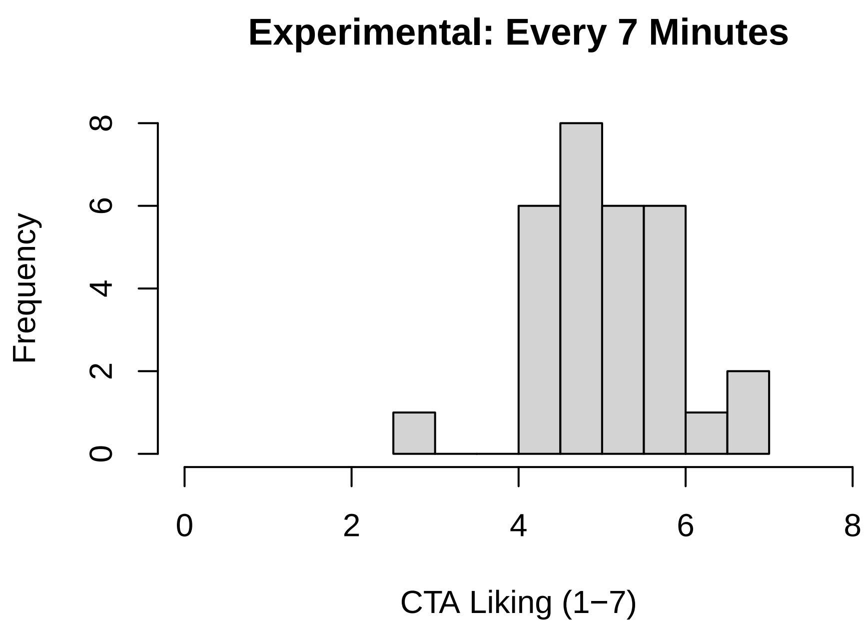
Where:

- μ_1 = Experimental (7-minute trains)
- μ_2 = Control (regular timing)

Under the null, $\mu_1 - \mu_2 = 0$.

7.3 Step 2: collect the Data





Already you should see separation. But eyeballing histograms is not a statistical test (even if it feels persuasive).

7.4 Step 3: Compute the Standard Error

For **Welch's t-test**, which R uses by default, the standard error is:

$$S_{M_1-M_2} = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

Welch's test does not assume equal population variances. It uses a Welch–Satterthwaite approximation for the degrees of freedom, which R will calculate because we have suffered enough.

If we instead use the classical pooled-variance version, the standard error is:

$$S_{M_1-M_2} = \sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

This represents how much variability we expect in the **difference between means** due to sampling error. The pooled version combines the two sample variances:

$$S_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2}$$

7.5 Step 4: Compute the t Statistic

$$t = \frac{M_1 - M_2 - (\mu_1 - \mu_2)}{S_{M1-M2}}$$

For the pooled-variance test, the degrees of freedom are:

$$df = (n_1 - 1) + (n_2 - 1)$$

Now let R do the arithmetic. R uses **Welch's t-test** by default because it does not require the two populations to have equal variances. That is usually the safer choice. The pooled formula above is still useful because it makes the basic machinery visible, but we should not force equal variances merely because an ancient statistics ritual told us to.

```
CTA.t.test <- t.test(x = CTA.Experimental,
                     y = CTA.Control,
                     alternative = "two.sided",
                     conf.level = 0.95)

CTA.t.test
```

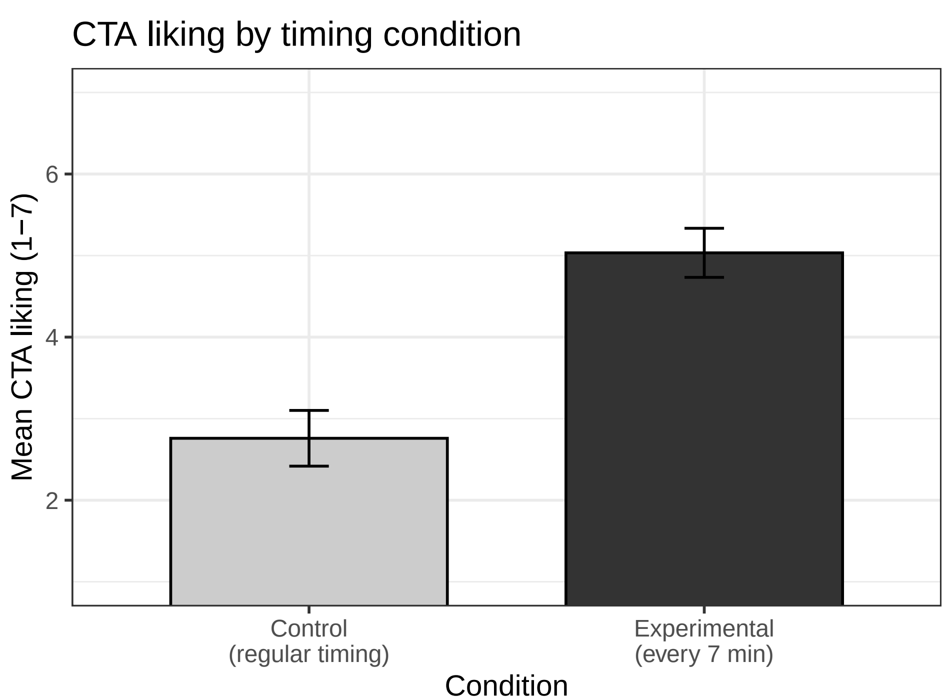
```

	Welch Two Sample t-test

data:  CTA.Experimental and CTA.Control
t = 10.213, df = 57.093, p-value = 1.693e-14
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 1.828420 2.720214
```

```
sample estimates:
mean of x mean of y
 5.033512  2.759195
```

If trains truly arrive every 7 minutes, we should see a statistically significant increase in liking.

7.6 Step 5: Visual Reporting - Means + 95% CI

```r
library(ggpubr)

ggbarplot(
  CTA.df,
  x = "Condition",
  y = "Liking",
  add = "mean_ci",        # mean + 95% CI
  fill = "Condition",
  palette = "grey",
  ylim = c(1, 7),
  xlab = NULL,
  ylab = "Mean CTA liking (1-7)",
  title = "CTA liking by timing condition"
) +
  theme_bw(base_size = 11) +
  theme(legend.position = "none")
```



7.7 Step 6: Extract the APA Values

Formatting packages can be convenient, but some do not support Welch's t-test. We can extract the values directly and avoid letting a package bully us back into an equal-variance assumption.

```
apa_values <- data.frame(
  t = unname(CTA.t.test$statistic),
  df = unname(CTA.t.test$parameter),
  p = CTA.t.test$p.value,
  CI_low = CTA.t.test$conf.int[1],
  CI_high = CTA.t.test$conf.int[2]
)

round(apa_values, 3)
```

```
       t     df p CI_low CI_high
1 10.213 57.093 0  1.828    2.72
```

7.8 Step 7: Cohen's d

Statistical significance tells us whether the effect is likely non-zero.

Effect size tells us **how big it is**.

For independent samples with equal variances:

$$d = \frac{M_1 - M_2}{S_p}$$

Where:

$$S_p = \sqrt{\frac{SS_1 + SS_2}{df_1 + df_2}}$$

Let's calculate it manually so you see where it comes from.

```
M1 <- mean(CTA.Experimental)
M2 <- mean(CTA.Control)
S1 <- sd(CTA.Experimental)
S2 <- sd(CTA.Control)

Sp <- sqrt(((n-1)*S1^2 + (n-1)*S2^2) / (2*n - 2))
cohens_d <- (M1 - M2) / Sp

c(M_experimental = M1, M_control = M2, Sp = Sp, d = cohens_d)
```

```
M_experimental      M_control             Sp              d
     5.0335115      2.7591945      0.8624439      2.6370608
```

The value tells us how far apart the two means are in pooled standard-deviation units. Context still matters. A conventional adjective cannot tell us whether the change matters for actual riders, budgets, access, or the continued emotional survival of anyone waiting for a bus in February.

Because let's be honest - trains every 7 minutes would change lives.

7.9 Step 8: APA Results Paragraph

Participants in the every-7-minutes condition ($M = 5.03$, $SD = 0.81$) reported greater liking of the CTA than those in the regular-timing control condition ($M = 2.76$, $SD = 0.92$), Welch's $t(57.09) = 10.21$, $p < .001$, 95% CI [1.83, 2.72], $d = 2.64$.

In this simulated experiment, increasing train frequency produced a very large difference in reported attitudes. In real life, the CTA has declined our invitation to be manipulated with `set.seed(343)`.

> **!** Welch by Default; Effect Size Beside It
>
> Welch's t-test does not require the two population variances to be equal, so it is a sensible default for independent groups. Report the mean difference and its confidence interval, then add an effect size. The p-value cannot tell the reader how far apart the groups are.

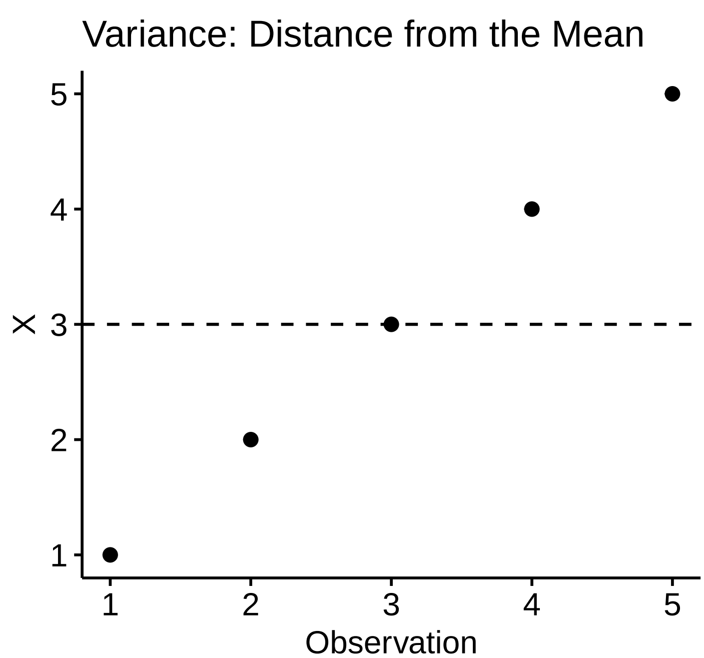
8 Covariance and Correlation

8.1 A Visual Intuition: Variance, Covariance, and Correlation

We'll start with a very simple dataset and visualize three situations:

- One variable by itself (variance)
- Two variables moving together (positive covariance)
- Two variables moving in opposite directions (negative covariance)

8.1.1 Variance: Spread Around a Mean

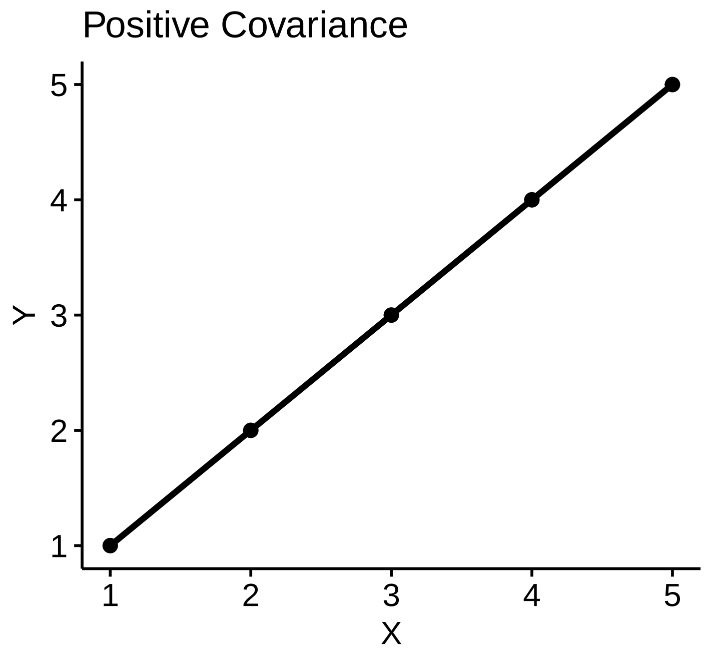


- Each point deviates from the mean of X
- Squaring those deviations and averaging them gives us variance
- Variance does not care about relationships with other variables

8.1.2 Do Two Variables Move Together?

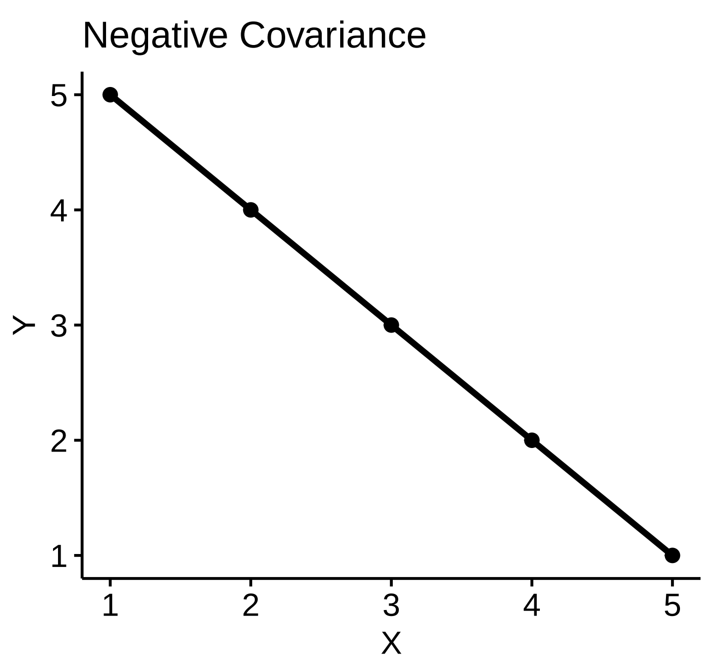
Covariance answers a two-variable question:

- When X is above its mean, is Y also above its mean (or below)?



- High X pairs with high Y
- Low X pairs with low Y
- Cross-products $(X - M_X)(Y - M_Y)$ are positive
- "co-" variance is positive

8.1.3 Negative Covariance



- High X pairs with low Y
- Low X pairs with high Y
- Cross-products $(X - M_X)(Y - M_Y)$ are negative
- "co-" variance is negative

8.1.4 What is Correlation?

Covariance has one big problem: *Its size depends on the units of measurement.*

Correlation fixes this by standardizing covariance:

- Variance $\rightarrow$ standardized $= 1$
- Covariance $\rightarrow$ standardized between -1 and $+1$

Correlation (r) tells us:

- Direction (positive vs negative)
- Strength (how tightly the points cluster)
- Scale-free comparison

You'll see shortly that:

- Perfect positive covariance yields a $r = +1$
- Perfect negative covariance yields a $r =$ *-1
- No systematic covariance yields a $r =$ *-0

8.1.5 Big Picture (Preview)

Concept	What it measures	How many variables
Variance	Spread around a mean	1
Covariance	Shared movement	2
Correlation	Scaled shared movement	2

- Variance tells us how much *one* variable moves.

- Covariance tells us whether *two* variables move together.

- Correlation rescales covariance so we can compare relationships across studies.

Now that we have the picture, we can formalize each idea.

8.2 Variance

- Variance is our primary measure of spread.
- It is defined as the sum of squared variation around a mean.
  - It is further very important for distributional theory.
- Here are several equivalent definitions:
  - we'll use s for sample sd and s^2 for sample variances

$$s^2 = Var(x) = \frac{\sum SS_X}{n-1} = \frac{\sum (X - M_X)^2}{n-1} = \frac{\sum (X - M_X)(X - M_X)}{n-1}$$

8.3 Covariance

- Variance is a summary of univariate distance from a mean.
- Covariance is summary of bivariate distance from a pair of means.
- It describes association between pairs of variables:
- What was a sum of squares is now a sum of products:

$$Cov_{xy} = \frac{\sum(X - M_X)(Y - M_Y)}{n-1} = \frac{SP_{xy}}{n-1}$$

8.3.1 Perfect Positive Relationship

- Why would we do this? Let's start with an illustrative example.
  - $X = 1, 2, 3, 4, 5$
- Imagine we took the variance of that variable:

```
X = c(1,2,3,4,5)
M_x<-mean(X)
SS_x <- sum((X-M_x)^2)
Var_X=SS_x/(length(X)-1) #or just do var(X)
Var_X
```

```
[1] 2.5
```

8.3.2 Identical Variables

- Now imagine two variables that are exact copies of one another:
  - Imagine we're measuring a very stable variable twice.
  - $X = 1, 2, 3, 4, 5$
  - $Y = 1, 2, 3, 4, 5$
  - What's Var(x)? What's Var(y)?

```
X = c(1,2,3,4,5)
Y = c(1,2,3,4,5)
var(X)
```

```
[1] 2.5
```

```
var(Y)
```

```
[1] 2.5
```

What's Cov(xy)?

```
cov(X,Y)
```

```
[1] 2.5
```

Why is it the same number?

8.3.3 Perfect Negative Relationship

- Now imagine two variables that are exact opposites:
  - Like and hate for stats, etc.
  - $X = 1, 2, 3, 4, 5$
  - $Z = 5, 4, 3, 2, 1$
  - Variances haven't changed?

```
X = c(1,2,3,4,5)
Z = c(5,4,3,2,1)
var(X)
```

```
[1] 2.5
```

```
var(Z)
```

```
[1] 2.5
```

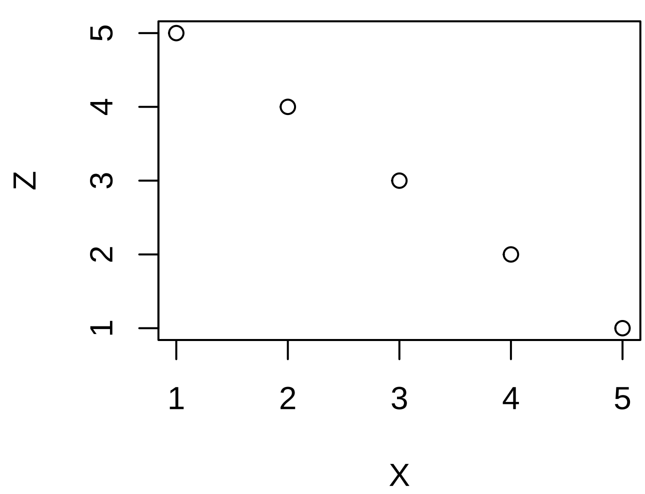
What's Cov(xz)?

```
cov(X,Z)
```

```
[1] -2.5
```

- What does this mean?

```
plot(X, Z)
```



8.3.4 Why the Sign Flips

- When there was a perfect positive relationship, every cross-product was positive.
- Now, they're all negative:
  - $-2 * 2, -1 * 1, 0, 0, 1 * -1, 2 * (-2)$
- When we add them up, we get a negative number.
- That negative sign indicates a negative relationship.

8.3.5 No Relationship

- Now imagine two variables that aren't strongly related. -Like for stats/like for Alex -$X = 1, 2, 3, 4, 5$ -$A = 2, 5, 3, 1, 4$ -Variances haven't changed?

```
X = c(1,2,3,4,5)
A = c(2,5,3,1,4)
var(X)
```

```
[1] 2.5
```

```
var(Z)
```

```
[1] 2.5
```

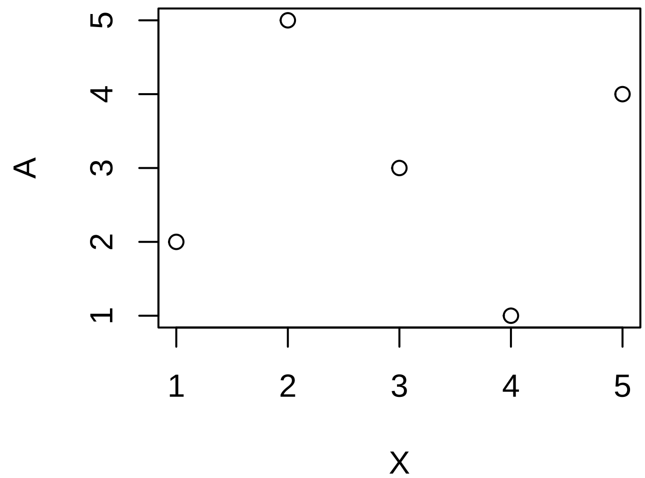
What's Cov(XA)?

```
cov(X,A)
```

```
[1] 0
```

- What does this mean?

```
plot(X, A)
```



8.3.6 Why Zero?

- Turn them into deviations from their means:
  - Old: X=1,2,3,4,5, Y=2,5,3,1,4.
  - Now: X=-2,-1,0,1,2, Y=-1,2,0,-2,1.
- Get the sum of cross products.
  - $(-2 * -1) + (-1 * 2) + (0 * 0) + (1 * -2) + (2 * 1)$
  - $2 + (-2) + 0 + (-2) + 2 = 0$
- The covariance is zero. There is no relationship between X and Y.

8.3.7 Covariance Review

- Assume that Var(x)=Var(y) for now.
- When there was a perfect relationship between x and y, $Cov(xy) = Var(x) = Var(y)$.
  - When the variances of X and Y aren't equal, $Cov(xy) = \sqrt{Var(x) * Var(y)}$
- When there was a perfect inverse relationship, $-Cov(xy) = \sqrt{Var(x) * Var(y)}$.
- When there was no relationship, $Cov(XY) = 0$.
  - We added up a bunch of negative and positive products.
  - The signs canceled out.

8.4 Correlations

- Correlation is a broad term for a scaled measure of association.
- Different correlations answer different questions about different kinds of variables.
- Pearson's r and Spearman's r_s range from -1 to $+1$.

- Zero means no **linear** association for Pearson and no **monotonic** association for Spearman. A strong U-shaped relationship can still produce zero, because correlation is very talented at missing questions it was never asked.
- Universal small, medium, and large labels are rarely useful. Interpret magnitude in the research context.

Everything correlates with everything, which Paul Meehl calls the "crud factor" (also called ambient correlational noise) (Meehl 1990a, 1990b). Our goal is to determine how much, and we will deal with two variables at a time before exploring the problems created by three or more variables.

8.4.1 Correlation Types

A few popular correlations between two variables:

- Pearson's r (interval by interval) [for population = ρ, for sample = r]
- Spearman's rho (interval by ordinal) [for population = ρ_s, for sample = r_s]

Other types we will not cover:

- Kendall's tau (τ), a rank-based measure that handles pair concordance and ties differently from Spearman's correlation
- Point-biserial correlation (quantitative by genuinely dichotomous)
- Polychoric (ordinal vs ordinal) [used more in psychometrics or factor analysis of ordinal by ordinal]
- Tetrachoric correlation (dichotomous vs. dichotomous, under a latent-continuous-variable model)

These measures do **not** all share one assumption set. Classical inference for Pearson's correlation commonly assumes independent pairs and, for its exact small-sample test, bivariate normality. Bivariate normality describes the joint distribution of two variables; it does not mean they become normal when "added together," and it certainly does not mean the two variables must be independent. If they were independent, their population correlation would be zero, which would make this a remarkably short chapter.

8.4.2 Pearson's Correlation

Most common type you will encounter and is a parametric method.

$$r_{xy} = \frac{\sum(X - M_X)(Y - M_Y)}{\sqrt{\sum(X - M_X)^2 \sum(Y - M_Y)^2}}$$

remember $SS = (X - M)^2$, so thus,

$$r_{xy} = \frac{SP}{\sqrt{SS_X SS_Y}}$$

- Numerator = How much they vary together (covariance)
- Denominator = The product of how much they vary alone (variance)
- Values are bounded between -1 and 1

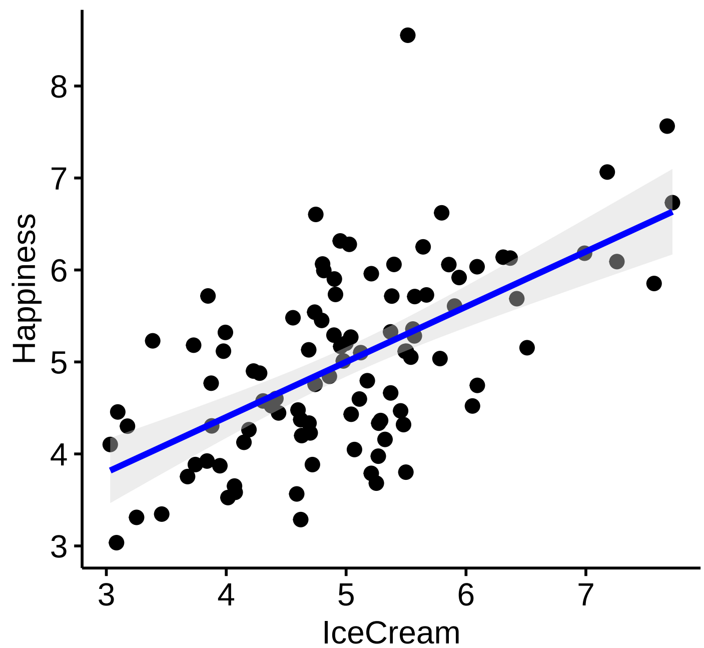
or more simply (see Cohen's textbook for the derivation):

$$r_{xy} = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

8.4.3 Simulate Data

We will use the `mvrnorm` function (multivariate normal distribution) from the *MASS* package, but to do this we need to make a **covariance** matrix with a $r = .60$ and set the mean values for each variable (which I will set to 5 for each)

8.4.4 Plot the data



8.4.5 Run Pearson's r

The `cor.test` function runs Pearson's correlation. **Note:** You will notice that I have attached the data frame to each variable.

```
Corr.Result.1<-cor.test(CorrData$Happiness, CorrData$IceCream,
         method = c("pearson"))
Corr.Result.1
```

```

    Pearson's product-moment correlation

data:  CorrData$Happiness and CorrData$IceCream
t = 7.4246, df = 98, p-value = 4.193e-11
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.4574985 0.7124547
sample estimates:
cor
0.6
```

You can also call the function via a formula command.

```
Corr.Result.1<-cor.test(~Happiness + IceCream, data= CorrData,
         method = c("pearson"))
```

8.4.6 P-value for Pearson's Correlation

The classical p-value for Pearson's correlation is adapted from the t-distribution. We will return to it when we cover linear regression.

8.4.7 Report them in APA format

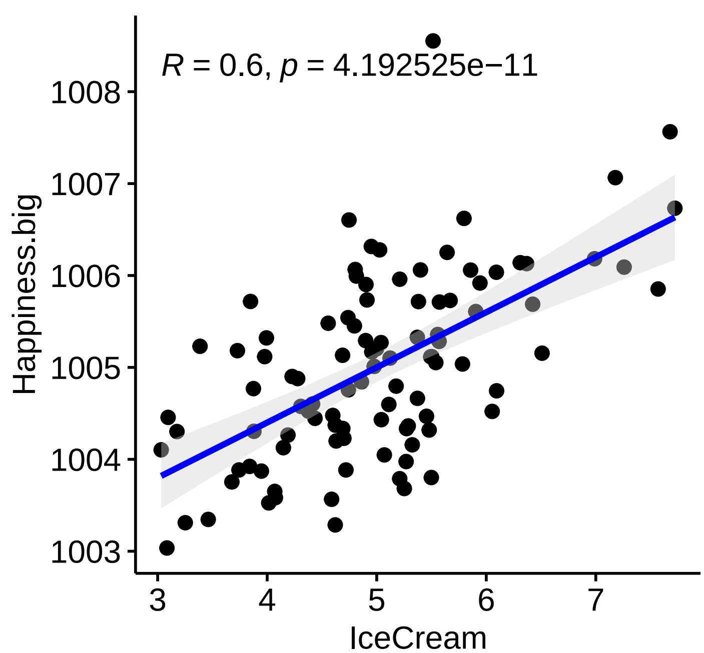
The `cor_apa` function in the APA package will report it in APA format for you. Here, $r(df)$ gives Pearson's r and its degrees of freedom. The degrees of freedom equal $N-2$ because two variables are involved. The function has several output-format options.

$r(98) = .60$, $p < .001$

8.4.8 Pearson's correlation is scale-independent!

No matter the mean differences or range of scores, the Pearson's r will give the same results. We can also z-score the data and get the same result. However, if they are scaled non-linearly (sqrt, ^2, log,...) the correlation will change.

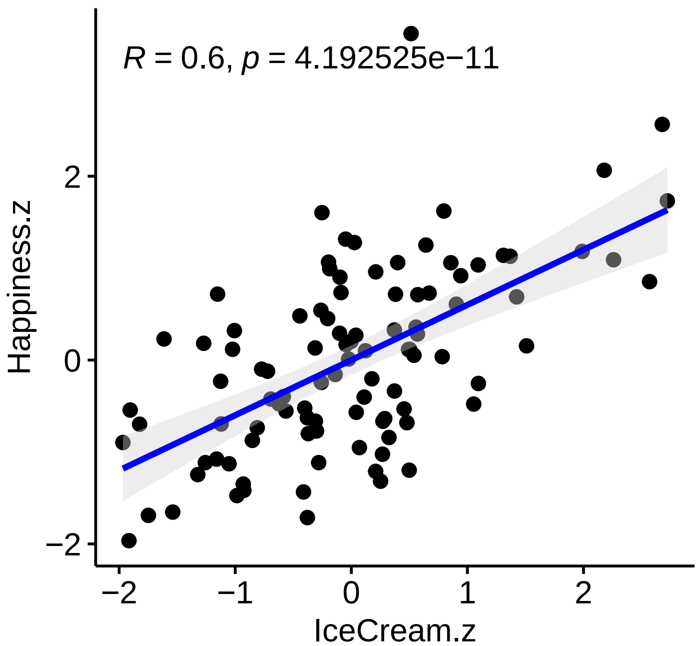
8.4.9 Scale change: Add a constant (shift the mean)




```
r(98) = .60, p < .001
```

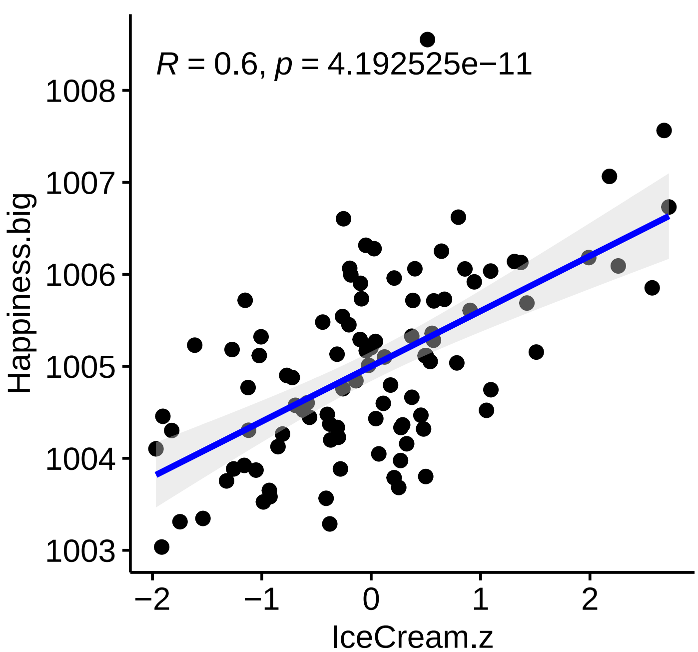
8.4.10 Scale change: Z-scored

Remember that, $Z = \frac{X-M}{S}$



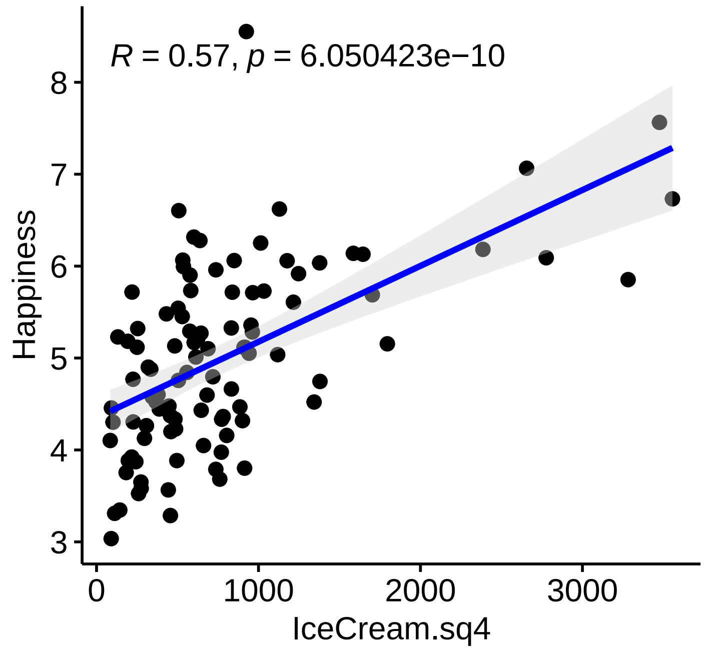
r(98) = .60, p < .001

8.4.11 Scale change: Linear (mixed scales)



r(98) = .60, p < .001

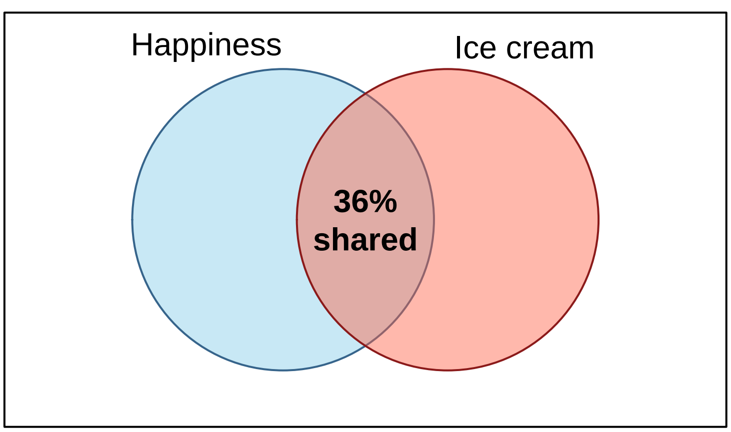
8.4.12 Scale change: Non-linear (breaks it)




```
r(98) = .57, p < .001
```

8.4.13 Pearson's: Let visualize our result

The overlap between the two variables is defined by r^2



The picture is conceptual rather than drawn to exact area scale. Its job is to keep the shared piece visible, not to audition for an engineering degree.

This means that 36% of the variance of happiness overlaps with ice cream consumption. Can we conclude ice cream causes happiness? No, because we did not design a study with a control group. We can not infer causation. It seems to make sense to say "eating ice cream causes me to be happy", but the opposite could be true as well "happiness causes me to eat ice cream". How do I know which causes which? We cannot without an experiment.

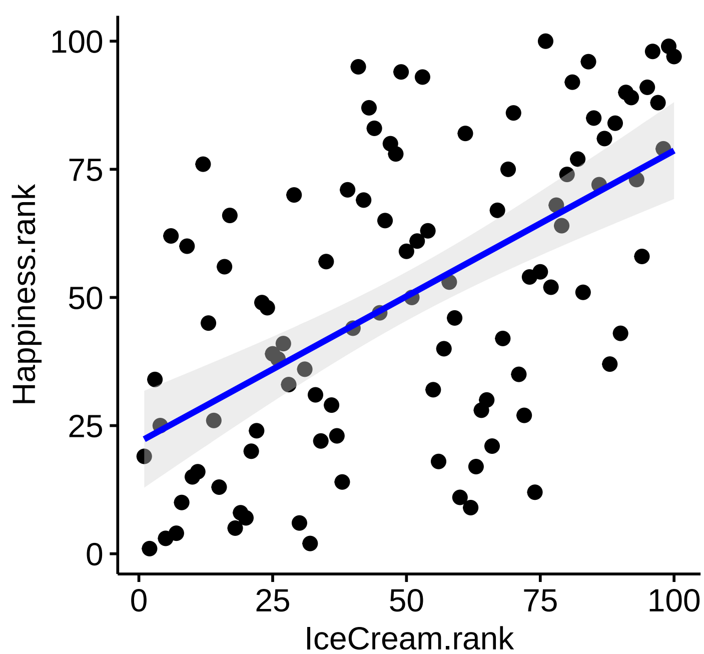
8.4.14 Non-Parametric Correlations

Spearman and Kendall correlations are useful for ordinal data and for **monotonic** relationships: as one variable increases, the other generally keeps moving in one direction, even if the pattern bends. They do not rescue every nonlinear relationship. A U-shaped relationship can have a correlation near zero because the upward and downward pieces cancel each other while laughing at you.

8.4.15 Spearman's Correlation

Spearman is a Pearson Correlation on rank-ordered data. Let's rank order our random correlated data. You must rank each variable independently first (where ties are averaged).

```
CorrData$Happiness.rank<-rank(CorrData$Happiness)
CorrData$IceCream.rank<-rank(CorrData$IceCream)

ggscatter(CorrData, x = "IceCream.rank", y = "Happiness.rank",
   add = "reg.line",  # Add regressin line
   add.params = list(color = "blue", fill = "lightgray"), # Customize reg. line
   conf.int = TRUE, # Add confidence interval
   )
```



```
cor_apa(cor.test(CorrData$IceCream.rank, CorrData$Happiness.rank, method = c("pearson")))
```

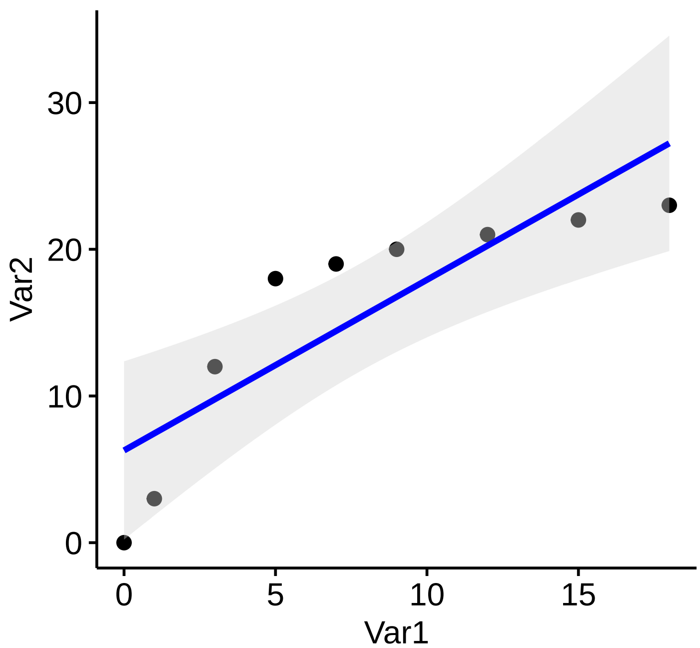
```
r(98) = .57, p < .001
```

Use the built-in Spearman correlation with `cor.test(..., method = "spearman")`. It ranks the raw data automatically and calculates the appropriate p-value.

```
r_s = .57, p < .001
```

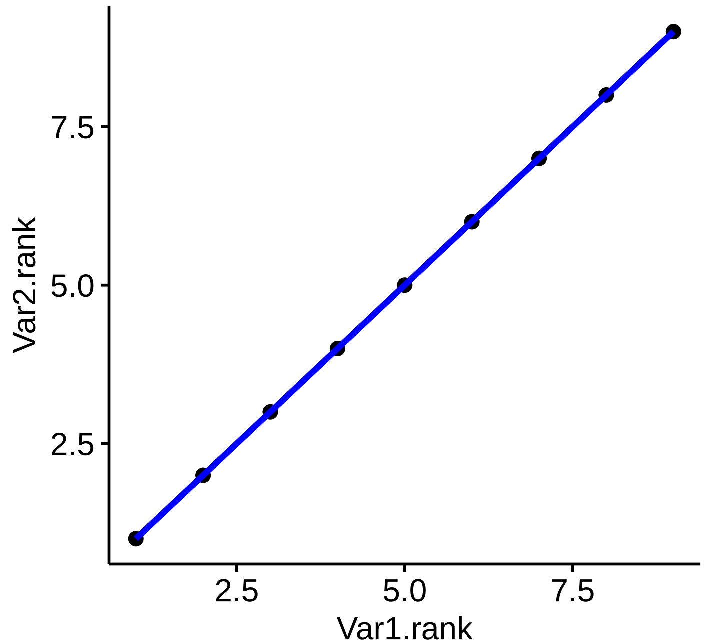
8.4.16 Pearson vs Spearman for Nonlinear Data

Let's say you get some data and clearly there is a slight nonlinearity in the relationship between the two variables. Pearson is designed for linear relationships and we can see the problem in our fitted line below.




```
r(7) = .86, p = .003
```

If we switch to a Spearman correlation the data are converted to ranks and the "bump" is now gone and our correlation gets stronger.

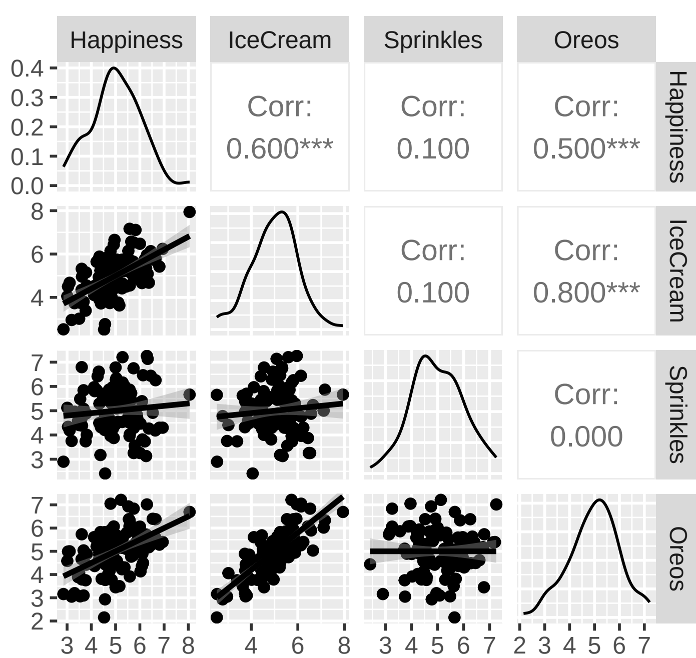



```
r_s = 1.00, p < .001
```

8.4.17 Correlation Matrices

When we have multiple variables we can compare them all to each other at once. First we will simulate a 4 variables and all their bivariate correlations using the mvrnorm function again.

We can use the `GGally` package to plot Pearson correlations quickly in an easy to visualize format once the data are in a data frame.



8.5 The Short Story

- Covariance tells us whether two variables move together, but its size depends on their units.
- Correlation standardizes that shared movement.
- Pearson's r summarizes a linear relationship; Spearman's r_s summarizes a monotonic relationship using ranks.
- r^2 is the proportion of variance shared in a simple two-variable linear setting.
- Correlation does not establish causation. Ice cream may produce happiness, happiness may produce ice-cream consumption, or heat may produce both while quietly stealing credit.
- Plot the data. A single coefficient can hide curves, outliers, separate groups, and other tiny statistical monsters.

Correlation Is Not a Tiny Causal Arrow

An r describes the direction and strength of a relationship. It does not tell us whether X causes Y, Y causes X, or a third variable is backstage operating both puppets. Design and theory must do that work.

9 Linear Regression

9.1 Introduction to Regression

Many psychological studies ask: how does X *predict* Y? How does ice cream (IV) predict happiness (DV)?

Unlike the t-test (which compares group averages), regression looks at the relationship between **two continuous variables**. *Predicting* just means making an educated guess — if I know how much ice cream you ate, I can guess how happy you are. The guess won't be perfect, but it should be reasonably close.

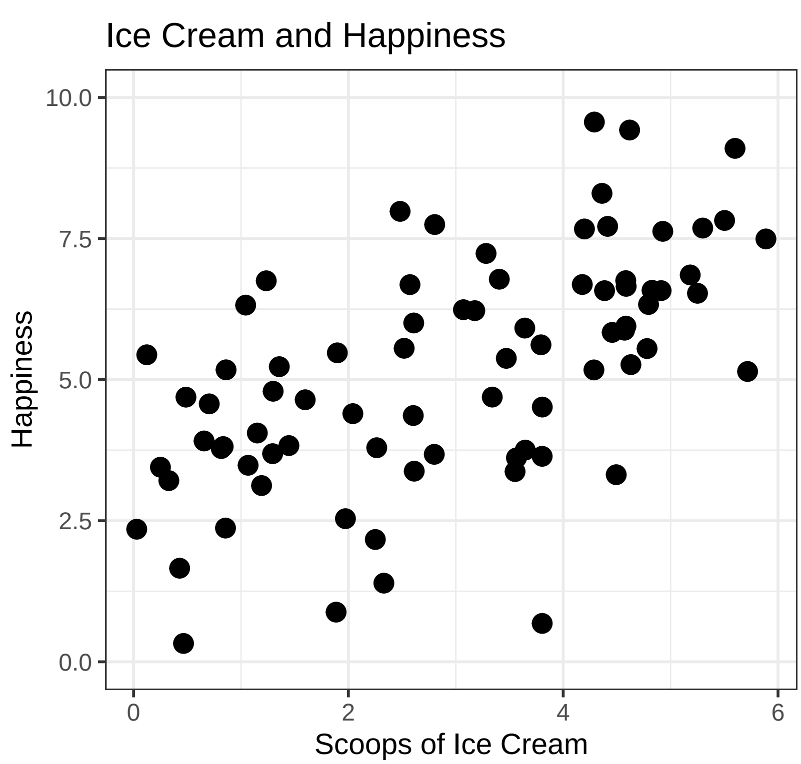
Regression tells us **how close those guesses tend to be**.

Regression predicts. It does not automatically explain causes. If people who eat more ice cream report greater happiness, ice cream may help, happier people may seek ice cream, hot weather may increase both, or the entire pattern may be driven by one billionaire buying all the pistachio. Design and theory decide which causal stories remain plausible.

Note: Regression assumes interval or ratio scale variables. In practice, psychologists use Likert scales all the time anyway.

9.1.1 The Experiment

I bought several tubs of vanilla ice cream, scooped out different amounts, and handed them to students. After eating, they rated their happiness on a 0–10 "feelings thermometer." Nobody got zero scoops, but some got barely a teaspoon.



9.1.2 Why not just use correlation?

You can — and correlation is often the first step. But correlation only handles **two variables at a time**. Regression lets us add more predictors: ice cream *plus* hot fudge *plus* cookie dough. In other words, regression scales up in ways correlation can't.

Here's the correlation:

```r
library(apa)
apa(cor.test(dat$Scoops, dat$Happiness))
```

```
[1] "*r*(78) = .60, *p* < .001"
```

Strong positive correlation, as expected!

9.2 Regression Theory

Correlation and regression describe the **same linear relationship** — just differently.

- **Simple linear regression** = 1 DV and 1 IV. The relationship is a straight line representing the **expected mean of Y** for each value of X.
- **Multiple regression** = 1 DV and 2+ IVs. (Coming later.)

9.2.1 The Regression Equation

You probably remember from middle school: $y = Mx + b$

The modern version:

$$Y = B_0 + B_1X + e$$

Where:

- Y = predicted value
- B_1 = slope (effect of X)
- B_0 = intercept
- e = residual (observed − predicted)

9.3 The Ice Cream Model

We fit regression models with `lm()` (linear model). Formula structure: **DV ~ IV**.

```r
Happy.Model.1 <- lm(Happiness ~ Scoops, data = dat)
summary(Happy.Model.1)
```

```
Call:
lm(formula = Happiness ~ Scoops, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.1621 -0.9242  0.0332  1.0608  3.3565 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   2.9799     0.3863   7.715 3.35e-11 ***
Scoops        0.7524     0.1127   6.675 3.25e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.681 on 78 degrees of freedom
Multiple R-squared:  0.3636,	Adjusted R-squared:  0.3554 
F-statistic: 44.56 on 1 and 78 DF,  p-value: 3.246e-09
```

9.3.1 Intercept

The intercept is the predicted happiness **when ice cream = 0**.

```r
range(dat$Scoops)
```

```
[1] 0.02927577 5.88743768
```

Nobody had zero scoops, so we can't actually interpret this directly. **Never predict outside the range of your data.** (We can fix this by centering X — more on that later.)

9.3.2 Slope

The slope is 0.7524359. For each additional scoop, happiness changes by that amount.

9.3.3 Making a prediction

The equation: $Y = B_0 + B_1X$

What's the predicted happiness for someone who eats **5 scoops**?

- $Y = 2.98 + (0.752 \times 5)$
- $Y = 6.742$

9.4 Predicted Values and Residuals

The line gives one predicted happiness score for every amount of ice cream. Real people do not sit perfectly on that line because humans are noisy, stubborn, and occasionally having a terrible day for reasons unrelated to dessert.

A **residual** is the distance between the observed score and the model's prediction:

$$e_i = Y_i - \widehat{Y}_i.$$

- A positive residual means the person was happier than the model predicted.
- A negative residual means the person was less happy than the model predicted.
- A residual of zero means the prediction was exactly right, which is suspiciously cooperative behavior from a human.

R stores both pieces inside the fitted model:

```
dat$Predicted <- fitted(Happy.Model.1)
dat$Residual <- residuals(Happy.Model.1)

head(dat[c("Scoops", "Happiness", "Predicted", "Residual")])
```

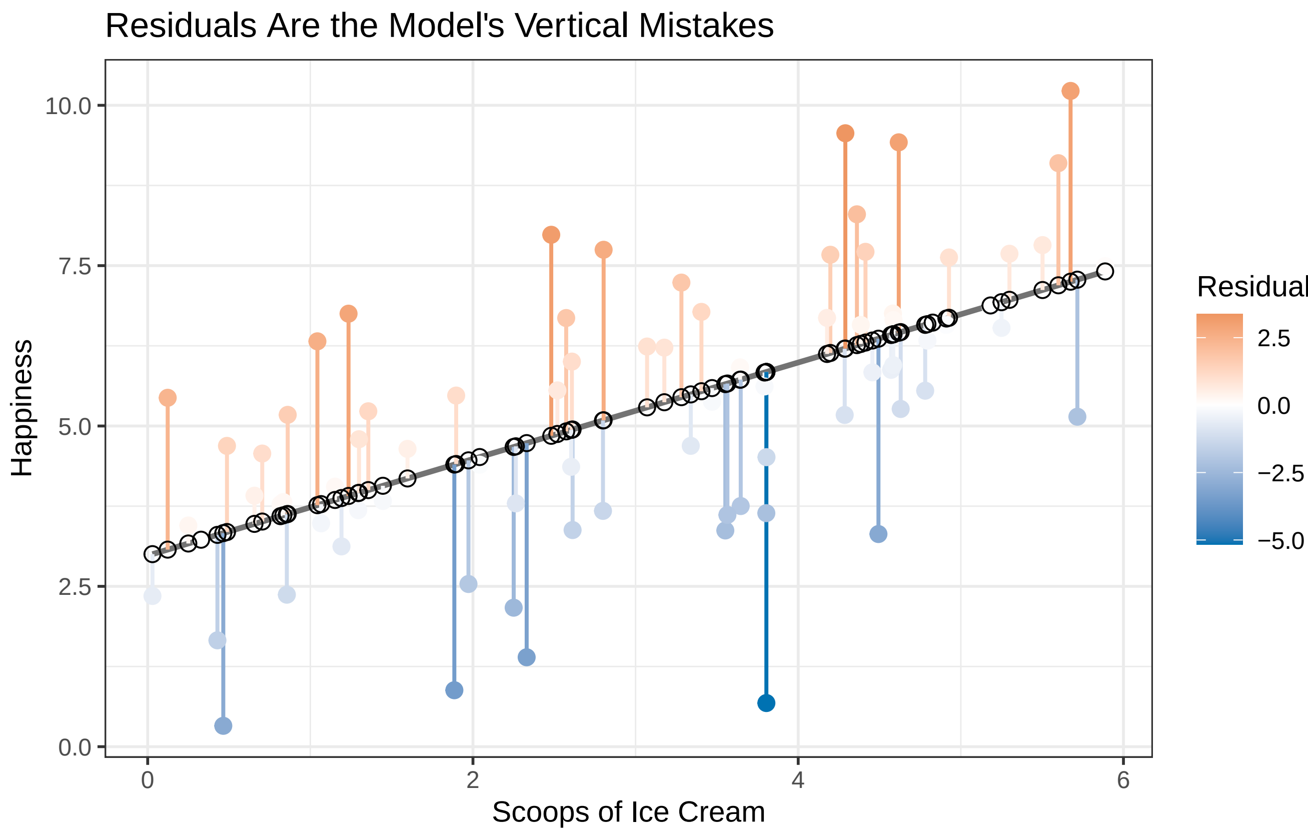
```
     Scoops Happiness Predicted   Residual
1 0.4874424  4.689709  3.346669  1.3430399
2 2.7996025  3.676425  5.086421 -1.4099958
3 4.1977287  7.669921  6.138422  1.5314991
4 4.7945917  6.331497  6.587523 -0.2560254
5 3.3393869  4.690570  5.492574 -0.8020041
6 5.2990277  7.685625  6.967079  0.7185465
```

```
mean(dat$Residual)
```

```
[1] 1.283102e-17
```

The residuals have a mean of approximately zero because ordinary least squares balances the line through the data. Positive and negative mistakes cancel out.

```
ggplot(dat, aes(x = Scoops, y = Happiness)) +
  geom_smooth(method = "lm", se = FALSE, color = "gray45") +
  geom_segment(
    aes(xend = Scoops, yend = Predicted, color = Residual),
    linewidth = 0.7
  ) +
  geom_point(aes(color = Residual), size = 2.5) +
  geom_point(aes(y = Predicted), shape = 1, size = 2.5) +
  scale_color_gradient2(low = "#0072B2", mid = "white", high = "#D55E00") +
  labs(
    title = "Residuals Are the Model's Vertical Mistakes",
    x = "Scoops of Ice Cream",
    y = "Happiness"
  ) +
  theme_bw()
```



The filled points are observed happiness. The hollow points are predicted happiness. Each vertical segment is one residual. This is the error the model did not explain.

9.5 The Assumptions: What Could Make the Line Lie?

Regression assumptions are claims about the errors left after fitting the line. They are not personality tests that the raw variables must pass before R permits them into the building.

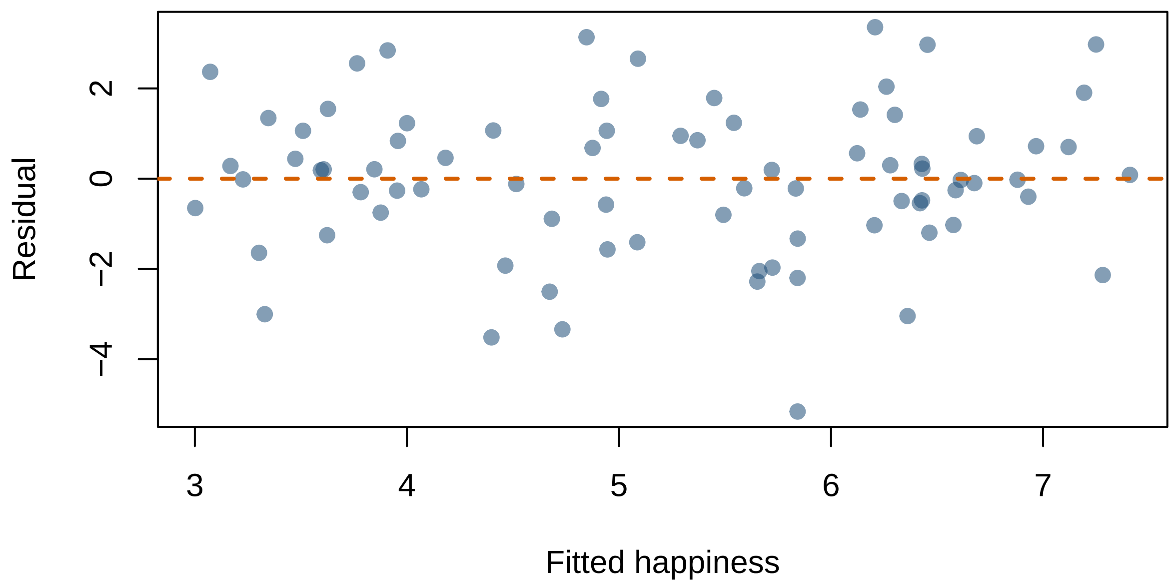
For now, focus on five questions:

1. **Linearity:** Is a straight line a reasonable summary of the average relationship?
2. **Independence:** Are the residuals from different observations reasonably independent?
3. **Constant variance:** Is the vertical spread of residuals roughly similar across fitted values?
4. **Approximately normal residuals:** Are the residuals reasonably bell-shaped, especially for small-sample tests and intervals?
5. **Influential observations:** Is one strange person steering the entire line while everyone else watches?

9.5.1 Residuals Versus Fitted Values

```
plot(
  fitted(Happy.Model.1),
  residuals(Happy.Model.1),
  xlab = "Fitted happiness",
```

```r
  ylab = "Residual",
  pch = 19,
  col = grDevices::adjustcolor("#1F4E79", alpha.f = 0.55)
)
abline(h = 0, lty = 2, col = "#D55E00", lwd = 2)
```



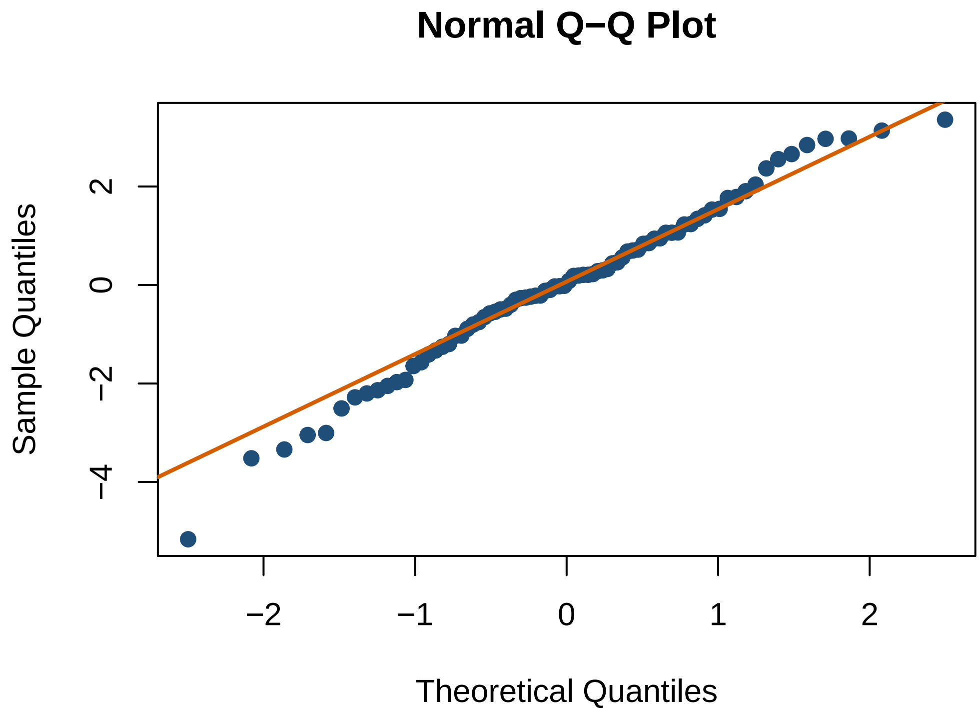
We want a fairly boring cloud centered around zero. A curve suggests the straight-line model missed a nonlinear pattern. A funnel suggests **heterogeneity of variance**, also called **heteroscedasticity**: the model's errors are much more variable for some predicted values than for others.

Why care? Ordinary regression calculates standard errors as if the residual variance is reasonably constant. Strong heteroscedasticity can make those standard errors and confidence intervals inaccurate. It can also reveal something psychologically interesting. Perhaps ice cream affects people similarly at low doses but produces wildly different reactions after the fourth scoop. Some people achieve joy. Others remember lactose intolerance.

> ⚠ Heterogeneity May Be a Clue, Not Merely a Violation
>
> Changing residual spread can mean the uncertainty differs across people or conditions. Do not immediately transform the outcome until the graph stops complaining. First ask what process could create the pattern.

9.5.2 The Q-Q Plot

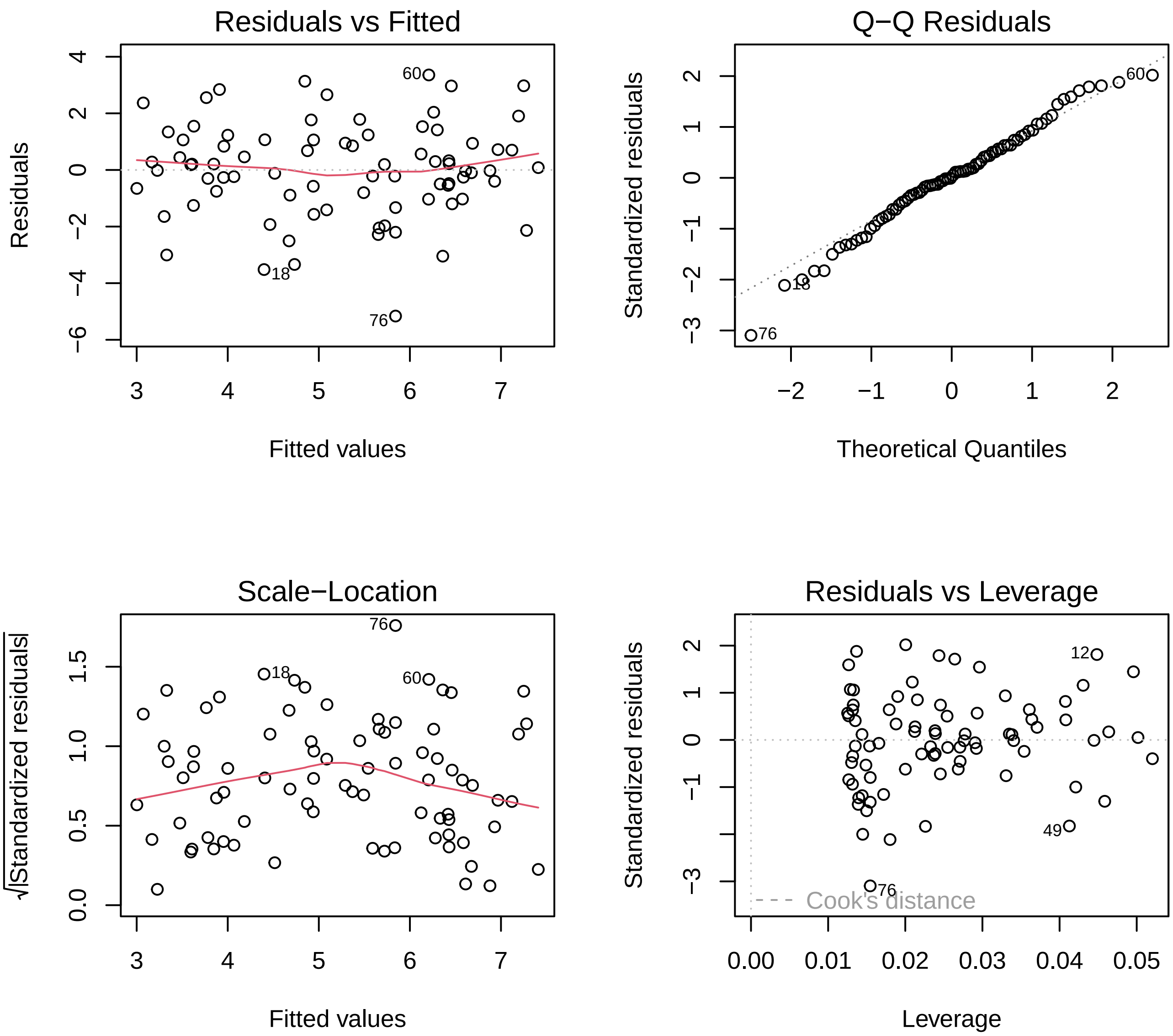
```r
qqnorm(residuals(Happy.Model.1), pch = 19, col = "#1F4E79")
qqline(residuals(Happy.Model.1), col = "#D55E00", lwd = 2)
```



A **Q-Q plot** compares the ordered residuals with values expected from a normal distribution. Points reasonably near the line are fine. Severe curves, separated tail points, or one observation escaping toward Canada deserve investigation. Minor imperfection is normal because the data were collected from humans, not manufactured by a normal-distribution factory.

R can produce four standard diagnostic plots at once:

```
par(mfrow = c(2, 2))
plot(Happy.Model.1)
par(mfrow = c(1, 1))
```



The first two are the residual-versus-fitted and Q-Q plots we just learned. The other two help identify unequal spread and influential observations. You do not need to master them yet. The advanced diagnostics chapter returns with more detail and several additional ways for data to misbehave.

9.6 Residual Variance and R^2

Before using ice cream to predict happiness, our best simple guess for everyone was the mean. The total variance describes how far scores spread around that mean. After fitting the model, residual variance describes how far scores spread around the regression line.

```
TotalVariance <- var(dat$Happiness)
ResidualVariance <- var(residuals(Happy.Model.1))
```

```r
c(TotalVariance = TotalVariance,
  ResidualVariance = ResidualVariance,
  R_squared_from_variance = 1 - ResidualVariance / TotalVariance,
  R_squared_from_model = summary(Happy.Model.1)$r.squared)
```

```
          TotalVariance        ResidualVariance R_squared_from_variance 
              4.3817090               2.7886391               0.3635727 
   R_squared_from_model 
              0.3635727 
```

R^2 describes the proportion of observed outcome variance explained by the fitted model. It is an effect-size measure for the model, not proof that the model is causal, correct, or ready to be tattooed onto your arm.

9.7 Relationship to Correlation

Why did we start with correlation if regression is so great? To show they're the same thing, just scaled differently. Run a regression on **standardized** variables:

```r
dat$Happiness.z <- scale(dat$Happiness)
dat$Scoops.z    <- scale(dat$Scoops)

std.reg <- lm(Happiness.z ~ Scoops.z, data = dat)
summary(std.reg)
```

```

Call:
lm(formula = Happiness.z ~ Scoops.z, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.46608 -0.44152  0.01588  0.50675  1.60349 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -2.037e-16  8.976e-02   0.000        1    
Scoops.z     6.030e-01  9.033e-02   6.675 3.25e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8029 on 78 degrees of freedom
Multiple R-squared:  0.3636,	Adjusted R-squared:  0.3554 
F-statistic: 44.56 on 1 and 78 DF,  p-value: 3.246e-09
```

9.7.1 What didn't change?

The **t-statistic and p-value** for the slope. These don't change with any linear rescaling, so you can transform or rescale variables whenever you want without affecting statistical significance.

9.7.2 What changed?

The intercept — it's now **zero**. When all variables are standardized (mean = 0), the regression line must pass through [0, 0]. And the slope changed numerically:

```r
print(summary(std.reg), digits = 6)
```

```

Call:
lm(formula = Happiness.z ~ Scoops.z, data = dat)

Residuals:
      Min        1Q    Median        3Q       Max 
-2.466080 -0.441520  0.015876  0.506753  1.603487 

Coefficients:
                Estimate   Std. Error t value   Pr(>|t|)    
(Intercept) -2.03715e-16  8.97626e-02 0.00000          1    
Scoops.z     6.02970e-01  9.03290e-02 6.67527 3.2461e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.802862 on 78 degrees of freedom
Multiple R-squared:  0.363573,  Adjusted R-squared:  0.355413 
F-statistic: 44.5592 on 1 and 78 DF,  p-value: 3.24612e-09
```

```r
cor.test(dat$Happiness, dat$Scoops)
```

```

    Pearson's product-moment correlation

data:  dat$Happiness and dat$Scoops
t = 6.6753, df = 78, p-value = 3.246e-09
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.4417814 0.7264454
sample estimates:
      cor 
0.6029699 
```

The standardized slope equals the correlation. Not a coincidence.

When everything is standardized, one unit = one SD. So the correlation tells us: every time X increases by 1 SD, Y increases by r SDs. Standardized regression slopes are called **beta weights** (**β**).

9.7.3 Converting slope back to raw units

$$B_1 = r\frac{sd_y}{sd_x} = \frac{Cov(x,y)}{Var(x)}$$

Your slope is just the correlation times the ratio of SDs. Makes sense — slope is rise over run:

- Double Y → twice as much rise → B_1 doubles
- Double X → twice as much run → B_1 is cut in half

9.7.4 What about the intercept?

Once you know B_1, you know the angle of the line. But the line must also pass through **[mean(X), mean(Y)]**:

```
mean(dat$Scoops)
```

```
[1] 2.993878
```

```
mean(dat$Happiness)
```

```
[1] 5.232601
```

$$Mean(Y) = B_0 + B_1 \times Mean(X)$$

Solve for B_0 and you're done.

9.8 The Short Story

- Simple linear regression predicts a quantitative outcome from one predictor.
- The intercept is the predicted outcome when $X = 0$.
- The slope is the expected change in Y for a one-unit increase in X.
- Residuals are observed outcomes minus predicted outcomes: the model's leftovers.
- R^2 is the proportion of observed outcome variance explained by the fitted linear model.
- With one predictor, the standardized slope equals Pearson's r.
- Prediction is not causation. A straight line cannot randomize participants, control confounds, or stop Reviewer 2 from asking whether heat explains the ice cream.

> **!** A Slope Always Has Units
>
> The slope is the expected change in Y for a one-unit increase in X. Before deciding that it is large, small, heroic, or pathetic, say what one unit of X and one unit of Y actually mean.

10 Linear Regression with Categorical Variables

10.1 Distracted Driving

Driving requires constant attention. When something unexpected happens—such as a car braking suddenly—you must react quickly. Psychologists studying attention have shown that distractions slow reaction time. One of the most dangerous distractions is texting while driving.

The data in this chapter are simulated. Do not reproduce the study by handing undergraduates phones, placing them in actual traffic, and chasing the car with a chainsaw to improve ecological validity. Use a simulator and an ethics-approved protocol. The Institutional Review Board enjoys fewer surprises than the Probability Gods.

In this example we examine **braking reaction time** (seconds) under different driving conditions

Reaction time is measured in seconds. Larger values indicate slower reactions.

10.1.1 Dummy Coding

When a predictor is categorical, regression cannot use the labels directly.
Regression requires numbers, so the categories must be translated into numeric variables.

This process is called **dummy coding**.

The most common coding scheme is **dummy coding**, which is also the **default used by R** when it detects a factor.

10.1.2 Dummy Coding Details

Dummy coding compares each group to a **reference group**.

The reference group serves as the **baseline** or **zero point** for the model.
All other groups are interpreted relative to that baseline.

In other words, the model asks: *How much does each group deviate from the reference group?*

10.2 Two-Level Example

Suppose we conduct a study comparing two driving conditions: **No distraction** vs **Texting while driving**

We collect data for: $n = 30$ drivers per group ($dataframe = dat2$)

We have two driving conditions (and predict):

Condition	Reaction Time
No distraction	faster
Texting	slower

To represent this in regression, we create a dummy variable.

Variable	C1
No Distraction	0
Texting	1

Here:

- **0 represents the reference group**
- **1 represents the comparison group**

So the dummy variable indicates whether a driver is texting.

10.2.1 How This Connects to Regression

The regression model becomes

$$RT = b_0 + b_1 C_1$$

Interpretation:

- b_0 = the mean reaction time for the **reference group**
- b_1 = how much the **Texting** group differs from that reference group

10.2.2 Predicted Values

For **No Distraction**:

$C_1 = 0$

$$RT = b_0$$

For **Texting**:

$C_1 = 1$

$$RT = b_0 + b_1$$

So the regression coefficients directly reproduce the group means.

10.2.3 Important

In R, you usually **do not need to create dummy variables yourself**.

When a predictor is stored as a **factor**, R automatically creates the dummy-coded variables internally when fitting the model.

The **first level of the factor becomes the reference group**.

For example:

```
dat2$Condition <- factor(dat2$Condition,
                         levels = c("No Distraction", "Texting"))
```

Because `"No Distraction"` appears first, it becomes the **reference group**.

This means:

- the **intercept equals the mean reaction time for No Distraction**
- the **ConditionTexting coefficient represents the difference between Texting and No Distraction**

10.2.4 Comparing the Groups with a t-test

We first test the hypothesis that texting increases reaction time.

First we load the packages needed for this example.

- **tidyverse**: used for data manipulation and summarizing the data

- **apa**: used to produce APA-style statistical output

```
library(apa)
library(tidyverse)
```

10.2.5 Calculate Group Means and Standard Deviations

Before running a statistical test, it is always good practice to examine the descriptive statistics.

The code below calculates the **mean (M)** and **standard deviation (SD)** of reaction time for each condition.

```
dat2 %>%
  group_by(Condition) %>%
  summarise(
    M = mean(RT),
    SD = sd(RT)
  )
```

```
# A tibble: 2 x 3
  Condition          M    SD
  <fct>          <dbl> <dbl>
1 No Distraction 0.664 0.137
2 Texting        1.01  0.121
```

This gives us a quick summary of how reaction time differs across the two driving conditions.

10.2.6 Run the Independent Samples t-test

Next we test whether the difference between groups is statistically significant.

```
texting.ttest <- t.test(RT ~ Condition,
                        data = dat2,
                        var.equal = TRUE)
```

Reminder:

- `RT ~ Condition` tells R we are predicting **reaction time** from the **driving condition**
- `data = dat2` specifies which dataset to use
- `var.equal = TRUE` assumes equal variances across groups (the standard Student t-test)

10.2.7 Print the Results in APA Format

Finally, we format the results in **APA style** using the `apa` package.

```
apa(texting.ttest)
```

```
[1] "*t*(58) = -10.21, *p* < .001, *d* = -2.64"
```

10.2.8 APA write up

Drivers who texted while driving had significantly slower reaction times ($M = 1.01$, $SD = 0.12$) than drivers with no distraction ($M = 0.66$, $SD = 0.14$), $t(58) = -10.21$, $p < .001$, $d = -2.64$, indicating that texting substantially increased reaction time.

Small note: the negative sign on t simply reflects the order of subtraction (No Distraction minus Texting).

10.2.9 The Same Analysis Using Regression

Now we run the equivalent regression model.

```
model2 <- lm(RT ~ Condition, data = dat2)
summary(model2)
```

```
Call:
lm(formula = RT ~ Condition, data = dat2)

Residuals:
     Min       1Q   Median       3Q      Max
-0.31427 -0.09282 -0.00903  0.08735  0.34813
```

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       0.66388    0.02362   28.11  < 2e-16 ***
ConditionTexting  0.34115    0.03340   10.21 1.41e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1294 on 58 degrees of freedom
Multiple R-squared:  0.6427,    Adjusted R-squared:  0.6365
F-statistic: 104.3 on 1 and 58 DF,  p-value: 1.412e-14
```

The regression equation is

$$RT = b_0 + b_1 X$$

Interpretation:

- b_0 = mean reaction time for **No Distraction**
- b_1 = difference between **Texting** and **No Distraction**

Because the dummy variable for No Distraction is 0:

For No Distraction

$$RT = b_0$$

For Texting

$$RT = b_0 + b_1$$

So the intercept is simply the mean reaction time for the reference group.

```r
coef(model2)
```

```
     (Intercept) ConditionTexting
       0.6638792        0.3411475
```

The model returns something like:

Coefficient	Value
Intercept	0.664
ConditionTexting	0.341

Now we can reconstruct the means.

10.2.10 No Distraction

$$RT = b_0$$

$$RT = 0.664$$

So the predicted mean reaction time is **0.664 seconds**.

10.2.11 Texting

$$RT = b_0 + b_1$$

$$RT = 0.664 + 0.341$$

$$RT = 1.005$$

So the predicted mean reaction time is **1.005 seconds** for texting while driving.

These values match the observed group means from the data.

Regression coefficients are simply another way of expressing the group means and their differences.

10.2.12 Extracting the Means with emmeans

We can also extract the model-estimated means so we do not have to reconstruct them manually from the regression equation.

```
library(emmeans)

emmeans(model2, ~ Condition)
```

```
 Condition      emmean     SE df lower.CL upper.CL
 No Distraction  0.664 0.0236 58    0.617    0.711
 Texting         1.005 0.0236 58    0.958    1.052

Confidence level used: 0.95
```

10.2.13 What the Code Does

`library(emmeans)` loads the **emmeans** package. This package is designed to extract and compare means from statistical models such as regression.

`emmeans(model2, ~ Condition)` tells R:

- use the regression model **model2**
- estimate the means for each level of **Condition**

The `~ Condition` part means:
compute the means for the variable Condition.

10.2.14 What "Estimated Marginal Means" Means

The name **emmeans** stands for **Estimated Marginal Means**.

Estimated marginal means are the **means predicted by the statistical model**.

Why *marginal*? Before computers did all this instantly, people wrote the means in the margins of tables using actual pencils. They were Number 2 pencils, naturally—the same pencils you never remember to bring to my exams. iPads are new. Most of the old people who write R packages or statistics books were born in the 1900s and are older than the internet.

In this simple example, they correspond directly to the group means:

Condition	Predicted Mean
No Distraction	b_0
Texting	$b_0 + b_1$

So `emmeans()` is simply doing the same math we just performed manually:

- extracting the intercept
- adding the appropriate regression coefficients
- reporting the predicted group means

10.2.15 Why This Is Useful

For simple models like this one, calculating the means by hand is straightforward.

However, as models become more complex (for, with multiple predictors or interactions), manually reconstructing the means becomes tedious.

`emmeans` provides a convenient way to:

- extract the model-based means
- compare groups
- obtain confidence intervals
- perform additional tests

All while keeping the results tied directly to the regression model.

10.2.16 Testing the Difference with `emmeans`

We can also test the difference between groups directly from the regression model using `emmeans`.

```r
pairs(emmeans(model2, ~ Condition))
```

```
 contrast                 estimate     SE df t.ratio p.value
 No Distraction - Texting   -0.341 0.0334 58 -10.213 <0.0001
```

10.2.17 What the Code Does

The function `emmeans(model2, ~ Condition)` first computes the **estimated marginal means** for each level of the factor **Condition** based on the regression model.

The function `pairs()` then compares those means to one another.

In this case, there are only two conditions, so the function tests the difference between:

- **Texting**
- **No Distraction**

10.2.18 How This Connects to the Regression Model

Recall that the regression equation is

$$RT = b_0 + b_1 X$$

where

- b_0 = mean reaction time for **No Distraction**
- b_1 = difference between **Texting** and **No Distraction**

The comparison performed by `pairs()` is therefore testing whether:

$$b_1 = 0$$

If b_1 is significantly different from zero, the two groups have different mean reaction times.

10.2.19 Comparison to t-test

Even though the output may look similar to a t-test, the comparison is now being computed **within the regression framework**.

The means and the group comparison are both derived directly from the fitted regression model.

10.2.20 APA Regression Write up

Drivers who texted while driving had significantly slower reaction times than drivers with no distraction. Relative to this reference group (no distractions), the Texting condition increased reaction time by 0.34 seconds, $b = 0.34, SE = 0.03, p < .001$. The model explained a substantial proportion of variance in reaction time, indicating that differences in driving condition accounted for about 64% of the variability in reaction times across drivers

10.2.21 Plotting the Model-Estimated Means

We can also visualize the model-estimated means returned by `emmeans`.

First we store the estimated means in an object.

```
means <- emmeans(model2, ~ Condition)

means
```

```
 Condition      emmean     SE df lower.CL upper.CL
 No Distraction  0.664 0.0236 58    0.617    0.711
 Texting         1.005 0.0236 58    0.958    1.052

Confidence level used: 0.95
```

To make plotting easier, we convert the results into a data frame.

```
means_df <- as.data.frame(means)

means_df
```

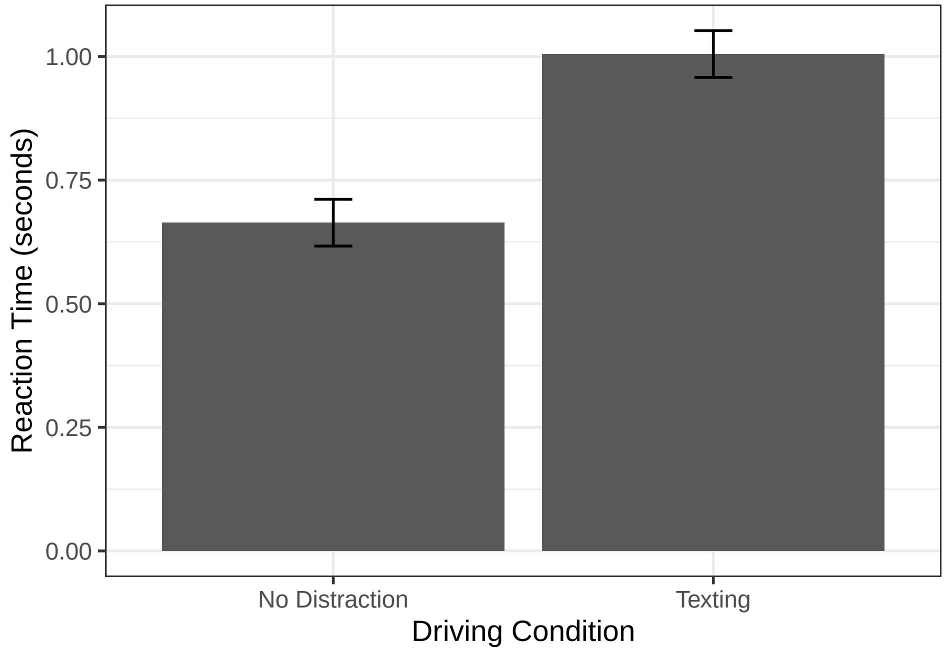
```
 Condition         emmean       SE df  lower.CL  upper.CL
 No Distraction 0.6638792 0.023619 58 0.6166006 0.7111577
 Texting        1.0050267 0.023619 58 0.9577482 1.0523053

Confidence level used: 0.95
```

The data frame contains:

- `emmean` = the estimated mean
- `lower.CL` = lower confidence interval
- `upper.CL` = upper confidence interval

Now we can quickly plot these values using **ggplot**.

```
library(ggplot2)

ggplot(means_df, aes(x = Condition, y = emmean)) +
  geom_col() +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = .1) +
  ylab("Reaction Time (seconds)") +
  xlab("Driving Condition")+theme_bw()
```



This plot shows the **model-estimated means and their 95% confidence intervals**.

Because the means come from `emmeans`, the figure directly represents the regression model we fit earlier.

10.3 Three-Level Example

Now suppose we repeat the study but add a third condition:

Listening to a statistics lecture while driving to class.

10.3.1 Factoring the Three-Level Predictor

Again we convert the predictor to a factor.

```
dat3$Condition <- factor(dat3$Condition,
                         levels = c("No Distraction",
                                    "Texting",
                                    "Lecture"))

levels(dat3$Condition)
```

```
[1] "No Distraction" "Texting"        "Lecture"
```

The first level becomes the reference group.

So again:

```
NoDistraction = reference (0)
```

R now creates **two dummy variables** internally.

Conceptually this looks like:

Condition	Texting	Lecture
No distraction	0	0
Texting	1	0
Lecture	0	1

10.3.2 Multiple t-tests Problem

If we used t-tests, we would need three comparisons:

1. No distraction vs texting

2. No distraction vs lecture

3. Lecture vs texting

Running multiple tests increases the probability of a **Type I error**.

If $\alpha = .05$, then the probability of at least one false positive $(1 - (1 - \alpha)^j)$ [$j = \#$ of t-tests]:

$$1 - (1 - .05)^3$$

```
1 - (1 - .05)^3
```

```
[1] 0.142625
```

This is about **14%**, much larger than the intended 5%. Using regression means we will not increase our type I error in the same way!

10.3.3 Regression with Three Groups

Regression allows us to test all comparisons within a single model.

```
model3 <- lm(RT ~ Condition, data = dat3)
summary(model3)
```

```

Call:
lm(formula = RT ~ Condition, data = dat3)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.31427 -0.09670 -0.00662  0.07988  0.34813 

Coefficients:
```

```
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       0.66388    0.02380  27.898  < 2e-16 ***
ConditionTexting  0.34115    0.03365  10.137  < 2e-16 ***
ConditionLecture  0.19229    0.03365   5.714 1.52e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1303 on 87 degrees of freedom
Multiple R-squared:  0.5429,    Adjusted R-squared:  0.5323
F-statistic: 51.66 on 2 and 87 DF,  p-value: 1.63e-15
```

The model is

$$RT = b_0 + b_1(\text{Texting}) + b_2(\text{Lecture})$$

Interpretation:

- b_0 = mean reaction time for **No Distraction**
- b_1 = difference between **Texting** and **No Distraction**
- b_2 = difference between **Lecture** and **No Distraction**

10.3.4 Connecting the Model to the Means

10.3.5 Reconstructing the Group Means from the Regression Model

Recall that the regression model is

$$RT = b_0 + b_1(\text{Texting}) + b_2(\text{Lecture})$$

From the model output we obtained the following coefficients:

Coefficient	Value
b_0 (Intercept)	0.664
b_1 (Texting)	0.341
b_2 (Lecture)	0.192

Again, the intercept represents the **mean reaction time for the reference group**, which is **No Distraction**.

```r
coef(model3)
```

```
     (Intercept) ConditionTexting ConditionLecture
       0.6638792        0.3411475        0.1922929
```

10.3.6 No Distraction

For the reference group, both dummy variables equal 0.

$$RT = b_0$$

Substituting the coefficient:

$$RT = 0.664$$

So the predicted mean reaction time for **No Distraction** is **0.664 seconds**.

10.3.7 Texting

For the texting group:

Texting $= 1$
Lecture $= 0$

$$RT = b_0 + b_1$$

Substituting the coefficients:

$$RT = 0.664 + 0.341$$

$$RT = 1.005$$

So the predicted mean reaction time for **Texting** is **1.005 seconds**.

10.3.8 Lecture

For the lecture group:

Texting $= 0$
Lecture $= 1$

$$RT = b_0 + b_2$$

Substituting the coefficients:

$$RT = 0.664 + 0.192$$

$$RT = 0.856$$

So the predicted mean reaction time for **Lecture** is **0.856 seconds**.

10.3.9 Prediction summarized

The regression coefficients allow us to **reconstruct the group means directly**.

Condition	Model Equation	Predicted Mean
No Distraction	b_0	0.664
Texting	$b_0 + b_1$	1.005
Lecture	$b_0 + b_2$	0.856

Regression is simply expressing the group means and their differences using an equation.

10.3.10 Extracting the Means

```r
emmeans(model3, ~ Condition)
```

```
 Condition      emmean     SE df lower.CL upper.CL
 No Distraction  0.664 0.0238 87    0.617    0.711
 Texting         1.005 0.0238 87    0.958    1.052
 Lecture         0.856 0.0238 87    0.809    0.903

Confidence level used: 0.95
```

10.3.11 APA Write-Up

A regression model was used to predict reaction time from driving condition.
Because **No Distraction** was entered as the first factor level, it served as the **reference group**, and the intercept represented the mean reaction time for that condition ($M = 0.66$, $SE = 0.02$, 95% CI [0.62, 0.71]).

Relative to this reference group, texting significantly increased reaction time, $b = 0.34$, $SE = 0.03$, $p < .001$. Listening to a statistics lecture also increased reaction time relative to the no-distraction condition, $b = 0.19$, $SE = 0.03$, $p < .001$.

Overall, the model explained a substantial proportion of variance in reaction time, indicating that driving condition accounted for about **54% of the variability in reaction times**.

10.3.12 Extracting the Model-Estimated Means

Just like before, we can extract the model-estimated means using **emmeans**.

```r
library(emmeans)
means3 <- emmeans(model3, ~ Condition)
means3
```

```
 Condition      emmean     SE df lower.CL upper.CL
 No Distraction  0.664 0.0238 87    0.617    0.711
 Texting         1.005 0.0238 87    0.958    1.052
 Lecture         0.856 0.0238 87    0.809    0.903

Confidence level used: 0.95
```

These values represent the **means predicted by the regression model**.

10.3.13 Testing Differences Between Groups

We can also test the differences between all pairs of conditions.

```
pairs(means3, adjust="none")
```

```
 contrast                  estimate     SE df t.ratio p.value
 No Distraction - Texting   -0.341 0.0337 87 -10.137 <0.0001
 No Distraction - Lecture   -0.192 0.0337 87  -5.714 <0.0001
 Texting - Lecture           0.149 0.0337 87   4.423 <0.0001
```

This compares the estimated means for:

- No Distraction vs Texting
- No Distraction vs Lecture
- Lecture vs Texting

All comparisons are derived directly from the regression model. This contains the comparison we did not test earlier (texting vs. lecture). You can add that to your APA report as well.

10.3.14 APA for follow up test

In addition, pairwise comparisons of the estimated marginal means showed that texting produced significantly slower reaction times than listening to a statistics lecture, $b = 0.15$, $SE = 0.03$, $p < .001$. Thus, although both distractions impaired reaction time relative to the no-distraction condition, texting was associated with the greatest slowing of reactions.

10.3.15 Plotting the Model-Estimated Means

We can also visualize the results using a simple plot.

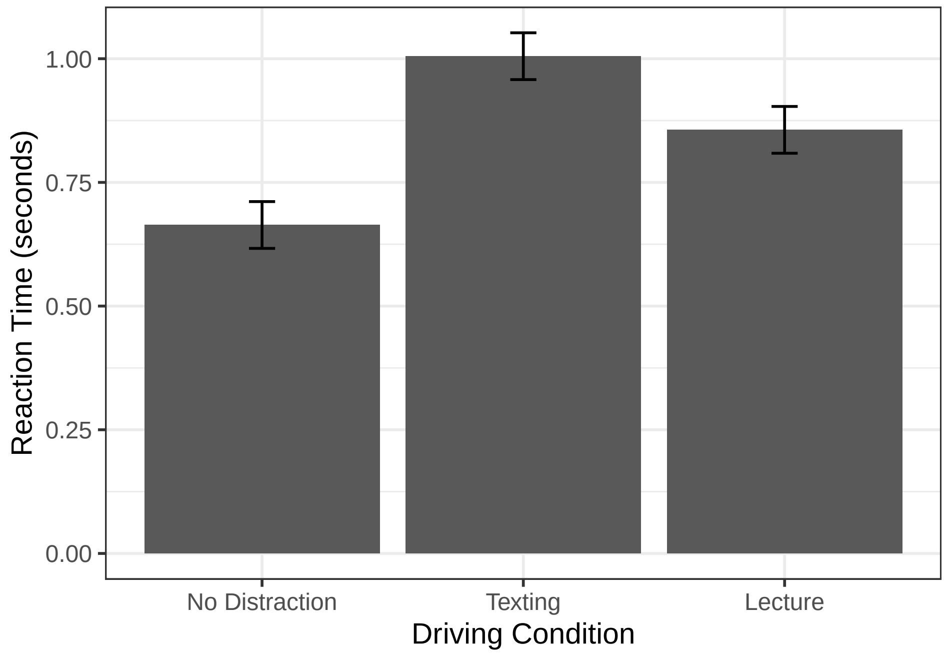
First we convert the estimated means to a data frame.

```
means3_df <- as.data.frame(means3)
means3_df
```

```
 Condition          emmean         SE df  lower.CL  upper.CL
 No Distraction 0.6638792 0.02379701 87 0.6165800 0.7111783
 Texting        1.0050267 0.02379701 87 0.9577276 1.0523259
 Lecture        0.8561720 0.02379701 87 0.8088729 0.9034712

Confidence level used: 0.95
```

Now we plot the estimated means and their 95% confidence intervals.

```r
library(ggplot2)

ggplot(means3_df, aes(x = Condition, y = emmean)) +
  geom_col() +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = .1) +
  ylab("Reaction Time (seconds)") +
  xlab("Driving Condition")+theme_bw()
```



This figure shows the **model-estimated mean reaction time for each driving condition** along with the 95% confidence intervals.

10.4 The Short Story

- R represents a categorical predictor with numeric contrast variables.
- Under ordinary treatment/dummy coding, the intercept is the reference-group mean.
- Each category coefficient is that group's difference from the reference group.
- With two groups, the regression test is equivalent to an independent-samples t-test under the matching variance assumptions.
- With three or more groups, the overall model asks whether the group means are all equal; follow-ups ask which contrasts differ.
- Changing the reference group changes the coefficient labels, not the fitted group means or the underlying data.
- Set factor levels deliberately. Otherwise R may select a reference group alphabetically, because the alphabet has apparently been granted authority over your scientific question.

> **ℹ Coding Changes the Coefficients, Not the People**
>
> Changing the reference group changes which comparison appears in each coefficient. It does not change the fitted group means, residuals, R^2, or the participants—who remain blissfully unaware that R renamed the argument.

11 Multiple Regression: Statistical Control

11.1 Introduction: Why One Predictor Is Never Enough

People are complicated. When a student does well on an exam, is it because they studied? Because they weren't anxious? Because they found the material interesting? It's all of the above — and more.

So far we've used **simple linear regression** (1 DV, 1 IV). But real prediction involves multiple factors at once. **Multiple regression** lets us predict an outcome from two or more predictors simultaneously.

The key concept: **statistical control** — estimating the association between X and Y while *holding other measured variables constant*. This lets us ask not just "does studying predict performance?" but "does studying predict performance **above and beyond** what anxiety alone explains?"

Without statistical control, you can't untangle those two things. With multiple regression, you can.

11.2 A Study on Exam Performance

Let's ground this in a specific example. Imagine you are a researcher interested in what predicts student exam performance. After reading the literature, you identify two strong candidates:

1. **Study Hours** (X1) — how many hours per week a student spends studying for the exam. More studying should lead to better scores.
2. **Test Anxiety** (X2) — how anxious a student feels about the exam (measured on a 0–100 scale, where 0 = no anxiety and 100 = extreme anxiety). More anxiety is expected to hurt performance.

You recruit 100 college students and collect these measures before a major exam. After the exam, you record their score (0–100).

11.2.1 The Complication: Our Two Predictors Are Correlated

Here is the twist. It turns out that Study Hours and Test Anxiety are **not independent of each other**. Students who study more tend to feel *less* anxious — perhaps because preparation builds confidence, or because anxious students have trouble sitting down to study in the first place.

This means:

- Students with high Study Hours tend to have low Test Anxiety.
- Students with low Study Hours tend to have high Test Anxiety.

In other words, Study Hours and Test Anxiety are **negatively correlated** with each other.

This creates a problem. If you only look at Study Hours → Exam Score, some of the apparent benefit of studying might actually be coming from lower anxiety — not from the studying itself. And if you only look at Anxiety → Exam Score, some of the apparent harm of anxiety might be coming from studying *less*, not from the anxiety per se.

Multiple regression lets us disentangle these overlapping influences.

11.2.2 Expected Correlations

Here are the correlations we expect to find in our data:

	Study Hours	Test Anxiety	Exam Score
Study Hours	—	$-.35$	$+.45$
Test Anxiety	$-.35$	—	$-.40$
Exam Score	$+.45$	$-.40$	—

- Study Hours *positively* predicts exam scores ($r = .45$): more studying, better scores.
- Test Anxiety *negatively* predicts exam scores ($r = -.40$): more anxiety, lower scores.
- Study Hours and Test Anxiety are *negatively correlated* ($r = -.35$): students who study more tend to be less anxious.

That third correlation is the key complication — and the reason we need multiple regression.

11.2.3 Simulating the Data

Always look at your data before running any analysis. Let's check the first few rows and the summary statistics.

```r
head(dat)
```

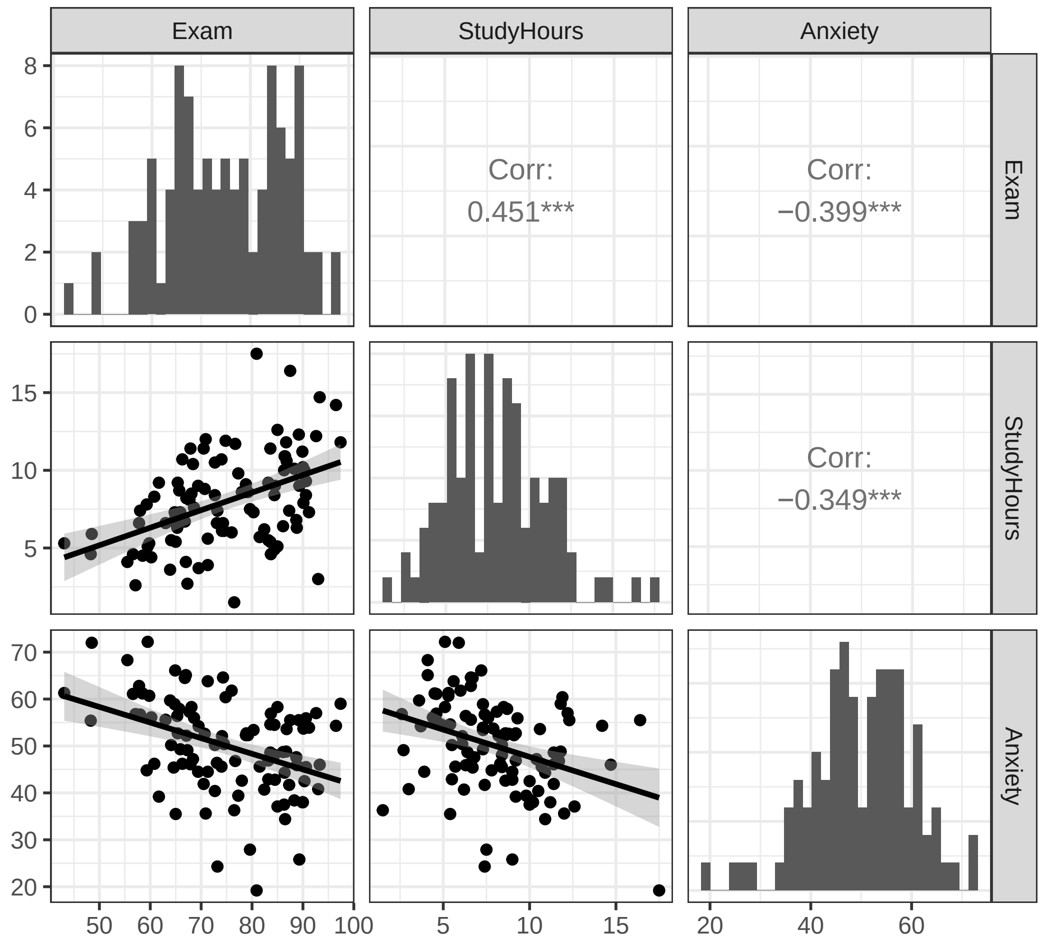
```
  Exam StudyHours Anxiety
1 43.1        5.3    61.3
2 72.7       10.5    40.4
3 88.8        6.3    45.9
4 73.2        7.4    24.3
5 66.8        6.7    64.5
6 65.7        8.7    57.9
```

```r
summary(dat)
```

```
      Exam         StudyHours        Anxiety
 Min.   :43.10   Min.   : 1.500   Min.   :19.20
 1st Qu.:65.85   1st Qu.: 5.850   1st Qu.:44.45
 Median :74.40   Median : 7.700   Median :50.30
 Mean   :75.00   Mean   : 7.999   Mean   :50.00
 3rd Qu.:85.28   3rd Qu.:10.025   3rd Qu.:56.73
 Max.   :97.40   Max.   :17.500   Max.   :72.20
```

And here is a pairs plot showing all three pairwise relationships simultaneously. Each panel in the lower triangle shows a scatterplot, and the diagonal shows histograms for each variable.

```
GGally::ggpairs(
  dat,
  lower = list(continuous = "smooth"),
  diag  = list(continuous = "barDiag")
) + theme_bw()
```



Here is what that code asks R to do:

- `GGally::ggpairs()` builds a matrix containing every pair of variables. The `GGally::` prefix tells R exactly which package owns the function.
- `dat` is the data frame whose variables enter the matrix.
- `lower = list(continuous = "smooth")` places scatterplots and fitted smooth lines in the lower triangle.
- `diag = list(continuous = "barDiag")` places a histogram for each variable on the diagonal.
- `theme_bw()` removes some visual clutter so the relationships are easier to inspect.

The plot is compact because each variable takes a turn as both an x-axis and a y-axis. Read one panel at a time. Do not stare at the whole matrix until a causal theory appears.

You can see all three correlations in the scatterplots:

- **Bottom left:** Study Hours and Test Anxiety slope downward — students who study more tend to be less anxious.

- **Bottom middle:** Test Anxiety and Exam Score slope downward — more anxious students tend to score lower.
- **Bottom right:** Study Hours and Exam Score slope upward — more studying is associated with higher scores.

11.3 Understanding Overlapping Variance

Before we run the multiple regression, we need to understand *why* having two correlated predictors creates the problem we described above. The key concept is **overlapping variance**.

11.3.1 The Venn Diagram Approach

When you use simple regression with a single predictor, things are straightforward: X either predicts Y or it does not. We can visualize this as two circles — one for X and one for Y — where the *overlap* represents the shared variance between them. As you already know, that overlap, when squared, gives us R^2.

When we add a second predictor, things get more interesting. Now we have three circles — X1, X2, and Y — and some of the circles overlap with each other:

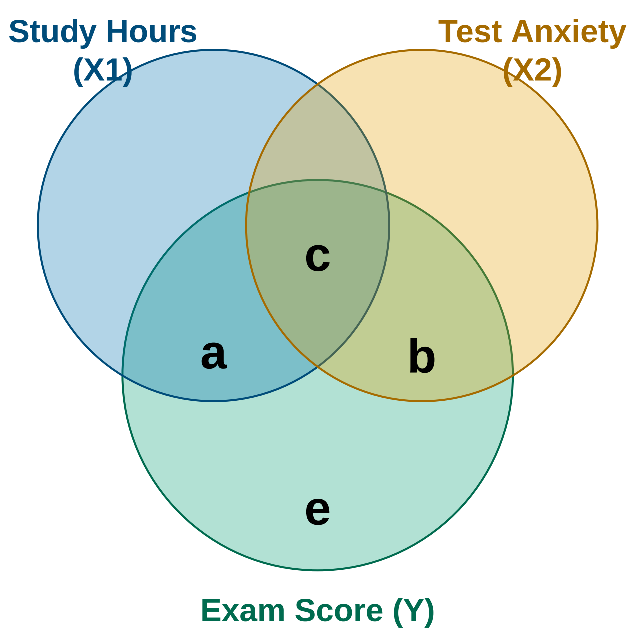


Figure 11.1: A Ballantine for Study Hours, Test Anxiety, and Exam Score. The areas are conceptual, not drawn to scale.

In this diagram, we can label four distinct regions:

- **Region a** = variance in Y that is **uniquely explained by X1** (Study Hours), above and beyond X2.
- **Region b** = variance in Y that is **uniquely explained by X2** (Test Anxiety), above and beyond X1.
- **Region c** = variance in Y that **both X1 and X2 share** — neither predictor can claim exclusive credit for this part.
- **Region e** = variance in Y that is **unexplained** by either predictor (residual error).

The total variance in Y explained by the model is $a + b + c$.

11.3.2 The Problem with Region c

Region c is the heart of the matter. It represents variance in exam scores that both Study Hours and Anxiety are *jointly* associated with, because the two predictors are correlated with each other.

Here is the problem that region c creates:

- If you only look at **Study Hours → Exam** (simple regression), the observed relationship includes region a *and* region c. Part of what looks like a "Study Hours effect" is actually shared with Anxiety, because students who study more tend to be less anxious, and less anxious students score higher.
- If you only look at **Anxiety → Exam** (simple regression), the observed relationship includes region b *and* region c. Part of what looks like an "Anxiety effect" is actually shared with Study Hours, for the same reason.

So the bivariate slopes mix unique and shared information. In this particular example, each slope is larger in absolute value when examined alone because it also has access to variance shared with the other predictor.

Multiple regression solves this by estimating each predictor's unique contribution. In the multiple regression model, Study Hours gets credit only for region a, and Anxiety gets credit only for region b. Region c is acknowledged as shared variance and included in R^2, but it is not falsely assigned to either predictor.

In these simulated data, the multiple-regression slopes will be **smaller** than the corresponding simple-regression slopes. That is not a universal rule. Depending on the correlation pattern, adding a control can make a slope shrink, grow, or reverse. The controlled slopes answer a different question: what does each predictor contribute conditionally, given the other predictor in the model?

11.4 The Multiple Regression Equation

You already know the simple regression equation:

$$Y = B_0 + B_1 X + e$$

Multiple regression extends this to two (or more) predictors:

$$Y = B_0 + B_1 X_1 + B_2 X_2 + e$$

Where:

- B_0 = intercept (predicted Y when all predictors = 0)
- B_1 = slope for X1 (Study Hours)
- B_2 = slope for X2 (Test Anxiety)

- e = residual error (observed − predicted)

In our example:

$$\widehat{Exam} = B_0 + B_1 \cdot StudyHours + B_2 \cdot Anxiety$$

The equation looks very similar to simple regression. But the **interpretation of the slopes has changed in an important way**.

11.4.1 What "Controlling For" Means

In simple regression, B_1 tells you: *"For each one-unit increase in X, how much does Y change?"*

In multiple regression, B_1 tells you something more precise: *"For each one-unit increase in X1, how much does Y change — **among people who have the same value of X2**?"*

This is what **statistical control** means. The slope for Study Hours in our multiple regression model describes the Study Hours–Exam association *as if we were comparing students at the same anxiety level.* We are estimating that association after holding anxiety **constant** — or equivalently, after **controlling for** anxiety.

Think of it this way: imagine sorting all 100 students into groups based on their anxiety score. Within the low-anxiety group, every student has roughly the same anxiety, so exam differences cannot be attributed to measured anxiety differences inside that group. They may be associated with study hours, or with other variables we did not measure. Now imagine doing the same thing for the medium-anxiety group and the high-anxiety group. Multiple regression performs a continuous version of this comparison for every value of Anxiety to isolate the unique predictive contribution of Study Hours.

The key insight: a slope in multiple regression is the association between that predictor and Y after accounting for the predictor's linear relationship with the other predictors in the model. It is the unique predictive contribution of X1 above and beyond what X2 already explains. It is not automatically a causal effect.

11.5 Running the Analysis in R

Let's run the analysis step by step. We will start with simple regression for each predictor alone, then combine them. This approach lets you see exactly what changes — and why — when you add the second predictor.

11.5.1 Step 1: Study Hours Alone

```r
Model.1 <- lm(Exam ~ StudyHours, data = dat)
summary(Model.1)
```

```
Call:
lm(formula = Exam ~ StudyHours, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-27.041  -8.274   1.174   8.799  26.996 
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  60.6078     3.0717  19.731  < 2e-16 ***
StudyHours    1.7987     0.3597   5.001  2.5e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.76 on 98 degrees of freedom
Multiple R-squared:  0.2033,    Adjusted R-squared:  0.1952
F-statistic: 25.01 on 1 and 98 DF,  p-value: 2.501e-06
```

So, looking at Study Hours alone:

- For every additional hour of studying per week, exam scores increase by about 1.8 points.
- This is a statistically significant positive effect.
- Study Hours alone explains about 20.3% of the variance in exam scores ($R^2 = 0.203$).

11.5.2 Step 2: Test Anxiety Alone

```r
Model.2 <- lm(Exam ~ Anxiety, data = dat)
summary(Model.2)
```

```
Call:
lm(formula = Exam ~ Anxiety, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-26.482  -8.321  -1.005   8.369  26.716 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  98.9535     5.6637  17.472  < 2e-16 ***
Anxiety      -0.4792     0.1111  -4.313 3.84e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.06 on 98 degrees of freedom
Multiple R-squared:  0.1595,    Adjusted R-squared:  0.151
F-statistic:  18.6 on 1 and 98 DF,  p-value: 3.844e-05
```

Looking at Test Anxiety alone:

- For every 1-point increase in test anxiety, exam scores decrease by about 0.48 points.
- This is also statistically significant.
- Anxiety alone explains about 16% of the variance ($R^2 = 0.16$).

11.5.3 Step 3: Both Predictors Together

Now for the key step — entering both predictors simultaneously. In R, we add a second predictor using the **+** sign in the formula:

```r
Model.3 <- lm(Exam ~ StudyHours + Anxiety, data = dat)
summary(Model.3)
```

```

Call:
lm(formula = Exam ~ StudyHours + Anxiety, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-24.3420  -7.8649  -0.3942   7.4378  22.0359 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  80.2059     7.2176  11.112  < 2e-16 ***
StudyHours    1.4149     0.3693   3.831 0.000226 ***
Anxiety      -0.3306     0.1111  -2.976 0.003684 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.36 on 97 degrees of freedom
Multiple R-squared:   0.27, Adjusted R-squared:  0.2549 
F-statistic: 17.94 on 2 and 97 DF,  p-value: 2.351e-07
```

11.6 Interpreting the Multiple Regression Output

Let's work through what changed when we added both predictors.

11.6.1 The Slopes Have Shrunk — Why?

Compare the slope estimates across all three models:

Predictor	Simple Regression (Alone)	Multiple Regression (Controlling for the Other)
Study Hours	1.799	1.415
Test Anxiety	-0.479	-0.331

Both slopes are smaller in absolute value when the two predictors are in the model together. This is exactly what the Venn diagram predicts. Here is why:

When Study Hours was the *only* predictor (Model 1), its prediction included region a *and* region c — its unique association with exam performance plus variance shared with Anxiety. Once both predictors enter Model 3, the Study Hours coefficient is based on Study Hours variation that Anxiety does not linearly predict. Region c still contributes to the model's overall R^2, but neither coefficient can claim it as uniquely its own.

The same logic applies in reverse: Anxiety's slope shrinks from -0.479 to -0.331 because some of its bivariate association with Exam Score overlaps with lower Study Hours.

This shrinkage is **not a problem**, nor is the model cold. Model 3 is answering the conditional question we asked. Its slopes estimate each predictor's unique association with Exam Score given the other measured predictor. They are not automatically "truer," more causal, or less biased; that depends on the research design, measurement quality, and whether controlling for that variable makes theoretical sense.

11.6.2 Interpreting Each Controlled Slope

Here is how to interpret each slope in Model 3 in plain language:

Study Hours ($B_1 = 1.41$):

> "For students with the *same* level of test anxiety, each additional hour of weekly study is associated with a 1.41-point increase in exam score."

Test Anxiety ($B_2 = -0.33$):

> "For students who study the *same* number of hours per week, each one-point increase in test anxiety is associated with a 0.33-point decrease in exam score."

The phrase **"for students with the same [other variable]"** is the key to interpreting any slope in multiple regression. When you see a slope reported in a multiple regression model, it always carries that "holding everything else constant" qualifier — even when it is not stated explicitly.

11.6.3 The Intercept

The intercept ($B_0 = 80.21$) is the predicted exam score when **both** Study Hours = 0 **and** Anxiety = 0. Since no student in our sample studied for zero hours *and* had zero anxiety simultaneously, this is not a prediction we would ever actually make. Like in simple regression, the intercept just anchors the equation mathematically. It becomes more meaningful when predictors are mean-centered (but we will save that for another time).

11.6.4 Making a Prediction

Now that we have the full equation, we can make predictions. For a student who studies 10 hours per week and has an anxiety score of 45:

```
# Predicted exam score for a student with 10 study hours and anxiety = 45
B0 <- Model.3$coefficients[1]
B1 <- Model.3$coefficients[2]
B2 <- Model.3$coefficients[3]

predicted <- B0 + B1 * 10 + B2 * 45
```

This student would be predicted to score about 79.5 on the exam.

11.7 How Well Does the Model Fit? R²

Let's look at how the model's explanatory power (R^2) changes across our three models.

```r
r2_model1 <- summary(Model.1)$r.squared
r2_model2 <- summary(Model.2)$r.squared
r2_model3 <- summary(Model.3)$r.squared
```

```
Model 1 (Study Hours only): R-squared = 0.203
```

```
Model 2 (Anxiety only):     R-squared = 0.16
```

```
Model 3 (Both predictors):  R-squared = 0.27
```

The full model ($R^2 = 0.27$) explains more variance than either predictor alone. But notice something: the combined model does **not** explain as much as the simple sum of the two individual R^2 values ($0.203 + 0.16 = 0.363$). Why not?

Go back to the Venn diagram. When Study Hours was the only predictor, it explained regions $a + c$. When Anxiety was the only predictor, it explained regions $b + c$. Together, they explain regions $a + b + c$. But notice that region c was already counted in Model 1's R^2. Adding Anxiety on top of Study Hours can only add region b to what was already explained — it cannot double-count region c.

This is why R^2 does not add up neatly when predictors are correlated. The combined model explains more than either predictor alone, but less than the naive sum of their individual R^2 values.

11.7.1 Breaking Down the Venn Diagram Numerically

We can actually estimate the size of each region in the Venn diagram using simple subtraction:

```r
# Unique variance of Study Hours (region a)
region_a <- r2_model3 - r2_model2

# Unique variance of Anxiety (region b)
region_b <- r2_model3 - r2_model1

# Shared variance (region c)
region_c <- r2_model1 + r2_model2 - r2_model3
```

```
Region a - unique to Study Hours: 0.11
```

```
Region b - unique to Anxiety:     0.067
```

```
Region c - shared variance:       0.093
```

```
Total R-squared:                  0.27
```

In our data:

- Study Hours has about 11% unique variance in exam scores — its unique predictive contribution after accounting for anxiety (region a).
- Anxiety has about 6.7% unique variance — its unique predictive contribution after accounting for study hours (region b).
- About 9.3% of the variance in exam scores is shared between the two predictors — variance that both contribute to together, but neither can claim exclusively (region c).

This numerical breakdown puts the Venn diagram concept in concrete terms. The **squared semi-partial correlations** correspond directly to regions a and b. The regression slopes are based on the same unique predictor variation, but express the expected outcome change in Exam Score units.

11.7.2 Adding Predictors Always Increases R²

There is an important quirk about R^2: it will **always increase** when you add another predictor to the model — even if that predictor is completely useless random noise. This is simply because any variable, no matter how irrelevant, will share at least a tiny sliver of variance with Y by chance in any given sample.

This means if you kept piling more and more predictors into a model, R^2 would keep creeping upward — not because those predictors were genuinely meaningful, but just because of sampling variability.

11.7.3 Adjusted R²

Adjusted R^2 corrects for this problem by applying a penalty for each additional predictor:

$$\text{Adjusted } R^2 = 1 - (1 - R^2)\,\frac{n-1}{n-k-1}$$

Where n = sample size and k = number of predictors in the model. With a small sample and many predictors, the penalty is larger. With a large sample and few predictors, the penalty is small.

Raw R^2 describes how well the model fits **this sample**. Adjusted R^2 asks a slightly less flattering question: after penalizing us for the number of predictors we used, how much variance does the model appear to explain? Because raw R^2 capitalizes on accidental relationships in the current sample, adjusted R^2 is usually closer to what we might expect in the population or a new sample. It is not a prophecy, but it is less easily impressed.

Adjusted R^2 can increase when a new predictor improves the model enough to overcome the penalty. It can decrease when the added predictor contributes too little. This makes it a fairer comparison tool when models contain different numbers of predictors.

```r
adj_r2_model1 <- summary(Model.1)$adj.r.squared
adj_r2_model3 <- summary(Model.3)$adj.r.squared
```

```
Model 1 Adjusted R-squared: 0.195
```

```
Model 3 Adjusted R-squared: 0.255
```

Our adjusted R^2 increased from 0.195 to 0.255 when we added Anxiety. This confirms that Anxiety is genuinely useful — after the penalty for adding a predictor, the model still explains meaningfully more variance.

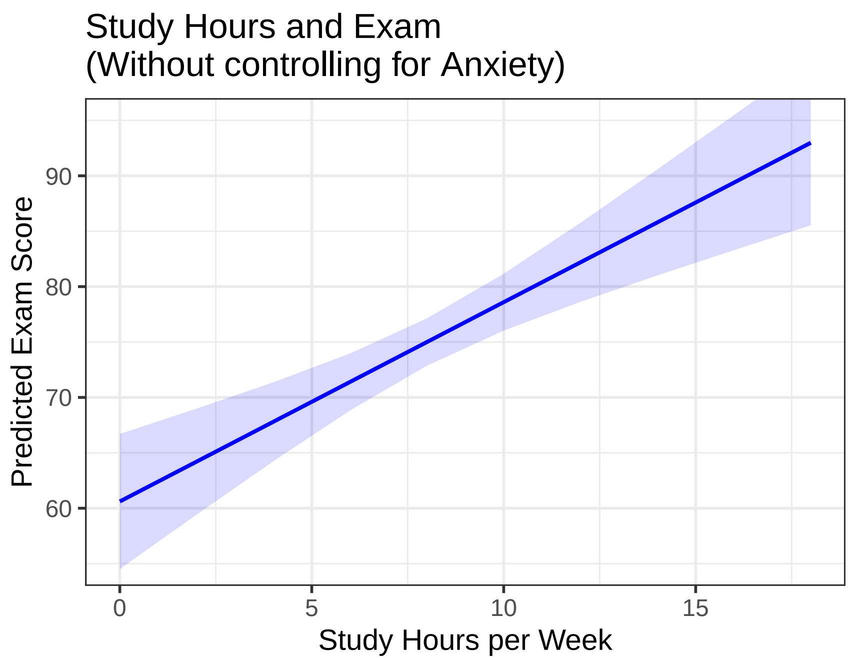
People commonly report **both** R^2 and adjusted R^2. Raw R^2 describes the variance explained in the observed sample. Adjusted R^2 provides the more conservative estimate after accounting for model size and is often the better guess about how much explanatory power will survive beyond this particular group of exploited undergraduates.

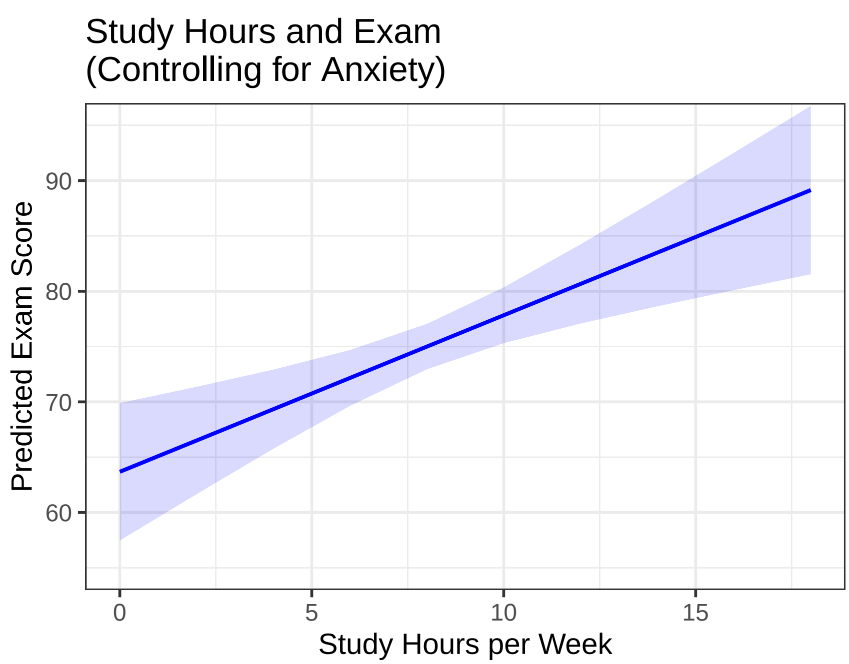
11.8 Visualizing Multiple Regression

Tables of numbers are useful, but plots help build intuition. Let's visualize what is happening.

11.8.1 The Study Hours Slope — Before and After Controlling

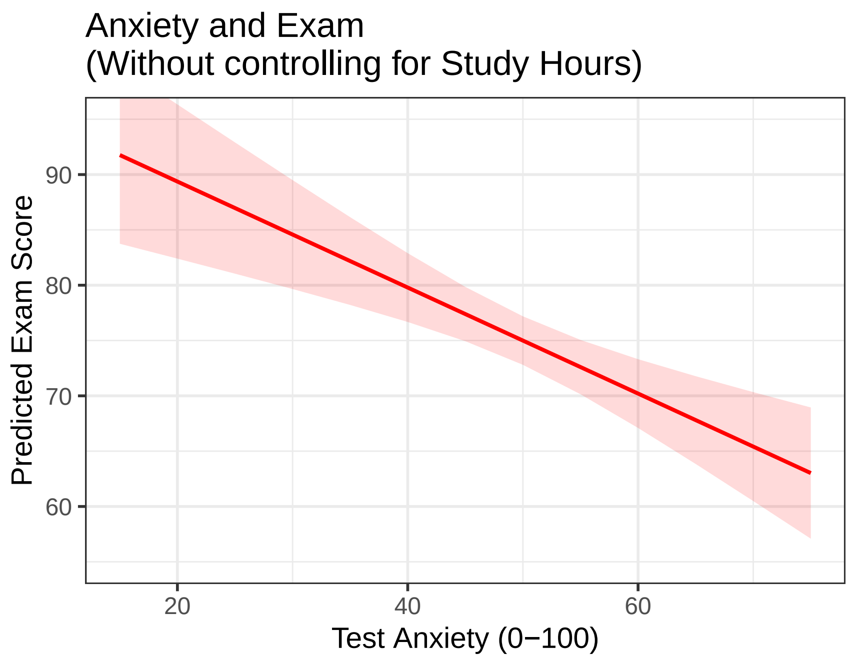
This first pair of plots shows the predicted relationship between Study Hours and Exam Score in the two models. The key question is: how does the slope change when we control for Anxiety?

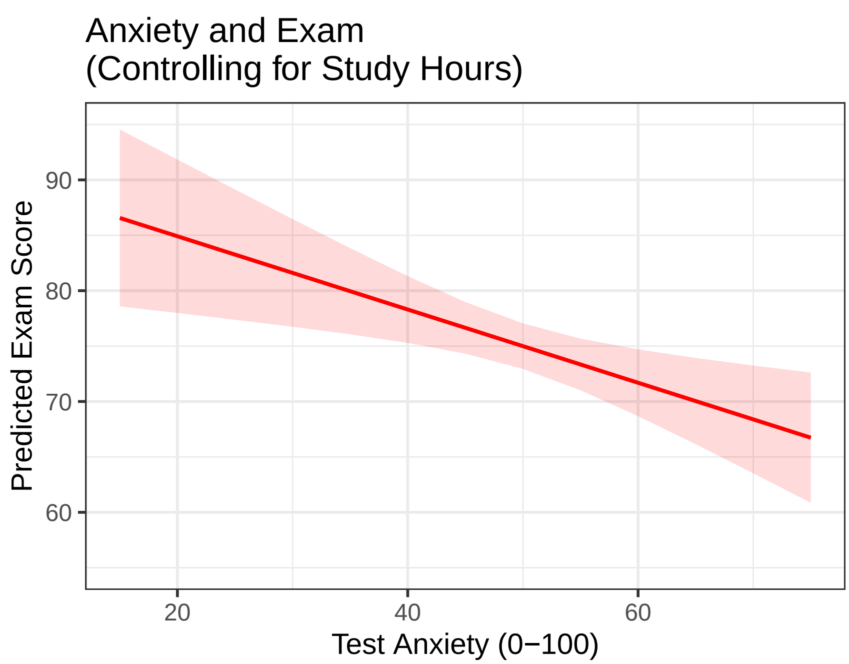




The slope becomes slightly shallower after controlling for Anxiety. This is statistical control in action: some of the original Study Hours effect was shared with (and partially driven by) lower anxiety levels in high-studying students. The controlled slope is more precise — it captures only what studying contributes, independent of anxiety.

11.8.2 The Anxiety Slope — Before and After Controlling

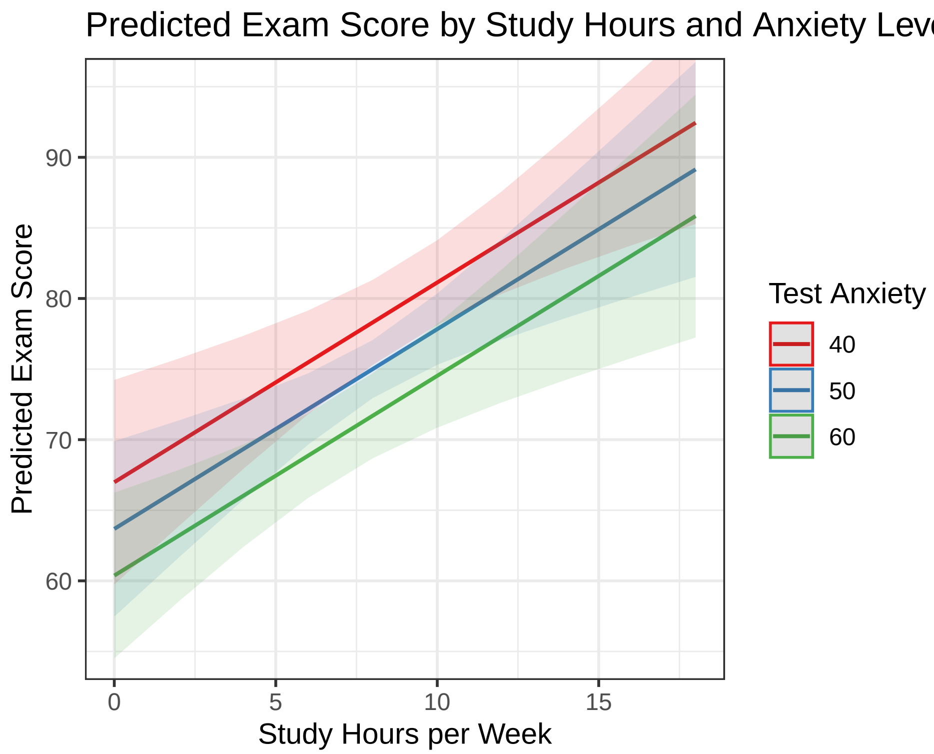




The same pattern appears in reverse: the negative slope for Anxiety becomes slightly smaller after we account for the fact that highly anxious students also tend to study less. The controlled slope describes Anxiety's conditional association with performance while Study Hours is held constant.

11.8.3 The Full Model: Both Predictors Together

We can also visualize both predictors simultaneously. The plot below shows how predicted exam scores change with Study Hours, for students at three different anxiety levels (low, average, and high).



Each line represents students at a different anxiety level: low anxiety (40), average anxiety (50), and high anxiety (60). Notice:

- **More studying always helps** — all three lines slope upward, regardless of anxiety level.
- **Higher anxiety always hurts** — the lines are shifted downward for higher-anxiety students.
- **The lines are parallel** — the predicted difference associated with each additional study hour is the same at every anxiety level. This is what it means for the terms to be **additive** in our model: the model contains separate Study Hours and Anxiety slopes but no interaction between them.

This plot gives you a visual intuition for what "controlling for" means: the slope of each line within one anxiety group represents the Study Hours association holding Anxiety constant — exactly what the multiple regression coefficient estimates.

11.9 A Note on What Did Not Change

In Model 3, **both predictors remain statistically significant** after controlling for each other. The fact that both slopes shrank does not mean either predictor is unimportant. Both retain unique predictive associations with Exam Score: Study Hours contributes above and beyond Anxiety (region a), and Anxiety contributes above and beyond Study Hours (region b).

The meaning: **a slope that shrinks when you add a control variable is not automatically a sign of a weak finding.** It means the controlled and uncontrolled models answer different questions. In this example, the controlled slope is based on predictor variance not linearly explained by the other predictor.

Adding a control variable can also make a slope grow, disappear, reverse, or change statistical significance. Those patterns can reflect confounding, suppression, collinearity, measurement problems, or a poorly chosen control. The next chapter takes apart what statistical control is actually doing.

11.10 APA Write-Up

Here is how you would write up these results in APA format:

Prior to the main analysis, zero-order correlations were examined. Study hours positively correlated with exam performance ($r = .45$, $p < .001$), and test anxiety negatively correlated with exam performance ($r = -.40$, $p < .001$). The two predictors were also significantly correlated with each other ($r = -.35$, $p < .001$), indicating that students who studied more tended to report lower anxiety.

A multiple linear regression was conducted with exam score as the outcome and study hours and test anxiety as predictors. The overall model was statistically significant, $F(2, 97) = 17.94$, $p < .001$, $R^2 = 0.27$, adjusted $R^2 = 0.255$.

After controlling for test anxiety, study hours significantly predicted exam performance, $b = 1.41$, $SE = 0.37$, $t(97) = 3.83$, $p < .001$. After controlling for study hours, test anxiety significantly predicted exam performance, $b =$ -0.33, $SE =$ 0.11, $t(97) =$ -2.98, $p < 0.004$.

Together, these results indicate that Study Hours and Test Anxiety each uniquely predict Exam Score in this model: students who study more and experience less anxiety tend to perform better, and both associations remain after accounting for the correlation between the predictors.

A few things to notice in this write-up:

- We tend to report the zero-order correlations first, so readers understand the bivariate relationships before seeing the controlled effects.
- We use the phrase **"after controlling for…"** (or equivalently, "above and beyond" or "holding constant") to signal that a multiple regression slope is being reported.
- We report F, R^2, and adjusted R^2 for the overall model, then report b, SE, t, and p for each individual predictor.
- We do not report Cohen's d here. In regression, b (the unstandardized slope) is the primary effect size. The overall model's R^2 tells us the total proportion of variance explained.

Next: We Take Control Apart

The next chapter, Statistical Control: What Are We Actually Removing?, uses these same students to show exactly how residuals, semi-partial correlation, partial correlation, standardized beta, and change in R^2 fit together. No new imaginary undergraduates will be recruited until the current ones have finished the exam.

12 Statistical Control: What Are We Actually Removing?

⚠ Advanced Topic: Your Brain May File a Complaint

This is difficult material. Graduate students routinely stare at partial and semi-partial correlations as if the correlations have personally betrayed them. If this chapter requires two readings, that does not mean you are bad at statistics. It means we are removing different pieces of relationships and then giving the leftovers nearly identical names.

Keep asking three questions:

1. What variable are we controlling?
2. Which variable or variables are we removing it from?
3. What does the resulting number mean?

12.1 Statistical Control Is Not a Statistics Jail

In the previous chapter, Multiple Regression: Statistical Control, we predicted **Exam Score** from **Study Hours** and **Test Anxiety**. Study Hours and Anxiety were related to one another, which meant their relationships with Exam Score overlapped.

Multiple regression gave us a Study Hours slope while *controlling for* Anxiety. That phrase sounds reassuringly official. It also hides nearly all the interesting work.

We did not put Anxiety in a tiny statistical jail. We did not force every student to experience precisely 50 units of Anxiety. We used a model to remove the **linear association** Anxiety had with another variable and then studied what remained.

This chapter takes that process apart. We will use the exact same example and simulated data as the multiple-regression chapter, because changing the variables while explaining an already difficult idea would be rude.

Our main question will remain:

> Do Study Hours predict Exam Score after accounting for Test Anxiety?

12.2 The Same Exam Study

The expected correlations are:

Relationship	Correlation	Translation
Study Hours with Exam Score	+.45	Students who study more tend to score higher.
Test Anxiety with Exam Score	−.40	Students who are more anxious tend to score lower.
Study Hours with Test Anxiety	−.35	Students who study more tend to be less anxious.

That last relationship creates the control problem. When Study Hours predicts Exam Score, part of that association may travel through the fact that students who study more also tend to be less anxious.

```r
library(MASS)

set.seed(42)

n   <- 100
rY1 <-  0.45  # Study Hours with Exam Score
rY2 <- -0.40  # Test Anxiety with Exam Score
r12 <- -0.35  # Study Hours with Test Anxiety

Sigma <- matrix(c(
   1,   rY1, rY2,
  rY1,   1,  r12,
  rY2,  r12,  1
), ncol = 3)

raw <- as.data.frame(
  mvrnorm(
    n,
    mu = c(0, 0, 0),
    Sigma = Sigma,
    empirical = TRUE
  )
)

colnames(raw) <- c("Exam", "StudyHours", "Anxiety")

dat <- data.frame(
  Exam       = round(raw$Exam * 12 + 75, 1),
  StudyHours = round(raw$StudyHours * 3 + 8, 1),
  Anxiety    = round(raw$Anxiety * 10 + 50, 1)
)
```

```r
round(cor(dat[c("StudyHours", "Anxiety", "Exam")]), 2)
```

```
           StudyHours Anxiety  Exam
StudyHours       1.00   -0.35  0.45
Anxiety         -0.35    1.00 -0.40
Exam             0.45   -0.40  1.00
```

12.3 The Ballantine, Now Made Fresh in R

The circles below represent variance. The overlapping portions represent shared variance. This diagram is called a **Ballantine** because apparently three overlapping circles needed a name, and statisticians had already used "Venn diagram" for something fun at parties.

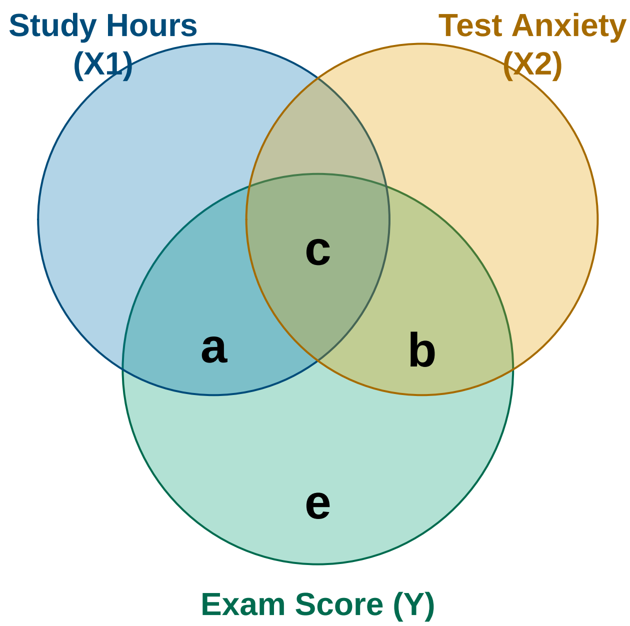


Figure 12.1: A Ballantine for Study Hours, Test Anxiety, and Exam Score. The areas are conceptual, not drawn to scale.

- **Region** a is Exam Score variance uniquely associated with Study Hours.
- **Region** b is Exam Score variance uniquely associated with Test Anxiety.
- **Region** c is Exam Score variance shared by both predictors. Neither predictor gets exclusive custody.
- **Region** e is Exam Score variance the full model does not explain.

The full regression model explains $a + b + c$. Our main target in this chapter is region a: what Study Hours contributes above and beyond Anxiety.

Circles Are a Metaphor

The areas are not drawn to scale, variables are not actually circular, and residuals do not fall onto the floor when we remove them. The diagram helps us track which variance belongs in each calculation. Do not demand anatomical accuracy from it.

12.4 Residuals: The Leftovers

Control begins with a familiar regression. First, predict Study Hours from Anxiety:

```
study_from_anxiety <- lm(StudyHours ~ Anxiety, data = dat)
summary(study_from_anxiety)
```

```
Call:
lm(formula = StudyHours ~ Anxiety, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-7.9379 -1.8870  0.1195  1.4007  8.9786

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 13.25031    1.45120   9.131 9.33e-15 ***
Anxiety     -0.10503    0.02847  -3.690 0.000369 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.833 on 98 degrees of freedom
Multiple R-squared:  0.122, Adjusted R-squared:  0.113
F-statistic: 13.61 on 1 and 98 DF,  p-value: 0.0003691
```

```
dat$ExpectedStudy <- fitted(study_from_anxiety)
dat$StudyResidual <- residuals(study_from_anxiety)
```

For every student:

$$\text{Study residual} = \text{Actual Study Hours} - \text{Expected Study Hours}$$

Suppose an entirely hypothetical student named Alex has an Anxiety score of 60, studies 10 hours, and earns a 78 on the exam. There is no connection between this fictional Alex and the author except the name, profession, and desperate wish that ten hours of studying always happened.

```
alex_anxiety <- 60
alex_actual_study <- 10
alex_actual_exam <- 78

alex_expected_study <- unname(
  predict(study_from_anxiety, newdata = data.frame(Anxiety = alex_anxiety))
)

alex_study_residual <- alex_actual_study - alex_expected_study

alex_table <- data.frame(
```

```r
  Anxiety = alex_anxiety,
  Expected_Study = round(alex_expected_study, 1),
  Actual_Study = alex_actual_study,
  Study_Residual = round(alex_study_residual, 1)
)

knitr::kable(alex_table)
```

Anxiety	Expected_Study	Actual_Study	Study_Residual
60	6.9	10	3.1

Given an Anxiety score of 60, the model expected Alex to study about 6.9 hours. Alex studied 10 hours, so the Study Hours residual is about +3.1. Alex studied that many hours **more than expected given Anxiety**.

A positive residual means "more than expected." A negative residual means "less than expected." A residual of zero means the model finally got one exactly right and will be insufferable about it.

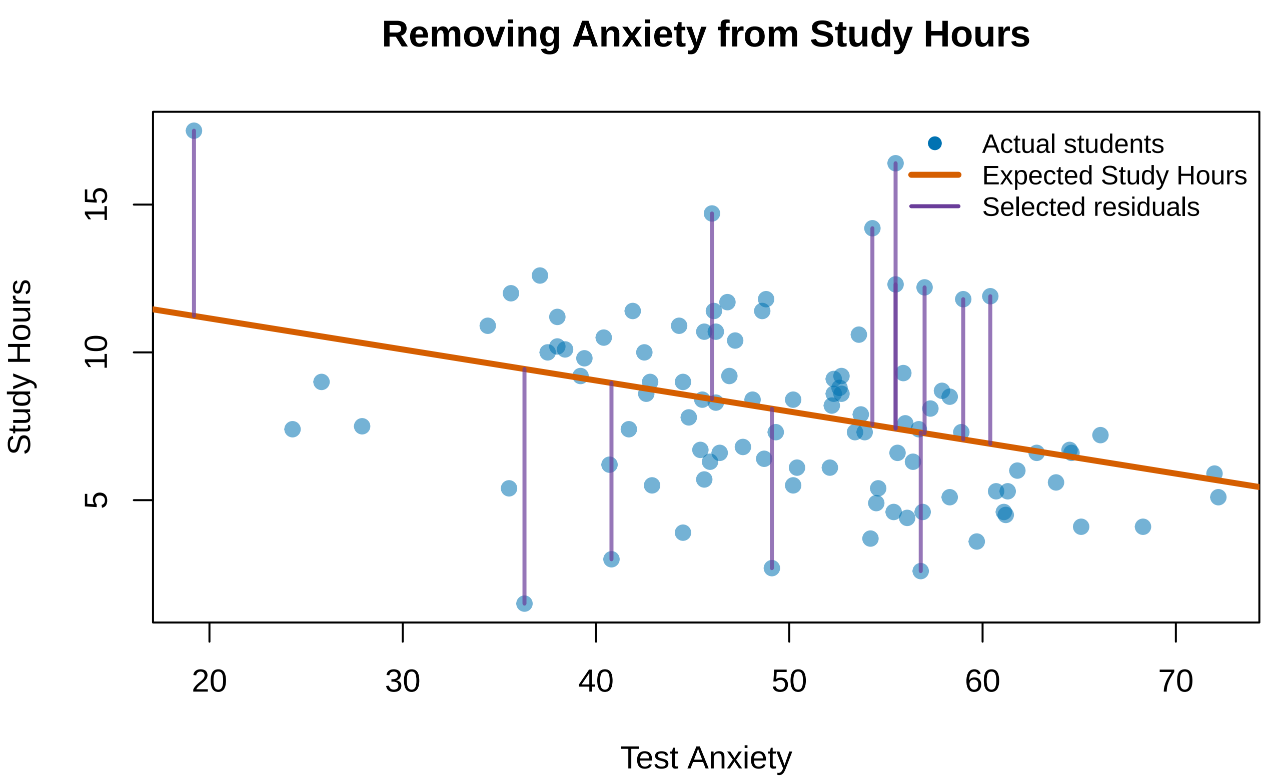


Figure 12.2: Study Hours predicted from Test Anxiety. Selected vertical lines show residuals: the distance between each student's actual and expected Study Hours.

Anxiety has now explained whatever linear association it can explain in Study Hours. The Study Hours residuals are the leftovers. Notice what happens if we correlate those leftovers with Anxiety:

```r
cor(dat$Anxiety, dat$StudyResidual)
```

```
[1] 3.582582e-18
```

It is essentially zero. That is not luck. Least-squares regression creates residuals that are uncorrelated with the predictors used to create them.

12.5 Semi-Partial Correlation: Remove Anxiety from One Side

We will learn **semi-partial correlation first** because it connects directly to multiple regression.

To find the semi-partial relationship between Study Hours and Exam Score while controlling for Anxiety:

1. Predict Study Hours from Anxiety.
2. Save the Study Hours residuals.
3. Leave Exam Score completely alone.
4. Correlate residualized Study Hours with the original Exam Score.

```r
sr_study <- cor(dat$Exam, dat$StudyResidual)
sr_study
```

```
[1] 0.3323602
```

The semi-partial correlation is $sr = 0.332$.

Its plain-language question is:

> Do students who study more than we would expect from their Anxiety tend to earn higher Exam Scores?

The answer is yes. More of the Study Hours that Anxiety cannot explain is associated with better Exam performance.

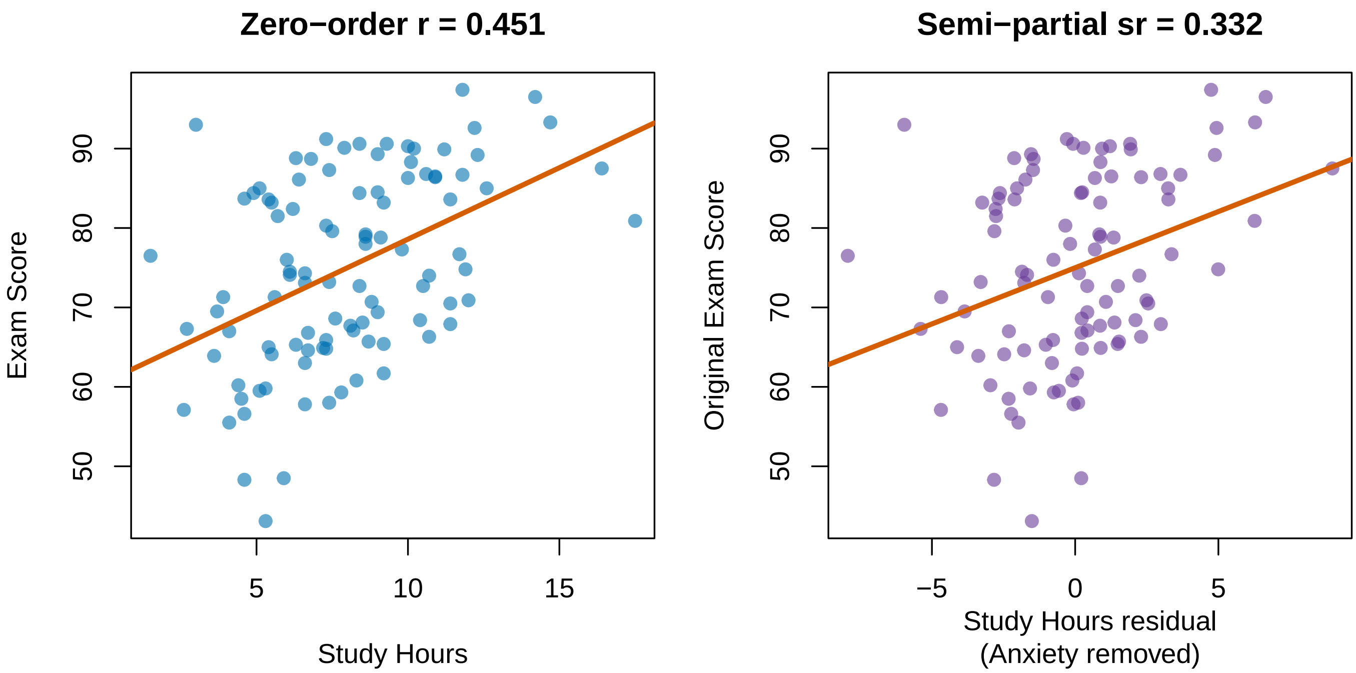


Figure 12.3: Left: the zero-order Study Hours–Exam Score relationship. Right: the semi-partial relationship after removing Anxiety from Study Hours only.

12.5.1 Why Squaring sr Matters

Squaring a semi-partial correlation gives the unique proportion of **total outcome variance** associated with that predictor:

```
sr2_study <- sr_study^2
sr2_study
```

```
[1] 0.1104633
```

Therefore, Study Hours uniquely accounts for about 11% of the total variance in Exam Score after the Anxiety-only model has explained everything it can, including the shared region c. In the Ballantine, the additional contribution is region a.

This also equals the change in R^2 when Study Hours is added to a model that already contains Anxiety.

```
anxiety_model <- lm(Exam ~ Anxiety, data = dat)
full_model <- lm(Exam ~ Anxiety + StudyHours, data = dat)

r2_anxiety <- summary(anxiety_model)$r.squared
r2_full <- summary(full_model)$r.squared
delta_r2_study <- r2_full - r2_anxiety

data.frame(
  Quantity = c(
    "R-squared: Anxiety only",
```

```r
    "R-squared: Anxiety + Study Hours",
    "Change in R-squared",
    "Squared semi-partial correlation"
  ),
  Value = round(c(r2_anxiety, r2_full, delta_r2_study, sr2_study), 3)
)
```

```
                          Quantity Value
1          R-squared: Anxiety only  0.16
2 R-squared: Anxiety + Study Hours  0.27
3              Change in R-squared  0.11
4 Squared semi-partial correlation  0.11
```

The last two values match:

$$sr^2 = \Delta R^2$$

That equation is the main reason semi-partial correlation matters in multiple regression. It tells us exactly how much total explanatory power Study Hours adds **above and beyond** Anxiety.

We can formally compare the nested models with `anova()`:

```r
anova(anxiety_model, full_model)
```

```
Analysis of Variance Table

Model 1: Exam ~ Anxiety
Model 2: Exam ~ Anxiety + StudyHours
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)
1     98 11978
2     97 10404  1    1574.3 14.678 0.0002263 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The test asks whether adding Study Hours improves prediction enough that we should take the improvement seriously, rather than blaming the Probability Gods for a tiny accidental increase.

12.6 Partial Correlation: Remove Anxiety from Both Sides

Partial correlation takes one additional step. We remove Anxiety from **both** Study Hours and Exam Score:

1. Predict Study Hours from Anxiety and save the residuals.
2. Predict Exam Score from Anxiety and save those residuals too.
3. Correlate the two sets of residuals.

```r
exam_from_anxiety <- lm(Exam ~ Anxiety, data = dat)
dat$ExpectedExam <- fitted(exam_from_anxiety)
dat$ExamResidual <- residuals(exam_from_anxiety)

pr_study <- cor(dat$StudyResidual, dat$ExamResidual)
pr_study
```

```
[1] 0.362534
```

The partial correlation is $pr = 0.363$.

Its plain-language question is:

> Among the portions of Study Hours and Exam Score that Anxiety cannot explain, how strongly are the remaining portions related?

Return to Hypothetical Alex. Based on Anxiety alone, the model predicts an Exam Score of about 70.2. Alex actually earned a 78. Therefore, Alex scored about 7.8 points **higher than expected given Anxiety**.

Partial correlation asks whether students who study more than Anxiety predicts also score higher than Anxiety predicts.

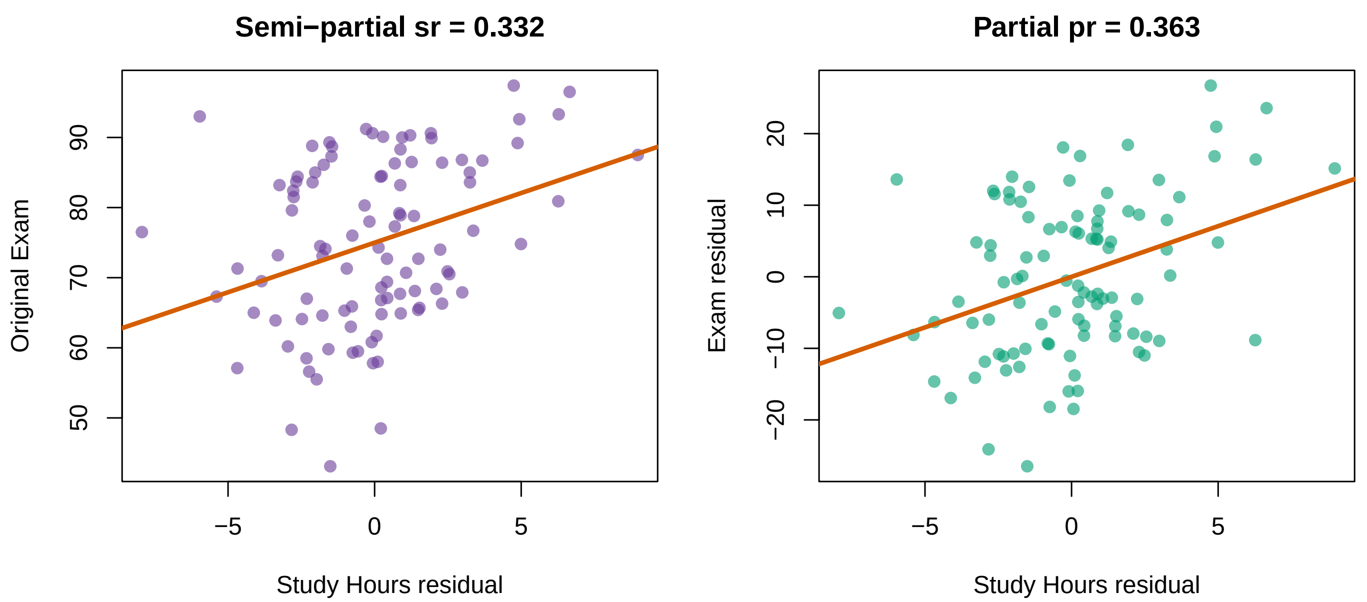


Figure 12.4: Semi-partial correlation leaves Exam Score untouched. Partial correlation removes Anxiety from both Study Hours and Exam Score.

12.6.1 Why Partial Correlation Is Larger

For these data:

```r
pr2_study <- pr_study^2
pr2_study
```

```
[1] 0.1314309
```

The squared partial correlation is about 13.1%. This does **not** mean Study Hours suddenly explains 13.1% of total Exam Score variance. Semi-partial correlation already told us that the unique contribution to total variance was 11%.

Partial correlation uses a smaller denominator. Anxiety has already removed the Exam Score variance it can explain. Of the Exam Score variance **remaining after Anxiety**, Study Hours is associated with about 13.1%.

In Ballantine language:

$$sr_1^2 = a$$

$$pr_1^2 = \frac{a}{a+e}$$

The numerator is the same unique Study Hours region. Partial correlation looks larger because it compares region a with only the Exam variance left after Anxiety is removed. It did not discover a stronger effect. It changed the denominator.

12.7 Semi-Partial Versus Partial

Quantity	Remove Anxiety from Study Hours?	Remove Anxiety from Exam Score?	What does the square mean?
Zero-order r	No	No	Total variance shared by Study Hours and Exam Score
Semi-partial sr	Yes	No	Unique Study Hours contribution to **total** Exam Score variance; ΔR^2
Partial pr	Yes	Yes	Study Hours association with Exam Score variance **remaining after Anxiety**

If you remember only one distinction, remember this:

> **Semi-partial leaves Y alone. Partial residualizes both sides.**

"Semi" does not mean the analysis tried less hard. It means the control variable was removed from only one member of the target relationship.

12.8 The Moderately Mathy Version

The residual method is the method I want you to understand. The formulas show why the two correlations differ.

Let:

- r_{Y1} be the Exam Score–Study Hours correlation,

- r_{Y2} be the Exam Score–Anxiety correlation, and
- r_{12} be the Study Hours–Anxiety correlation.

The Study Hours semi-partial correlation controlling for Anxiety is:

$$sr_{Y(1.2)} = \frac{r_{Y1} - r_{Y2}r_{12}}{\sqrt{1 - r_{12}^2}}$$

The partial correlation is:

$$pr_{Y1.2} = \frac{r_{Y1} - r_{Y2}r_{12}}{\sqrt{(1 - r_{Y2}^2)(1 - r_{12}^2)}}$$

Notice that the numerators are identical. Both calculations isolate the same unique Study Hours–Exam Score relationship. The partial-correlation denominator also removes the outcome variance associated with Anxiety.

```r
r_y1 <- cor(dat$Exam, dat$StudyHours)
r_y2 <- cor(dat$Exam, dat$Anxiety)
r_12 <- cor(dat$StudyHours, dat$Anxiety)

sr_formula <- (r_y1 - r_y2 * r_12) / sqrt(1 - r_12^2)

pr_formula <- (r_y1 - r_y2 * r_12) /
  sqrt((1 - r_y2^2) * (1 - r_12^2))

round(c(
  Residual_sr = sr_study,
  Formula_sr = sr_formula,
  Residual_pr = pr_study,
  Formula_pr = pr_formula
), 3)
```

```
Residual_sr  Formula_sr Residual_pr  Formula_pr
      0.332       0.332       0.363       0.363
```

The residual and formula routes agree. Statistics has briefly behaved itself.

12.9 How This Becomes the Multiple-Regression Slope

Here is the important connection. Compare three slopes:

1. The Study Hours slope from the full multiple regression.
2. The slope predicting original Exam Score from residualized Study Hours.
3. The slope predicting residualized Exam Score from residualized Study Hours.

```r
semipartial_lm <- lm(Exam ~ StudyResidual, data = dat)
partial_lm <- lm(ExamResidual ~ StudyResidual, data = dat)

round(c(
  Full_multiple_regression = coef(full_model)["StudyHours"],
  Original_Y_on_residualized_X = coef(semipartial_lm)["StudyResidual"],
  Residualized_Y_on_residualized_X = coef(partial_lm)["StudyResidual"]
), 6)
```

```
           Full_multiple_regression.StudyHours
                                      1.414884
    Original_Y_on_residualized_X.StudyResidual
                                      1.414884
Residualized_Y_on_residualized_X.StudyResidual
                                      1.414884
```

All three slopes equal about 1.41 Exam Score points per Study Hour.

This is not a coincidence. The Study Hours coefficient in multiple regression is built from the Study Hours variation that Anxiety cannot linearly explain. This result is sometimes called the **Frisch–Waugh–Lovell theorem**, which is an impressive name for "residualize the relevant pieces and you recover the same slope."

The practical interpretation is:

> Among students at the same Anxiety level, one additional Study Hour is associated with about 1.41 additional Exam Score points.

That is a conditional association. It is not automatically a causal effect.

12.10 Where Standardized Beta Fits

Now we have several numbers describing the Study Hours relationship. This is where students reasonably begin asking whether statistics could have used fewer symbols.

```r
dat_z <- as.data.frame(scale(dat[c("Exam", "StudyHours", "Anxiety")]))
standardized_model <- lm(Exam ~ StudyHours + Anxiety, data = dat_z)
beta_study <- coef(standardized_model)["StudyHours"]

comparison <- data.frame(
  Quantity = c(
    "Zero-order correlation (r)",
    "Standardized regression slope (beta)",
    "Semi-partial correlation (sr)",
    "Partial correlation (pr)",
    "Squared semi-partial (sr-squared)",
    "Squared partial (pr-squared)"
  ),
  Value = round(c(
    r_y1,
```

```
    beta_study,
    sr_study,
    pr_study,
    sr2_study,
    pr2_study
  ), 3),
  Meaning = c(
    "Uncontrolled Study Hours-Exam association",
    "Conditional slope in standard-deviation units",
    "Unique Study association scaled by total Exam variance",
    "Unique Study association scaled by remaining Exam variance",
    "Unique contribution to total Exam variance; change in R-squared",
    "Proportion of residual Exam variance associated with Study"
  )
)

knitr::kable(comparison)
```

Quantity	Value	Meaning
Zero-order correlation (r)	0.451	Uncontrolled Study Hours-Exam association
Standardized regression slope (beta)	0.355	Conditional slope in standard-deviation units
Semi-partial correlation (sr)	0.332	Unique Study association scaled by total Exam variance
Partial correlation (pr)	0.363	Unique Study association scaled by remaining Exam variance
Squared semi-partial (sr-squared)	0.110	Unique contribution to total Exam variance; change in R-squared
Squared partial (pr-squared)	0.131	Proportion of residual Exam variance associated with Study

The standardized beta is $\beta = 0.355$. It tells us how many standard deviations Exam Score changes for a one-standard-deviation increase in Study Hours while Anxiety is held constant.

Beta, *sr*, and *pr* have the same sign here and test the same unique predictor contribution, but they are **not interchangeable**. They standardize the relationship using different variance. Never report one while calling it another because the decimals looked lonely.

12.11 `ppcor`: A Shortcut

For a strict correlation question involving one outcome, one target predictor, and a control variable, the `ppcor` package can verify our work:

```
# Semi-partial: remove Anxiety from the SECOND variable, StudyHours
ppcor::spcor.test(
  x = dat$Exam,
  y = dat$StudyHours,
  z = dat$Anxiety
)

# Partial: remove Anxiety from both Exam and StudyHours
ppcor::pcor.test(
```

```
  x = dat$Exam,
  y = dat$StudyHours,
  z = dat$Anxiety
)
```

The order matters for `spcor.test()` because the control variable is removed from the **second** variable. Order matters less conceptually for partial correlation because the control is removed from both target variables.

`ppcor` can accept more than one control variable, but it remains a specialized correlation shortcut. Real analyses quickly acquire several predictors, categorical variables, interactions, missing observations, model comparisons, diagnostics, and Reviewer 2. Use `lm()` as your default. It scales to those problems and shows exactly which model you estimated. Use `ppcor` as a quick check when the question is genuinely simple.

12.12 Statistical Control Is Not Experimental Control

We removed linear associations from numbers. We did **not** randomly assign Anxiety, repair measurement error, eliminate variables we forgot to measure, or reach backward through time.

After controlling for Anxiety, the Study Hours coefficient describes the association between Study Hours and Exam Score conditional on Anxiety. Calling it "the true causal effect of studying" would require a research design and assumptions much stronger than this regression alone provides.

Adding a control variable does not guarantee that a slope will shrink. Depending on the correlation pattern, a slope can:

- shrink,
- grow,
- reverse direction,
- become more precise,
- become less precise, or
- stare into the abyss and become a suppression effect.

Control variables must be selected using theory and research design. Throwing every available column into a regression is not rigor. It is making the software eat the entire refrigerator because you could not decide what to cook.

12.13 Four Questions Before You Escape

12.13.1 1. Caffeine, Sleep, and Exam Scores

You want to know how much **total Exam Score variance** Caffeine uniquely explains after Sleep Hours are already in the model. Do you want a partial or semi-partial correlation?

12.13.2 2. The Entirely Ethical Chainsaw Study

You want the relationship between Fear and Heart Rate after removing Running Speed from **both** variables. Participants are not actually chased with chainsaws because the IRB has once again ruined science. Partial or semi-partial?

12.13.3 3. Reviewer 2 Predicts Rejection

You want to know how much R^2 increases when Formatting Errors are added after Manuscript Length. Which squared correlation should equal that change in R^2?

12.13.4 4. The Causal Caffeine Disaster

You observe that Caffeine predicts Exam Score after controlling for Sleep. May you conclude that forcing students to drink six espressos will improve their grades?

Answers, If You Have Suffered Enough

1. **Semi-partial.** You want Caffeine's unique contribution to total Exam Score variance.
2. **Partial.** Running Speed is removed from both Fear and Heart Rate.
3. **Squared semi-partial correlation.** $sr^2 = \Delta R^2$ when Formatting Errors are added last.
4. **Absolutely not.** Statistical control does not establish causality, and six espressos may merely establish a new relationship with campus medical services.

12.14 The Short Story

- **Residual**: what remains after subtracting a model's predicted value from the observed value.
- **Semi-partial correlation**: remove the control variable from one target variable; sr^2 is the unique addition to total R^2.
- **Partial correlation**: remove the control variable from both target variables; pr^2 describes the relationship within the outcome variance that remains.
- **Multiple-regression slope**: can be recovered by predicting Y from the residualized target predictor.
- **Standardized beta**: the controlled regression slope in standard-deviation units.
- **Statistical control**: adjustment for measured associations, not a tiny machine that produces causal truth.

If you can identify what was removed, what it was removed from, and what denominator remains, you understand statistical control. That places you ahead of many graduate students and at least several published papers.

13 Hierarchical Regression: Testing Whether Your Model Improves

13.1 Where We Left Off

In the previous chapter, we used **multiple regression** to predict exam performance from two predictors simultaneously: Study Hours and Test Anxiety. We learned that each predictor has a *unique* effect on exam scores — an effect that persists even after statistically controlling for the other predictor.

By the end of that chapter, we had fit three models:

- **Model 1:** Exam ~ Study Hours (simple regression, one predictor)
- **Model 2:** Exam ~ Test Anxiety (simple regression, one predictor)
- **Model 3:** Exam ~ Study Hours + Test Anxiety (multiple regression, both predictors)

We showed that Model 3 explained more variance in exam scores (R^2) than either predictor alone, and we compared the slopes before and after controlling. But we never formally asked: **does adding Test Anxiety to the model produce a statistically significant improvement?** Or put differently, is the improvement in R^2 when we go from Model 1 to Model 3 bigger than we would expect from chance alone?

That question is the heart of **hierarchical regression** — and it is the focus of this chapter.

13.2 What Is Hierarchical Regression?

Hierarchical regression (also called **sequential regression**) is a strategy for building and comparing regression models in a deliberate, theory-driven order. Instead of putting all your predictors in at once, you add them in meaningful *steps* — and at each step, you ask whether the new addition genuinely improved your ability to predict the outcome.

The word "hierarchical" refers to the hierarchy of models, not a hierarchy of people or importance. Each model in the sequence is more complex than the one before it, and you evaluate whether that added complexity is worth it.

This approach is useful any time you have a specific question about the *incremental value* of a predictor. In our example, the question is precise and testable:

> **Does Test Anxiety explain variance in exam performance above and beyond what Study Hours already explains?**

Notice the phrase "above and beyond." That is the language of hierarchical regression. You are not just asking whether Test Anxiety predicts exam scores (it does — we know this from the simple regression). You are asking whether it adds *something new* to a model that already includes Study Hours.

13.3 Nested Models

To compare two models formally, they need to be **nested**. Two models are nested when the predictors in the simpler model are a *subset* of the predictors in the more complex model. In other words, the simpler model is contained inside the more complex one.

Here is how nesting works in our example:

Model	Predictors	Is it nested in...?
Model 1	Study Hours	—
Model 3	Study Hours + Test Anxiety	Yes, contains Model 1

Model 1 only has Study Hours. Model 3 has Study Hours **plus** Test Anxiety. Because Model 1's predictor (Study Hours) is fully contained within Model 3, these two models are nested. The difference between them is exactly one predictor: Test Anxiety.

This is what makes the comparison meaningful. When we compare these two models, we are directly asking: **what does Test Anxiety add, given that Study Hours is already in the model?** We are not comparing apples to oranges — we are asking a precise, surgical question about one specific addition.

If the models were *not* nested — for example, if Model A had Study Hours and Model B had Test Anxiety — we could not make a direct statistical comparison, because the difference between them involves both removing one predictor *and* adding a different one. The formal test we will use below only works for nested models.

13.4 Setting Up the Hierarchical Comparison

In our example, the research question drives the ordering of the models:

Step 1 (Block 1): Enter Study Hours. This is our primary predictor — the one we already know matters, and the one that came first in our theoretical thinking about what drives exam performance.

Step 2 (Block 2): Add Test Anxiety. This is the incremental question: does anxiety explain variance in exam scores *above and beyond* what study behavior already tells us?

We are deliberately **not** starting from scratch with each model. We are building on what came before, which is why this is called hierarchical or sequential entry.

Let's run both models:

```r
# Block 1: Study Hours only
Model.1 <- lm(Exam ~ StudyHours, data = dat)

# Block 2: Add Test Anxiety
Model.3 <- lm(Exam ~ StudyHours + Anxiety, data = dat)
```

And let's recall what R^2 looks like for each:

```r
r2_m1 <- summary(Model.1)$r.squared
r2_m3 <- summary(Model.3)$r.squared

cat("Model 1 (Study Hours only): R-squared =", round(r2_m1, 3), "\n")
```

```
Model 1 (Study Hours only): R-squared = 0.203
```

```r
cat("Model 3 (Both predictors):  R-squared =", round(r2_m3, 3), "\n")
```

```
Model 3 (Both predictors):  R-squared = 0.27
```

```r
cat("Change in R-squared: Delta R-squared =", round(r2_m3 - r2_m1, 3), "\n")
```

```
Change in R-squared: Delta R-squared = 0.067
```

The R^2 went up when we added Test Anxiety. But R^2 **always** goes up when you add a predictor — even a useless one will explain a tiny sliver of variance just by chance. So we need a formal test to decide whether this improvement is meaningful.

13.5 The Change in R²: What Are We Measuring?

Before we run the test, let's understand what ΔR^2 (read: "delta R-squared," or "change in R-squared") actually represents.

Recall the Venn diagram from the previous chapter. When Study Hours is the only predictor, it explains regions $a + c$ — its unique share of the variance in exam scores, plus the variance it shares with Test Anxiety. When we add Test Anxiety to the model, we pick up region b: the variance in exam scores that Test Anxiety explains *uniquely*, after Study Hours has already claimed its portion.

$$\Delta R^2 = R^2_{\text{Model 3}} - R^2_{\text{Model 1}} = \text{Region } b$$

So ΔR^2 is the proportion of variance in exam scores that Test Anxiety explains *uniquely* — the additional predictive power it brings to the table after Study Hours has already been accounted for. This is precisely what we want to know.

The question is: is this ΔR^2 of 0.067 (6.7% of variance) large enough to conclude that Test Anxiety is a genuinely useful addition? Or could a difference this big arise just from sampling variability, even if Test Anxiety had no real predictive power?

13.6 The F-Test for Model Comparison

The tool we use to answer that question is an **F-test for model comparison**. It tests the null hypothesis that the added predictor(s) explain *zero additional variance in the population* — that the improvement in R^2 is nothing more than noise.

$$H_0 : \Delta R^2 = 0 \text{ (Test Anxiety adds nothing beyond Study Hours)}$$

$$H_1 : \Delta R^2 > 0 \text{ (Test Anxiety adds meaningful predictive power)}$$

The F-statistic for this test is computed as:

$$F = \frac{\Delta R^2 / \Delta df_{\text{model}}}{(1 - R^2_{\text{full}}) / df_{\text{residual, full}}}$$

Where:

- ΔR^2 = the improvement in R^2 from the simpler to the more complex model
- Δdf_{model} = the number of new predictors added (how many degrees of freedom were "used up" by the new predictors)
- R^2_{full} = the R^2 of the more complex (full) model
- $df_{\text{residual, full}}$ = residual degrees of freedom in the full model ($n - k - 1$, where k is the total number of predictors)

The numerator captures how much new variance the added predictor explains *per additional parameter*. The denominator captures the average unexplained variance per residual degree of freedom — essentially, the baseline "noise level" of the full model. If the ratio is large, the new predictor is explaining more variance than you would expect from noise. If the ratio is close to 1, the new predictor is not doing much better than chance.

You do not need to compute this by hand. In R, the `anova()` function computes it for you when you feed it two nested models.

13.6.1 Running the Comparison in R

```r
anova(Model.1, Model.3)
```

```
Analysis of Variance Table

Model 1: Exam ~ StudyHours
Model 2: Exam ~ StudyHours + Anxiety
  Res.Df   RSS Df Sum of Sq      F   Pr(>F)
1     98 11354
2     97 10404  1     950.1 8.8581 0.003684 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Let's read this output carefully. The `anova()` output when comparing two models shows:

- **RSS** (Residual Sum of Squares): the total unexplained variance in each model. A lower RSS means a better-fitting model.
- **Df**: the residual degrees of freedom for each model, and the change in degrees of freedom between models.
- **F**: the F-statistic for the comparison.
- **Pr(>F)**: the p-value for that F-statistic.

The key row to focus on is the second one — that is where the comparison lives. The F-statistic tells you whether the drop in residual sum of squares (i.e., the gain in explained variance) between Model 1 and Model 3 is statistically significant.

13.6.2 What to Look For

When you run a hierarchical model comparison, here is what you check:

1. Is the F-test significant? If $p < .05$, the added predictor(s) explain a statistically significant amount of additional variance. The improvement in model fit is unlikely to be due to chance. In our case:

```
F( 1 , 97 ) = 8.86
```

```
p = 0.0037
```

```
Delta R-squared = 0.067
```

The F-test is statistically significant ($F(1, 97) = 8.86$, $p = 0.0037$). This tells us that adding Test Anxiety to a model that already contains Study Hours produces a statistically significant improvement in fit. The 6.7% additional variance explained is more than we would expect from chance.

2. How large is ΔR^2? Statistical significance tells you the improvement is real, but it does not tell you how *big* it is. Always look at ΔR^2 to judge practical importance:

Our $\Delta R^2 = 0.067$, which means Test Anxiety explains an additional 6.7% of the observed variance in Exam Score above and beyond Study Hours. There are no universal small, medium, and large labels for raw ΔR^2. Whether this improvement matters depends on the outcome, measurement quality, theory, cost, and consequences. A tiny improvement in predicting a rare medical crisis may matter enormously; the same improvement in predicting which student will choose the blue pen may not deserve a parade. Report the number and interpret it in context instead of sending it to the Small/Medium/Large Sorting Hat.

3. What do the individual coefficients look like in the final model? After confirming that the model improved, you examine the full model to interpret the individual slopes. The conclusion is only complete when you can say both *that* the model improved and *how* each predictor contributes.

13.7 Interpreting the Full Model

Now that we have confirmed the model improved significantly, let's look at the final model in full:

```
summary(Model.3)
```

```
Call:
lm(formula = Exam ~ StudyHours + Anxiety, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-24.3420  -7.8649  -0.3942   7.4378  22.0359 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  80.2059     7.2176  11.112  < 2e-16 ***
StudyHours    1.4149     0.3693   3.831 0.000226 ***
Anxiety      -0.3306     0.1111  -2.976 0.003684 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.36 on 97 degrees of freedom
Multiple R-squared:   0.27, Adjusted R-squared:  0.2549 
F-statistic: 17.94 on 2 and 97 DF,  p-value: 2.351e-07
```

Reading this together with the hierarchical comparison gives us a complete picture:

- **Study Hours** (b = 1.41): Each additional hour of studying per week is associated with a 1.41-point increase in exam score, controlling for anxiety. This is statistically significant.
- **Test Anxiety** (b = -0.33): Each one-point increase in anxiety is associated with a 0.33-point *decrease* in exam score, controlling for study hours. This is also statistically significant, and it is the predictor whose incremental value we just confirmed with the F-test.

The overall model $R^2 = 0.27$: together, Study Hours and Test Anxiety explain 27% of the variance in exam performance.

13.8 Visualizing the Model Comparison

It helps to see the R^2 values side by side. Here is a simple summary of where we are:

```r
cat("Model 1 (Study Hours only):       R-squared =", round(r2_m1,  3),
    " | Adjusted R-squared =", round(summary(Model.1)$adj.r.squared, 3), "\n")
```

```
Model 1 (Study Hours only):       R-squared = 0.203  | Adjusted R-squared = 0.195
```

```r
cat("Model 3 (Study Hours + Anxiety):  R-squared =", round(r2_m3,  3),
    " | Adjusted R-squared =", round(summary(Model.3)$adj.r.squared, 3), "\n")
```

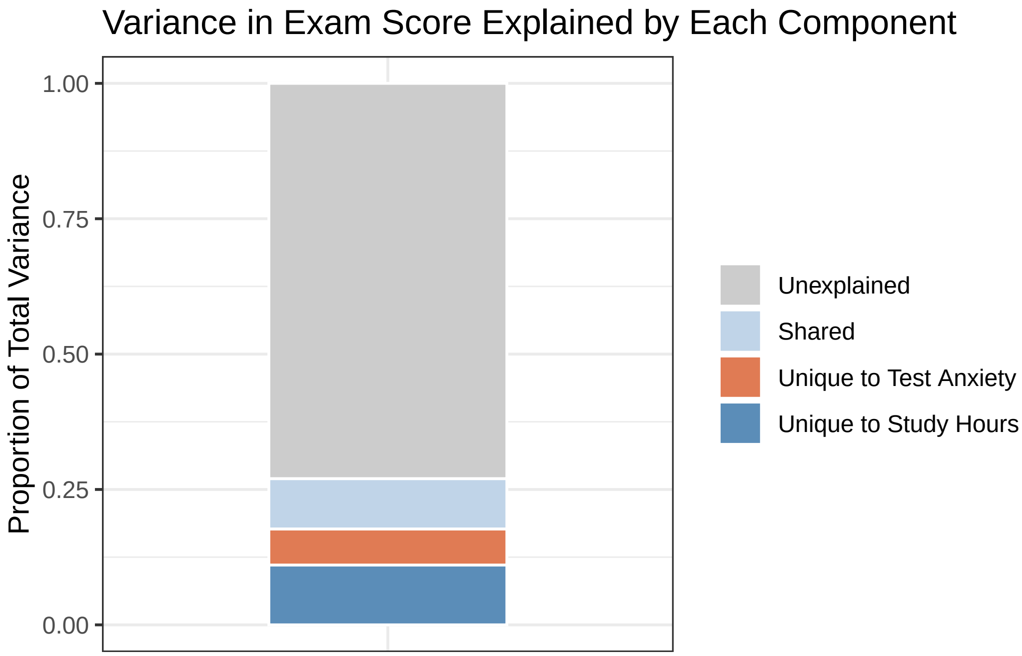
```
Model 3 (Study Hours + Anxiety):  R-squared = 0.27  | Adjusted R-squared = 0.255
```

```r
cat("Change: Delta R-squared =", round(r2_m3 - r2_m1, 3), "\n")
```

```
Change: Delta R-squared = 0.067
```

Notice that the **adjusted** R^2 also increased from 0.195 to 0.255. Recall that adjusted R^2 penalizes models for adding predictors. When adjusted R^2 goes up, it means the new predictor is earning its keep — the improvement in fit outweighs the cost of the extra parameter. This is consistent with the significant F-test.

We can also visualize this as a stacked bar showing how the explained variance is distributed:



The orange segment — Test Anxiety's unique contribution ($\Delta R^2 = 0.067$) — is what the F-test just confirmed is statistically significant. It represents the additional explanatory power Test Anxiety brings to a model that already contains Study Hours. Adding it was worth it.

13.9 The Logic in Plain Language

Let's step back from the numbers and make sure the core logic is clear.

Hierarchical regression is built on a simple and intuitive idea: **if a predictor truly matters, it should explain variance in the outcome that no other predictor in the model can already account for.** If it does not, it is just explaining the same territory as something already in the model — and we should not clutter up our model with redundant predictors.

Here is the sequence of reasoning:

1. **Start with what you already know.** We know Study Hours predicts exam scores. Build a model with it first.
2. **Ask: does the new predictor add something?** We want to know if Test Anxiety explains variance in exam scores *above and beyond* Study Hours. The ΔR^2 tells us the size of the gain.
3. **Test whether the gain is real.** The F-test evaluates whether that ΔR^2 could plausibly have arisen by chance. A significant F tells us: no, the improvement is larger than chance fluctuations in R^2 would predict.
4. **Interpret the final model.** If the gain is significant, include the new predictor and interpret its slope. If the gain is not significant, you have evidence that the new predictor does not earn its place in the model (at least, not above and beyond what is already there).

This four-step logic applies no matter how many predictors you are adding. You could be testing whether a single predictor improves fit, or whether a whole *block* of predictors (say, all demographic variables, or all anxiety-related measures) improves fit over a baseline model. The mechanics are the same.

13.10 Blocks of Predictors

One of the most powerful applications of hierarchical regression is adding **blocks** of predictors — groups of theoretically related variables — all at once. For example, imagine we had three anxiety-related measures instead of one. We could build:

- **Block 1:** Study Hours
- **Block 2:** Test Anxiety + Generalized Anxiety + Worry (three anxiety measures added simultaneously)

The F-test for Block 2 would then ask: do these three anxiety measures *together* explain additional variance in exam scores, above and beyond Study Hours? The degrees of freedom for the F-test would be 3 (one per added predictor), and the test is still a single, omnibus comparison — no need to test each anxiety measure separately at this stage.

This is particularly useful when you want to:

- **Control for covariates** (e.g., age, gender) by entering them in Block 1, then test your predictors of interest in Block 2
- **Test a theoretical construct** measured by multiple items, rather than testing each item individually
- **Evaluate competing theories** by building them into successive models and checking which block produces the largest gain in R^2

13.11 What Not to Do: The Danger of "Fishing"

Hierarchical regression is a tool for **theory-driven** model building. The order in which you enter predictors should be decided *before* you look at the results, based on your understanding of the literature and your hypotheses.

It is tempting to try many different orderings and report the one that looks best. This is a form of "fishing" — running multiple comparisons in the hope of finding a significant result — and it inflates the Type I error rate. If you run 20 model comparisons, you should expect one of them to be "significant" by chance at the $p < .05$ threshold, even if nothing is real.

The safe approach: decide on your model sequence based on theory, pre-register it if you can, and report all models you tested — not just the one that "worked."

13.12 APA Write-Up

Here is how to report a hierarchical regression analysis in APA format:

Prior to the main analysis, zero-order correlations were examined. Study hours positively correlated with exam performance ($r = .45$, $p < .001$), and test anxiety negatively correlated with exam performance ($r = -.40$, $p < .001$). The two predictors were also significantly correlated with each other ($r = -.35$, $p < .001$).

A hierarchical multiple regression was conducted to examine whether test anxiety explained variance in exam performance above and beyond study hours. Study hours was entered in Step 1, and test anxiety was added in Step 2.

In Step 1, study hours significantly predicted exam performance, $b = 1.8$, $SE = 0.36$, $t(98) = 5$, $p < .001$, $R^2 = 0.203$, adjusted $R^2 = 0.195$.

In Step 2, the addition of test anxiety produced a statistically significant improvement in model fit, $\Delta R^2 = 0.067$, $F(1, 97) = 8.86$, $p = 0.0037$. In the final model, both study hours ($b = 1.41$, $SE = 0.37$, $t(97) = 3.83$, $p < .001$) and test anxiety ($b =$ -0.33, $SE = 0.11$, $t(97) =$ -2.98, $p = 0.0037$) significantly predicted exam performance. The final model explained 27% of the variance in exam scores, $R^2 = 0.27$, adjusted $R^2 = 0.255$.

A few things to notice in this write-up:

- Report each *step* as a block, not just the final model. Readers need to see what each step added.
- Report ΔR^2 and the F-test for the change — these are the key hierarchical results.
- After reporting the model comparison, also report the individual slopes from the final model.
- Use the phrase **"above and beyond"** or **"produced a statistically significant improvement in model fit"** to signal that a hierarchical comparison is being reported.

! The Order Must Come from the Question

ΔR^2 belongs to a particular sequence of models. Enter predictors in an order justified by theory or design, not the order that made the Probability Gods finally hand you an asterisk.

14 Continuous-by-Categorical Interactions: When the Effect of One Variable Depends on Another

14.1 Where We Are and Where We Are Going

So far in this course, you have learned to build regression models where each predictor has its own independent, additive effect on the outcome. Study hours helps exam scores. Test anxiety hurts them. Each predictor does its job, and we can interpret the effects one at a time.

But human behavior is rarely that tidy. In the real world, the effect of one variable often *changes* depending on the value of another variable. A new medication might be effective for women but not men. Praise for effort might motivate some students while leaving others cold. Practice might matter enormously for novices but very little for experts.

These situations, where the effect of one predictor **depends on** another predictor, are called **interactions**, or equivalently, **moderation effects**. In this chapter, you will learn to detect, estimate, visualize, and interpret an interaction between a continuous variable and a categorical one.

Before we touch any data, we need to build the concept carefully. That is where we will start.

14.2 The Friendship Study

You have probably heard the expression "right place, right time." It turns out that this is not just a saying; it is backed by social psychological research. A classic study by Back, Schmukle, and Egloff (2008) found that the friendships people form in university are driven, in large part, by seating proximity. Literally, the people you end up sitting next to on the first day of class are the people most likely to become your close friends by the end of the semester. Chance determines who you sit near. Geography becomes destiny.

You decide to replicate and extend this work. You follow 200 first-semester students throughout their first semester, recording how close of a friendship they formed with one particular person (measured on a 0–100 feelings thermometer, where 0 = complete strangers and 100 = best friends). You also measure two things that might predict the quality of that friendship:

- **Being Nice:** each person is rated on their general agreeableness and warmth by a trained panel, on a scale from 0 to 10. This is a continuous variable.
- **Going Out:** whether the student regularly attended social events and parties throughout the semester. This is a binary variable: *Stay in* (0) vs. *Go out* (1).

The intuition behind your study is this: being a warm, friendly person should help you make a close friend. But being nice in your apartment, alone, does not do much for your social life. To turn niceness into friendship, you have to actually be around people. **The effect of being nice might depend on whether you go out.**

That is your hypothesis. And testing it is the focus of this chapter.

Back, M. D., Schmukle, S. C., & Egloff, B. (2008). Becoming friends by chance. *Psychological Science, 19*(5), 439–440.

14.3 What Is an Interaction? The Concept Before the Math

14.3.1 "It Depends" as a Scientific Statement

In everyday conversation, "it depends" can sound like an evasion, a way of refusing to commit to an answer. In statistics, it is a precise and testable claim. When we say "the effect of X on Y depends on Z," we mean something specific: **the slope of X on Y is not the same across different values of Z.** That is an interaction.

Let's make this concrete. There are two fundamentally different ways the world could work in our friendship study.

Possibility 1: Being nice always helps (no interaction). Whether you stay in or go out, being a nicer person leads to a closer friendship. The social opportunities provided by going out matter too: going out gives you more friends overall, but the *benefit of being nice* is the same either way.

Possibility 2: Being nice only pays off if you go out (interaction). For students who stay in all semester, niceness makes little difference; they simply do not have enough social contact for their personality to matter. For students who go out, however, being nice is powerful: it turns those social opportunities into genuine close friendships.

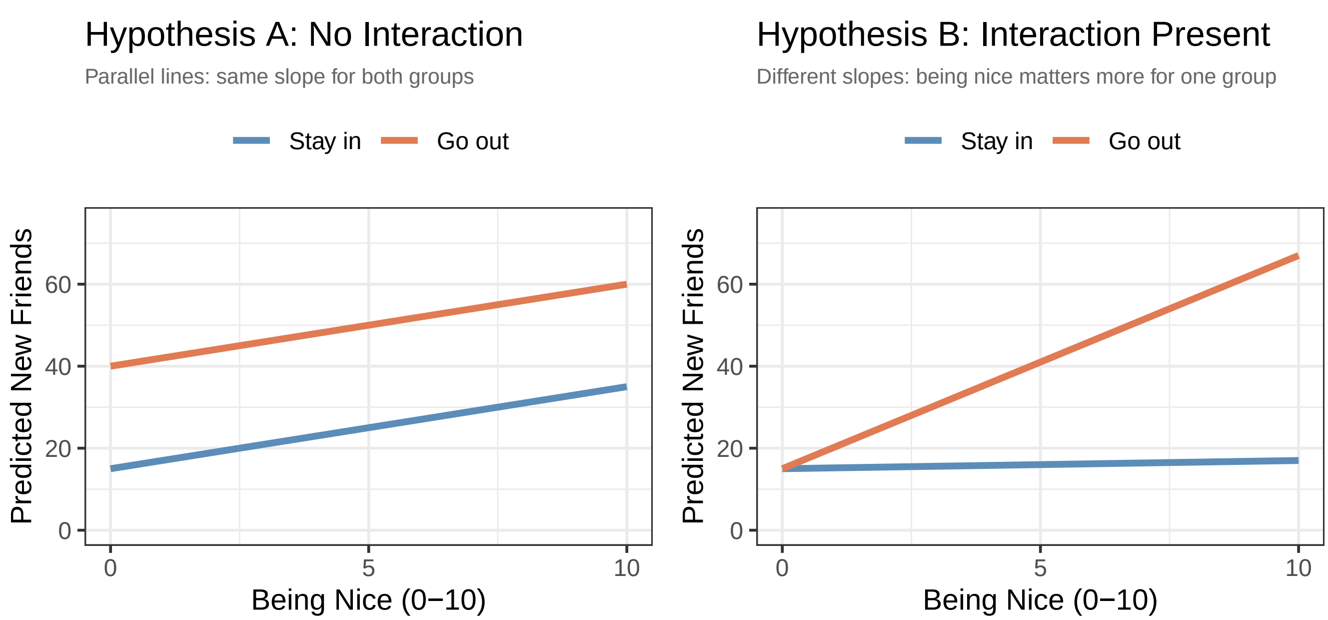
These two possibilities predict completely different patterns in the data. Before we analyze a single number, we can sketch out what each would look like.

14.3.2 Visual Hypotheses: Sketching What We Expect to See

One of the most useful habits in data analysis is to draw a picture of what you *expect* your results to look like before you run a single model. This keeps your hypotheses honest: you cannot accidentally convince yourself you predicted something after the fact. It also gives you an intuition for what you are looking for.

In our study, we want to predict *New Friends* (Y) from *Being Nice* (X) separately for people who *Stay in* and *Go out* (Z). This naturally produces two lines on a plot: one for each group. The question is what those two lines look like relative to each other.

Here are our two competing hypotheses, visualized:



Look at the difference between these two pictures:

- In **Hypothesis A**, the two lines are **parallel**. The "Go out" line is higher than the "Stay in" line (going out gives you more friends overall), but the *slope* of both lines is the same. Being nice helps by about the same amount whether you go out or not. This is what a **main effects only** world looks like: each variable has an independent, additive effect.
- In **Hypothesis B**, the two lines are **not parallel**. For people who *stay in*, the line is almost flat: being nice barely moves the needle. For people who *go out*, the line is steep: being nice strongly predicts closeness. The slope for "Go out" is much steeper than the slope for "Stay in." This is an **interaction**: the slope of Being Nice *depends on* whether you go out.

The key visual signature of an interaction is **non-parallel lines**. If you run an interaction model and your lines look parallel, the interaction is not doing anything. If the lines fan out, cross, or differ in steepness, you have an interaction.

We will use this visual intuition throughout this chapter to interpret the output.

14.3.3 The Two-Regressions Intuition

Here is another way to think about what an interaction model is doing, which many students find helpful.

Imagine, for a moment, that instead of running one regression on the full dataset, you split the data in half: one dataset with only the students who *stay in*, and one with only the students who *go out*. You then run a separate simple regression (Niceness $\rightarrow$ Friends) in each half.

- **Regression for Stay-in students:** You get a slope for Being Nice. Let's call it b_{stay}.
- **Regression for Go-out students:** You get a slope for Being Nice. Let's call it b_{go}.

If there is no interaction, $b_{stay} \approx b_{go}$: the two slopes are about the same. If there is an interaction, $b_{stay} \neq b_{go}$: the slopes differ.

This approach would *show* you the interaction, but it has a critical weakness: **you cannot formally test whether the two slopes are significantly different from each other.** You have two estimates, each with its own standard error, from two different analyses that are not connected. There is no built-in test of the gap between them.

This is exactly what an interaction model in regression solves. It fits both slopes simultaneously in a single model and includes a formal significance test for the **difference** between those slopes. That test is the interaction term. When the interaction coefficient is statistically significant, it means the slopes genuinely differ more than sampling variability would predict. When it is not significant, the data do not support the claim that the slopes differ.

So an interaction model is essentially a formalized, simultaneous, significance-tested version of running separate regressions for each group. Keep that image in mind as we work through the math.

14.4 Before We Run the Models: Two Concepts You Need First

14.4.1 Centering: Making Zero Meaningful

You have seen the intercept in every regression model we have run in this course, and you know it means "the predicted value of Y when X = 0." That definition is simple enough. The problem is that zero is not always a meaningful value.

In our friendship study, *Being Nice* is measured on a 0–10 scale. Zero means "not nice at all," a real value technically, but one that describes an essentially sociopathic person who presumably does not exist in our sample. When our regression model reports an intercept, it is making a prediction for a hypothetical person with zero niceness. That is not a very interesting or useful prediction.

Now consider what happens when we build an interaction model. The intercept will be the predicted friendship score for a person with zero niceness *who also stays in*. The coefficient for Going Out will be the gap between the two groups *at zero niceness*. These values are anchored to a strange and artificial point that nobody in our data actually occupies.

Centering solves this by shifting the scale of a continuous variable so that zero has a meaningful interpretation: **the average person in the sample.**

The process is simple: subtract the mean.

$$X_c = X - X_{Mean}$$

After centering:

- A score of $X_c = 0$ means this person is exactly average on Being Nice.
- A score of $X_c = 2$ means this person is 2 points *above* the average.
- A score of $X_c = -3$ means this person is 3 points *below* the average.

The scale is shifted, but nothing else changes. The **distances between people are identical**. The **slope of the regression line is identical**. The R^2 **is identical**. The only thing that changes is the meaning of the intercept (and the meaning of some of the other coefficients in the interaction model, as we will see).

Important distinction: centering is not the same as standardizing.

You may recall from earlier in the course that we can also *standardize* (or *z-score*) a variable by subtracting the mean *and* dividing by the standard deviation:

$$X_z = \frac{X - X_{Mean}}{S}$$

Standardizing changes the units of the variable: a standardized coefficient represents the change in Y for a one-standard-deviation increase in X. Centering does **not** change the units: a centered coefficient still represents the change in Y for a one-unit increase in X. In this chapter, we will only center, not standardize. We want to preserve the original units of our scale (a one-point increase in niceness = one point) while simply making zero mean "the average person."

14.4.2 How to Center in R

Centering in R is straightforward. We use the `scale()` function with `center = TRUE` and `scale = FALSE`. Setting `scale = FALSE` tells R to subtract the mean *without* dividing by the standard deviation, so we get the shift without the rescaling.

```
# Suppose BeingNice has a mean of 5.22
# After centering: a person with a raw score of 5.22 gets a centered score of 0

# The scale() function with center=TRUE and scale=FALSE does exactly this:
# data$BeingNice.C <- scale(data$BeingNice, center = TRUE, scale = FALSE)

# To check that it worked: the centered variable should have a mean of (essentially) zero
# mean(BeingNice.C)  # Should be ~0
```

When we center *Being Nice*, the intercept in our interaction model will represent the predicted friendship score for a person of **average niceness who stays in**. That is a real, interpretable, meaningful prediction; a much better anchor than "a person with zero niceness."

We will center our continuous variable before building any models. This is standard practice whenever you include an interaction term with a continuous variable.

14.4.3 Dummy Coding: A Quick Review

You already know how to dummy code a categorical variable for regression. When we use a binary categorical predictor like Going Out (Stay in vs. Go out), R automatically converts it into a dummy variable: one group is coded as 0 (the **reference group**) and the other as 1 (the **comparison group**). In our study, *Stay in* will be the reference group (coded 0), and *Go out* will be the comparison group (coded 1). We tell R which group comes first by setting the factor levels in the desired order.

With dummy coding in place:

- The **intercept** represents the predicted outcome for the reference group (Stay in) at whatever value the continuous predictor takes.
- The **coefficient for GoingOut** represents the gap between the two groups, holding the continuous predictor constant.

We will see exactly how this plays out, and how centering changes the interpretation, as we work through the models.

14.5 The Data

We collected data from 200 first-semester university students. Each student's friendship closeness with one person they met at the start of the semester was recorded at the end of the semester using a 0–100 feelings thermometer. Their agreeableness and warmth was rated by a trained panel on a 0–10 scale, and we noted whether they regularly attended social events and parties throughout the semester.

Here is a summary of the sample broken down by social engagement:

```
Friend.Data %>%
  group_by(GoingOut_) %>%
  summarise(
    n          = n(),
    M_Nice     = round(mean(BeingNice), 2),
    M_Friends  = round(mean(NewFriends), 2),
    SD_Friends = round(sd(NewFriends), 2)
  )
```

```
# A tibble: 2 x 5
  GoingOut_     n M_Nice M_Friends SD_Friends
  <fct>     <int>  <dbl>     <dbl>      <dbl>
1 Stay in     116   5.07      15.7       6.65
2 Go out       84   5.43      42.0      15.6
```

The "Go out" group has far more close friends on average. This makes sense: going out creates more social contact. But the standard deviations are large and similar across groups, meaning there is substantial individual variability within each group. Something beyond just whether you go out is driving that spread. That is what we will investigate.

The mean of *Being Nice* in our sample is:

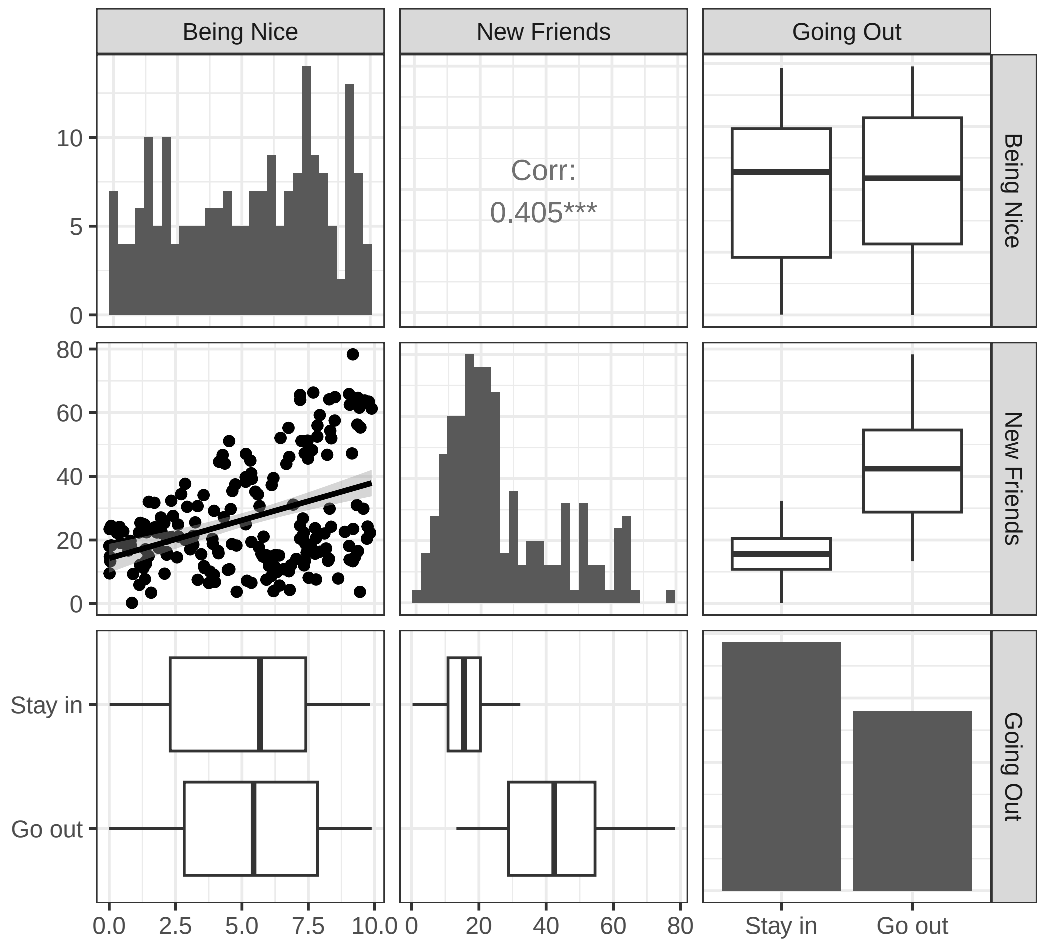
```
round(mean(Friend.Data$BeingNice), 2)
```

```
[1] 5.22
```

So when we center *Being Nice* (which we will do before running the models), a centered value of 0 will correspond to a person who scored approximately 5.22 on the original 0–10 scale.

14.6 A First Look at the Data

Before running any models, let's visualize the relationships in the data. This is always step one.



A few things stand out immediately:

- **Being Nice and New Friends** have a moderate positive correlation overall. More agreeable people tend to form closer friendships.

- **Going Out and New Friends** show a clear difference: students who go out have many more close friends on average.
- The **overall correlation between Being Nice and New Friends** is moderate, but look at the scatterplot: there is a lot of spread around the trend line. Something else might be going on.

The key question is whether the relationship between Being Nice and New Friends looks *the same* for both groups. The pairs plot makes this hard to see directly. That is what the interaction model is for.

14.7 Building the Models

14.7.1 Model 1: The Additive Model

We start with the simpler model: the one that says each variable has its own independent effect, with no interaction. This is our **main effects model** or **additive model**.

$$\widehat{NewFriends} = B_0 + B_1 \cdot BeingNice_C + B_2 \cdot GoingOut + \varepsilon$$

Note two things about how we set this up:

1. We use the **centered** version of Being Nice (`BeingNice.C`), so the intercept will be the predicted friendship score for an average-niceness person.
2. We use the **factored** GoingOut variable (`GoingOut_`), with *Stay in* as the reference group, so R handles the dummy coding automatically.

```
Model.1 <- lm(NewFriends ~ BeingNice.C + GoingOut_, data = Friend.Data)
summary(Model.1)
```

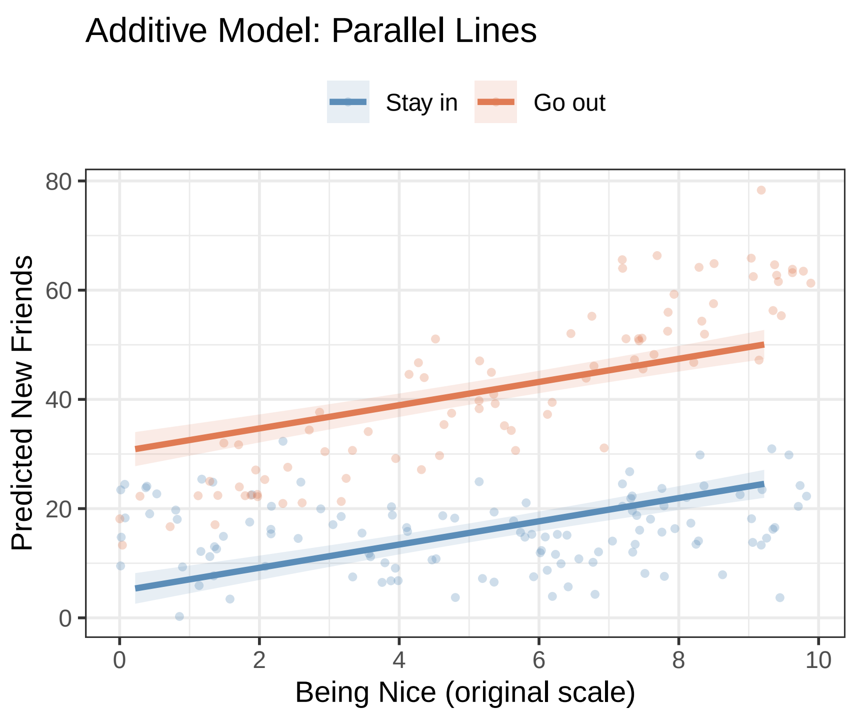
```
Call:
lm(formula = NewFriends ~ BeingNice.C + GoingOut_, data = Friend.Data)

Residuals:
    Min      1Q  Median      3Q     Max 
-21.320  -7.209  -1.063   5.883  28.356 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      16.0242     0.8813  18.182   <2e-16 ***
BeingNice.C       2.1281     0.2307   9.226   <2e-16 ***
GoingOut_Go out  25.5128     1.3613  18.741   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.485 on 197 degrees of freedom
Multiple R-squared:  0.6996,	Adjusted R-squared:  0.6966 
F-statistic: 229.4 on 2 and 197 DF,  p-value: < 2.2e-16
```

14.7.2 What do the coefficients mean?

- **Intercept:** The predicted number of new friends for a person of *average niceness* (BeingNice.C = 0, i.e., roughly a 5.2 on the original scale) who *stays in* (GoingOut = reference group). This is a real, interpretable prediction: someone who is neither particularly nice nor antisocial, and who mostly stays home.
- **BeingNice.C:** Controlling for whether the student goes out or not, each additional point of niceness (above the average) is associated with about 2.13 more friends. This is averaged across both groups.
- **GoingOut_Go out:** Controlling for how nice the student is, going out is associated with about 25.5 more friends than staying in. This is the overall main effect of social engagement.

14.7.3 Visualizing the additive model

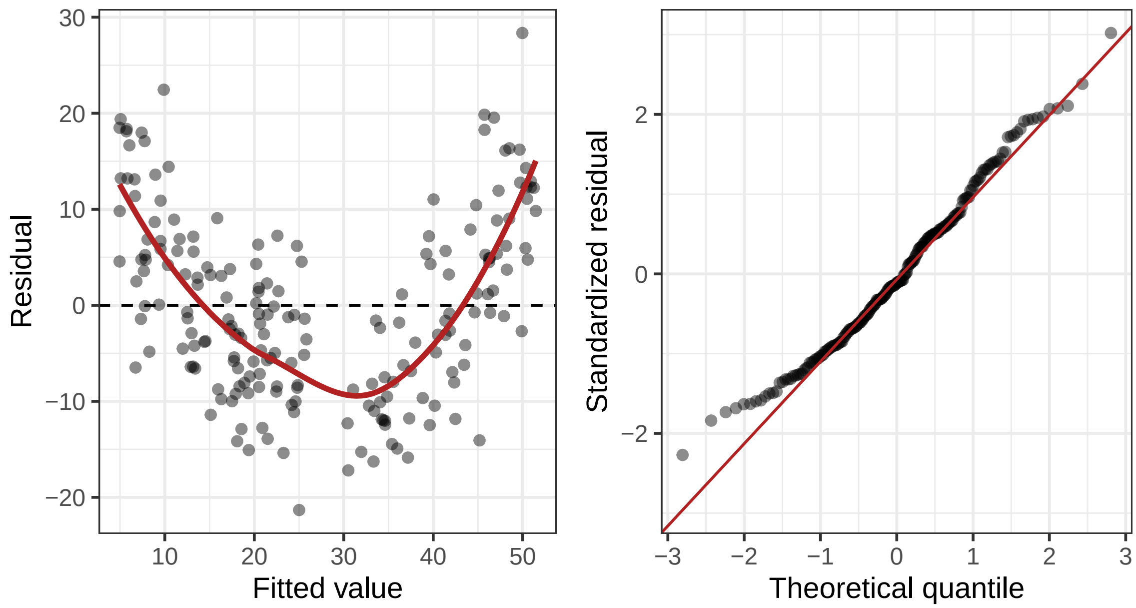
The critical test of the additive model is whether it produces parallel lines when we plot the predicted values.



The additive model forces the two lines to be parallel: that is built into the model's assumptions when there is no interaction term. Notice how the lines run at the same angle, just shifted up for the "Go out" group. This model says: being nice helps equally, no matter what.

But does this model actually fit the data? Look at the raw data points (the faint dots). The "Go out" points (orange) seem to fan upward much more steeply as niceness increases, while the "Stay in" points (blue) look relatively flat. The parallel lines do not seem to capture that pattern. We need a model that lets the slopes differ.

And there is one more diagnostic concern with the additive model. Let's look at the residuals:



The residual plot on the left shows a clear curved pattern, a classic sign that the model is missing something systematic. The data have a structure that a straight-line additive model cannot capture. This is exactly what you would expect when a real interaction exists and you have not modeled it. Time to add the interaction term.

14.7.4 Model 2: The Interaction Model

Now we add the interaction term. The equation becomes:

$$\widehat{NewFriends} = B_0 + B_1 \cdot BeingNice_C + B_2 \cdot GoingOut + B_3 \cdot BeingNice_C \times GoingOut + \varepsilon$$

The new term, $B_3 \cdot BeingNice_C \times GoingOut$, is the product of our two predictors. This is what allows the slope of Being Nice to be *different* for each group. In R, we use the * operator, which automatically includes both main effects and the interaction:

```
Model.2 <- lm(NewFriends ~ BeingNice.C * GoingOut_, data = Friend.Data)
summary(Model.2)
```

```
Call:
lm(formula = NewFriends ~ BeingNice.C * GoingOut_, data = Friend.Data)

Residuals:
     Min       1Q   Median       3Q      Max
-18.2429  -4.4894  -0.2029   4.3964  17.9396

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                  15.7363     0.6183  25.449   <2e-16 ***
BeingNice.C                   0.1998     0.2106   0.949    0.344
GoingOut_Go out              25.2284     0.9548  26.422   <2e-16 ***
```

```
BeingNice.C:GoingOut_Go out   4.7044     0.3289  14.304   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.651 on 196 degrees of freedom
Multiple R-squared:  0.853, Adjusted R-squared:  0.8508
F-statistic: 379.2 on 3 and 196 DF,  p-value: < 2.2e-16
```

Before we interpret these coefficients in detail, let's ask the most important question first: **does this more complex model actually fit the data better?**

14.7.5 Does the Interaction Model Explain More Variance?

Adding the interaction term uses up one additional degree of freedom. We need to check whether the improvement in model fit justifies this cost. We do this with an **F-test for model comparison**, the same logic as hierarchical regression. The null hypothesis is that the interaction explains zero additional variance:

$$H_0 : \Delta R^2 = 0 \quad \text{(the interaction adds nothing)}$$

```
Additive model R-squared:     0.7


Interaction model R-squared:  0.853


Change in R-squared:          0.153
```

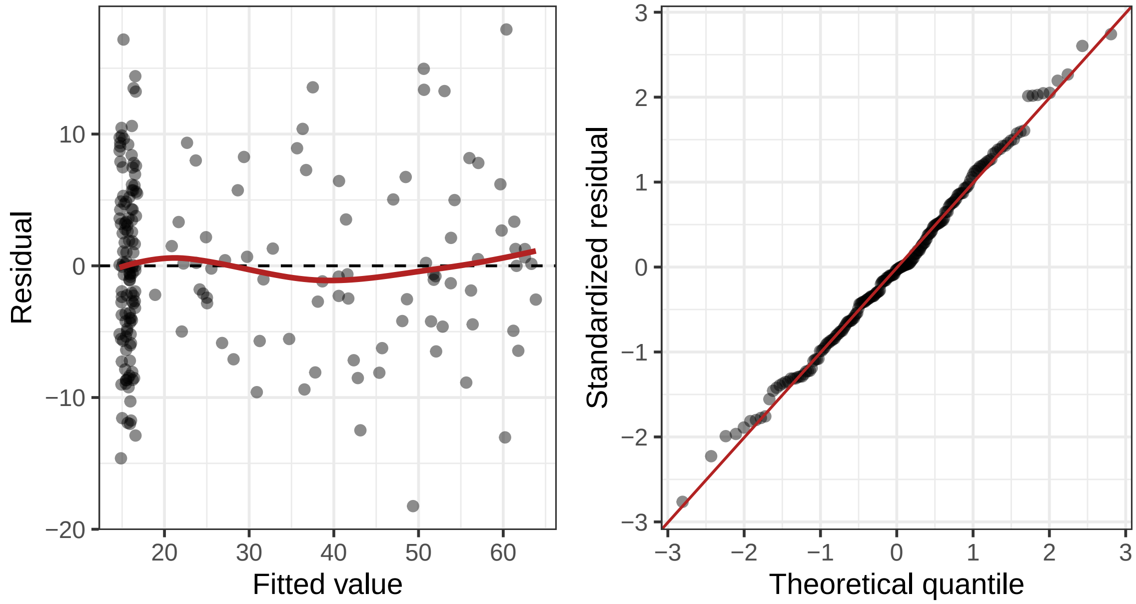
```r
knitr::kable(anova(Model.1, Model.2), digits = 3,
             caption = "Model Comparison: Additive vs. Interaction")
```

Table 14.1: Model Comparison: Additive vs. Interaction

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
197	17722.135	NA	NA	NA	NA
196	8670.551	1	9051.584	204.613	0

The interaction model explains substantially more variance than the additive model. The F-test is significant, meaning that the improvement in R^2 ($\Delta R^2 = 0.153$) is far larger than we would expect by chance. Adding the interaction term was worth it. The two lines are not parallel in the population.

Let's also check the residuals from the interaction model:

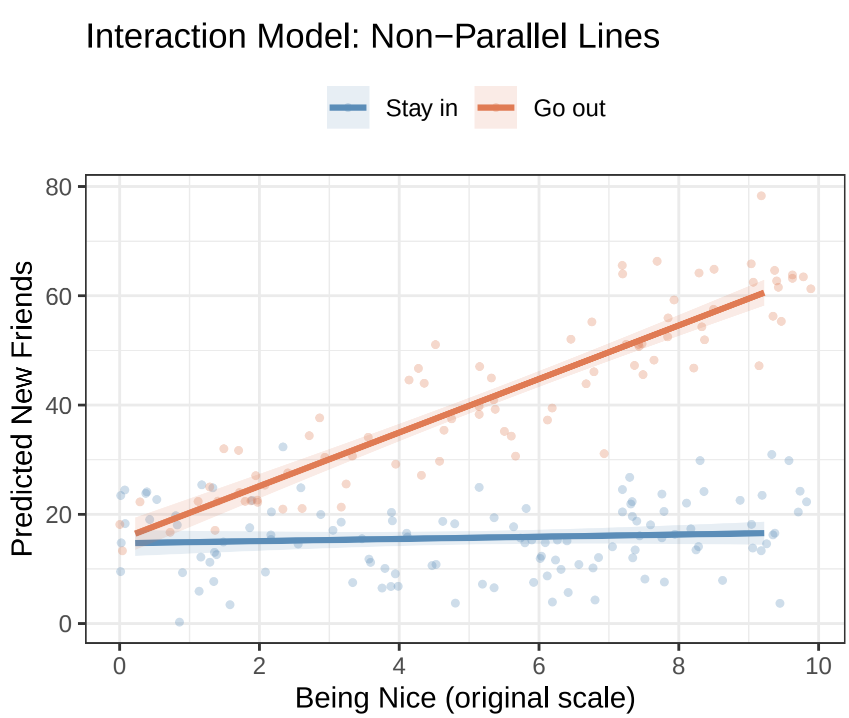


The residual pattern is now much healthier: the curved structure is gone, and the points are scattered more evenly around zero. This confirms that the interaction model is capturing what was actually going on in the data.

14.8 Interpreting the Interaction Model

14.8.1 The Plot First

Before reading the numbers, look at the picture. The plot below shows the predicted values from the interaction model:



This is **Hypothesis B** from our earlier sketch come to life in real data. The "Stay in" line (blue) is essentially flat: being nice does not seem to predict friendship closeness for students who spend the semester at home. The "Go out" line (orange) has a strong positive slope: among students who regularly attend social events, being a warmer and more agreeable person leads to substantially closer friendships. The two lines start at roughly the same point on the left and diverge dramatically as niceness increases.

Compare this to the parallel-lines plot from the additive model. The two models tell completely different stories. The additive model said: being nice helps equally for everyone. The interaction model says: being nice only really pays off if you are actually around people to be nice *to*.

Now let's connect this picture to the numbers.

14.8.2 Walking Through Each Coefficient

Let's go back to the model output and interpret each term carefully, using the plot as our guide.

```
Call:
lm(formula = NewFriends ~ BeingNice.C * GoingOut_, data = Friend.Data)

Residuals:
     Min       1Q   Median       3Q      Max 
-18.2429  -4.4894  -0.2029   4.3964  17.9396 

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)    
(Intercept)                  15.7363     0.6183  25.449   <2e-16 ***
BeingNice.C                   0.1998     0.2106   0.949    0.344    
GoingOut_Go out              25.2284     0.9548  26.422   <2e-16 ***
BeingNice.C:GoingOut_Go out   4.7044     0.3289  14.304   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.651 on 196 degrees of freedom
Multiple R-squared:  0.853,	Adjusted R-squared:  0.8508 
F-statistic: 379.2 on 3 and 196 DF,  p-value: < 2.2e-16
```

14.8.3 The Intercept

The intercept (B_0) is the predicted friendship score when *both* predictors equal zero.

With our setup:

- `BeingNice.C = 0` means a person of **average niceness** (the mean of 5.22 on the original scale)
- `GoingOut = "Stay in"` is the reference group (value $= 0$)

So the intercept is the predicted number of new friends for **an average-niceness person who stays in**. Looking at our model output, the intercept is approximately 15.7.

On the plot, this is the point on the blue “Stay in” line directly above the center of the x-axis (average niceness). It should be around 15–16 friends, enough to form one decent friendship even without going out, if you happen to live near someone.

The p-value on the intercept tests whether this value is significantly different from zero (i.e., do average-niceness, stay-at-home students form *any* friendship at all?). Yes: they form about 15 friends on average, significantly more than zero.

14.8.4 B_1: The Slope of Being Nice for the Stay-In Group

`BeingNice.C` ($B_1 = 0.2$) tells you the slope of Being Nice *for the reference group*, that is, the slope of the blue line for students who *stay in*.

This is called a **simple effect** rather than a main effect, because it is not the effect of Being Nice *on average across groups*: it is the effect of Being Nice *within one specific group*. Specifically, it is the rate of change along the blue line.

The p-value for `BeingNice.C` tests: **is the slope of the “Stay in” line significantly different from zero?**

Looking at the output, this slope is small and non-significant ($p \approx 0.344$). This is exactly what the flat blue line in our plot tells you: for students who stay in, being nicer is not significantly predictive of closer friendships. There is essentially no relationship there.

14.8.5 B_2: The Group Difference at Average Niceness

`GoingOut_Go out` ($B_2 = 25.2$) tells you the predicted difference between the “Go out” and “Stay in” groups **at the centering point of Being Nice**, that is, at average niceness (BeingNice.C $= 0$).

This is a **main effect** in the sense that it is evaluated at the mean of the continuous variable (thanks to centering). It tells you: for a person of average niceness, how many more friends do they get by going out versus staying in?

On the plot, this is the vertical gap between the two lines at the center of the x-axis (average niceness). Students who go out have about 25 more close friends than students who stay in, when both groups are at average niceness. This gap is highly significant.

14.8.6 B_3: The Interaction Term

`BeingNice.C:GoingOut_Go out` ($B_3 = 4.7$) is the heart of the analysis. This coefficient represents **how much steeper the “Go out” slope is compared to the “Stay in” slope.**

Think of it as the answer to: *by how many additional friends, per niceness point, does the “Go out” group outpace the “Stay in” group?*

- The slope for the “Stay in” group is $B_1 = 0.2$ (essentially flat)
- The slope for the “Go out” group is $B_1 + B_3 = 0.2 + 4.7 = 4.9$

That is a very steep slope for the “Go out” group. The p-value for the interaction tests whether this difference in slopes is statistically significant. It is, emphatically.

Here is a summary table to tie everything together:

Coefficient	Value	What it represents	What it tests
Intercept	15.74	Predicted friends for average-niceness, Stay-in student	Is this > 0?
BeingNice.C	0.2	Slope of Being Nice *for Stay-in group* (simple effect)	Is the Stay-in slope $\neq 0$?
GoingOut (Go out)	25.23	Group gap at average niceness (main effect)	Is there a group difference at the mean?
BeingNice.C × GoingOut	4.7	Difference in slopes between groups	Are the two slopes significantly different?

14.9 Simple Slopes: Testing Each Group's Slope Separately

The interaction term tells us that the two slopes *differ significantly*. But we also want to know, for each group separately: **is the relationship between Being Nice and New Friends significantly different from zero?**

We already have the answer for the "Stay in" group: the `BeingNice.C` coefficient is the "Stay in" slope, and it is non-significant. Being nice does not predict friendship closeness for students who stay in.

But what about the "Go out" group? We know the "Go out" slope ($B_1 + B_3 \approx 4.9$) is large and positive, but we need a formal significance test.

We use the **`emtrends()`** function from the **`emmeans`** package, which extracts and tests the slope of Being Nice *separately within each level* of Going Out:

```
simple.slopes <- emtrends(Model.2, "GoingOut_", var = "BeingNice.C")
# infer = TRUE shows both confidence intervals and significance tests in one table
summary(simple.slopes, infer = TRUE)
```

```
 GoingOut_ BeingNice.C.trend    SE  df lower.CL upper.CL t.ratio p.value
 Stay in                 0.2 0.211 196   -0.215    0.615   0.949  0.3439
 Go out                  4.9 0.253 196    4.406    5.402  19.412 <0.0001

Confidence level used: 0.95
```

This output gives us what we need:

- **Stay in group:** The slope of Being Nice is approximately 0.2, which is *not* significantly different from zero ($p = 0.344$). For students who stay in, being nicer does not predict how close a friendship they form.
- **Go out group:** The slope of Being Nice is approximately 4.9, which is *significantly* positive ($p < .001$). For students who go out, every additional point of niceness is associated with about 5 more close friends.

This completes the picture. The effect of Being Nice on friendship closeness is **conditional**: it only emerges among students who are out there meeting people. For students who stay home, personality differences do not seem to matter much for friendship formation. Going out is what activates the effect of being a warm and agreeable person.

14.10 APA Write-Up

Here is how you would write up these results following APA style. Notice the structure: descriptive statistics first, overall model fit, the interaction test, then the simple slopes to unpack what the interaction means.

Prior to the main analysis, descriptive statistics were examined by group. Students who stayed in formed fewer close friendships on average ($M = 15.7$, $SD = 6.7$) compared with students who went out regularly ($M = 42$, $SD = 15.6$).

A hierarchical multiple regression was conducted to examine whether going out moderated the relationship between agreeableness and friendship closeness. Being Nice (centered at the sample mean) and Going Out (dummy coded, *Stay in* as reference group) were entered in Step 1 as main effects; their product (the interaction term) was entered in Step 2. The additive model (Step 1) was statistically significant, $F(2, 197) = 229.42$, $p < .001$, $R^2 = 0.7$. Adding the interaction term (Step 2) produced a statistically significant improvement in model fit, $\Delta R^2 = 0.153$, $F(1, 196) = 204.61$, $p < .001$, indicating that the effect of agreeableness on friendship closeness differed significantly between students who stayed in and those who went out.

In the full interaction model ($R^2 = 0.853$, adjusted $R^2 = 0.851$), the interaction between agreeableness and going out was statistically significant, $b = 4.7$, $SE = 0.33$, $t(196) = 14.3$, $p < .001$. To interpret the interaction, simple slopes for agreeableness were examined separately for each group. Among students who stayed in, agreeableness did not significantly predict friendship closeness, $b = 0.2$, $SE = 0.21$, $t(196) = 0.95$, $p = 0.344$. Among students who went out, agreeableness was a strong positive predictor of friendship closeness, $b = 4.9$, $SE = 0.25$, $t(196) = 19.41$, $p < .001$. These results indicate that the benefit of agreeableness for friendship formation is conditional on social engagement: being a warm and agreeable person leads to closer friendships only among students who are regularly out meeting others.

A few things to notice about this write-up:

- We report the **model comparison** (the ΔR^2 and its F-test) as the primary test of whether an interaction exists.
- We then report the **interaction coefficient** (b, SE, t, p) from the full model.
- We follow up with **simple slopes** (the slope of the continuous variable separately for each group) to explain *what* the interaction looks like. The interaction tells us the slopes differ; the simple slopes tell us how.
- We use the phrase **"the effect of X was conditional on Z"** to describe an interaction in plain language.
- We **do not** emphasize or interpret the main effects from the interaction model in isolation, because in the presence of a significant interaction, the main effects represent the effect at one specific value of the other variable (in this case, the reference group or the mean) and can be misleading to report without that context.

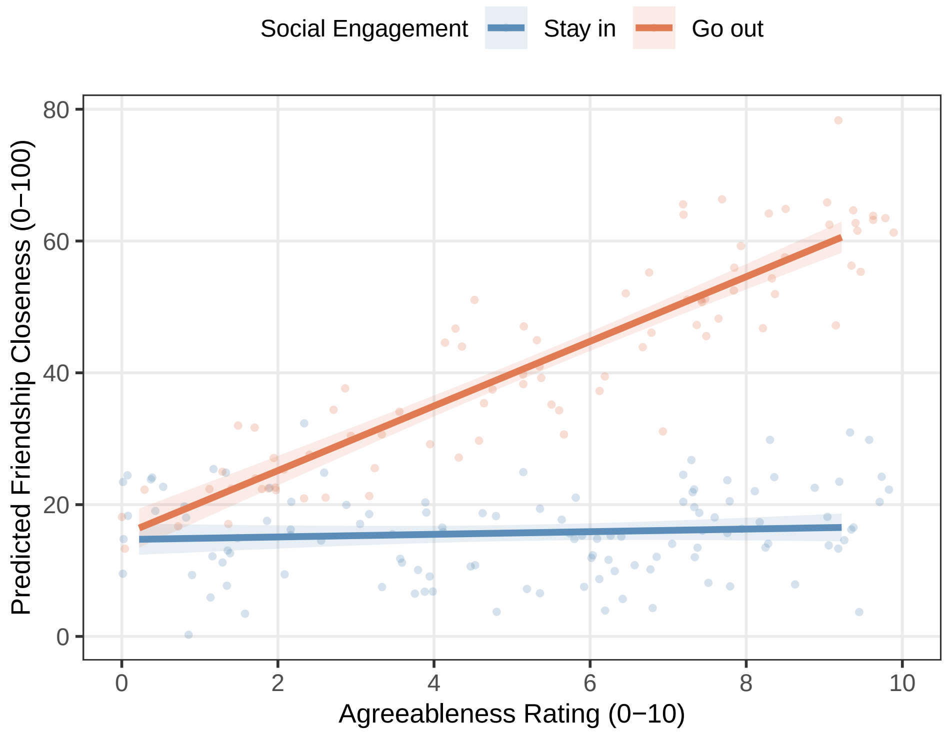
14.11 APA-Style Figure

A well-constructed figure is often the clearest way to communicate an interaction. Below is a publication-ready interaction plot that you can adapt for your own write-ups. This figure shows the predicted values from the interaction model, with 95% confidence bands, and the raw data displayed in the background for transparency.

Figure 1. Predicted friendship closeness as a function of agreeableness and social engagement. Lines represent predicted values from the interaction model; shaded bands represent 95% confidence intervals; points represent raw data. Among students who regularly went out (*Go out*; orange), agreeableness was a strong positive predictor of friendship closeness. Among students who stayed in (*Stay in*; blue), agreeableness was not significantly associated with friendship closeness.

```
# --- APA-Style Interaction Plot ---

# Step 1: Get predicted values from the model using emmeans
Inter.Em.Fig <- emmeans(Model.2,
                        ~ BeingNice.C | GoingOut_,
```

```
                        at = list(BeingNice.C = seq(Min.BN, Max.BN, 0.25)))
Fig.Data <- as.data.frame(Inter.Em.Fig)

# Step 2: Transform centered x-axis back to original scale (0-10)
mean.BN <- mean(Friend.Data$BeingNice)
Fig.Data$BeingNice.orig <- Fig.Data$BeingNice.C + mean.BN

# Step 3: Build the plot
ggplot(Fig.Data, aes(x     = BeingNice.orig,
                     y     = emmean,
                     color = GoingOut_,
                     fill  = GoingOut_,
                     group = GoingOut_)) +
  # Raw data points (faint, for context)
  geom_point(data  = Friend.Data,
             aes(x = BeingNice, y = NewFriends,
                 color = GoingOut_, group = GoingOut_),
             alpha = 0.25, size = 0.8) +
  # Confidence ribbons
  geom_ribbon(aes(ymin = lower.CL, ymax = upper.CL),
              alpha = 0.15, color = NA) +
  # Regression lines
  geom_line(linewidth = 1.2) +
  # Formatting
  scale_color_manual(values = c("Stay in" = "#5b8db8",
                                "Go out"  = "#e07b54"),
                     labels = c("Stay in", "Go out")) +
  scale_fill_manual(values  = c("Stay in" = "#5b8db8",
                                "Go out"  = "#e07b54"),
                    labels = c("Stay in", "Go out")) +
  scale_x_continuous(breaks = seq(0, 10, 2), limits = c(0, 10)) +
  labs(x     = "Agreeableness Rating (0-10)",
       y     = "Predicted Friendship Closeness (0-100)",
       color = "Social Engagement",
       fill  = "Social Engagement") +
  theme_bw() +
  theme(legend.position   = "top",
        legend.title      = element_text(size = 9),
        legend.text       = element_text(size = 9),
        axis.title        = element_text(size = 10),
        panel.grid.minor  = element_blank())
```



⚠ Do Not Diagnose an Interaction by Staring at Two P-Values

One simple slope being significant while the other is not does **not** prove that the slopes differ. The interaction coefficient is the direct test of that difference. The plot and simple slopes explain the interaction after it has been tested.

15 Categorical-by-Categorical Interactions: The 2-by-2 Design

15.1 Where We Are and Where We Are Going

In the prior chapter you learned how to detect, estimate, and interpret an interaction between a *continuous* predictor and a *categorical* one. In the friendship study, Being Nice (0–10 scale) was continuous and Going Out (Stay in vs. Go out) was categorical. The interaction told you that the **slope** of Being Nice on friendship closeness was different for students who went out compared to those who stayed in.

That kind of interaction is very common in psychology research. But so is a second kind: interactions where **both predictors are categorical**. You have probably already encountered this design — it is the foundation of the **two-way factorial ANOVA**. A 2 × 2 design (two predictors, each with two levels) is the simplest and most common version.

This chapter bridges the two frameworks. We will keep the same friendship study and the same hypothesis, but we will make one important change: instead of treating Being Nice as a full continuous variable, we will focus on **just the extremes** — students whose niceness scores are near 0 (the least agreeable people in the sample) and students whose scores are near 10 (the most agreeable). This turns our continuous variable into a two-level categorical variable: **Low Nice** vs. **High Nice**. Combined with Going Out (Stay in vs. Go out), we have a **2 × 2 design**.

This chapter assumes you have already read the prior chapter material. You know what an interaction is, you know how to interpret the interaction model equation, and you know how to use `anova()` to compare a main effects model to an interaction model. What is new here is what changes — and what stays the same — when both predictors are categorical.

15.1.1 What Changes from prior chapter

The biggest conceptual change is this: **there is no continuous variable to center.** In the prior chapter centering was necessary because the intercept needed a meaningful zero point on the continuous scale. When both predictors are categorical, every predictor already takes on a small set of discrete values, and the "zero point" is simply the reference category of each variable — a specific cell in the design, not an abstract mean.

Everything else follows from that:

- No centering is needed. Both variables just need dummy coding.
- The interaction coefficient is no longer a "difference in slopes." There are no slopes here. Instead, B_3 is the **difference in group differences** — sometimes called the "difference of differences." We will unpack this carefully.
- The follow-up to the interaction is **simple effects analysis**: testing, within each level of one variable, whether the levels of the other variable differ significantly. We use `pairs()` from the `emmeans` package instead of `emtrends()`.
- Effect sizes for pairwise comparisons are expressed as **Cohen's *d***, not unstandardized slopes.

What **stays the same**: the interaction is still about "it depends." The visual signature is still non-parallel lines (or unequal gaps in a bar chart). The model comparison is still done with `anova()` and ΔR^2. The APA write-up still follows the same structure.

15.1.2 What the 2×2 Looks Like as a Table

Before we touch any data, it helps to lay out the full design as a 2×2 table. Each cell represents a unique combination of the two categorical predictors, and we expect a separate (estimated) mean outcome in each cell.

	Stay in	**Go out**
Low Nice	μ_{11}: Low Nice × Stay in	μ_{12}: Low Nice × Go out
High Nice	μ_{21}: High Nice × Stay in	μ_{22}: High Nice × Go out

The interaction question is: **is the difference between High Nice and Low Nice the same in both columns (Going Out groups)?** Or equivalently: **is the difference between Go out and Stay in the same in both rows (Niceness groups)?**

Both questions ask the same thing. The answer is the interaction.

15.2 The Core Concept: The "Difference of Differences"

15.2.1 What It Means When Both Predictors Are Categorical

In the prior chapter, the interaction coefficient B_3 told you how much steeper the slope of Being Nice was for the "Go out" group compared to the "Stay in" group. Slopes are a continuous concept: the rate at which friendship closeness increases as niceness increases by one unit.

When Being Nice is categorical (Low vs. High), there are no slopes. Instead, there is a **gap**: the difference in predicted friendship closeness between High Nice and Low Nice students. The interaction now asks whether that gap is the same size in both Going Out groups.

Here is the idea in formula form. Define:

- **Effect of Niceness for Stay-in students** $= \mu_{21} - \mu_{11}$ (High Nice – Low Nice, among those who stay in)
- **Effect of Niceness for Go-out students** $= \mu_{22} - \mu_{12}$ (High Nice – Low Nice, among those who go out)

The **interaction** is the difference between these two effects:

$$B_3 = (\mu_{22} - \mu_{12}) - (\mu_{21} - \mu_{11})$$

This is the "difference of differences." If there is no interaction, the effect of Niceness is the same size in both Going Out groups, and $B_3 = 0$. If there is an interaction, the effect of Niceness is larger in one group than the other, and $B_3 \neq 0$.

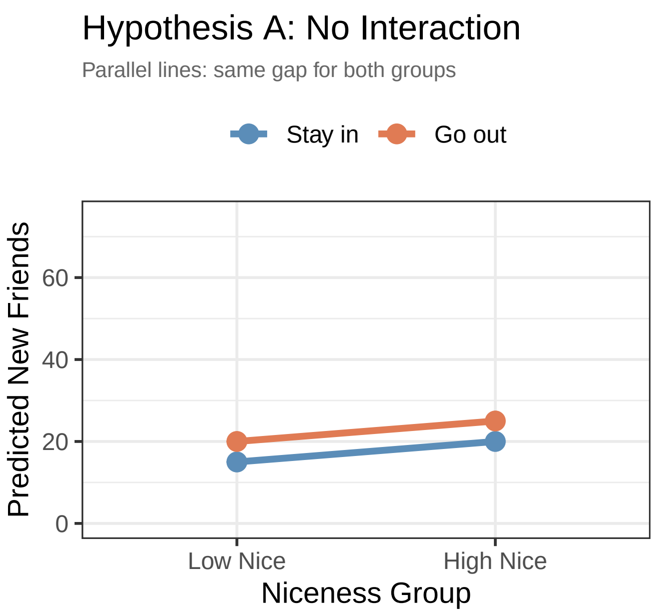
Equivalently (and this is worth pausing on), you can write it the other way:

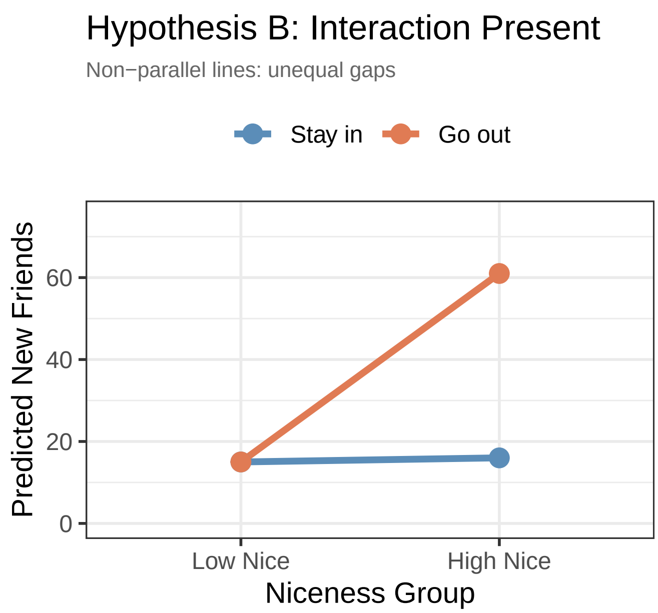
$$B_3 = (\mu_{22} - \mu_{21}) - (\mu_{12} - \mu_{11})$$

This version asks: is the effect of Going Out (Go out – Stay in) the same for High Nice and Low Nice students? The two formulations are mathematically identical and represent the same interaction. Which framing you use depends on which story you want to tell. In our study, we are primarily interested in whether the **benefit of being nice depends on whether you go out** — so we will usually use the first framing.

15.2.2 Visual Hypotheses: What We Expect to See

As in the prior chapter, we can sketch out our two competing possibilities before running a single model.





Notice the visualization has changed from the prior chapter. Instead of regression lines across a continuous X-axis, we now have two points connected by a line for each Going Out group. The key visual signature of an interaction is still **non-parallel lines** — but "non-parallel" now means the **gaps between the two groups are different sizes** across the niceness levels, rather than different slopes.

- In **Hypothesis A**, the two lines are parallel: the gap between "Go out" and "Stay in" is the same for Low Nice and High Nice students. Being nice adds a fixed amount regardless of going out.
- In **Hypothesis B**, the lines fan out: the gap between "Go out" and "Stay in" is tiny for Low Nice students but enormous for High Nice students. This is our interaction hypothesis: being nice only pays off substantially if you are also out there meeting people.

15.3 Before We Run the Models: Dummy Coding Both Variables

15.3.1 No Centering Required

In the prior chapter, you had to center the continuous predictor (Being Nice) before building the interaction model. Centering shifted the scale so that zero represented the average person, making the intercept interpretable.

When both predictors are categorical, there is nothing to center. Each predictor already takes on discrete values. The "zero point" for each predictor is simply its **reference category** — whichever group we designate as the baseline.

This is actually simpler than the prior chapter. The only setup step is making sure both variables are properly factored in R, with the reference groups ordered first.

15.3.2 Dummy Coding Both Variables

We have two categorical predictors, each with two levels:

Variable	Levels	Reference group (coded 0)	Comparison group (coded 1)
NiceGroup	Low Nice, High Nice	Low Nice	High Nice
GoingOut	Stay in, Go out	Stay in	Go out

With this coding, the **four coefficients** in the interaction model have the following meanings:

Coefficient	What it represents
B_0 (Intercept)	Predicted friends for **Low Nice** × **Stay in** students — the baseline cell
B_1 (NiceGroup: High Nice)	High Nice – Low Nice **among Stay-in students** (simple effect of niceness for stay-in group)
B_2 (GoingOut: Go out)	Go out – Stay in **among Low Nice students** (simple effect of going out for low-nice group)
B_3 (NiceGroup × GoingOut)	The "difference of differences": does the effect of niceness differ by going-out group?

Notice that B_1 and B_2 are no longer main effects in the traditional sense — they are **simple effects**, each representing the effect of one variable *holding the other variable at its reference level.* This is the same logic as in the prior chapter, where B_1 was the slope for the reference group only. Here, B_1 is the niceness gap for the reference going-out group only.

15.3.3 Recovering All Four Cell Means

One of the useful features of the 2 × 2 interaction model is that you can reconstruct **every cell mean** directly from the coefficients. Here is the formula for each cell:

Cell	Formula	Plain meaning
Low Nice × Stay in	B_0	Baseline
High Nice × Stay in	$B_0 + B_1$	Baseline + niceness effect (Stay in)
Low Nice × Go out	$B_0 + B_2$	Baseline + going-out effect (Low Nice)
High Nice × Go out	$B_0 + B_1 + B_2 + B_3$	All four terms

The final cell is the most important: for High Nice students who also go out, the prediction is the baseline plus the simple effect of niceness, plus the simple effect of going out, **plus the interaction term**. The interaction term B_3 adjusts the prediction for the cell where both variables are at their non-reference level. If there is no interaction ($B_3 = 0$), the effects of the two variables add up independently and the model is purely additive.

15.4 The Data

We return to our 200 first-semester students. Recall that we measured friendship closeness (0–100 feelings thermometer), agreeableness (0–10), and whether they went out regularly.

For this analysis, we focus on the **extremes of the niceness distribution**: students who scored below 2 on the agreeableness scale (near 0 = **Low Nice**) and students who scored above 8 (**High Nice**). The 120 or so students in the middle of the scale are not used in this analysis — we are focusing on the contrast between the most and least agreeable people in the sample.

The resulting dataset has 81 participants spread across four cells:

```
Friend.Data %>%
  group_by(NiceGroup_, GoingOut_) %>%
  summarise(n = n(),
            M_Friends  = round(mean(NewFriends), 2),
            SD_Friends = round(sd(NewFriends), 2),
            .groups = "drop")
```

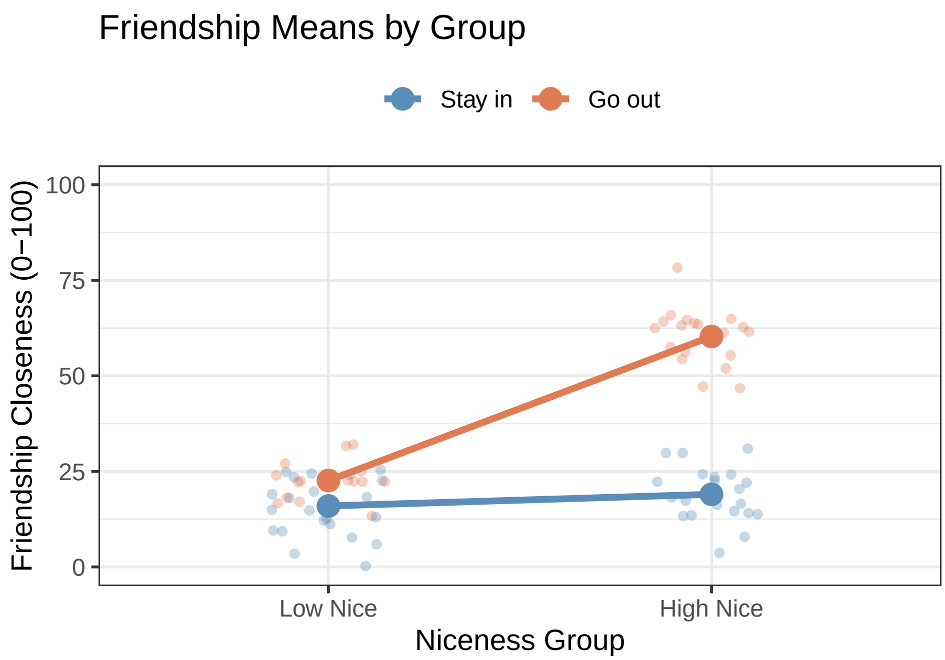
```
# A tibble: 4 x 5
  NiceGroup_ GoingOut_     n M_Friends SD_Friends
  <fct>      <fct>     <int>     <dbl>      <dbl>
1 Low Nice   Stay in      25      16.0       7.27
2 Low Nice   Go out       16      22.6       4.97
3 High Nice  Stay in      21      19         7.05
4 High Nice  Go out       19      60.3       7.32
```

The cell counts are not equal — this is typical when you extract extremes from a continuous distribution rather than randomly assigning participants to conditions. In regression (unlike ANOVA), this unequal cell size is handled automatically. The estimates are still valid.

The cell means already tell an interesting story: Low Nice students have similar friendship scores whether they go out or not (both around 15), while High Nice students who go out have dramatically higher scores than High Nice students who stay in. This pattern matches Hypothesis B.

15.5 A First Look at the Data

Before running any models, let's visualize the four cells directly.



The pattern is striking. The "Stay in" line (blue) is almost flat: Low Nice and High Nice students who stay in have virtually the same friendship scores. The "Go out" line (orange) rises sharply: High Nice students who go out form far closer friendships than Low Nice students who go out. The two lines are clearly **non-parallel**, which is the visual signature of an interaction. This picture already suggests Hypothesis B.

15.6 Building the Models

15.6.1 Model 1: The Additive Model

We begin with the main effects (additive) model. This model says the two variables each have an independent effect on friendship closeness, and those effects add up without interacting.

$$\widehat{NewFriends} = B_0 + B_1 \cdot NiceGroup + B_2 \cdot GoingOut + \varepsilon$$

```
Model.1 <- lm(NewFriends ~ NiceGroup_ + GoingOut_, data = Friend.Data)
summary(Model.1)
```

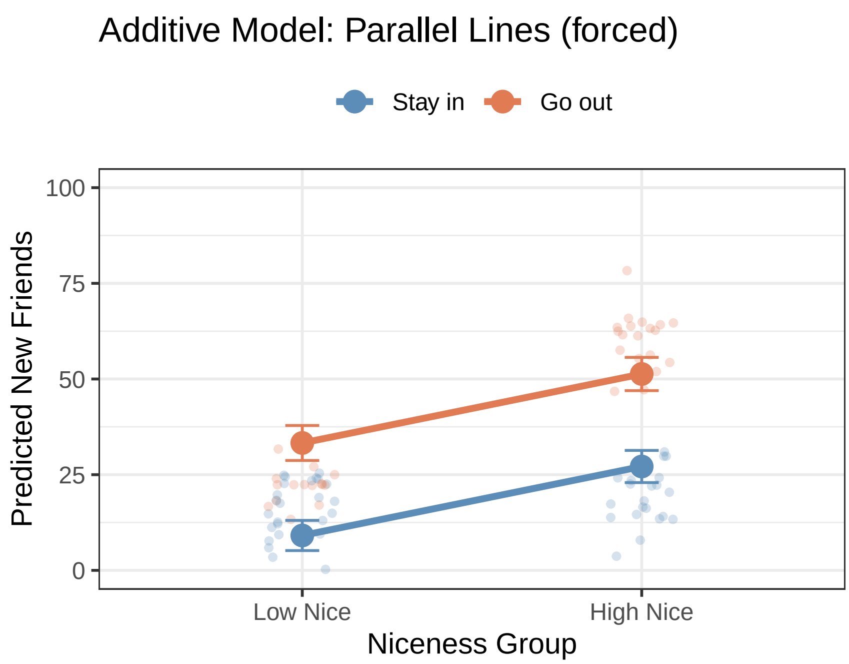
```
Call:
lm(formula = NewFriends ~ NiceGroup_ + GoingOut_, data = Friend.Data)

Residuals:
     Min       1Q   Median       3Q      Max 
-23.4417  -9.7999   0.2044   9.9769  27.0074 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)            9.112      1.980   4.602 1.59e-05 ***
```

```
NiceGroup_High Nice   18.025      2.465   7.314 1.98e-10 ***
GoingOut_Go out       24.173      2.487   9.718 4.40e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.05 on 78 degrees of freedom
Multiple R-squared:  0.674, Adjusted R-squared:  0.6657
F-statistic: 80.64 on 2 and 78 DF,  p-value: < 2.2e-16
```

The additive model estimates a single effect of NiceGroup (averaged across both Going Out groups) and a single effect of GoingOut (averaged across both Niceness groups). These are conventional main effects.

Notice that the R^2 here (0.674) is already reasonably high, driven primarily by the Going Out main effect. But as we will see, forcing the effects to be additive is a poor fit for the actual pattern in the data.

15.6.2 Visualizing the Additive Model



The additive model forces the two lines to be parallel. The gap between "Go out" and "Stay in" is identical for Low Nice and High Nice students. But look at the raw data: the "Go out" students in the High Nice group (orange, right side) are far higher than what the parallel-line model predicts for them, and the "Stay in" students in the same group are not as high as the model suggests. The parallel assumption is clearly wrong.

15.6.3 Model 2: The Interaction Model

Now we add the interaction term:

$$\widehat{NewFriends} = B_0 + B_1 \cdot NiceGroup + B_2 \cdot GoingOut + B_3 \cdot NiceGroup \times GoingOut + \varepsilon$$

```r
Model.2 <- lm(NewFriends ~ NiceGroup_ * GoingOut_, data = Friend.Data)
summary(Model.2)
```

```

Call:
lm(formula = NewFriends ~ NiceGroup_ * GoingOut_, data = Friend.Data)

Residuals:
     Min       1Q   Median       3Q      Max 
-15.7001  -4.7327  -0.0055   4.3511  18.0093 

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)                           15.950      1.367  11.670  < 2e-16 ***
NiceGroup_High Nice                    3.045      2.023   1.505  0.13632    
GoingOut_Go out                        6.649      2.188   3.039  0.00324 ** 
NiceGroup_High Nice:GoingOut_Go out   34.663      3.077  11.265  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.834 on 77 degrees of freedom
Multiple R-squared:  0.8769,	Adjusted R-squared:  0.8721 
F-statistic: 182.8 on 3 and 77 DF,  p-value: < 2.2e-16
```

15.6.4 Does the Interaction Model Explain More Variance?

```r
knitr::kable(anova(Model.1, Model.2), digits = 3,
             caption = "Model Comparison: Additive vs. Interaction")
```

Table 15.1: Model Comparison: Additive vs. Interaction

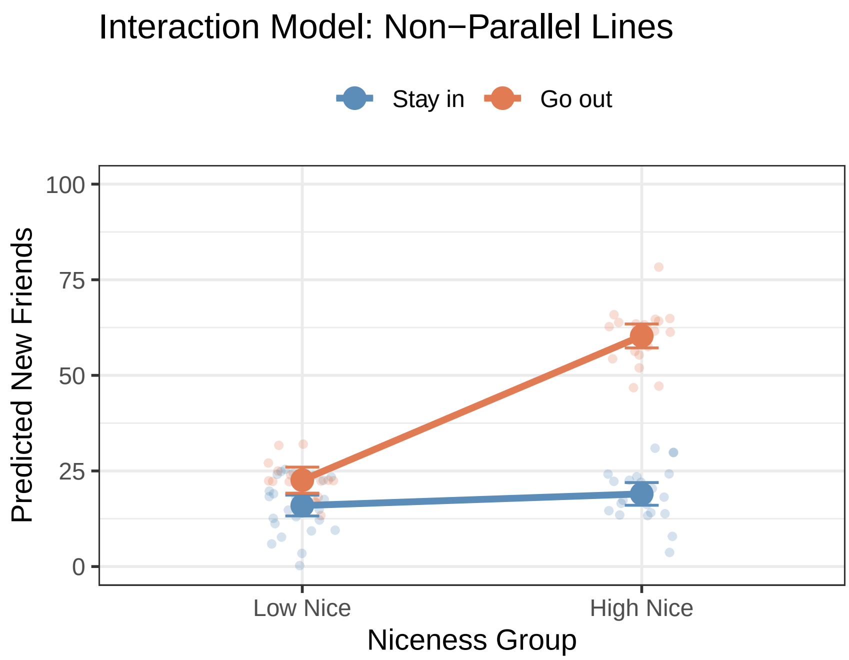
Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
78	9522.081	NA	NA	NA	NA
77	3595.951	1	5926.13	126.896	0

```
Additive model R-squared:     0.674

Interaction model R-squared:  0.877

Change in R-squared:          0.203
```

The interaction model explains substantially more variance. The F-test ($\Delta R^2 = 0.203$, $F(1, 77) = 126.9$, $p < .001$) confirms that this improvement is far larger than would be expected by chance. Adding the interaction term was worth it.

15.7 Interpreting the Interaction Model

15.7.1 The Plot First



This is the interaction model's predicted means for each cell. Notice the non-parallel lines: the "Stay in" line is nearly flat (very small gap between Low Nice and High Nice), while the "Go out" line rises steeply (a large gap between Low Nice and High Nice). Compare this to the additive model's forced parallel lines — the story told by these two models is completely different.

15.7.2 Walking Through Each Coefficient

```
Call:
lm(formula = NewFriends ~ NiceGroup_ * GoingOut_, data = Friend.Data)

Residuals:
     Min       1Q   Median       3Q      Max
-15.7001  -4.7327  -0.0055   4.3511  18.0093

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)
(Intercept)                           15.950      1.367  11.670  < 2e-16 ***
NiceGroup_High Nice                    3.045      2.023   1.505  0.13632
GoingOut_Go out                        6.649      2.188   3.039  0.00324 **
NiceGroup_High Nice:GoingOut_Go out   34.663      3.077  11.265  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.834 on 77 degrees of freedom
```

```
Multiple R-squared:  0.8769,    Adjusted R-squared:  0.8721
F-statistic: 182.8 on 3 and 77 DF,  p-value: < 2.2e-16
```

15.7.3 The Intercept ($B_0 = 15.95$)

The intercept is the predicted friendship score for the **baseline cell**: Low Nice students who Stay in (both predictors at 0). This is a real, specific group of people in the dataset. A completely disagreeable person who never leaves home still forms about 15.95 points on the friendship thermometer — presumably because they live near someone and physical proximity does the work even for unsociable people.

15.7.4 B_1: Effect of High Niceness Among Stay-in Students ($B_1 = 3.05$)

B_1 is the difference in predicted friendship between High Nice and Low Nice students **within the Stay-in group**. Looking at the "Stay in" line in the plot, this gap is very small — and indeed, $B_1 = 3.05$, which is essentially zero and non-significant.

For students who stay home, being highly agreeable provides virtually no additional benefit for friendship formation compared to being low in agreeableness. Without social exposure, personality differences do not matter much.

This is a **simple effect**: it is the effect of NiceGroup *within one specific level* of GoingOut (the reference level, Stay in). It is not a main effect.

15.7.5 B_2: Effect of Going Out Among Low Nice Students ($B_2 = 6.65$)

B_2 is the difference in predicted friendship between Go-out and Stay-in students **within the Low Nice group**. Students low in agreeableness who go out have about 6.65 more friends than their equally disagreeable counterparts who stay in.

This is also a **simple effect**: the effect of GoingOut *within one specific level* of NiceGroup (the reference level, Low Nice).

15.7.6 B_3: The Interaction Term ($B_3 = 34.66$)

This is the heart of the analysis. B_3 tells you how much **larger** the effect of niceness is for the Go-out group compared to the Stay-in group.

We can reconstruct the simple effect of Niceness for each Going Out group:

Group	Simple effect of Niceness	Formula
Stay in	3.05	B_1
Go out	37.71	$B_1 + B_3 = 3.05 + 34.66$

The difference between these two simple effects is $B_3 = 34.66$. For Go-out students, being highly agreeable is associated with about 37.71 more friends than being low in agreeableness. For Stay-in students, the same comparison yields only 3.05 more friends. The interaction coefficient captures this dramatic difference.

Here is the full cell mean summary:

Cell	Predicted mean	Derivation
Low Nice × Stay in	15.95	B_0
High Nice × Stay in	19	$B_0 + B_1$
Low Nice × Go out	22.6	$B_0 + B_2$
High Nice × Go out	60.31	$B_0 + B_1 + B_2 + B_3$

15.8 Following Up the Interaction: Simple Effects

The interaction term tells us the two niceness gaps are significantly different between Going Out groups. But we also want to know: **within each Going Out group, is the difference between High Nice and Low Nice statistically significant?**

We use `pairs()` from the `emmeans` package to test each pairwise comparison within each level of the other variable.

15.8.1 Simple Effects of Niceness within Each Going-Out Group

```
Fit.2 <- emmeans(Model.2, ~ NiceGroup_ | GoingOut_)
pairs(Fit.2, adjust = "none") %>% summary(infer = TRUE)
```

```
GoingOut_ = Stay in:
 contrast             estimate   SE df lower.CL upper.CL t.ratio p.value
 Low Nice - High Nice    -3.05 2.02 77    -7.07    0.983  -1.505  0.1363

GoingOut_ = Go out:
 contrast             estimate   SE df lower.CL upper.CL t.ratio p.value
 Low Nice - High Nice   -37.71 2.32 77   -42.33  -33.091 -16.262 <0.0001

Confidence level used: 0.95
```

- **Stay-in group:** The difference between High Nice and Low Nice students is -3.05 points, which is **not** statistically significant ($p = 0.136$). For students who stay in, agreeableness does not significantly predict friendship closeness.
- **Go-out group:** The difference between High Nice and Low Nice students is -37.71 points, which is **highly** significant ($p < .001$). Among students who go out, being highly agreeable is associated with substantially closer friendships than being low in agreeableness.

15.8.2 Effect Sizes: Cohen's *d*

Pairwise comparisons in this framework can be accompanied by Cohen's d as an effect size measure. We use the model's residual standard deviation as the pooled standard deviation:

```
eff_size(Fit.2,
         sigma = sigma(Model.2),
         edf   = df.residual(Model.2),
         method = "pairwise")
```

```
GoingOut_ = Stay in:
 contrast             effect.size    SE df lower.CL upper.CL
 Low Nice - High Nice      -0.446 0.298 77    -1.04    0.148

GoingOut_ = Go out:
 contrast             effect.size    SE df lower.CL upper.CL
 Low Nice - High Nice      -5.518 0.559 77    -6.63   -4.404

sigma used for effect sizes: 6.834
Confidence level used: 0.95
```

The Cohen's d values confirm the pattern:

- **Stay in:** $d = 0.45$ — a negligible effect. Being nice does not meaningfully distinguish friendships for students who stay home.
- **Go out:** $d = 5.52$ — a large effect. Among students who go out, highly agreeable people form dramatically closer friendships than their less agreeable counterparts.

15.9 APA Write-Up

Here is how you would report these results following APA style. The structure parallels the prior chapter: model comparison, interaction coefficient, then simple effects.

Cell means and standard deviations are presented in Table 1. Among students who stayed in, friendship closeness was similar regardless of agreeableness group (Low Nice: $M = 16$, $SD = 7.3$; High Nice: $M = 19$, $SD = 7.1$). Among students who went out, High Nice students formed substantially closer friendships than Low Nice students (Low Nice: $M = 22.6$, $SD = 5$; High Nice: $M = 60.3$, $SD = 7.3$).

A hierarchical multiple regression was conducted to examine whether social engagement (Going Out) moderated the relationship between agreeableness group (Low vs. High Nice) and friendship closeness. Niceness group and Going Out (both dummy coded with Low Nice and Stay in as reference groups) were entered in Step 1 as main effects; their product was entered in Step 2. The additive model (Step 1) was statistically significant, $F(2, 78) = 80.64$, $p < .001$, $R^2 = 0.674$. Adding the interaction term (Step 2) produced a statistically significant improvement in model fit, $\Delta R^2 = 0.203$, $F(1, 77) = 126.9$, $p < .001$, indicating that the effect of agreeableness on friendship closeness differed significantly depending on whether students went out.

In the full interaction model ($R^2 = 0.877$, adjusted $R^2 = 0.872$), the interaction between Niceness Group and Going Out was statistically significant, $b = 34.66$, $SE = 3.08$, $t(77) = 11.26$, $p < .001$. To interpret the interaction, simple effects of Niceness Group were examined separately within each level of Going Out. Among students who stayed in, agreeableness group did not significantly predict friendship closeness, $b =$ -3.05, $SE = 2.02$, $t(77) =$ -1.51, $p = 0.136$, $d = 0.45$. Among students who went out, High Nice students formed significantly closer friendships than Low Nice students, $b =$ -37.71, $SE = 2.32$, $t(77) =$ -16.26, $p < .001$, $d = 5.52$. These results indicate that the benefit of high agreeableness for friendship formation is conditional on social engagement: being a warm and agreeable person leads to substantially closer friendships only among students who are regularly out meeting others.

Things to notice about this write-up:

- We report **Cohen's *d*** alongside the simple effects, since the comparisons are between two categorical groups. This is the appropriate effect size for a two-group mean difference.
- The **model comparison** (ΔR^2, F-test) is still the primary test of the interaction, exactly as in the prior chapter.

- We describe the interaction in plain language: *"the benefit of high agreeableness … is conditional on social engagement."*
- The **main effects** from the interaction model are not heavily emphasized. In the presence of a significant interaction, they represent effects at the reference level of the other variable (simple effects), not population-level averages.

15.10 APA-Style Figure

```
# APA interaction plot: categorical × categorical
Fig.Data <- as.data.frame(emmeans(Model.2, ~ NiceGroup_ | GoingOut_))

ggplot(Fig.Data, aes(x = NiceGroup_, y = emmean,
                     color = GoingOut_, group = GoingOut_)) +
  # Raw data (faint)
  geom_point(data = Friend.Data,
             aes(x = NiceGroup_, y = NewFriends, color = GoingOut_),
             alpha = 0.20, size = 1,
             position = position_jitter(width = 0.12)) +
  # Confidence intervals
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL),
                width = 0.08, linewidth = 0.8) +
  # Connecting lines
  geom_line(linewidth = 1.2) +
  # Cell mean points
  geom_point(size = 3.5) +
  scale_color_manual(values = c("Stay in" = "#5b8db8",
                                "Go out"  = "#e07b54"),
                     labels = c("Stay in", "Go out")) +
  scale_y_continuous(limits = c(0, 100),
                     breaks = seq(0, 100, 20)) +
  labs(x     = "Agreeableness Group",
       y     = "Predicted Friendship Closeness (0-100)",
       color = "Social Engagement") +
  theme_bw() +
  theme(legend.position   = "top",
        legend.title      = element_text(size = 9),
        legend.text       = element_text(size = 9),
        axis.title        = element_text(size = 10),
        panel.grid.minor  = element_blank())
```

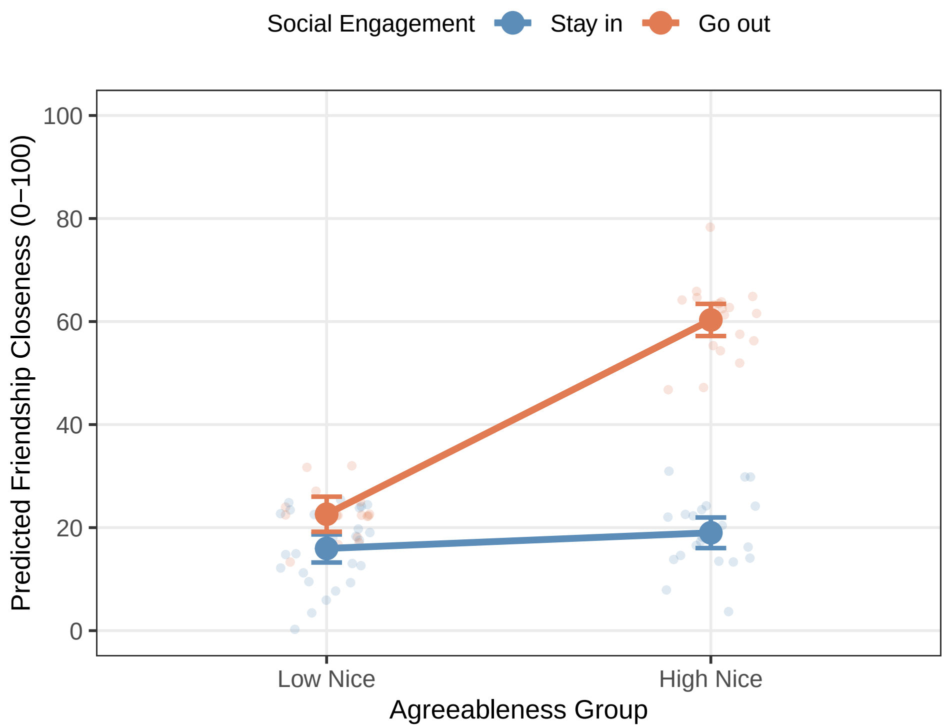


Figure 1. Predicted friendship closeness as a function of agreeableness group and social engagement. Points represent model-estimated cell means; error bars represent 95% confidence intervals; faint points represent raw data. Among students who went out (*Go out*; orange), highly agreeable students formed substantially closer friendships than less agreeable students ($d = 5.52$). Among students who stayed in (*Stay in*; blue), agreeableness group was not significantly associated with friendship closeness ($d = 0.45$).

15.11 Comparison: Continuous × Categorical (the prior chapter) vs. Categorical × Categorical (This chapter)

It is worth pausing to compare the two approaches we have now seen.

Feature	Prior chapter (Continuous × Categorical)	This chapter (Categorical × Categorical)
Being Nice	Continuous (0–10)	Categorical (Low vs. High)
Centering	Required	Not needed
Interaction coefficient	Difference in slopes	Difference of differences
Follow-up	`emtrends()` — simple slopes	`pairs()` — simple effects
Effect size	Unstandardized slope (b)	Cohen's d
Sample used	All 200 participants	~80 (extremes only)

The categorical version uses fewer participants (only the extremes of the niceness distribution), which means less statistical power. The continuous version squeezes more information out of the data because it uses every participant and models the full gradient of niceness effects.

However, the categorical × categorical design is conceptually simpler and maps directly onto the two-way factorial ANOVA framework that many researchers in psychology use. Understanding both is essential.

! An Interaction Is a Difference of Differences

First compare the groups inside one condition. Then compare them inside the other condition. The interaction asks whether those two differences are different. If you skip that last comparison, you have described two results but have not tested an interaction.

16 Paired-Samples t-Test

> **The Same People Have Returned**
>
> Use a paired-samples t-test when the **same people are measured twice** or when observations form meaningful pairs. The two scores from one person know each other. Pretending they are independent is statistical identity theft.
>
> The central idea is simple: turn each pair into one difference score, then ask whether the average difference is zero. That is the whole test. Nobody needs a random effect yet. The random effects are waiting in the next chapter with paperwork.

16.1 Why Pair the Scores?

An **independent-samples** design compares different people. Half the students receive chocolate during a lecture and half receive nothing except the lecture itself, which already seems cruel. Each student contributes one score.

A **repeated-measures** design measures the same students twice. Every student attends one lecture without chocolate and one with chocolate. Each student is now their own control.

A **matched-pairs** design uses different people who were deliberately matched on something important, such as age or baseline ability. This can help, but matched strangers are not magical clones. If you place two undergraduates beside each other because they have the same GPA, one may still be a morning person while the other communicates only through exhausted grunting.

Repeated scores tend to be correlated because stable characteristics follow people into both conditions. A student who generally looks at the screen for a long time may do that with or without chocolate. Another student may stare out the window in both conditions while contemplating escape. Pairing lets us remove some of those stable person-to-person differences.

16.2 A Chocolate-Fixation Study

Suppose we ask whether eating a small piece of chocolate during a lecture changes **fixation time**: the average number of seconds a student's eyes remain on the screen per glance. We measure 48 students twice:

- once with no chocolate;
- once with chocolate.

This example is simulated. Do not distribute mystery candy to human participants and announce that a textbook authorized it. Allergies, informed consent, and research ethics remain annoyingly real.

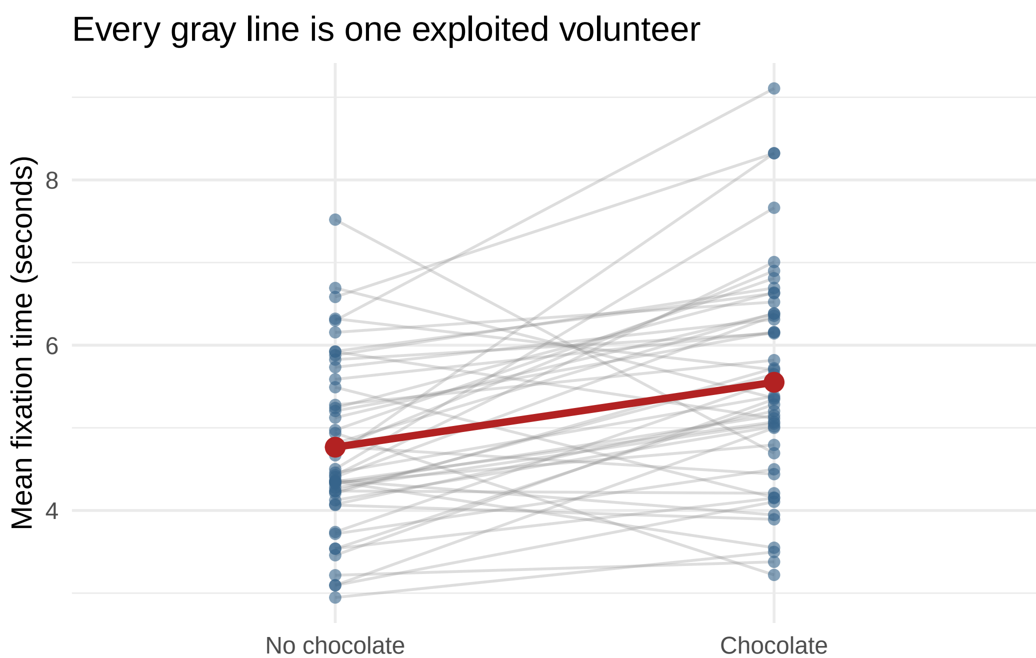
```r
set.seed(343)

n <- 48
participant_effect <- rnorm(n, mean = 0, sd = 1.20)

paired_wide <- data.frame(
  participant = factor(seq_len(n)),
  no_chocolate = 5.00 + participant_effect + rnorm(n, 0, 0.80),
  chocolate = 5.75 + participant_effect + rnorm(n, 0, 0.80)
)

paired_wide$difference <-
  paired_wide$chocolate - paired_wide$no_chocolate

paired_long <- tidyr::pivot_longer(
  paired_wide,
  cols = c(no_chocolate, chocolate),
  names_to = "condition",
  values_to = "fixation"
)

paired_long$condition <- factor(
  paired_long$condition,
  levels = c("no_chocolate", "chocolate"),
  labels = c("No chocolate", "Chocolate")
)
```

The **wide data** have one row per person and one column per condition. This shape makes the pairing obvious and works nicely for a paired t-test. The **long data** have two rows per person. We use that shape for the picture below and will need it again when we reach mixed regression.

16.3 Look at the Pairs Before Testing Them

```r
library(ggplot2)

ggplot(
  paired_long,
  aes(x = condition, y = fixation, group = participant)
) +
  geom_line(alpha = 0.30, color = "gray55") +
  geom_point(alpha = 0.60, color = "steelblue4") +
  stat_summary(
    aes(group = 1), fun = mean, geom = "line",
    color = "firebrick", linewidth = 1.3
  ) +
  stat_summary(
    aes(group = 1), fun = mean, geom = "point",
```

```
    color = "firebrick", size = 3
  ) +
  labs(
    x = NULL,
    y = "Mean fixation time (seconds)",
    title = "Every gray line is one exploited volunteer"
  ) +
  theme_minimal(base_size = 11)
```



The red line shows the mean change. The gray lines show something the means hide: some people change more than others, and a few may move in the opposite direction. If you plot only two bars, all those people are compressed into rectangular coffins.

16.4 The Paired t-Test Is a One-Sample t-Test in Disguise

For each person, compute a difference score:

$$D_i = Y_{i,\text{chocolate}} - Y_{i,\text{no chocolate}}$$

Then ask whether the population mean difference is zero:

$$t = \frac{M_D - 0}{S_D/\sqrt{n}}$$

This is the one-sample t-test from the earlier chapter. The paired t-test does not compare two unrelated piles of scores. It reduces every pair to one difference and tests the set of differences.

```r
paired_result <- t.test(
  paired_wide$chocolate,
  paired_wide$no_chocolate,
  paired = TRUE
)

paired_result
```

```

	Paired t-test

data:  paired_wide$chocolate and paired_wide$no_chocolate
t = 4.3366, df = 47, p-value = 7.597e-05
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.4209697 1.1495099
sample estimates:
mean difference 
      0.7852398 
```

16.4.1 R Has Become Picky About the Formula Method

The formula method of `t.test()` does not accept `paired = TRUE`, so this will fail:

```r
t.test(fixation ~ condition, data = paired_long, paired = TRUE)
```

Use the two paired vectors, as above. Even better, make the pairing visible by calculating the differences yourself:

```r
difference_result <- t.test(paired_wide$difference, mu = 0)
difference_result
```

```

	One Sample t-test

data:  paired_wide$difference
t = 4.3366, df = 47, p-value = 7.597e-05
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.4209697 1.1495099
sample estimates:
mean of x 
0.7852398 
```

The t statistic, degrees of freedom, and p-value match. R has not discovered two separate truths. We asked the same question twice.

! The Unit of Analysis Is the Pair

We began with 96 scores, but the test has 48 difference scores. Therefore, $df = n - 1 = 47$. Counting every measurement as an independent person would double the apparent sample size and create confidence unsupported by the design. The spreadsheet may have 96 rows. The study still has 48 humans.

16.5 Effect Size

A common standardized effect for paired data is Cohen's d_z:

$$d_z = \frac{M_D}{S_D}$$

```
d_z <- mean(paired_wide$difference) /
  sd(paired_wide$difference)

d_z
```

```
[1] 0.6259363
```

This standardizes the mean change by the standard deviation of the **difference scores**. Say which version of d you used because repeated-measures effect sizes have several possible denominators. Calling all of them "Cohen's d" and wandering away is how future readers begin sharpening office supplies.

16.6 Why Pairing Can Help

The variance of a difference score is:

$$S_D^2 = S_1^2 + S_2^2 - 2rS_1S_2$$

When the two measurements are positively correlated, the last term subtracts variability. Stable person differences cancel: the consistently attentive student is compared with themself, and the student planning an elaborate escape is also compared with themself.

This often gives a smaller standard error than an independent-samples analysis. Pairing does not guarantee more power, however. If the repeated scores barely correlate, little stable variation is removed. Pairing also cannot repair a terrible measure. Measuring attention by counting how often someone blinks at the ceiling remains a terrible plan in both conditions.

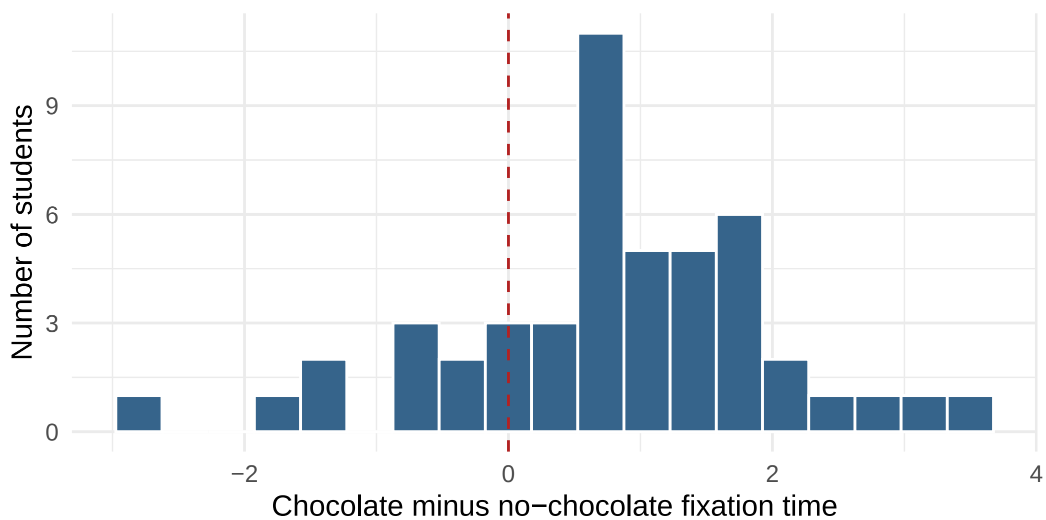
16.7 Assumptions That Actually Matter

For the paired-samples t-test:

- the observations form genuine pairs;
- different pairs are independent of one another;
- the outcome is quantitative;
- the **difference scores** have no severe outliers;
- for small samples, the population distribution of difference scores is reasonably normal.

The test does **not** require the two condition variances to be equal. It operates on one set of differences. Inspect those differences, not two separate normality tests performed as ritual offerings.

```r
ggplot(paired_wide, aes(x = difference)) +
  geom_histogram(binwidth = 0.35, color = "white", fill = "steelblue4") +
  geom_vline(xintercept = 0, linetype = 2, color = "firebrick") +
  labs(
    x = "Chocolate minus no-chocolate fixation time",
    y = "Number of students"
  ) +
  theme_minimal(base_size = 11)
```



16.8 Reporting the Result

For the present study, a concise report could be:

> Students showed longer fixation times during the chocolate condition ($M = 5.55$, $SD = 1.33$) than during the no-chocolate condition ($M = 4.77$, $SD = 1.05$). The mean paired difference was 0.79 seconds, 95% CI [0.42, 1.15], $t(47) = 4.34$, $p < .001$, $d_z = 0.63$.

The statistic cannot prove that chocolate *caused* better attention unless the study was properly randomized and alternative explanations were controlled. It certainly cannot prove that replacing every lecture with a bucket of candy is sound educational policy.

16.9 When Two Measurements Are Not Enough

One difference score works beautifully when each person has two measurements and the question is about their average change. With three or more occasions, missing observations, both within-person and between-person predictors, or people who change at different rates, one difference score cannot represent the whole design.

That is where mixed regression begins. It keeps the repeated observations, tells the model which rows belong to the same human, and allows people to differ without creating one indicator variable for every person who wandered into the study.

16.10 The Short Story

- Use a paired-samples t-test when observations form genuine pairs.
- The same person measured twice is the most common paired design.
- Compute one difference score for each pair.
- A paired-samples t-test is a one-sample t-test on those differences.
- The unit of analysis is the pair, not the individual measurement.
- Inspect the distribution and outliers of the **differences**.
- Report the mean change, confidence interval, p-value, and the specific paired effect size you used.
- When the design grows beyond one clean pair of scores, continue to the mixed-regression chapter.

The grand lesson is not "memorize another command." It is "keep the two scores attached to the human who produced them." The humans already know they are the same person. R needs paperwork.

17 Mixed Regression: Between People and Within People

> ⚠ Advanced Chapter: Several Things Are Happening at Once
>
> A mixed design combines a **between-subjects** variable with a **within-subjects** variable. That creates fixed effects, an interaction, repeated observations, and random effects. It is not impossible. It is merely statistics arriving with all its relatives and expecting you to provide chairs.
>
> Our strategy is to ask one question at a time: Where are the repeated observations? What does each fixed coefficient mean? How are people allowed to differ? What comparison answers the research question?

The paired-samples t-test chapter handled one clean comparison between two measurements from the same person. This chapter begins when the design grows up and acquires complications: more occasions, a between-person group, individual trajectories, missing observations, or several of these problems arriving together in a trench coat.

17.1 The Design

Suppose researchers compare cognitive behavioral therapy (**CBT**) with a control condition for reducing depression symptoms. Participants are randomly assigned to CBT or control, then complete a depression inventory at baseline and after 2, 4, and 6 weeks.

- **Group** is between subjects: each person is in CBT *or* control.
- **Week** is within subjects: each person is measured four times.
- **Depression score** is the quantitative outcome.

The central question is not merely whether the groups differ or whether scores change. It is whether they change **differently**:

> Does the slope of depression over time depend on treatment group?

That is a Week × Group interaction.

This is simulated data. Real treatment studies require clinical expertise, ethical oversight, careful monitoring, informed consent, and substantially more planning than "Alex found `rnorm()`."

```
set.seed(3436)

n_per_group <- 50
people <- data.frame(
  participant = factor(seq_len(2 * n_per_group)),
  group = factor(
    rep(c("Control", "CBT"), each = n_per_group),
    levels = c("Control", "CBT")
  ),
  person_intercept = rnorm(2 * n_per_group, 0, 5.0),
```

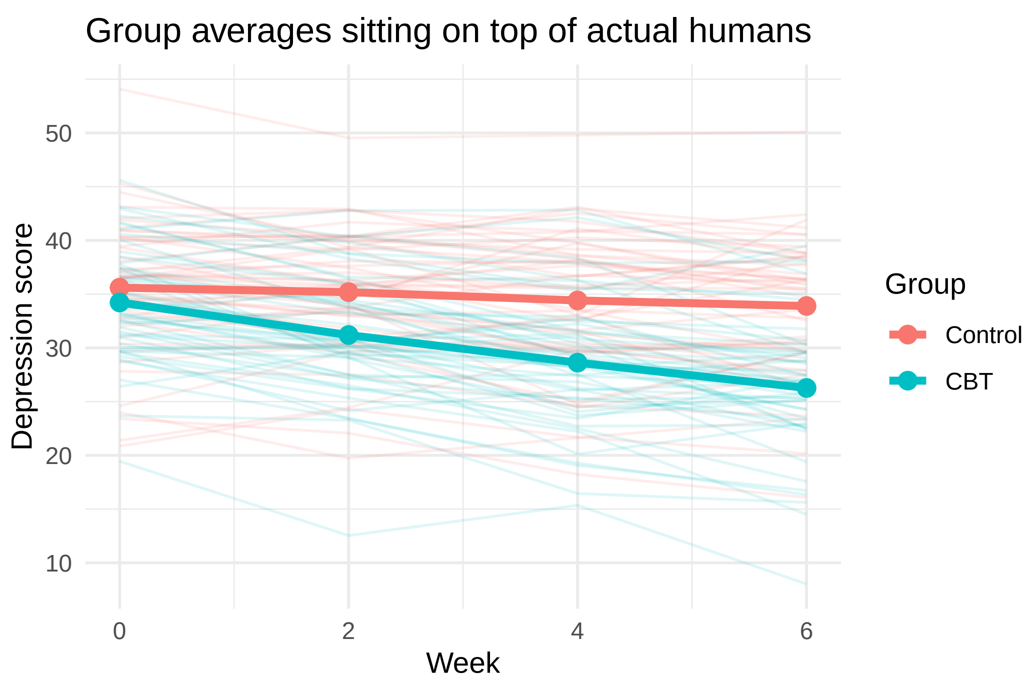
```r
  person_slope = rnorm(2 * n_per_group, 0, 0.45)
)

mixed_data <- merge(
  people,
  data.frame(week = c(0, 2, 4, 6)),
  by = NULL
)

mixed_data <- mixed_data[order(mixed_data$participant,
                               mixed_data$week), ]

mixed_data$depression <-
  35 +
  mixed_data$person_intercept +
  (-0.20 * mixed_data$week) +
  ifelse(mixed_data$group == "CBT",
         -1.20 * mixed_data$week, 0) +
  mixed_data$person_slope * mixed_data$week +
  rnorm(nrow(mixed_data), 0, 2.0)

mixed_data$occasion <- factor(
  mixed_data$week,
  levels = c(0, 2, 4, 6),
  labels = c("Baseline", "Week 2", "Week 4", "Week 6")
)
```

The data are **long**: one row per person per occasion. The participant identifier repeats because the person repeats. If your participant column disappears, stop. Do not proceed directly to the ritual sacrifice.

17.2 First Look: Means and People

```r
library(ggplot2)

ggplot(
  mixed_data,
  aes(x = week, y = depression, group = participant, color = group)
) +
  geom_line(alpha = 0.13) +
  stat_summary(
    aes(group = group), fun = mean, geom = "line",
    linewidth = 1.4
  ) +
  stat_summary(
    aes(group = group), fun = mean, geom = "point",
    size = 2.8
  ) +
```

```r
  scale_x_continuous(breaks = c(0, 2, 4, 6)) +
  labs(
    x = "Week",
    y = "Depression score",
    color = "Group",
    title = "Group averages sitting on top of actual humans"
  ) +
  theme_minimal(base_size = 11)
```



The bold lines show group means. The faint lines show individual trajectories. The CBT mean declines faster, but people do not all begin at the same score or change at the same rate. Some improve quickly, some slowly, and some appear to have received therapy from a motivational poster in a dentist's office.

Never decide whether an effect is statistically significant by staring at whether confidence intervals overlap. CI overlap is not the direct test of a difference, and interaction tests are about **differences between differences**. Fit the model that asks the question.

17.3 Why Ordinary Regression Is Not Enough

An ordinary regression such as this is incomplete:

```r
ordinary_model <- lm(depression ~ week * group, data = mixed_data)
```

It treats all 400 rows as independent. They are not. Four rows belong to each participant, and observations from the same person tend to resemble one another.

This structure is called **nesting**:

occasions nested within participants

The levels are:

- Level 1: repeated occasions;
- Level 2: participants.

Group belongs to Level 2 because it does not change within a person. Week belongs to Level 1 because it does.

17.4 The Fixed Effects

We begin with the part that resembles ordinary regression:

$$Y_{ij} = b_0 + b_1(\text{Week}_{ij}) + b_2(\text{CBT}_i) + b_3(\text{Week}_{ij} \times \text{CBT}_i) + \text{random part} + e_{ij}$$

Because Week 0 is baseline and Control is the reference group:

- b_0: estimated baseline score for Control;
- b_1: expected change per week for Control;
- b_2: CBT-minus-Control difference at baseline;
- b_3: difference between the CBT and Control slopes.

The CBT slope is $b_1 + b_3$. The interaction b_3 is the main research target. If it is negative, CBT scores decline faster than Control scores.

Notice why coding matters. A "main effect of Group" is not a universal average group difference here. It is the group difference when Week equals zero. We chose zero to mean baseline, so the coefficient has a useful interpretation instead of describing the difference during Week Zero Hundred and Twelve, a period known only to the Probability Gods.

17.5 The Random Effects

The repeated observations require us to describe how people differ.

17.5.1 Random Intercepts

A random-intercept model allows each participant to have their own starting level:

$$Y_{ij} = b_0 + b_1\text{Week}_{ij} + b_2\text{CBT}_i + b_3(\text{Week}_{ij} \times \text{CBT}_i) + u_{0i} + e_{ij}$$

```r
random_intercept_model <- glmmTMB::glmmTMB(
  depression ~ week * group + (1 | participant),
  data = mixed_data,
  REML = TRUE
)

summary(random_intercept_model)
```

```
 Family: gaussian  ( identity )
Formula:          depression ~ week * group + (1 | participant)
Data: mixed_data

      AIC       BIC    logLik -2*log(L)  df.resid
   2095.3    2119.3   -1041.7    2083.3       394

Random effects:

Conditional model:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 30.95    5.564
 Residual                 4.67    2.161
Number of obs: 400, groups:  participant, 100

Dispersion estimate for gaussian family (sigma^2): 4.67

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   35.66006    0.82731   43.10  < 2e-16 ***
week          -0.29454    0.06833   -4.31 1.63e-05 ***
groupCBT      -1.61122    1.16999   -1.38    0.168
week:groupCBT -1.02796    0.09664  -10.64  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The expression `(1 | participant)` means "give participants different intercepts." It does not estimate 100 unrelated participant coefficients. Instead, it estimates the variance of the participant intercepts.

17.5.2 Random Slopes

People may also change at different rates. A random-slope model allows each person to have their own Week slope:

```r
random_slope_model <- glmmTMB::glmmTMB(
  depression ~ week * group + (1 + week | participant),
  data = mixed_data,
  REML = TRUE
)

summary(random_slope_model)
```

```
 Family: gaussian  ( identity )
Formula:          depression ~ week * group + (1 + week | participant)
Data: mixed_data

      AIC       BIC    logLik -2*log(L)  df.resid
   2091.2    2123.1   -1037.6    2075.2       392

Random effects:
```

```
Conditional model:
 Groups      Name        Variance Std.Dev. Corr
 participant (Intercept) 29.3952  5.4217
             week         0.1046  0.3233   0.07
 Residual                 3.9820  1.9955
Number of obs: 400, groups:  participant, 100

Dispersion estimate for gaussian family (sigma^2): 3.98

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   35.66006    0.80228   44.45  < 2e-16 ***
week          -0.29454    0.07793   -3.78 0.000157 ***
groupCBT      -1.61122    1.13459   -1.42 0.155584
week:groupCBT -1.02796    0.11021   -9.33  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The expression `(1 + week | participant)` estimates:

- variance in participants' intercepts;
- variance in participants' Week slopes;
- correlation between participant intercepts and slopes.

Random slopes are often substantively reasonable in longitudinal psychology. People do not all change identically merely because the spreadsheet has aligned their rows. They are humans, not synchronized microwave clocks.

The random-effects structure should reflect the design and theory, while remaining supported by the data. If a model is singular or refuses to converge, do not repeatedly click Run while shouting encouragement. Inspect the data, scaling, model specification, number of clusters, and whether the random structure is estimable.

17.6 Read the Model Without Panic

```r
fixed_effects <- summary(random_slope_model)$coefficients$cond
round(fixed_effects, 3)
```

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)     35.660      0.802  44.448    0.000
week            -0.295      0.078  -3.780    0.000
groupCBT        -1.611      1.135  -1.420    0.156
week:groupCBT   -1.028      0.110  -9.327    0.000
attr(,"ddf")
[1] "asymptotic"
```

Read the rows in order:

1. `(Intercept)` estimates the Control group's baseline mean.
2. `week` estimates weekly change in Control.

3. `groupCBT` estimates the CBT-minus-Control baseline difference.
4. `week:groupCBT` estimates how much the CBT slope differs from the Control slope.

Suppose the interaction estimate is about -1.2. That means the CBT group's depression score declines roughly 1.2 points **more per week** than the Control group's score, on average. It does not mean CBT lowers every participant's score by exactly 1.2 points each week. Fixed effects describe an average pattern while the random effects document the inconvenient existence of individuals.

`glmmTMB` reports Wald z statistics and p-values for the fixed effects. These rely on the fitted likelihood and a large-sample approximation; they do not create an infinite number of participants merely because the output prints a z. Report the method and inspect confidence intervals. If a field specifically requires ANOVA-style denominator degrees of freedom for a **Gaussian** mixed model, the later inference chapter shows the deliberate `lmerTest` detour. "R produced a number and I felt hopeful" is not an inferential method.

17.7 Test the Interaction Directly

To test whether the interaction improves the model, compare two models with the same random-effects structure:

- a reduced model without Week $\times$ Group;
- a full model with Week $\times$ Group.

When comparing **fixed effects**, refit both models with maximum likelihood (`REML = FALSE`).

```r
reduced_ml <- glmmTMB::glmmTMB(
  depression ~ week + group + (1 + week | participant),
  data = mixed_data,
  REML = FALSE
)

full_ml <- glmmTMB::glmmTMB(
  depression ~ week * group + (1 + week | participant),
  data = mixed_data,
  REML = FALSE
)

anova(reduced_ml, full_ml)
```

```
Data: mixed_data
Models:
reduced_ml: depression ~ week + group + (1 + week | participant), zi=~0, disp=~1
full_ml: depression ~ week * group + (1 + week | participant), zi=~0, disp=~1
           Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
reduced_ml  7 2148.9 2176.9 -1067.5   2134.9
full_ml     8 2087.4 2119.3 -1035.7   2071.4 63.539      1  1.572e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This likelihood-ratio test asks whether allowing the groups to have different slopes improves model fit. It is not the only defensible test; coefficient tests and other approximations are also used. The important point is to match the test to the question and identify the method, not to collect p values like cursed trading cards.

After model comparison, report estimates from the final model fit with REML when appropriate.

17.8 Follow Up the Interaction

The interaction says the slopes differ. We still want the slope within each group.

```
group_slopes <- emmeans::emtrends(
  random_slope_model,
  specs = ~ group,
  var = "week"
)

group_slopes
pairs(group_slopes)
```

```
 group   week.trend     SE  df asymp.LCL asymp.UCL
 Control     -0.295 0.0779 Inf    -0.447    -0.142
 CBT         -1.322 0.0779 Inf    -1.475    -1.170

Confidence level used: 0.95
 contrast      estimate   SE  df z.ratio p.value
 Control - CBT     1.03 0.11 Inf   9.327 <0.0001
```

`emtrends()` estimates the Week slope separately for Control and CBT. `pairs()` tests their difference, which corresponds to the interaction.

We can also estimate model-implied group means at the observed weeks:

```
estimated_means <- emmeans::emmeans(
  random_slope_model,
  specs = ~ group | week,
  at = list(week = c(0, 2, 4, 6))
)

estimated_means
```

```
week = 0:
 group   emmean    SE  df asymp.LCL asymp.UCL
 Control   35.7 0.802 Inf      34.1      37.2
 CBT       34.0 0.802 Inf      32.5      35.6

week = 2:
 group   emmean    SE  df asymp.LCL asymp.UCL
 Control   35.1 0.794 Inf      33.5      36.6
 CBT       31.4 0.794 Inf      29.8      33.0

week = 4:
 group   emmean    SE  df asymp.LCL asymp.UCL
```

```
 Control   34.5 0.816 Inf      32.9      36.1
 CBT       28.8 0.816 Inf      27.2      30.4

week = 6:
 group   emmean    SE  df asymp.LCL asymp.UCL
 Control   33.9 0.866 Inf      32.2      35.6
 CBT       26.1 0.866 Inf      24.4      27.8

Confidence level used: 0.95
```

Do not automatically test every possible pair at every possible week because the software makes it easy. That is how a sensible research question becomes 28 tests and a dead salmon. Choose comparisons that answer the theory, and adjust for multiplicity when you conduct a family of tests.

17.9 Should Time Be Continuous or Categorical?

Treating Week as numeric assumes a straight-line average trajectory. That is a strong but efficient summary: one slope per group.

If change may be nonlinear—perhaps treatment helps rapidly and then levels off—treat occasion as categorical:

```r
categorical_time_model <- glmmTMB::glmmTMB(
  depression ~ occasion * group + (1 + week | participant),
  data = mixed_data,
  REML = TRUE
)

summary(categorical_time_model)$coefficients$cond
```

```
                         Estimate Std. Error   z value     Pr(>|z|)
(Intercept)            35.6044655  0.8172028 43.568704 0.000000e+00
occasionWeek 2         -0.4164512  0.4114638 -1.012121 3.114801e-01
occasionWeek 4         -1.1898064  0.4402741 -2.702422 6.883630e-03
occasionWeek 6         -1.7057961  0.4844990 -3.520742 4.303404e-04
groupCBT               -1.3749344  1.1556992 -1.189699 2.341647e-01
occasionWeek 2:groupCBT -2.6145688  0.5818977 -4.493176 7.016878e-06
occasionWeek 4:groupCBT -4.4122466  0.6226416 -7.086335 1.377106e-12
occasionWeek 6:groupCBT -6.2538495  0.6851850 -9.127242 7.026650e-20
attr(,"ddf")
[1] "asymptotic"
```

Categorical time estimates separate deviations at Weeks 2, 4, and 6. It is flexible but uses more parameters. Numeric time is compact but imposes shape. Neither coding is morally superior. The choice depends on the research question, plausible change process, timing, and amount of data.

Other options include polynomial terms, splines, or nonlinear models. Do not add a seventh-degree polynomial merely because R accepts it. R will also accept many bad ideas with the serene indifference of a vending machine.

17.10 Missing Repeated Observations

Mixed models can use participants who are missing some occasions; they do not require deleting every person with one missing row. This is a major advantage over older complete-case repeated-measures procedures.

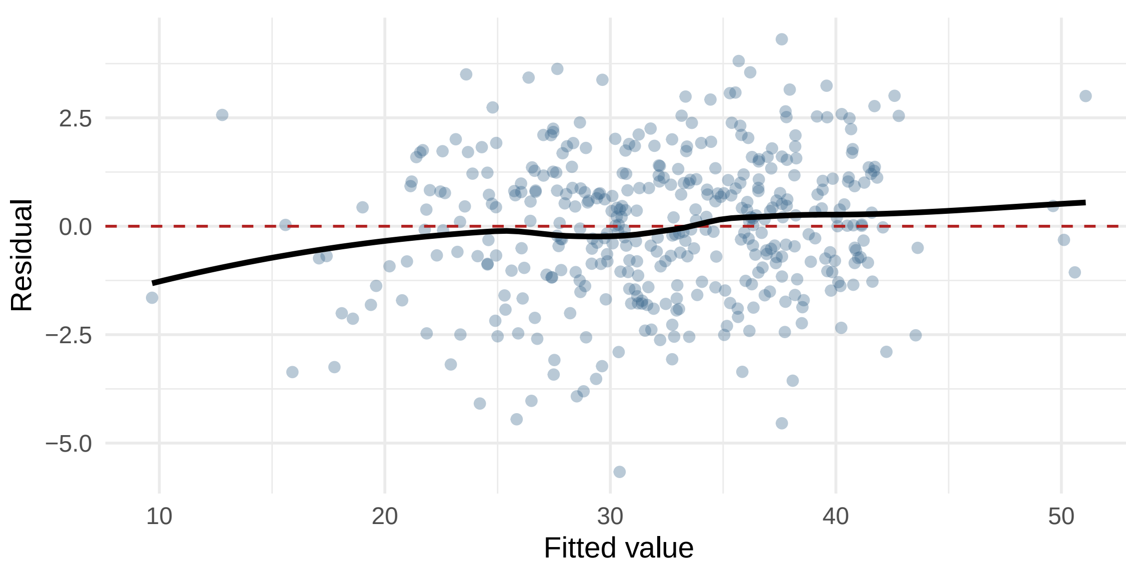
However, "the model ran" does not mean missing data ceased to matter. Standard likelihood-based mixed-model inference commonly relies on a **missing at random** assumption conditional on variables in the model. If people with worsening symptoms leave the study for reasons the model does not capture, the missingness can bias results. Investigate dropout, include relevant predictors when justified, and conduct sensitivity analyses in serious work.

17.11 Check the Model

At minimum, examine:

- whether the mean trajectory is represented sensibly;
- residual-versus-fitted patterns;
- severe residual outliers;
- the distribution and influence of clusters;
- convergence and singular-fit warnings;
- whether variance changes across time or groups;
- whether enough participants—not merely enough rows—support the model.

```
diagnostic_data <- data.frame(
  fitted = fitted(random_slope_model),
  residual = resid(random_slope_model)
)

ggplot(diagnostic_data, aes(x = fitted, y = residual)) +
  geom_point(alpha = 0.35, color = "steelblue4") +
  geom_hline(yintercept = 0, linetype = 2, color = "firebrick") +
  geom_smooth(method = "loess", se = FALSE, color = "black") +
  labs(x = "Fitted value", y = "Residual") +
  theme_minimal(base_size = 11)
```



Traditional repeated-measures ANOVA emphasizes **sphericity**. A mixed model does not require that single sphericity assumption, but it does require a defensible covariance/random-effects structure. We did not escape assumptions. We traded one rigid assumption for a model that must be specified thoughtfully.

17.12 Reporting the Result

A compact results paragraph could read:

> Depression scores were analyzed with a linear mixed-effects model containing fixed effects of Week, Group, and their interaction, plus correlated participant random intercepts and Week slopes. The Week × Group interaction was negative, b = -1.03, SE = 0.11, z = -9.33, p < .001, indicating that depression declined more rapidly in CBT than in the control condition. The p-value used the Wald z approximation reported by `glmmTMB`.

A real report should also provide confidence intervals, descriptive/model-implied means, the exact random-effects structure, software, missing-data handling, and theoretically planned follow-ups. The paragraph above is the skeleton. Please give it organs before submitting it to Reviewer 2.

17.13 The Short Story

- A mixed design contains at least one between-subjects predictor and one within-subjects predictor.
- Repeated occasions are nested within people, so ordinary independent-error regression is incomplete.
- Fixed effects describe average intercepts, slopes, group differences, and interactions.
- Random intercepts allow people to start at different levels.
- Random slopes allow people to change at different rates.
- With baseline coded as Week 0, the Group coefficient is the baseline group difference.
- The Week × Group coefficient is the difference between group slopes.
- Compare fixed-effects structures using models fit with maximum likelihood, not REML.
- Continuous time tests a compact trajectory; categorical time permits separate occasion patterns.
- Mixed models can retain partially observed participants, but missingness can still bias the answer.
- Never count rows and forget to count people. Four hundred observations from 100 people are not 400 independent humans, no matter how urgently the grant deadline approaches.

The model is "mixed" because it combines fixed and random effects—not because we placed several statistical procedures into a blender and hoped the lid stayed on.

18 Pearson's Chi-Square

18.1 Parametric vs. Non-Parametric

Statistical tests are often divided into parametric and nonparametric families. The difference concerns what the model assumes and what part of the data it compares.

18.1.1 Parametric Tests (coming later)

Parametric tests describe populations with parameters such as means and variances. Particular tests make particular distributional and error assumptions; they do not all demand that the raw scores themselves be perfectly normal. They are powerful when their assumptions are reasonable.

You already know the ingredients (mean, SD) — later we'll formalize how they connect to tests like t and F.

18.1.2 Nonparametric Tests (this chapter and the next)

Nonparametric tests make fewer or different distributional assumptions and often ask questions about counts, signs, or ranks:

- They usually do not require normally distributed raw scores
- Work with ordinal or nominal data
- Also useful when parametric assumptions are violated
- Instead of means, they compare counts/frequencies or rank observations

They are not assumption-free. Independence, sampling, measurement, and the design still matter. The next chapter examines that entire nonparametric family in more detail.

18.1.3 Chi-Square

Pearson's chi-square is often placed under the "nonparametric" umbrella because it does not model means with a normal error distribution. More precisely, it is a test for **categorical counts**. It is not the emergency version of a t-test to use whenever normality looks annoyed.

Rather than asking: "How far is this group mean from another group mean?"

Chi-square asks: "Do the observed counts differ from what we would expect by chance?"

In technical terms, chi-square hypothesis testing asks whether an observed pattern is consistent with a chance-based null hypothesis.

- We compare observed data to expected values under null hypothesis (H_0), which means we expect no difference from chance.
- Large discrepancies lead us to reject H_0 and provide support for the alternative hypothesis (H_1).

Note: We will return the logic of hypothesis testing in more detail later in the semester.

Chi-square is especially useful when:

- Outcomes are categories (yes/no, group A/B/C, correct/incorrect)
- Means and standard deviations either don't exist or don't make sense

There are two types of chi-square:

- One-Way Pearson's Chi-Square [Goodness of Fit]
- Two-Way Pearson's Chi-Squared [Test of Independence]

18.2 One-Way Pearson's Chi-Square [Goodness of Fit]

Have you been told: "Stay with your first answer on a multiple-choice test." *So is changing answers more likely to be harmful?*

> Best studied the responses of 261 students in an introductory psychology course (Best 1979). He recorded the number of right-to-wrong (27), wrong-to-right (195), and wrong-to-wrong (39) answer changes for each student. Note: We will ignore wrong-to-wrong as he did in his first analysis.

Frequency	Right-to-Wrong	Wrong-to-Right
Observed	27	195

Total of 222 observations

We will use the **One-Way Chi-Squared** [Goodness of fit]. It asks whether the relative frequencies observed in the categories of a sample frequency distribution are in agreement with the relative frequencies hypothesized to be true in the population.

18.2.1 Hypothesis for Goodness of fit

There are two versions (but it does not change the math, just how you talk about it):

- No preference: there will be no preference between different categories (e.g., do people prefer Coke or Pepsi)
- No difference from a known population: This type hypothesis compares two populations (or representative samples)

If people were making chance guesses from a population of people who don't have any knowledge about material on the test (version 2 of the hypothesis):

- $H_0 : P_{right-to-wrong} = .5, P_{wrong-to-right} = .5$
- $H_1 : H_0$ is False

Note: The probability you set can be whatever you want, but across all cells, they must add to 1. When you do not know what probability should be from a theoretical standpoint, you can simply do 1/Number of cells.

18.2.2 Find Expected Frequencies

If we expect by chance people would have gotten p = .50 on wrong-to-right and right to wrong, $f_e = pn$

$F_{e_{right-to-wrong}} = .5 * 222 = 111$ $F_{e_{wrong-to-right}} = .5 * 222 = 111$

Frequency	Right-to-Wrong	Wrong-to-Right
Observed	27	195
Expected	111	111

Note: These tables are often shown in papers (sometimes as proportions).

18.2.3 Goodness of Fit Calculation by Hand

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

$\chi^2 = \frac{(27-111)^2}{111} + \frac{(195-111)^2}{111} = 127.14$

Note: Never enter proportions into the chi-square. You must work with frequencies.

18.2.4 Test Against Chance Distribution

χ^2 values close to zero are probably "chance," and values that are larger are farther from chance. Wait—how do I know that?

Look at our f_o values. If the f_o values were close to the f_e values, then the total value we calculate would be very close to zero.

For example, if we actually found this pattern instead:

Frequency	Right-to-Wrong	Wrong-to-Right
Observed	111	111
Expected	111	111

$\chi^2 = \frac{(111-111)^2}{111} + \frac{(111-111)^2}{111} = 0$

While it's very rare for the observed frequencies to be exactly what we expect by chance, it's often really close to zero.

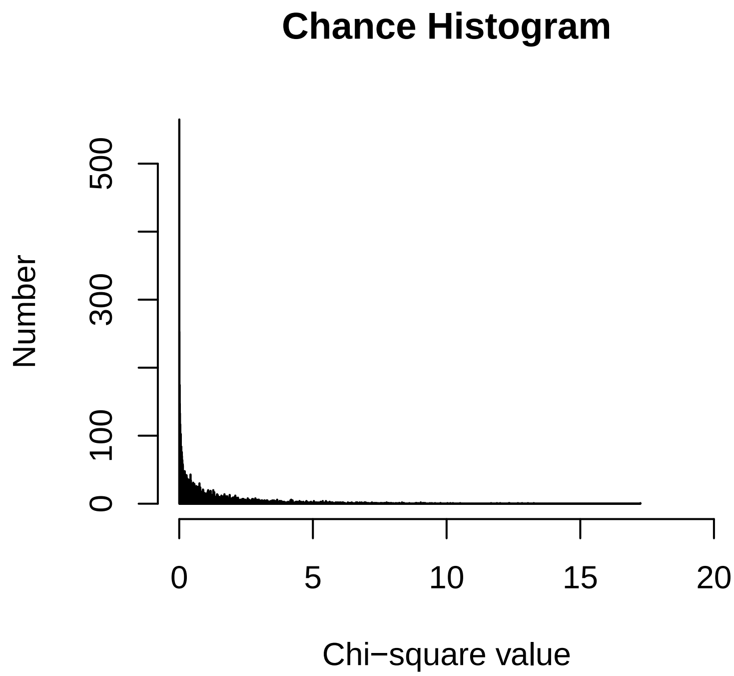
To get a sense of how often it would be close to zero, imagine we created a new test and the multiple-choice question said:

> "Please select an answer, then erase it and select something else. There is one right answer."

That's all you get—no actual knowledge being tested. It's a pure game of chance! In other words, people are totally guessing and are forced to change their answer.

Sometimes, purely by chance, they will go from Right-to-Wrong or Wrong-to-Right (and we will ignore all the Wrong-to-Wrong cases). What would that look like?

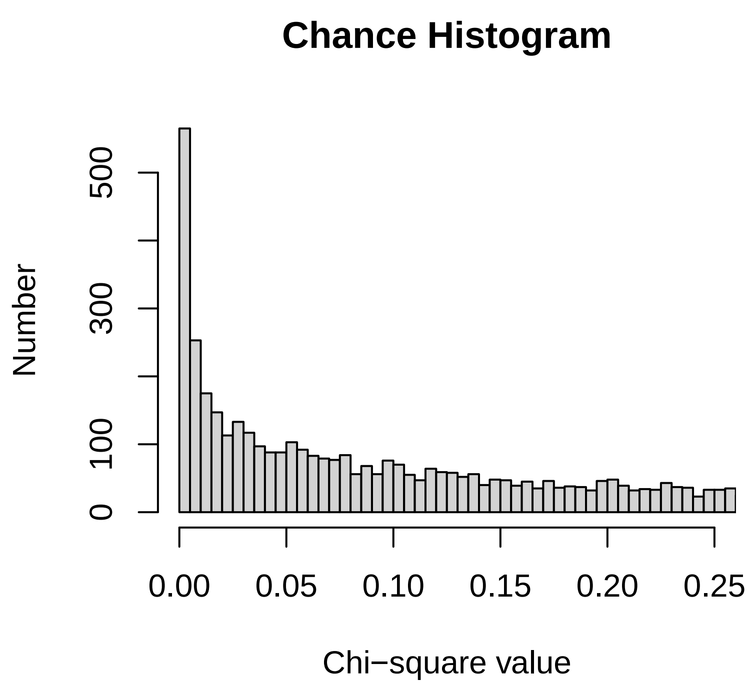
I can run 10,000 subjects to get a sense of what that might look like.



It looks like we get zero and values close to zero a lot—but also, purely by chance, we can get values all the way out to about 17.25. So even by chance, we can get a value that's "far" from zero.

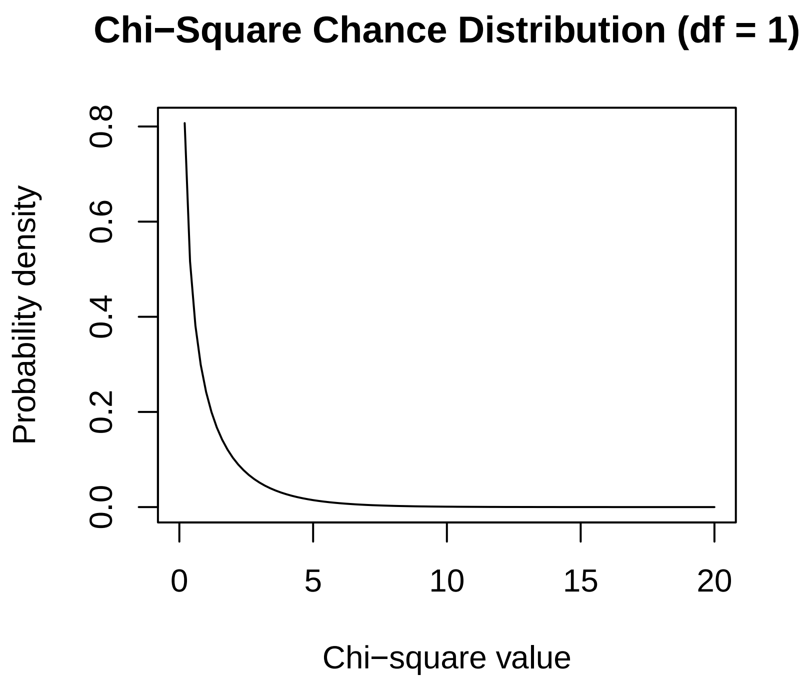
But does that happen often?

Before we answer that question, let's zoom in closer to zero so we can see what's going on more clearly.



So how far from zero is still "chance"?

To answer that question, we would need more than 10,000 people. In fact, we would need the entire population. That would be impossible, but luckily, someone has done the math-magic to figure out what that would look like:



Earlier, we looked at frequencies—how often different chi-square values occurred in a simulation.

Here, we switch to a probability density, which describes how likely different chi-square values are under the null hypothesis.

- The x-axis shows possible chi-square values.
- The y-axis shows relative likelihood (how often it will occur), not counts.

The important idea is that:

- Values near zero are very likely under chance
- Larger values are increasingly unlikely under chance

The total area under the curve equals 1, representing all possible outcomes under the null hypothesis.

18.2.5 How do we set chance?

How extreme does a chi-square value need to be before we stop believing it's just chance?

One very common rule is the 95% / 5% rule.

> We say: "If the null hypothesis is true, then 95% of the time our chi-square values should fall below some cutoff."

That cutoff defines the critical value. Any value beyond that point is rare enough that we start to doubt the chance explanation.

This does not mean:

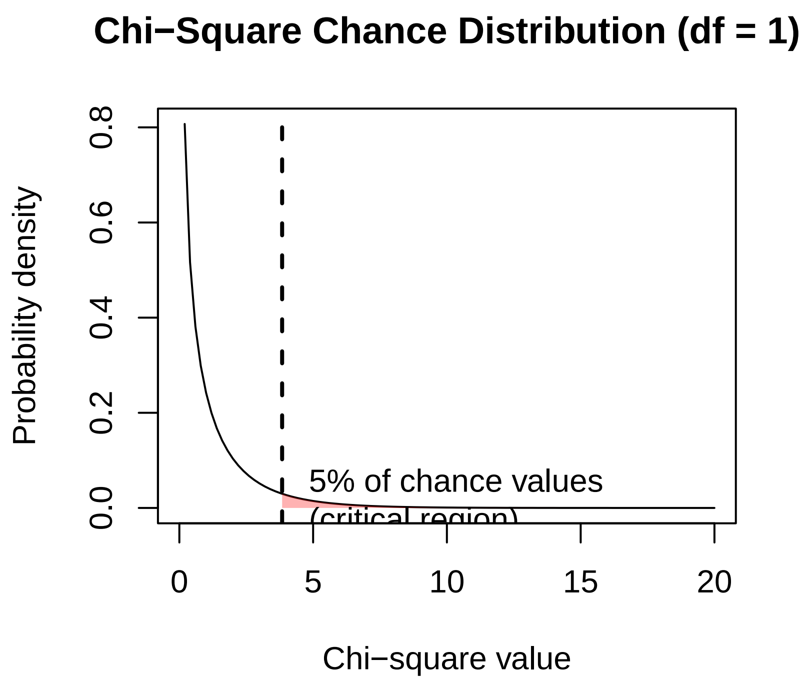
- That chance is impossible past that value
- Or that we are 100% certain the result is “real”

It means:

- About 5% of the time, even when the null hypothesis is true, we will still see values this extreme by chance alone
- We accept that risk in exchange for having a decision rule

So when we say a result is “unlikely due to chance,” what we really mean is:

> “If this were just chance, we would expect to see something this extreme only about 5% of the time.”



The distribution changes as a function of the degrees of freedom [df is the number of cells - 1]. Degrees of freedom tell us how much independent information is available to estimate variability in the data. If we have diferent number of conditions, we have a different distribution to compare it.

We get the crit = 3.8414588. Any χ^2 we calculate above that value means it has only a 5% probability of being a chance result.

18.2.6 Our results?

We ask where is our chi-square value relative to this chance distribution given our $df = 1$. We calculated 127.14, and that is in the tail (the small part, well above chance value). We can use R or a table from a book to get the critical values for alpha (p) =.05. We don't need to make the graph every time as someone did this already using calculus and has given us all the values with a simple function in R called `qchisq`.

This function answers the question:

- "What chi-square value cuts off a certain percentage of chance outcomes?"
  - p = .05
- We are defining the 5% region — the part of the distribution that happens only about 5% of the time under chance.
- df = 1
  - This tells R which chi-square distribution to use.
    - With two categories, the degrees of freedom is 1.
- lower.tail = FALSE
  - This is important:
    - FALSE means we want the upper tail of the distribution

We get the crit = 3.8414588 < 124.14 (experimental value we got), so we reject the null hypothesis (H_0) and provide evidence for the alternative (H_1).

18.2.7 Report in APA

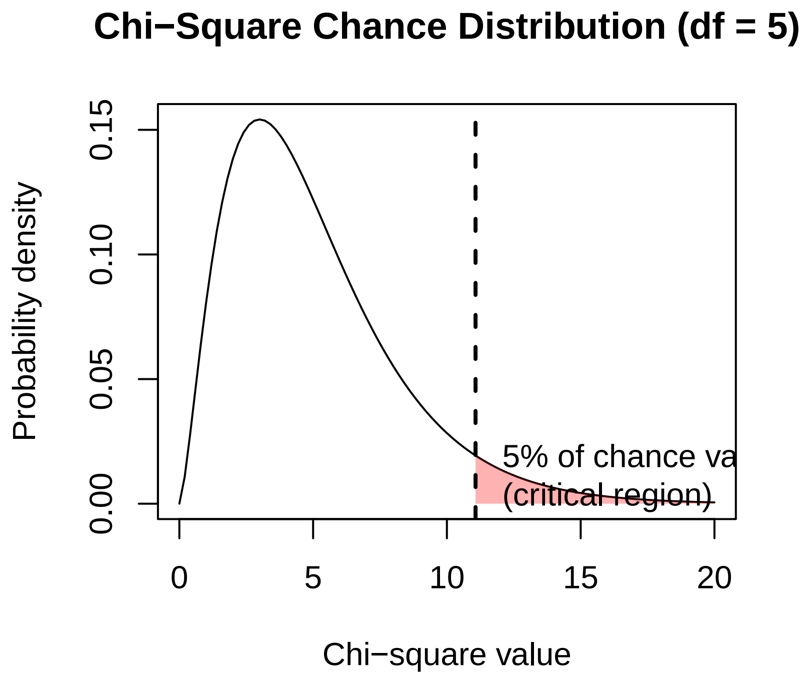
$\chi^2(df, N) = X.XX, p < or >= .XX$

Since our result was p-value < .05, as rejected our null.

Students were significantly more likely to change an answer from wrong to right (87.83%) than from right to wrong (12.16%), $\chi^2(1, N = 222) = 124.14, p < .05$.

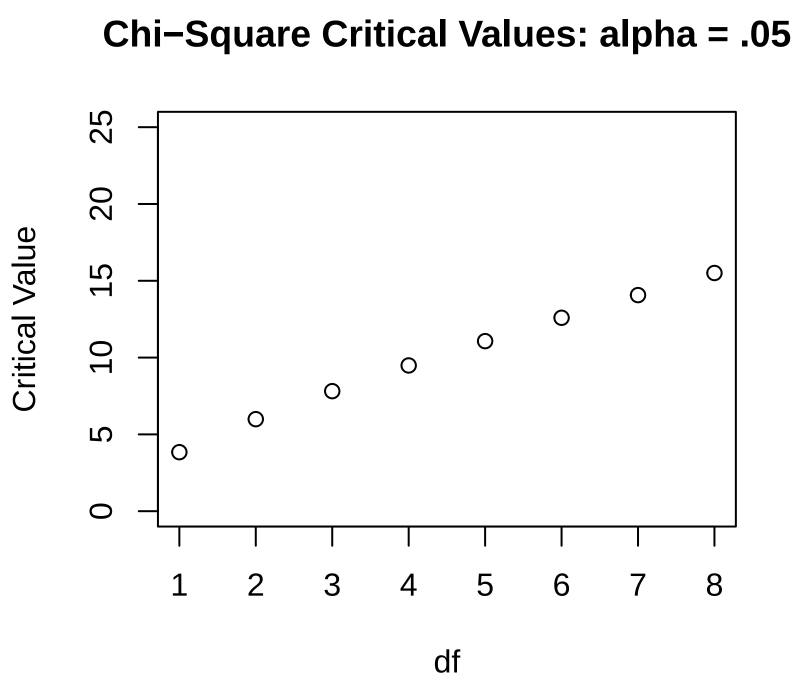
18.2.8 Impact of higher DF

Note: shape of chance distribution changes as we have more df (say 6 cells to compare)



As you can see the distribution gets more spread out. So with larger DF our critical value increases

```
ChiCrit<-qchisq(.05, 1:8, lower.tail = FALSE)
plot(ChiCrit, main="Chi-Square Critical Values: alpha = .05",
  xlab="df", ylab="Critical Value", xlim=c(1,8), ylim=c(0,25))
```



18.2.9 Goodness of Fit Calculation by Code

The way above if the hard way and shown you understand what the computer is doing. Instead, doing one-way chi-Squares in R is really easy!

18.2.10 Step 1: Build your Table

```
MCQ.Study <- as.table(rbind(c(27, 195)))
dimnames(MCQ.Study) <- list(Freq = c("Observed"),
                            Action = c("RtoW", "WtoR"))
MCQ.Study
```

```
          Action
Freq       RtoW WtoR
  Observed   27  195
```

18.2.11 Step 2: Calculate Chi-squared

```
MCQ.chi <- chisq.test(MCQ.Study, p=c(.5, .5))
MCQ.chi
```

```

    Chi-squared test for given probabilities

data:  MCQ.Study
X-squared = 127.14, df = 1, p-value < 2.2e-16
```

Again, in APA format, we would say,

Students were significantly more likely to change an answer from wrong to right (87.83%) than from right to wrong (12.16%), $\chi^2(1, N = 222) = 124.14, p < .05$.

18.3 Two-Way Pearson's Chi-Squared [Test of Independence]

"Maybe we should only change our answers on easy items, but on hard items, we should trust ourselves."

> Best (1979) also examined the 1670 total number of responses across the 261 students and divided the items into easy/difficult in an introductory psychology course. He recorded the number of right-to-wrong, wrong-to-right, and wrong-to-wrong (39) answer changes for each student. Again we will ignore wrong-to-wrong (17% of the responses) for simplicity.

Frequency	Right to Wrong	Wrong to Right
Easy	97	411
Difficult	251	620

We need to do a Two-Way Chi-Square [Test of Independence]. It asks whether observed frequencies reflect the independence of two qualitative variables. Compares the actual observed frequencies of some phenomenon (in our sample) with the frequencies we would expect if there were no relationship at all between the two variables in the larger (sampled) population. We ask if the two variables are independent: knowledge of the value of one variable provides no information about the value of another variable.

18.3.1 Hypothesis for Test of Independence

The null hypothesis is that for each, the value obtained for one variable is not related to or influenced by the second variable. This idea can be expressed in two versions:

- Version 1: A single sample is measured on two separate variables. For example, knowing your personality will help predict your color preference (alternative). Knowing your personality will NOT help you predict your color preference (null) [Personality and color preference are measured per person]
  - Null: The variables are not related.
  - Alternative: The variables are related.
- Version 2: Two (or more) samples are measured and compared to see if there are differences. For example, the same proportions of extroverts (group 1) and introverts (group 2) prefer the same color (null). If the proportions are different for the two groups, then you have the alternative hypothesis
  - Null: The samples are independent
  - Alternative: The samples are not independent

Here we are working with Version 1 (one group taking two types of items)

18.3.2 Null Hypothesis (H_0)

- Answer-changing behavior is independent of item difficulty.
- The proportion of right-to-wrong and wrong-to-right changes is the same for easy and difficult items

Knowing whether an item is easy or difficult provides no information about how answers are changed

18.3.3 Alternative Hypothesis (H_1)

Answer-changing behavior depends on item difficulty.

- Students are more likely to benefit from changing answers on easy items
- On difficult items, answer changes are less likely to improve performance

Knowing whether an item is easy or difficult does provide information about how answers are changed

Simply put:

- H_0 : The item difficulty is not related to how a person changes their answer
- H_1 : The item difficulty is related to how a person changes their answer

18.3.4 Find Expected Frequencies

This is little more complex than the goodness of fit test

18.3.5 Find row/col sums

Frequency	Right to Wrong	Wrong to Right	Sum
Easy	97	411	**508**
Difficult	251	620	**871**
Sum	**348**	**1031**	**1379**

18.3.6 Step 2: Expected frequencies per cell

$$f_e = \frac{col\, total * row\, total}{overall\, total}$$

Frequency	Right to Wrong	Wrong to Right	Sum
Easy	97 (128.1972)	411 (379.8028)	**508**
Difficult	251 (219.8028)	620 (651.1972)	**871**
Sum	**348**	**1031**	**1379**

18.3.7 Test of Independence Calculation by Hand

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

$\chi^2 = \frac{(97-128.1972)^2}{128.1972} + \frac{(411-379.8028)^2}{379.8028} + \frac{(251-219.8028)^2}{219.8028} + \frac{(620-651.1972)^2}{651.1972}$

$\chi^2 = 16.08$

18.3.8 Test Against Distribution

$df = (Row - 1)(Col - 1)$ $df = (2 - 1)(2 - 1) = 1$

Given our $df = 1$. We calculated 16.08 we can use R or a table to get the critical values for alpha = .05

We get the crit = 3.8414588 < 16.08, so we reject the null.

18.3.9 Report in APA

$\chi^2(df, N) = X.XX, p < or >= .XX, V = .XX$

In APA format:

Item difficulty was related to how students changed their answers, $\chi^2(1, N = 1379) = 16.08, p < .05$.

[That is all we can say statistically. Normally people present a table in percentages to unpack this result, **but** You have to decide if you want to do row or column percentages. People do not present both]. I think the row percent make more sense in this case we want to compare across item difficulty.

% Row	Right to Wrong	Wrong to Right	%
Easy	19.1%	80.9%	100%
Difficult	28.8%	71.2%	100%
Col Total	25.2%	74.8%	100%

18.3.10 Test of Independence Calculation by Code

Again the code for R is very easy, so I will show you how it works:

18.3.11 Step 1: Build Your table

```
MCQ.Study.2 <- as.table(rbind(c(97, 411), c(251, 620)))
dimnames(MCQ.Study.2) <- list(Social = c("Easy","Diff"),
                              Change = c("RtoW", "WtoR"))

MCQ.Study.2
```

```
      Change
Social RtoW WtoR
  Easy   97  411
  Diff  251  620
```

18.3.12 Step 2: Calculate chi-square

In tests of independence, R calculates the expected frequencies for you. For a 2×2 table, `chisq.test()` applies **Yates's continuity correction** by default when `correct = TRUE`. The correction tries to improve a continuous chi-square approximation to discrete counts, but it can be overly conservative. It is not a ritual that must always be performed. Use it deliberately, report whether you used it, and consider Fisher's exact test when expected counts are sparse.

$$\chi^2_{Yates} = \sum \frac{(|f_o - f_e| - .5)^2}{f_e}$$

Note: You can view the observed and expected frequencies. For larger sparse tables, `chisq.test(..., simulate.p.value = TRUE)` can estimate a p-value by simulation.

```r
MCQ.chi.2 <- chisq.test(MCQ.Study.2,correct = TRUE)
MCQ.chi.2
```

```
	Pearson's Chi-squared test with Yates' continuity correction

data:  MCQ.Study.2
X-squared = 15.566, df = 1, p-value = 7.968e-05
```

18.3.13 Row Proportion Code

```r
prop.table(MCQ.Study.2,1)
```

```
      Change
Social      RtoW      WtoR
  Easy 0.1909449 0.8090551
  Diff 0.2881745 0.7118255
```

18.3.14 Col Proportion Code

```r
prop.table(MCQ.Study.2,2)
```

```
      Change
Social      RtoW      WtoR
  Easy 0.2787356 0.3986421
  Diff 0.7212644 0.6013579
```

18.3.15 Effect Size: Because an Asterisk Is Not a Personality

The chi-square test tells us whether the observed table is difficult to explain under the null hypothesis. It does **not** tell us whether the association is large enough to matter. With enough observations, a microscopic association can earn an asterisk and begin acting important at faculty meetings.

For a 2×2 table, **phi** (ϕ) is:

$$\phi = \sqrt{\frac{\chi^2}{N}}$$

For a table of any size, use **Cramer's V**:

$$V = \sqrt{\frac{\chi^2}{N \min(r-1, c-1)}}$$

Here, r and c are the numbers of rows and columns. Cramer's V ranges from 0, meaning no association, to 1, meaning perfect association. Do not automatically translate `.10`, `.30`, and `.50` into small, medium, and large without thinking about the design and research area. Those labels are rough landmarks, not sacred commandments delivered on stone tablets by Cohen.

```
chi_uncorrected <- chisq.test(MCQ.Study.2, correct = FALSE)
n_total <- sum(MCQ.Study.2)

phi <- sqrt(as.numeric(chi_uncorrected$statistic) / n_total)
cramers_v <- sqrt(
  as.numeric(chi_uncorrected$statistic) /
    (n_total * min(dim(MCQ.Study.2) - 1))
)

c(phi = phi, cramers_v = cramers_v)
```

```
      phi cramers_v 
0.1079744 0.1079744 
```

For these data, $V = .11$: there is an association, but it is not a gigantic one. The asterisk has not transformed into a flaming sword.

18.3.16 Odds Ratios for a 2-by-2 Table

An **odds ratio** compares the odds of an outcome between two groups. In our table, the odds of changing an easy item from right to wrong rather than wrong to right are 97/411. For difficult items, those odds are 251/620.

$$OR = \frac{97/411}{251/620} = 0.58$$

An odds ratio of 1 would mean equal odds. Here, the odds of a right-to-wrong rather than wrong-to-right change were about 42% lower for easy items than for difficult items. If you reverse the comparison, the odds for difficult items are about $1/0.58 = 1.71$ times those for easy items. Odds ratios are direction-sensitive little creatures, so always say which group and outcome are in the numerator.

Base R gives us the odds ratio and its confidence interval with Fisher's exact test:

```
fisher_result <- fisher.test(MCQ.Study.2)
fisher_result$estimate
fisher_result$conf.int
```

```
odds ratio 
 0.5831962 
[1] 0.4421001 0.7654197
attr(,"conf.level")
[1] 0.95
```

The estimated odds ratio is 0.58, 95% CI [0.44, 0.77].

18.3.17 Power for Chi-Square

Power asks how often a study would detect an association of a specified size if that association really exists. Small samples give fickle chance more room to cause trouble. A weak association plus twelve people and one undergraduate who clicked every response while watching TikTok is not a sturdy research program.

For chi-square tests, Cohen's w is commonly used in power calculations. In a 2×2 table it is equivalent to phi and Cramer's V. The function below uses only base R. It finds the total sample size needed for 80% power when $w = .1062$, approximately the association observed here.

```
chisq_power <- function(n, w, df, alpha = .05) {
  critical_value <- qchisq(1 - alpha, df = df)
  1 - pchisq(critical_value, df = df, ncp = n * w^2)
}

required_n <- uniroot(
  function(n) chisq_power(n, w = .1062, df = 1) - .80,
  interval = c(2, 10000)
)$root

ceiling(required_n)
```

```
[1] 696
```

The answer is about 696 total observations. That number is not permission to choose an effect size after seeing the results. Plan power using the smallest effect worth detecting, prior evidence, or a defensible range of possible effects.

18.3.18 Reporting the Result

A compact report can include the test, effect size, and a table of interpretable percentages:

> Item difficulty was associated with answer-change direction, $\chi^2(1, N = 1379) = 15.57$, $p < .001$, $V = .11$. Compared with difficult items, easy items had lower odds of being changed from right to wrong rather than wrong to right, $OR = 0.58$, 95% CI [0.44, 0.77].

The chi-square value above uses Yates's correction, matching the test already run in this chapter. The effect-size calculation uses the ordinary Pearson chi-square. State your choices so readers do not have to conduct forensic accounting on your results section.

18.3.19 Chi-Square Issues

- Chi-square works with **counts**, not percentages, means, or the emotional intensity with which you typed the numbers.
- The observations must be independent. One person should not quietly appear in several cells unless the design and model explicitly handle repeated observations.
- The approximation becomes unreliable when expected counts are very small. A common guideline is that no expected count should be below 1 and no more than 20% should be below 5, but this is a guideline rather than a commandment. For a small 2×2 table, use Fisher's exact test. For larger sparse tables, a simulated p-value may be useful.

- Chi-square cannot be negative because every discrepancy is squared. It can equal zero when every observed count equals its expected count, an event normally witnessed immediately before the Probability Gods demand another sacrifice.
- A cell contributes more when its observed and expected counts differ by a large amount **relative to the expected count**.
- A significant omnibus test says that some association exists. It does not identify every cell responsible for it. Follow-up comparisons create a multiple-testing problem, which the advanced chapter later in this book handles before the asterisks multiply and form a union.

! Counts First; Interpretation Second

A chi-square test asks whether the observed **counts** differ from the counts expected under the null model. After that test, inspect the percentages, residuals, and effect size to learn where the association lives and whether anybody should care.

19 Nonparametric Tests: When the Data Refuse to Behave

19.1 What Makes a Test Nonparametric?

Many familiar tests describe data using parameters such as means and variances and make assumptions about how scores behave. **Nonparametric tests** make fewer or different assumptions, often by analyzing counts, signs, or ranks instead of the original scores.

This does **not** mean they are assumption-free. No statistical test arrives free of assumptions, calories, or consequences. Nonparametric tests still care about the design, independence, what was measured, and what question the test actually answers.

They are especially useful when:

- The outcome is nominal or ordinal.
- A few extreme scores are dragging the mean across the room by its ankles.
- The sample is too small to trust an approximation that needs large samples.
- Ranks or directions of change answer the research question better than raw-score differences.

Do not select a nonparametric test merely because a normality test produced an asterisk. With large samples, normality tests detect tiny, harmless departures. With small samples, they may miss serious ones. Look at the data, consider the design, and decide what feature of the outcome you actually want to compare.

19.2 The Very Short Decision Guide

Research question	Design	Useful test
Is one category more common than a specified probability predicts?	One binary variable	Exact binomial test
Are two binary variables associated in a small table?	Independent observations, usually 2×2	Fisher's exact test
Do two independent groups tend to have different scores?	Two separate groups	Wilcoxon rank-sum / Mann–Whitney test
Did paired scores tend to change?	Two scores from each person	Sign test or Wilcoxon signed-rank test
Do three or more independent groups differ?	Separate groups	Kruskal–Wallis test
Do three or more repeated conditions differ?	Same people in every condition	Friedman test

The table is a map, not an oracle. A map can get you to the correct neighborhood. It cannot stop you from driving into a volcano because Google said the road continued.

19.3 Tests for Nominal Outcomes

19.3.1 Fisher's Exact Test

Pearson's chi-square uses an approximation that works well when expected counts are reasonably large. When a 2×2 table is tiny or sparse, **Fisher's exact test** calculates an exact p-value under fixed margins.

Suppose sixteen students volunteer for an escape-room study. Half receive calm-breathing instructions. Half are pursued by a research assistant holding a foam chainsaw and repeatedly yelling, "This was approved by the IRB!" The outcome is whether they escape before the timer expires.

```r
escape_table <- matrix(
  c(1, 7,
    6, 2),
  nrow = 2,
  byrow = TRUE,
  dimnames = list(
    Training = c("Calm breathing", "Foam chainsaw"),
    Escaped = c("Yes", "No")
  )
)

escape_table
```

```
                Escaped
Training         Yes No
  Calm breathing   1  7
  Foam chainsaw    6  2
```

```r
chisq.test(escape_table)$expected
```

```
                Escaped
Training         Yes  No
  Calm breathing 3.5 4.5
  Foam chainsaw  3.5 4.5
```

```r
fisher.test(escape_table)
```

```

    Fisher's Exact Test for Count Data

data:  escape_table
p-value = 0.04056
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.0009187486 0.9194607531
sample estimates:
odds ratio
0.06173059
```

Fisher's test is not merely "chi-square for small samples," but that is its most common job in an introductory workflow. Unlike Pearson's large-sample approximation, it obtains the p-value from the possible tables with the same margins. It also returns an odds-ratio estimate and confidence interval.

Do not automatically use Fisher's test for every table. For larger tables it can become computationally expensive, and the exact question depends on how the margins are treated. When expected counts are healthy, Pearson's chi-square is usually perfectly happy doing its job.

19.3.2 The Exact Binomial Test

The **binomial test** handles a binary outcome when trials are independent and the probability under the null hypothesis is specified.

Suppose 18 of 24 students report hearing colors after consuming a legally and ethically questionable quantity of caffeine. If "yes" and "no" were equally likely, the null probability would be .50.

```r
binom.test(
  x = 18,
  n = 24,
  p = .50,
  alternative = "two.sided"
)
```

```
	Exact binomial test

data:  18 and 24
number of successes = 18, number of trials = 24, p-value = 0.02266
alternative hypothesis: true probability of success is not equal to 0.5
95 percent confidence interval:
 0.5328872 0.9022696
sample estimates:
probability of success 
                  0.75 
```

The p-value asks how unusual 18 or a result at least this far from the expected 12 would be if the true probability were .50. The confidence interval estimates the probability of hearing colors. Neither number establishes that caffeine **caused** chromatic audition. Perhaps these students already heard mauve every Thursday. Design still matters.

19.3.3 The Sign Test

The **sign test** is a binomial test applied to paired differences. It ignores how large each change is and records only its direction:

- `+` means the score moved in the predicted direction.
- `-` means it moved in the opposite direction.
- Ties are removed because they have no direction.

Suppose students rate exam panic before and after receiving a one-page study guide. Lower scores are better.

```r
panic_before <- c(8, 7, 9, 6, 8, 7, 9, 5, 8, 6, 7, 9, 8, 6, 7, 8)
panic_after  <- c(5, 6, 7, 6, 4, 5, 6, 4, 7, 5, 7, 5, 6, 7, 5, 6)

change <- panic_before - panic_after
non_ties <- change[change != 0]
n_improved <- sum(non_ties > 0)

binom.test(n_improved, length(non_ties), p = .50)
```

```

	Exact binomial test

data:  n_improved and length(non_ties)
number of successes = 13, number of trials = 14, p-value = 0.001831
alternative hypothesis: true probability of success is not equal to 0.5
95 percent confidence interval:
 0.6613155 0.9981932
sample estimates:
probability of success 
             0.9285714 
```

The sign test is wonderfully stubborn: a one-point improvement and a seven-point improvement both count as one plus. That makes it resistant to extreme magnitudes, but it also throws away useful information. This is the statistical equivalent of judging a chainsaw and a butter knife only by whether each object is pointy.

19.4 Tests Based on Ranks

Ranking replaces the original scores with their ordered positions. The smallest score gets the lowest rank, the next score gets the next rank, and tied scores share ranks. Extreme raw values therefore lose much of their power to terrorize the analysis.

However, a rank test does not automatically become "a test of medians." Under additional assumptions—especially similarly shaped distributions—it may be interpreted as a location or median comparison. Without those assumptions, the safer question is whether scores from one condition tend to be higher or lower than scores from the other.

19.4.1 Two Independent Groups: Wilcoxon Rank-Sum

The **Wilcoxon rank-sum test**, also called the **Mann–Whitney U test**, compares two independent groups.

```r
fear_data <- data.frame(
  training = factor(rep(c("Breathing", "Foam chainsaw"), each = 10)),
  fear = c(
    3, 4, 2, 5, 4, 3, 6, 2, 4, 5,
    7, 8, 6, 9, 7, 10, 8, 6, 9, 7
  )
)
```

```r
aggregate(fear ~ training, data = fear_data, median)
```

```
       training fear
1     Breathing  4.0
2 Foam chainsaw  7.5
```

```r
wilcox.test(
  fear ~ training,
  data = fear_data,
  exact = FALSE,
  conf.int = TRUE
)
```

```

	Wilcoxon rank sum test with continuity correction

data:  fear by training
W = 1, p-value = 0.0002243
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -5.000009 -2.999983
sample estimates:
difference in location 
                    -4 
```

The groups are independent because each student appears in only one training condition. If Alex secretly reuses the same students because recruitment is annoying, the design becomes paired and the analysis must change. Laziness is not a covariance structure.

19.4.2 Two Paired Conditions: Wilcoxon Signed-Rank

The **Wilcoxon signed-rank test** uses both the direction and rank of the absolute paired differences. It is more informative than the sign test, but it assumes the distribution of paired differences is reasonably symmetric.

Use the two-vector interface in modern R:

```r
wilcox.test(
  x = panic_before,
  y = panic_after,
  paired = TRUE,
  exact = FALSE,
  conf.int = TRUE
)
```

```

	Wilcoxon signed rank test with continuity correction

data:  panic_before and panic_after
```

```
V = 102, p-value = 0.001881
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 0.9999786 2.4999692
sample estimates:
(pseudo)median
      1.500002
```

Do **not** write `wilcox.test(score ~ condition, paired = TRUE)`. The formula method does not safely identify which observation should be paired with which. Giving the two aligned vectors makes the pairing explicit and prevents R from responding with the statistical equivalent of "I have no idea which student is which."

19.5 Three or More Conditions

19.5.1 Independent Groups: Kruskal–Wallis

The **Kruskal–Wallis test** extends rank-based comparison to three or more independent groups. It is the common nonparametric relative of a one-way independent-groups ANOVA.

```r
coping_data <- data.frame(
  strategy = factor(rep(c("Breathing", "Pep talk", "Statistics ritual"), each = 8)),
  panic = c(
    4, 5, 3, 6, 4, 5, 2, 5,
    6, 7, 5, 8, 6, 7, 5, 6,
    8, 9, 7, 10, 8, 9, 7, 9
  )
)

kruskal.test(panic ~ strategy, data = coping_data)
```

```

    Kruskal-Wallis rank sum test

data:  panic by strategy
Kruskal-Wallis chi-squared = 17.343, df = 2, p-value = 0.0001714
```

A significant result says that the groups do not all behave alike. It does not say which pairs differ. Follow-up pairwise Wilcoxon tests need a multiple-comparison correction:

```r
pairwise.wilcox.test(
  x = coping_data$panic,
  g = coping_data$strategy,
  p.adjust.method = "holm",
  exact = FALSE
)
```

```
	Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  coping_data$panic and coping_data$strategy

                  Breathing Pep talk
Pep talk          0.0093    -
Statistics ritual 0.0026    0.0093

P value adjustment method: holm
```

Holm's method controls the familywise error rate and is generally preferable to applying the plain Bonferroni correction to every comparison with equal enthusiasm.

19.5.2 Repeated Conditions: Friedman

The **Friedman test** compares three or more repeated conditions. Each row below is one student; each column is something the student was asked to confront.

```r
fear_matrix <- matrix(
  c(
    2, 6, 9,
    3, 7, 8,
    1, 5, 9,
    4, 6, 10,
    2, 7, 9,
    3, 5, 8,
    2, 6, 10,
    4, 7, 9
  ),
  nrow = 8,
  byrow = TRUE,
  dimnames = list(
    Student = paste0("S", 1:8),
    Stimulus = c("Duck", "Reviewer 2", "Exam")
  )
)

fear_matrix
```

```
       Stimulus
Student Duck Reviewer 2 Exam
     S1    2          6    9
     S2    3          7    8
     S3    1          5    9
     S4    4          6   10
     S5    2          7    9
     S6    3          5    8
     S7    2          6   10
```

```
    S8    4          7    9
```

```r
friedman.test(fear_matrix)
```

```

	Friedman rank sum test

data:  fear_matrix
Friedman chi-squared = 16, df = 2, p-value = 0.0003355
```

The blocking variable is the student: every student contributes one score to every condition. Follow-up paired Wilcoxon tests can compare condition pairs, again with corrected p-values.

```r
pair_names <- combn(colnames(fear_matrix), 2, simplify = FALSE)

raw_p <- vapply(pair_names, function(pair) {
  wilcox.test(
    fear_matrix[, pair[1]],
    fear_matrix[, pair[2]],
    paired = TRUE,
    exact = FALSE
  )$p.value
}, numeric(1))

data.frame(
  comparison = vapply(pair_names, paste, collapse = " vs. ", FUN.VALUE = character(1)),
  raw_p = raw_p,
  holm_p = p.adjust(raw_p, method = "holm")
)
```

```
           comparison      raw_p     holm_p
1 Duck vs. Reviewer 2 0.01298403 0.03895209
2       Duck vs. Exam 0.01356012 0.03895209
3 Reviewer 2 vs. Exam 0.01356012 0.03895209
```

19.6 What These Tests Do Not Fix

Nonparametric tests cannot rescue:

- Dependent observations analyzed as independent.
- A confounded design.
- A measure that does not represent the construct.
- Selective deletion of inconvenient participants.
- Twenty-seven unplanned analyses followed by a sudden claim that the only significant one was "the hypothesis all along."

They can also have lower power than a well-chosen parametric model when that model's assumptions are reasonable. More importantly, rank tests change the question by discarding some information about raw distances. Sometimes that is exactly what you want. Sometimes it is like buying noise-canceling headphones and then complaining that you cannot hear the fire alarm.

19.7 Reporting Nonparametric Results

Report enough information for a reader to understand both the result and the data:

1. Name the test and the design.
2. Report the test statistic, degrees of freedom when applicable, and p-value.
3. Describe each condition with counts, proportions, medians, and interquartile ranges as appropriate.
4. Include an effect size and confidence interval when a defensible one is available.
5. State how ties, zero differences, exact p-values, and multiple comparisons were handled.

The test is only one sentence in the argument. The data, design, effect size, uncertainty, and actual psychological question still have to show up for work.

> ⚠ Nonparametric Does Not Mean Assumption-Free
>
> Rank and exact tests make different assumptions and often answer different questions from their parametric cousins. Choose one because it fits the design, scale, and scientific question—not because the data offended a normality test at 2:00 a.m.

Part II

Advanced Regression: A Practical Crash Course

20 Advanced: Regression Diagnostics and Influential Weirdos

20.1 Advanced Topic: Who Is This For?

This chapter is for graduate students and advanced undergraduates who already know how to fit and interpret a regression model. We now ask a more alarming question: **Should we trust the model we just fit?**

Regression will always give you coefficients if the mathematics permits it. It does not pause to ask whether one participant entered a reaction time of 94 years or whether the residuals resemble a startled squid.

> ⚠ A Significant Model Can Still Be a Bad Model
>
> A small p-value does not certify linearity, equal residual variance, independence, honest measurement, or sensible causal reasoning. It only answers a narrower question under the model's assumptions.

20.2 A Deeply Responsible Therapy Study

Suppose we predict symptom improvement from therapy attendance and baseline symptom severity. Most participants behave normally. Participant 60 attends nearly every session and reports a suspiciously enormous improvement, possibly because they misunderstood the scale or wanted the research assistant to stop calling.

```r
set.seed(343)
n <- 60
TherapyData <- data.frame(
  Sessions = runif(n, 0, 12),
  Baseline = rnorm(n, 60, 10)
)

TherapyData$Improvement <-
  1.8 * TherapyData$Sessions +
  0.18 * TherapyData$Baseline +
  rnorm(n, 0, 5)

TherapyData$Sessions[60] <- 12
TherapyData$Baseline[60] <- 45
TherapyData$Improvement[60] <- 60

TherapyModel <- lm(Improvement ~ Sessions + Baseline, data = TherapyData)
summary(TherapyModel)
```

```
Call:
lm(formula = Improvement ~ Sessions + Baseline, data = TherapyData)

Residuals:
     Min       1Q   Median       3Q      Max
-12.4882  -2.9666  -0.4928   2.5910  26.0396

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.77323    5.68723   2.070    0.043 *
Sessions     1.93638    0.21937   8.827 2.96e-12 ***
Baseline    -0.02332    0.09016  -0.259    0.797
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.865 on 57 degrees of freedom
Multiple R-squared:  0.5827,    Adjusted R-squared:  0.568
F-statistic: 39.79 on 2 and 57 DF,  p-value: 1.525e-11
```

20.3 Residuals: What the Model Failed to Explain

For person i,

$$e_i = Y_i - \widehat{Y}_i.$$

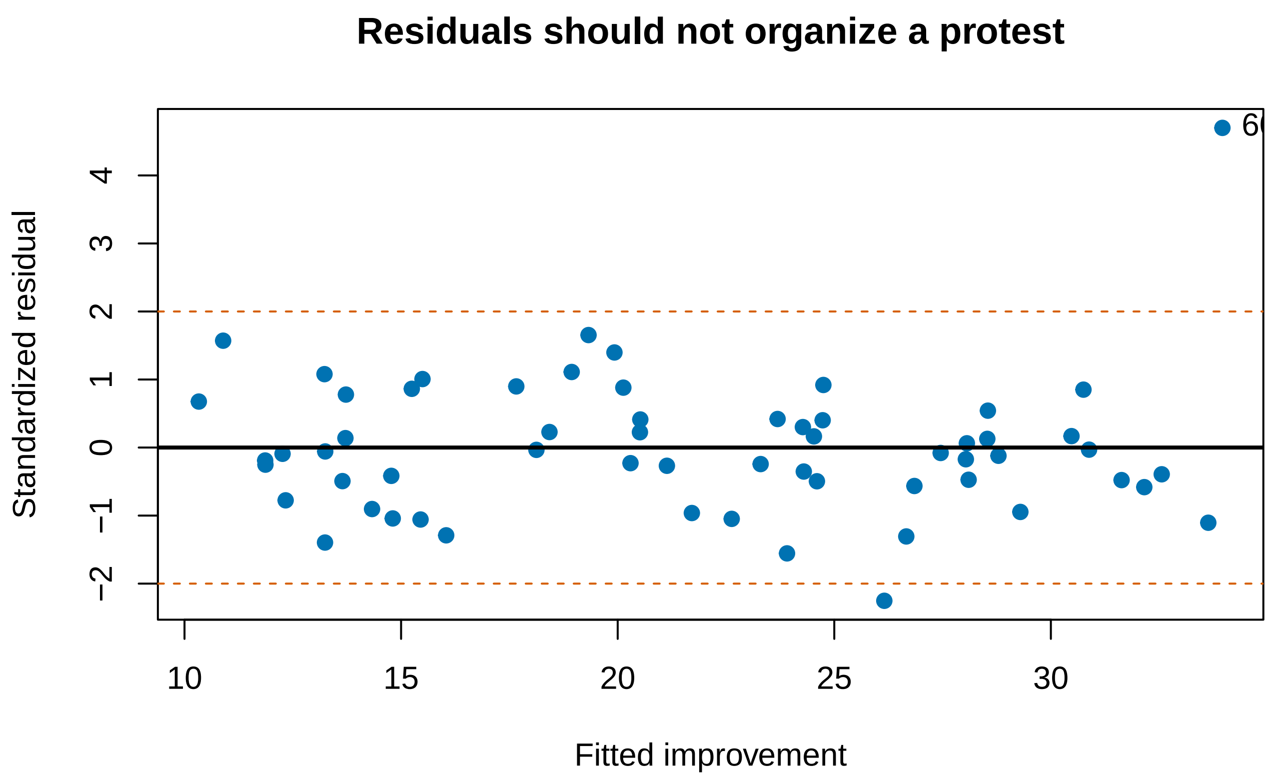
A residual is the observed outcome minus the model's prediction. Positive residuals mean the person scored above the prediction; negative residuals mean they scored below it.

The ordinary linear model assumes:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + \epsilon_i,$$

where the errors have an expected value of zero, are independent, and have approximately constant variance. Normality matters primarily for conventional small-sample inference, not because every raw variable must form a perfect bell.

```r
plot(
  fitted(TherapyModel), rstandard(TherapyModel),
  pch = 19, col = "#0072B2",
  xlab = "Fitted improvement",
  ylab = "Standardized residual",
  main = "Residuals should not organize a protest"
)
abline(h = 0, lwd = 2)
abline(h = c(-2, 2), lty = 2, col = "#D55E00")
text(fitted(TherapyModel)[60], rstandard(TherapyModel)[60], "60", pos = 4)
```



Look for curves, funnels, clusters, and isolated points. A random cloud around zero is comforting. A visible pattern means the model has left information lying on the floor.

> **ℹ Raw Residuals Versus Standardized Residuals**
>
> Raw residuals remain in the outcome's units. Standardized or studentized residuals divide by an estimated residual standard deviation, making unusually large residuals easier to identify. Values beyond roughly ± 2 deserve inspection. That is an investigation threshold, not an automatic deletion command.

20.4 Linearity

Linear regression assumes the mean outcome is a linear combination of the predictors. If the true relationship bends, a straight line may average over the bend and lie everywhere with great consistency.

Possible responses include transformations, polynomial terms, splines, or a model with a more appropriate link function. The graph should guide the question; repeatedly adding powers until an asterisk appears is polynomial fishing.

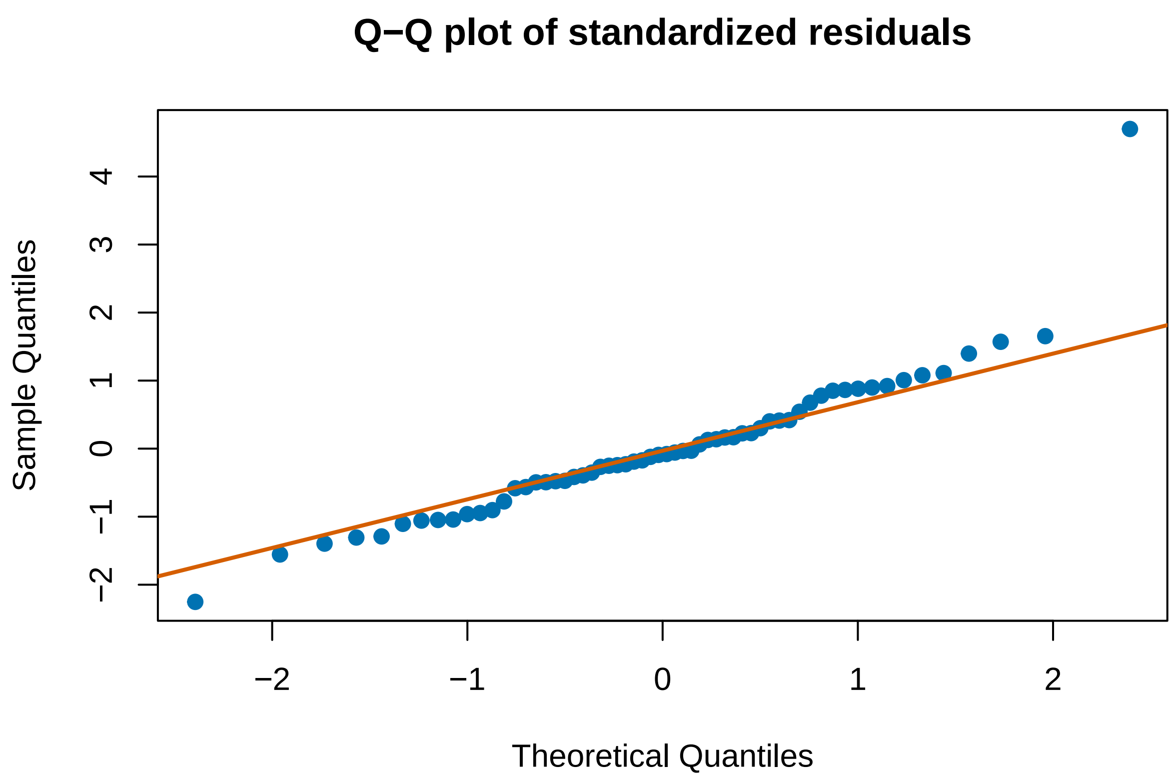
20.5 Homoscedasticity

Homoscedasticity means that residual spread is roughly constant across fitted values. A funnel shape indicates **heteroscedasticity**: the model predicts some portions of the outcome more precisely than others.

Heteroscedasticity does not automatically bias ordinary least-squares slopes, but it can make conventional standard errors and confidence intervals inaccurate. We can reconsider the outcome scale, use heteroscedasticity-consistent standard errors, or fit a model that explicitly represents changing variance.

20.6 Normality of the Errors

```
qqnorm(rstandard(TherapyModel), pch = 19, col = "#0072B2",
       main = "Q-Q plot of standardized residuals")
qqline(rstandard(TherapyModel), col = "#D55E00", lwd = 2)
```



A Q–Q plot compares residual quantiles with those expected under a normal distribution. Mild deviations are ordinary. Severe tail departures may signal outliers, a wrong outcome distribution, or a model that has ignored important structure.

> ⚠ Do Not Test Normality and Surrender
>
> With a large sample, a formal normality test can reject harmless deviations. With a small sample, it may miss serious ones. Inspect the residuals, consider the design, and ask whether the inferential procedure is robust enough for the observed problem.

20.7 Leverage, Influence, and Cook's Distance

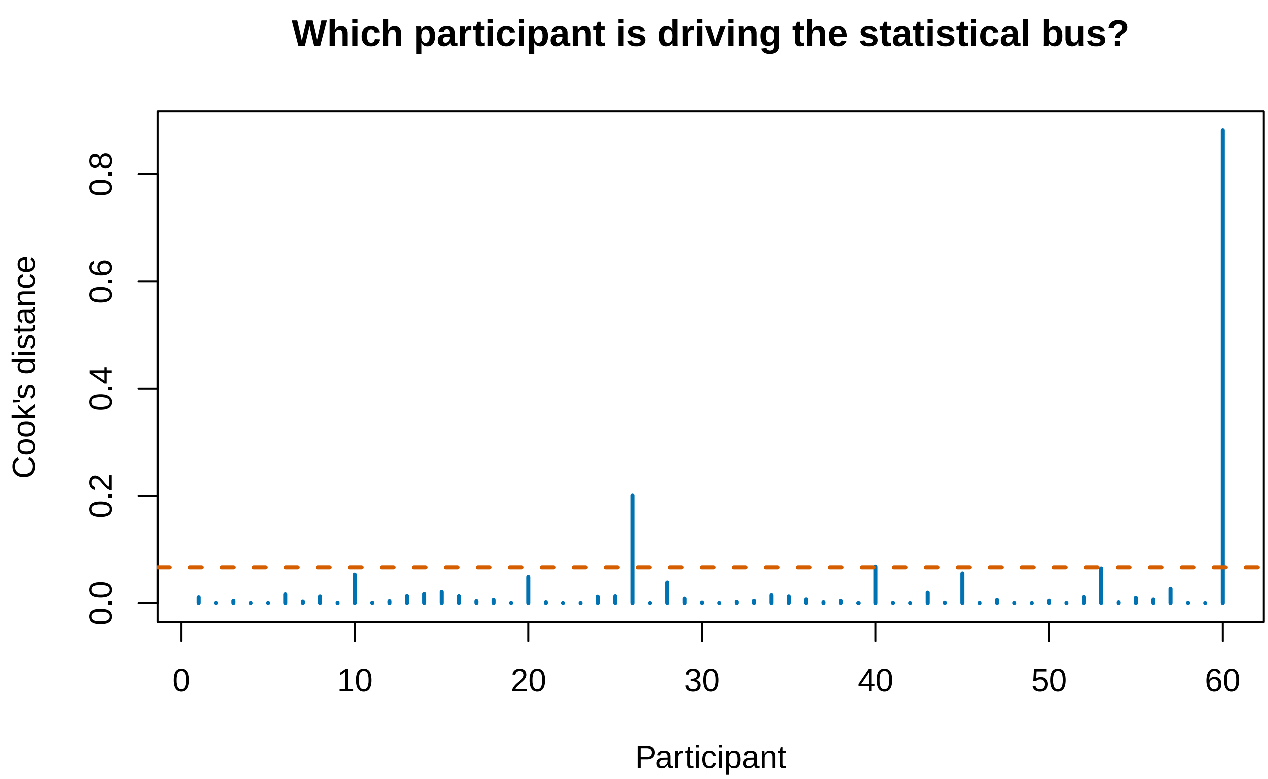
An observation has high **leverage** when its predictor values are unusual. It is **influential** when removing it meaningfully changes the fitted model. A person can have a large residual without much leverage, or high leverage while sitting obediently on the fitted line.

Cook's distance combines residual size and leverage. One common expression is:

$$D_i = \frac{e_i^2}{p\hat{\sigma}^2} \frac{h_{ii}}{(1 - h_{ii})^2},$$

where p is the number of fitted coefficients and h_{ii} is leverage.

```
CookD <- cooks.distance(TherapyModel)
plot(
  CookD, type = "h", lwd = 2, col = "#0072B2",
  xlab = "Participant", ylab = "Cook's distance",
  main = "Which participant is driving the statistical bus?"
)
abline(h = 4 / n, lty = 2, col = "#D55E00", lwd = 2)
text(60, CookD[60], "60", pos = 3)
```



Rules such as $D_i > 4/N$ are screening devices. They tell us where to look, not whom to throw into the statistical volcano.

20.8 Multicollinearity

When predictors are strongly related, the model struggles to separate their unique slopes. Coefficients may become unstable, standard errors inflate, and signs may change when another predictor enters.

For predictor j, the variance inflation factor is:

$$VIF_j = \frac{1}{1 - R_j^2},$$

where R_j^2 comes from predicting X_j with the other predictors. A large VIF means the other predictors nearly reconstruct that predictor.

Multicollinearity is not repaired by staring harder at the output. Possible solutions include improving the design, collecting more informative cases, combining redundant measures when theory supports it, or narrowing the question.

20.9 Independence and Clustering

Ordinary regression assumes residuals are independent. Repeated observations from one person, students within classrooms, siblings within families, and responses to the same stimulus are not independent simply because the spreadsheet gave them separate rows.

If residuals share a source, use a design-aware method such as a mixed-effects model. We will do that in the graduate mixed-model section.

20.10 A Diagnostic Workflow

1. Plot the raw data.
2. Fit the theoretically justified model.
3. Plot residuals against fitted values.
4. inspect Q–Q behavior and residual tails.
5. Examine leverage and influence.
6. Check predictor redundancy.
7. Ask whether observations are independent.
8. Investigate suspicious cases against the original records.
9. Report consequential problems and sensitivity analyses.

! Deleting the Problem Is Not Diagnosing the Problem

Never remove an observation merely because it weakens the hypothesis. Correct impossible values, explain defensible exclusions, and compare conclusions with and without influential cases. If one person reverses the entire result, that instability is part of the result.

20.11 The Mathematics Under the Diagnostic Pictures

Ordinary least squares produces fitted values through the **hat matrix**:

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'.$$

The diagonal value h_{ii} is observation i's leverage. Leverage comes from an unusual predictor pattern, not an unusual outcome by itself. A point can have high leverage and still sit obediently on the regression line. It becomes influential when its leverage combines with a meaningful residual.

Studentized residuals scale each residual by an estimate of its own uncertainty. Cook's distance then asks how much the fitted model would change if observation i vanished:

$$D_i \propto \frac{r_i^2}{p} \frac{h_{ii}}{1 - h_{ii}}.$$

The exact cutoffs are screening rules, not automatic deletion warrants. Inspect the case, verify the data, refit with and without it, and explain what changed.

20.12 When the Variance Changes

Heteroscedasticity does not usually bias the OLS slope when the mean model is correct, but it can damage the ordinary standard errors. Possible responses include modeling the variance, transforming the outcome when scientifically sensible, using a model family that matches the outcome, or reporting heteroscedasticity-consistent standard errors. The picture tells you a problem exists; theory tells you which response makes sense.

20.13 The Short Story

Diagnostics ask whether the model's leftovers behave as expected and whether a few observations control the conclusion. Plot residuals, inspect influence, respect clustering, and do not allow a significant coefficient to distract you from a visibly terrible model.

21 Advanced: Continuous and Higher-Order Interactions

21.1 Advanced Topic: It Depends, Again

The earlier interaction chapters asked whether a slope differs across categories. Here both predictors can be continuous. This produces a model that is compact to write and remarkably easy to explain badly (Aiken and West 1991).

Suppose caffeine predicts concentration differently depending on sleep. Caffeine may help a rested student focus while merely allowing an exhausted student to hear colors at a higher frame rate.

21.2 The Continuous-by-Continuous Model

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \epsilon.$$

The slope of X is:

$$\frac{\partial Y}{\partial X} = \beta_1 + \beta_3 Z.$$

That derivative is the entire interaction in one line: the slope of X changes as Z changes.

> **!** There Is No Single Main Effect of X
>
> When XZ is in the model, β_1 is the slope of X specifically when $Z = 0$. If zero is meaningless or outside the data, the coefficient answers a question nobody asked.

21.3 Simulating Sleep, Caffeine, and Questionable Judgment

```
set.seed(343)
n <- 240
InterData <- data.frame(
  Sleep = pmin(pmax(rnorm(n, 6.5, 1.3), 2.5), 10),
  Caffeine = pmin(pmax(rnorm(n, 220, 100), 0), 500)
)

InterData$Concentration <-
  40 +
  3.5 * InterData$Sleep +
  0.055 * InterData$Caffeine -
  0.012 * InterData$Sleep * InterData$Caffeine +
```

```r
  rnorm(n, 0, 5)

InterData$Sleep.C <- InterData$Sleep - mean(InterData$Sleep)
InterData$Caffeine.C <- InterData$Caffeine - mean(InterData$Caffeine)

AdditiveModel <- lm(Concentration ~ Sleep.C + Caffeine.C, data = InterData)
InteractionModel <- lm(Concentration ~ Sleep.C * Caffeine.C, data = InterData)
anova(AdditiveModel, InteractionModel)
```

```
Analysis of Variance Table

Model 1: Concentration ~ Sleep.C + Caffeine.C
Model 2: Concentration ~ Sleep.C * Caffeine.C
  Res.Df    RSS Df Sum of Sq      F   Pr(>F)
1    237 6940.9
2    236 6460.0  1     480.9 17.569 3.92e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(InteractionModel)
```

```

Call:
lm(formula = Concentration ~ Sleep.C * Caffeine.C, data = InterData)

Residuals:
     Min       1Q   Median       3Q      Max
-13.0288  -3.4584  -0.4433   3.6100  14.2160

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        57.313647   0.337748 169.693  < 2e-16 ***
Sleep.C             0.907775   0.286585   3.168  0.00174 **
Caffeine.C         -0.018760   0.003559  -5.271 3.06e-07 ***
Sleep.C:Caffeine.C -0.012262   0.002926  -4.191 3.92e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.232 on 236 degrees of freedom
Multiple R-squared:  0.2071,    Adjusted R-squared:  0.197
F-statistic: 20.54 on 3 and 236 DF,  p-value: 7.326e-12
```

Centering places zero at the sample mean. It changes the interpretation of the intercept and lower-order slopes. It does **not** change fitted values, residuals, R^2, or the interaction coefficient.

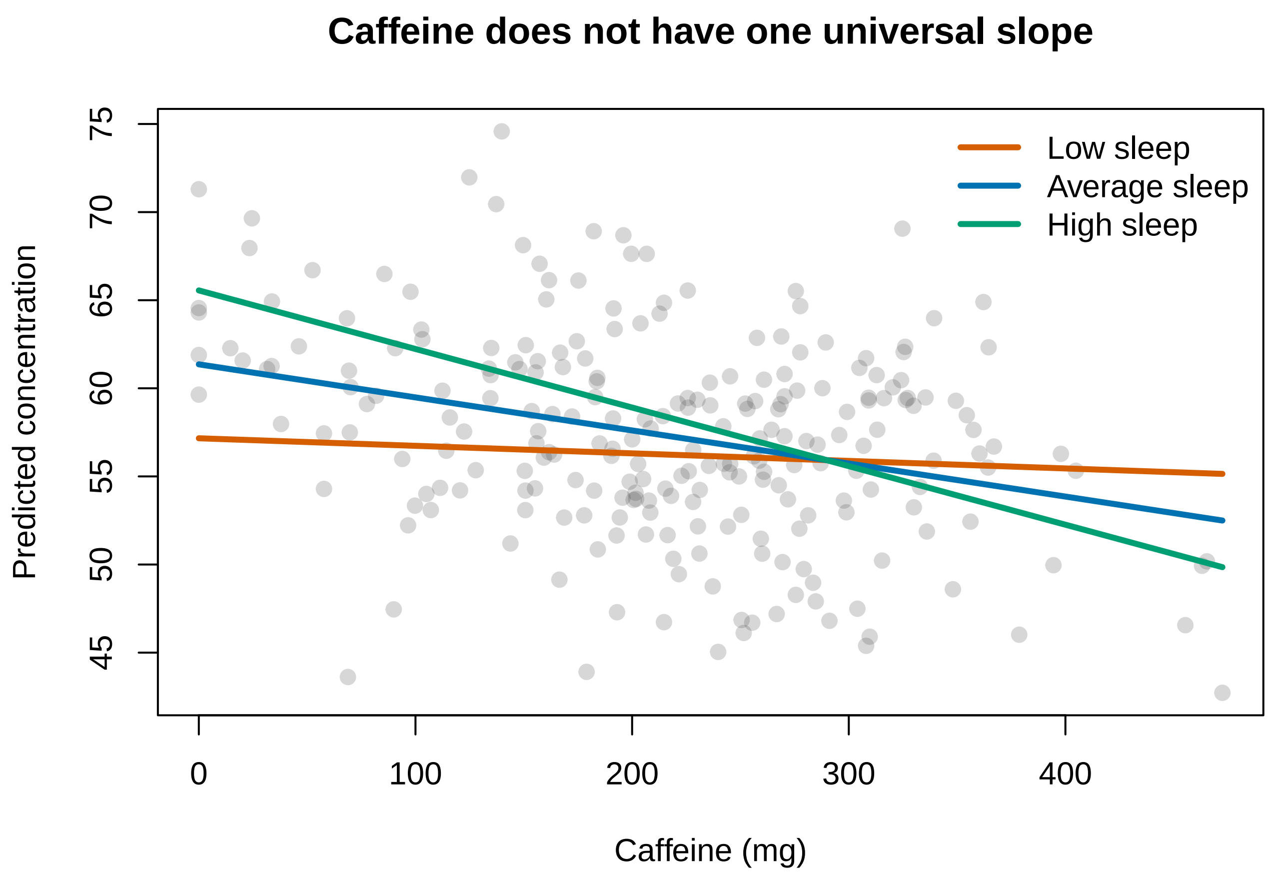
Centering Is Translation, Not Exorcism

Centering can make lower-order terms interpretable and reduce nonessential correlation between a product term and its components. It does not repair poor measurement, restricted range, or two predictors that are fundamentally redundant.

21.4 Plot the Model

We will draw the caffeine slope at low sleep, average sleep, and high sleep. The common $\pm 1 SD$ convention is useful because it gives three recognizable values. It is not a law delivered by the Moderation Gods.

```r
sleep_values <- c(
  Low = -sd(InterData$Sleep.C),
  Average = 0,
  High = sd(InterData$Sleep.C)
)

caffeine_grid <- seq(min(InterData$Caffeine.C), max(InterData$Caffeine.C), length.out = 100)
plot(
  InterData$Caffeine, InterData$Concentration,
  pch = 19, col = rgb(0.3, 0.3, 0.3, 0.22),
  xlab = "Caffeine (mg)", ylab = "Predicted concentration",
  main = "Caffeine does not have one universal slope"
)

line_colors <- c("#D55E00", "#0072B2", "#009E73")
for (i in seq_along(sleep_values)) {
  newdata <- data.frame(
    Sleep.C = sleep_values[i],
    Caffeine.C = caffeine_grid
  )
  lines(
    caffeine_grid + mean(InterData$Caffeine),
    predict(InteractionModel, newdata = newdata),
    col = line_colors[i], lwd = 3
  )
}
legend("topright", legend = paste(names(sleep_values), "sleep"),
       col = line_colors, lwd = 3, bty = "n")
```



21.5 Simple Slopes

For a chosen value z^* of the moderator, the simple slope of caffeine is:

$$b_{Caffeine|z^*} = b_1 + b_3 z^*.$$

Its standard error also depends on the covariance between b_1 and b_3:

$$SE(b_1 + b_3 z^*) = \sqrt{Var(b_1) + z^{*2} Var(b_3) + 2z^* Cov(b_1, b_3)}.$$

This is why we let software test simple slopes. Doing it by hand is possible, but so is churning butter.

```
emmeans::emtrends(
  InteractionModel,
  specs = "Sleep.C",
  var = "Caffeine.C",
  at = list(Sleep.C = unname(sleep_values))
)
```

```
 Sleep.C Caffeine.C.trend      SE  df lower.CL upper.CL
   -1.18         -0.00428 0.00528 236  -0.0147  0.00613
    0.00         -0.01876 0.00356 236  -0.0258 -0.01175
    1.18         -0.03324 0.00462 236  -0.0423 -0.02415

Confidence level used: 0.95
```

Use theoretically meaningful moderator values when possible. Quantiles may be better than $\pm 1SD$ for a skewed moderator. Never probe values outside the observed range merely because the resulting line looks exciting.

21.6 Reverse-Scoring Changes the Signs, Not the Story

If $Z^* = -Z$, the signs of its main effect and interactions reverse. Predictions remain identical after the coding is translated correctly. A negative interaction does not inherently mean antagonism, harm, or Reviewer 2. Its meaning depends on variable coding.

> ⚠ Interaction Labels Are Stories, Not New Statistics
>
> Terms such as *synergistic*, *antagonistic*, and *buffering* describe theoretical patterns. The model still estimates slopes and products. Decide which story applies only after inspecting coding, predictions, and the scientific context.

21.7 Three-Way Interactions

A three-way interaction asks whether a two-way interaction changes across a third variable:

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 W + \beta_4 XZ + \beta_5 XW + \beta_6 ZW + \beta_7 XZW + \epsilon.$$

If $\beta_7 \neq 0$, the X-by-Z interaction depends on W.

```r
InterData$Deadline <- factor(
  ifelse(seq_len(nrow(InterData)) %% 2 == 0, "Tomorrow", "Next week")
)

ThreeWayModel <- lm(
  Concentration ~ Sleep.C * Caffeine.C * Deadline,
  data = InterData
)
```

The `*` operator includes all main effects, all two-way interactions, and the three-way interaction. Keep those lower-order terms. Removing them because they are nonsignificant changes the model's meaning and usually creates interpretive debris.

To interpret a three-way interaction:

1. Plot the predicted two-way relationship at meaningful values of the third variable.
2. Test the two-way interaction within those values or groups.
3. Probe simple slopes only where theory requires them.
4. Report enough numbers that the reader does not have to reconstruct the model with candles and a Ouija board.

21.8 The Johnson-Neyman Question

Simple slopes at the mean and one standard deviation above and below the mean are convenient. They are not sacred locations handed down by Dorkaos. The **Johnson-Neyman** approach solves for the values of moderator Z where the conditional slope of X changes from statistically distinguishable from zero to not distinguishable from zero.

The conditional slope is

$$b_{X|Z} = b_1 + b_3 Z.$$

Its standard error depends on the variances and covariance of b_1 and b_3:

$$SE(b_{X|Z}) = \sqrt{Var(b_1) + Z^2 Var(b_3) + 2ZCov(b_1, b_3)}.$$

Johnson-Neyman regions can be useful when the moderator is truly continuous. Do not report a region far outside the observed values of Z. The equation is willing to extrapolate into space; your data did not go there.

21.9 Centering Cannot Remove the Interaction

Mean-centering changes the value represented by zero and therefore changes the lower-order coefficient labels. It does not change fitted values, residuals, R^2, or the interaction coefficient in an ordinary two-way product model. If an interaction disappears merely because you centered correctly, something else in the analysis changed too.

21.10 The Short Story

A continuous interaction means one slope changes with another variable. Center to make zero useful, plot predictions, test meaningful simple slopes, and remember that a three-way interaction is a changing two-way interaction—not merely a coefficient with extra punctuation.

22 Advanced: Contrasts and ANOVA as Regression Wearing an Older Hat

22.1 Why This Chapter Exists

Psychology reports many analyses as ANOVA even though the same questions can be expressed through regression. This chapter provides the translation and then returns you safely to regression.

Suppose students are randomly assigned to three study conditions:

- Silence
- Instrumental music
- A research assistant repeatedly whispering, "You are running out of time"

The third condition has been rejected by the ethics board, but the simulated data have fewer legal protections.

```
set.seed(343)
n_per_group <- 40
ContrastData <- data.frame(
  Condition = factor(
    rep(c("Silence", "Music", "Whispering"), each = n_per_group),
    levels = c("Silence", "Music", "Whispering")
  )
)

ContrastData$Recall <- c(
  rnorm(n_per_group, 72, 8),
  rnorm(n_per_group, 78, 8),
  rnorm(n_per_group, 60, 8)
)
```

22.2 Dummy Coding

With Silence as the reference group:

$$Y = \beta_0 + \beta_1 D_{Music} + \beta_2 D_{Whispering} + \epsilon.$$

```
DummyModel <- lm(Recall ~ Condition, data = ContrastData)
coef(DummyModel)
```

```
        (Intercept)      ConditionMusic ConditionWhispering
          70.679691            7.080421          -11.917088
```

```r
aggregate(Recall ~ Condition, data = ContrastData, mean)
```

```
   Condition   Recall
1    Silence 70.67969
2      Music 77.76011
3 Whispering 58.76260
```

- β_0 is the Silence mean.
- β_1 is Music minus Silence.
- β_2 is Whispering minus Silence.

Changing the reference group changes the coefficient labels, not fitted group means, residuals, R^2, or the scientific result.

> **Coding Changes the Questions Printed in the Table**
>
> The model predicts the same cell means under any full-rank contrast coding. Coding determines which comparisons appear directly as coefficients. It does not rearrange the participants or alter what happened to them.

22.3 The Omnibus F-Test

The omnibus null is:

$$H_0 : \mu_{Silence} = \mu_{Music} = \mu_{Whispering}.$$

```r
anova(DummyModel)
```

```
Analysis of Variance Table

Response: Recall
           Df Sum Sq Mean Sq F value    Pr(>F)
Condition   2 7374.1  3687.0   68.02 < 2.2e-16 ***
Residuals 117 6342.0    54.2
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The F ratio compares explained variability per model degree of freedom with residual variability:

$$F = \frac{MS_{Model}}{MS_{Error}}.$$

A significant omnibus test says at least one estimable group difference exists. It does not identify which one. The omnibus test is the statistical equivalent of yelling "Something happened!" and leaving the room.

22.4 Why $F = t^2$—Sometimes

When a hypothesis has exactly one numerator degree of freedom, its F statistic equals the corresponding squared t statistic:

$$F_{1,df} = t_{df}^2.$$

This applies to a two-group comparison or a single one-degree-of-freedom contrast. It does not mean every multi-group omnibus F can be replaced by squaring one coefficient's t value.

> ⚠ The Numerator Degrees of Freedom Matter
>
> If an effect uses more than one numerator degree of freedom, there is no single t whose square reproduces the omnibus F. The slogan $F = t^2$ requires the quiet but essential phrase "when $df_1 = 1$."

22.5 Planned Contrasts

A contrast is a weighted comparison of means:

$$L = \sum_{j=1}^{k} c_j \mu_j,$$

where the weights usually sum to zero.

Suppose our planned hypotheses are:

1. Music differs from Silence: $(-1, 1, 0)$.
2. Whispering differs from the average of the two reasonable conditions: $(.5, .5, -1)$.

```
EstimatedMeans <- emmeans::emmeans(DummyModel, ~ Condition)

PlannedTests <- emmeans::contrast(
  EstimatedMeans,
  method = list(
    "Music - Silence" = c(-1, 1, 0),
    "Reasonable conditions - Whispering" = c(.5, .5, -1)
  ),
  adjust = "none"
)

PlannedTests
```

```
 contrast                           estimate   SE  df t.ratio p.value
 Music - Silence                        7.08 1.65 117   4.301 <0.0001
 Reasonable conditions - Whispering    15.46 1.43 117  10.842 <0.0001
```

Planned contrasts should come from theory before inspecting every possible comparison. If you invented the contrast after seeing the means, it is exploratory. Calling it planned in the manuscript does not produce time travel.

22.6 Deviation or Effect Coding

Deviation coding compares levels with the grand mean rather than one reference group.

```
EffectData <- ContrastData
contrasts(EffectData$Condition) <- contr.sum(3)
EffectModel <- lm(Recall ~ Condition, data = EffectData)
coef(EffectModel)
```

```
(Intercept)  Condition1  Condition2
  69.067469    1.612222    8.692644
```

In a balanced design, the intercept becomes the unweighted grand mean. In an unbalanced design, "the grand mean" requires more care because observed, equally weighted, and model-estimated marginal means may differ.

22.7 Eta-Squared and Regression R-Squared

For a one-factor model,

$$\eta^2 = \frac{SS_{Effect}}{SS_{Total}}.$$

That equals the model's R^2 in this simple design.

```
ANOVA.Table <- anova(DummyModel)
EtaSquared <- ANOVA.Table$`Sum Sq`[1] / sum(ANOVA.Table$`Sum Sq`)
c(EtaSquared = EtaSquared, RSquared = summary(DummyModel)$r.squared)
```

```
EtaSquared   RSquared
 0.5376223  0.5376223
```

Partial eta-squared uses effect variance relative to that effect plus its error term:

$$\eta_p^2 = \frac{SS_{Effect}}{SS_{Effect} + SS_{Error}}.$$

Partial eta-squared values from several effects do not generally add to one. They each receive their own private denominator and therefore cannot be stacked like pieces of one variance pie.

> **!** Effect Sizes Need Names
>
> Report whether a value is R^2, η^2, partial η_p^2, generalized η_G^2, or Cohen's d. Writing "the effect size was .18" is like reporting that the trip was 18 without mentioning miles, minutes, or terrified undergraduates.

22.8 Contrast Weights Are a Hypothesis

A planned contrast is a weighted comparison of cell means:

$$L = \sum_{j=1}^{g} c_j \mu_j, \qquad \sum_{j=1}^{g} c_j = 0.$$

Weights summing to zero compare groups rather than estimating the grand level of the outcome. Multiplying every weight by the same constant changes the scale of L but not the hypothesis test. For example, $(-1, -1, 2)$ and $(-.5, -.5, 1)$ ask the same scientific question.

Two contrasts are orthogonal in a balanced design when

$$\sum_j c_{1j} c_{2j} = 0.$$

Orthogonality means the comparisons partition independent pieces of between-group information. With unequal cell sizes, the definition and software details require more care. This is one reason to state the contrast weights instead of reporting that an unnamed dropdown menu was selected with confidence.

22.9 Estimability

A contrast must be estimable from the design. Empty cells, perfect confounding, and redundant predictors can make some comparisons impossible no matter how strongly we believe in them. `emmeans` works from the fitted model's reference grid and will often warn when a requested comparison is not estimable. Believe the warning before inventing participants to fill the empty cell.

22.10 The Short Story

ANOVA and regression can express the same group comparisons. Dummy coding compares groups with a reference, effect coding compares with a grand-mean structure, and planned contrasts test theory-driven combinations. The omnibus F finds whether something differs; contrasts explain what.

23 Advanced: Bootstrapping and Resampling

23.1 The Basic Idea

Bootstrapping asks what would happen if we could repeatedly sample from a population that resembles the one we observed. Because we do not possess the population, we resample the observed data **with replacement** (Efron and Tibshirani 1993).

With replacement matters. After selecting a participant, we return them to the sample before the next draw. Some participants appear several times; others disappear entirely. This is statistically acceptable and ethically preferable to cloning the undergraduates.

> **i** What the Bootstrap Treats as the Population
>
> The empirical distribution of the observed sample stands in for the population. If the original sample is biased or misses important parts of the population, bootstrapping reproduces that problem thousands of times with impressive efficiency.

23.2 Why Use It?

Bootstrapping is useful when:

- A sampling distribution is difficult to derive.
- The statistic is a median, indirect effect, or other quantity with awkward standard-error formulas.
- Distributional assumptions are questionable.
- We want to understand sampling variability by simulation.

It does not repair dependence, bad measurement, tiny unrepresentative samples, or a design in which the control group accidentally received the treatment.

23.3 Bootstrapping a Skewed Median

Suppose students report how many minutes they procrastinated before opening the assignment. The distribution is strongly right-skewed because several students began procrastinating during the previous semester.

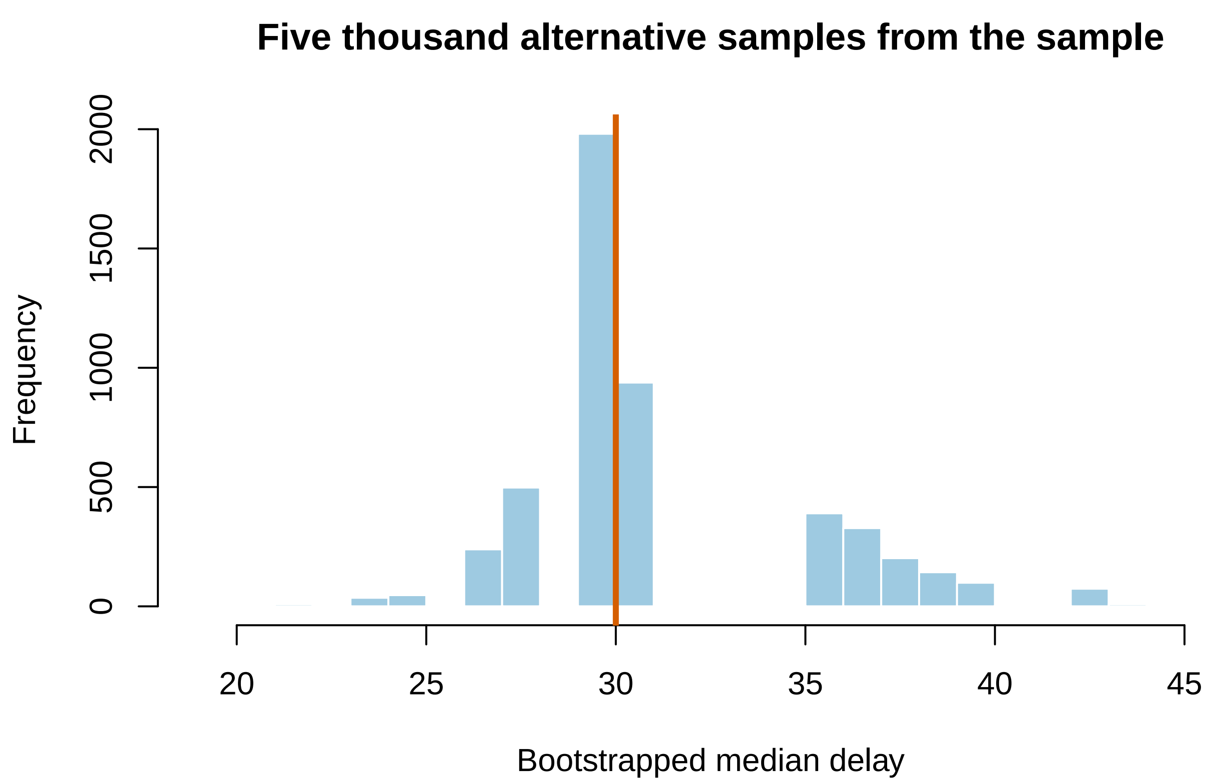
```r
set.seed(343)
Delay <- round(rlnorm(55, meanlog = log(35), sdlog = .8))
median(Delay)
```

```
[1] 30
```

```
mean(Delay)
```

```
[1] 40.61818
```

```
B <- 5000
BootMedian <- replicate(
  B,
  median(sample(Delay, size = length(Delay), replace = TRUE))
)

hist(
  BootMedian, breaks = 30,
  col = "#9ECAE1", border = "white",
  xlab = "Bootstrapped median delay",
  main = "Five thousand alternative samples from the sample"
)
abline(v = median(Delay), col = "#D55E00", lwd = 3)
```



The bootstrap standard error is the standard deviation of the bootstrapped estimates:

$$SE_{boot} = SD(\hat{\theta}^*_1, \dots, \hat{\theta}^*_B).$$

```r
BootSE <- sd(BootMedian)
PercentileCI <- quantile(BootMedian, c(.025, .975))
c(SE = BootSE, PercentileCI)
```

```
       SE      2.5%     97.5% 
 3.859208 27.000000 40.000000 
```

The percentile interval uses the 2.5th and 97.5th percentiles of the bootstrap distribution.

23.4 Percentile, Basic, and BCa Intervals

- **Percentile interval:** Uses bootstrap quantiles directly.
- **Basic interval:** Reflects bootstrap errors around the observed estimate.
- **BCa interval:** Adjusts for bias and skewness in the bootstrap distribution.

BCa intervals are often attractive for skewed statistics, but they require additional calculations and can behave poorly in very small or unstable samples. The `boot` package calculates them without requiring students to derive an acceleration constant during lunch.

```r
median_statistic <- function(data, indices) median(data[indices])
BootObject <- boot::boot(Delay, statistic = median_statistic, R = 5000)
boot::boot.ci(BootObject, type = c("perc", "bca"))
```

```
BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 5000 bootstrap replicates

CALL : 
boot::boot.ci(boot.out = BootObject, type = c("perc", "bca"))

Intervals : 
Level     Percentile            BCa          
95%   (27, 40 )   (17, 30 )  
Calculations and Intervals on Original Scale
Warning : BCa Intervals used Extreme Quantiles
Some BCa intervals may be unstable
```

> ⚠ A Bootstrap Distribution Is Not Automatically a Null Distribution
>
> Resampling the observed data estimates uncertainty around the observed statistic. A hypothesis test may require resampling under a null model, permuting labels, or imposing the null in another defensible way. Dividing a bootstrap estimate by its bootstrap SE and feeding it to an ordinary t table is not a universal bootstrap test.

23.5 Bootstrapping a Regression Slope

For independent observations, resample rows so each resampled participant retains all their variables.

```r
set.seed(344)
RegressionData <- data.frame(
  Sleep = runif(80, 3, 9)
)
RegressionData$Memory <- 42 + 4 * RegressionData$Sleep + rnorm(80, 0, 8)

OriginalModel <- lm(Memory ~ Sleep, data = RegressionData)
OriginalSlope <- coef(OriginalModel)["Sleep"]

BootSlopes <- replicate(3000, {
  rows <- sample(seq_len(nrow(RegressionData)), replace = TRUE)
  coef(lm(Memory ~ Sleep, data = RegressionData[rows, ]))["Sleep"]
})

c(
  OriginalSlope = OriginalSlope,
  BootstrapSE = sd(BootSlopes),
  quantile(BootSlopes, c(.025, .975))
)
```

```
OriginalSlope.Sleep         BootstrapSE                2.5%               97.5%
          4.2612862           0.5373217           3.2294410           5.3136534
```

For repeated or clustered data, resampling individual rows destroys the dependence structure. Resample at the appropriate unit, such as participants, classrooms, or clusters. Mixed-model bootstrap procedures can instead simulate from the fitted model; we revisit that in the graduate mixed-model overview.

> **!** Resample the Unit That Was Sampled
>
> If trials are nested within people, treating trials as independent resampling units manufactures a fictional sample size. Preserve the design when resampling. The bootstrap is flexible, not psychic.

23.6 How Many Replications?

A few hundred replications may be useful for teaching or debugging. Final interval estimates usually require thousands, especially for tail quantiles. Increase B until the reported result is stable enough for its purpose.

23.7 Bootstrap Bias and Standard Error

If $\hat{\theta}$ is the estimate from the original sample and $\hat{\theta}^*_b$ is the estimate from bootstrap sample b, the bootstrap bias estimate is

$$\widehat{Bias}_{boot} = \bar{\theta}^* - \hat{\theta},$$

and the bootstrap standard error is

$$SE_{boot} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\hat{\theta}_b^* - \bar{\theta}^*)^2}.$$

The number of replications B controls Monte Carlo noise in the bootstrap calculation. More replications do not repair a biased original sample or a resampling scheme that ignores the design. Ten thousand wrong resamples are wrong with excellent numerical stability.

23.8 Nonparametric, Parametric, and Residual Bootstraps

- A **nonparametric bootstrap** resamples observed cases or clusters.
- A **parametric bootstrap** simulates new outcomes from the fitted probability model.
- A **residual bootstrap** holds predictors fixed and resamples model residuals under assumptions about the error structure.

These procedures answer related but not identical questions. State which one you used, what unit was resampled, how many replications were run, and how the interval was constructed.

23.9 The Short Story

Bootstrap samples are drawn with replacement from the observed data. Their statistics approximate a sampling distribution. Preserve the study's dependence structure, choose an interval appropriate to the problem, and remember that resampling cannot add information the original study failed to collect.

24 Advanced: Missing Data and Multiple Imputation

24.1 Missing Is Information

Missing data are not empty spreadsheet cells waiting politely for deletion. They occurred through a process. Understanding that process matters more than discovering which button replaces `NA` with a number.

Suppose participants complete a stress survey:

- Some skip an item because the page failed to load.
- Some exhausted participants abandon the survey halfway through.
- Some refuse to report stress precisely because their stress is extreme.

Those are different missing-data mechanisms and may produce different biases.

24.2 MCAR, MAR, and MNAR

Let R indicate whether a value is observed and let Y_{mis} and Y_{obs} denote missing and observed data.

- **MCAR:** Missingness is unrelated to observed or missing values.

$$P(R \mid Y_{obs}, Y_{mis}) = P(R).$$

- **MAR:** After conditioning on observed variables, missingness no longer depends on the missing value.

$$P(R \mid Y_{obs}, Y_{mis}) = P(R \mid Y_{obs}).$$

- **MNAR:** Missingness still depends on the unobserved value after accounting for observed information.

> ⚠ MAR Does Not Mean Missingness Is Random Nonsense
>
> Under MAR, missingness may be strongly predictable from variables we observed. A participant may skip the pet-spider question because they already reported hating spiders. The missingness is systematic, but the observed spider-hatred variable can help model it.

24.3 Why Mean Imputation Is Not a Solution

Mean imputation replaces every missing value with the same number. It preserves the mean mechanically while shrinking variability, weakening relationships, and pretending invented values were observed with certainty.

```r
set.seed(343)
n <- 300
CompleteData <- data.frame(
  Sleep = rnorm(n, 7, 1.4)
)
CompleteData$Stress <- 70 - 5 * CompleteData$Sleep + rnorm(n, 0, 6)

MissingData <- CompleteData
miss <- runif(n) < plogis((MissingData$Stress - 38) / 5)
MissingData$Stress[miss] <- NA

MeanImputed <- MissingData
MeanImputed$Stress[is.na(MeanImputed$Stress)] <- mean(MeanImputed$Stress, na.rm = TRUE)

rbind(
  Complete = c(SD = sd(CompleteData$Stress),
             Slope = coef(lm(Stress ~ Sleep, CompleteData))["Sleep"]),
  CompleteCases = c(SD = sd(MissingData$Stress, na.rm = TRUE),
                  Slope = coef(lm(Stress ~ Sleep, MissingData))["Sleep"]),
  MeanImputed = c(SD = sd(MeanImputed$Stress),
                 Slope = coef(lm(Stress ~ Sleep, MeanImputed))["Sleep"])
)
```

```
                    SD Slope.Sleep
Complete      8.898153   -4.988459
CompleteCases 8.105092   -4.412445
MeanImputed   6.129435   -2.374779
```

Increasing the sample size does not make mean imputation principled. It can make the wrong answer more precise.

! Do Not Impute the Outcome Once and Pretend It Was Observed

Single imputation ignores uncertainty about the missing values. Multiple imputation creates several plausible completed datasets, analyzes each one, and combines the results so imputation uncertainty enters the standard errors.

24.4 Multiple Imputation

Multiple imputation proceeds in three stages:

1. **Impute:** Create m plausible completed datasets.
2. **Analyze:** Fit the same analysis in every dataset.
3. **Pool:** Combine coefficients and uncertainty using Rubin's rules (Rubin 1987).

If $\widehat{Q}_j$ is the estimate from imputation j, then:

$$\bar{Q} = \frac{1}{m}\sum_{j=1}^{m}\widehat{Q}_j.$$

The total variance combines average within-imputation variance $\bar{U}$ and between-imputation variance B:

$$T = \bar{U} + \left(1 + \frac{1}{m}\right) B.$$

The second term acknowledges that we do not know which plausible replacements were correct. This is uncertainty doing its job instead of being locked in a closet.

24.5 A Modern `mice` Workflow

```
library(mice)

# Inspect missingness and choose methods/predictors deliberately.
Imputations <- mice(
  MissingData,
  m = 20,
  method = "pmm",
  seed = 343,
  printFlag = FALSE
)

# Fit the model in every imputed dataset.
ImputedModels <- with(
  Imputations,
  lm(Stress ~ Sleep)
)

# Pool estimates and uncertainty across all 20 analyses.
PooledResults <- pool(ImputedModels)
summary(PooledResults, conf.int = TRUE)
```

```
         term  estimate std.error statistic       df      p.value     2.5 %
1 (Intercept) 63.367223 3.0119816  21.03838 40.01290 3.025475e-23 57.279843
2       Sleep -4.457738 0.3925451 -11.35599 49.56118 2.128704e-15 -5.246361
     97.5 %  conf.low conf.high
1 69.454604 57.279843 69.454604
2 -3.669115 -5.246361 -3.669115
```

The essential step is `pool()`. Calling `complete()` once and analyzing one completed dataset discards the entire reason for multiple imputation. The `mice` workflow implements chained-equation imputation and pooling in R (Buuren and Groothuis-Oudshoorn 2011).

Predictive mean matching (`pmm`) replaces missing continuous values with observed values from cases having similar predicted means. It can preserve plausible ranges and some nonnormality, but no default method is correct for every variable.

24.6 What Goes Into the Imputation Model?

Include:

- Variables in the final analysis.
- Variables that predict missingness.
- Variables that predict the incomplete variable.
- Design features such as interactions, nonlinear terms, clustering, or bounds when relevant.

The imputation model should generally be at least as rich as the analysis model. Imputing each variable as if the interaction, repeated structure, or group membership does not exist may quietly erase the effect you planned to test.

24.7 Mixed Models and Missing Outcomes

Likelihood-based mixed models can use participants with incomplete outcome records under assumptions related to MAR. This is a major advantage over classical repeated-measures ANOVA, which may discard an entire participant for one missing occasion.

That does not mean mixed models solve every missing-data problem. Missing predictors still require attention, dropout can be informative, and MNAR mechanisms require sensitivity analyses or stronger models.

> ⚠ MNAR Cannot Be Diagnosed from the Observed Data Alone
>
> If missingness depends on values we never observed, the data cannot reveal the mechanism by themselves. State assumptions, use substantive knowledge, and examine how conclusions change under plausible alternatives.

24.8 A Practical Workflow

1. Report the amount and pattern of missingness.
2. Ask why each variable may be missing.
3. Compare observed characteristics of complete and incomplete cases.
4. Choose an analysis and imputation strategy together.
5. Diagnose the imputation models and inspect completed values.
6. Pool results across imputations.
7. Conduct sensitivity analyses when conclusions depend on unverifiable assumptions.

24.9 The Short Story

Missing data arise through a process. Mean imputation invents certainty and damages variation. Multiple imputation creates several plausible datasets, fits every one, and pools the results. No method rescues an unexamined missingness mechanism.

25 Advanced: Generalized Linear Models

25.1 When Y Refuses to Be Normally Distributed

Ordinary linear regression expects a continuous outcome whose conditional mean can be modeled on the original scale with approximately constant residual variance. Psychology routinely produces outcomes that reject this lifestyle:

- Passed or failed treatment.
- Chose the frightening image or the boring image.
- Number of panic symptoms.
- Proportion of trials answered correctly.

A **generalized linear model** keeps the linear predictor but changes the outcome distribution and the function connecting that predictor to the outcome (McCullagh and Nelder 1989).

25.2 The Three Pieces

1. A response distribution from the exponential family.
2. A linear predictor:

$$\eta_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki}.$$

3. A link function:

$$g(\mu_i) = \eta_i,$$

where $\mu_i = E(Y_i \mid X_i)$.

> **The Model Is Linear on the Link Scale**
>
> Logistic regression is linear in log-odds, not probability. Poisson regression is linear in log expected counts, not counts. The transformed scale is not software decoration; it determines coefficient interpretation.

25.3 Logistic Regression

For a binary outcome $Y \in \{0, 1\}$,

$$Y_i \sim Bernoulli(p_i),$$

and the logit link is:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i.$$

Suppose we predict whether a student passes an exam from hours of sleep. No participants are chased with chainsaws because this outcome already contains enough anxiety.

```r
set.seed(343)
n <- 300
LogisticData <- data.frame(Sleep = runif(n, 2, 10))
LogOdds <- -3.5 + .62 * LogisticData$Sleep
LogisticData$Pass <- rbinom(n, size = 1, prob = plogis(LogOdds))

LogisticModel <- glm(Pass ~ Sleep, data = LogisticData, family = binomial())
summary(LogisticModel)
```

```

Call:
glm(formula = Pass ~ Sleep, family = binomial(), data = LogisticData)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.42274    0.43605  -7.849 4.18e-15 ***
Sleep        0.61030    0.07175   8.506  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 414.55  on 299  degrees of freedom
Residual deviance: 312.20  on 298  degrees of freedom
AIC: 316.2

Number of Fisher Scoring iterations: 4
```

25.3.1 Log-Odds, Odds Ratios, and Probabilities

The slope is the change in log-odds for a one-unit increase in X. Exponentiating gives an odds ratio:

$$OR = e^{\beta_1}.$$

```r
exp(coef(LogisticModel))
```

```
(Intercept)       Sleep 
 0.03262305  1.84097794 
```

An odds ratio of 1 means no change in odds. Values above 1 indicate increasing odds; values below 1 indicate decreasing odds.

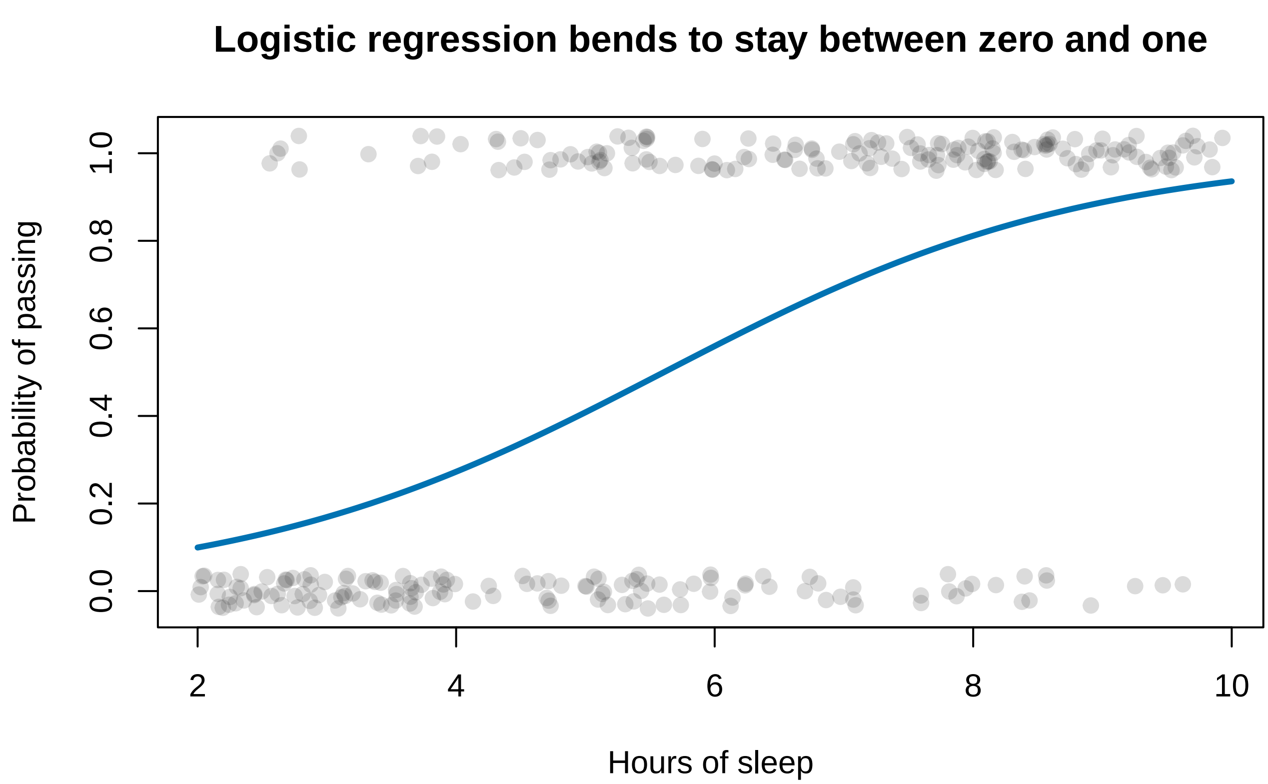
Odds are not probabilities:

$$Odds = \frac{p}{1-p}, \qquad p = \frac{Odds}{1+Odds}.$$

An Odds Ratio Is Not a Probability Ratio

An odds ratio of 2 does not generally mean the outcome became twice as probable. The difference is small for rare outcomes and can become substantial for common outcomes. Report predicted probabilities when readers live on the probability scale, which is nearly everyone.

```
SleepGrid <- data.frame(Sleep = seq(2, 10, length.out = 150))
SleepGrid$Probability <- predict(LogisticModel, newdata = SleepGrid, type = "response")

plot(
  jitter(LogisticData$Sleep), jitter(LogisticData$Pass, amount = .04),
  pch = 19, col = rgb(0.2, 0.2, 0.2, .18),
  xlab = "Hours of sleep", ylab = "Probability of passing",
  main = "Logistic regression bends to stay between zero and one"
)
lines(SleepGrid$Sleep, SleepGrid$Probability, lwd = 3, col = "#0072B2")
```



25.4 Testing Model Improvement

Generalized models use likelihood rather than ordinary least squares. For nested models,

$$LR = 2(\ell_{Full} - \ell_{Reduced}),$$

which is commonly compared with a chi-square distribution.

```
NullLogistic <- glm(Pass ~ 1, data = LogisticData, family = binomial())
anova(NullLogistic, LogisticModel, test = "Chisq")
```

```
Analysis of Deviance Table

Model 1: Pass ~ 1
Model 2: Pass ~ Sleep
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1       299     414.55
2       298     312.20  1   102.35 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is no single GLM R^2 identical to ordinary regression R^2. Pseudo-R^2 measures answer different questions and must be named rather than reported as if one number inherited the family business.

25.5 Count Outcomes

For counts, a Poisson model uses:

$$Y_i \sim Poisson(\mu_i), \qquad \log(\mu_i) = \eta_i.$$

The Poisson distribution assumes $Var(Y_i) = E(Y_i)$. Psychological counts are often **overdispersed**, meaning variance exceeds the mean. Negative-binomial models add a dispersion parameter and are often more realistic.

```
set.seed(344)
CountData <- data.frame(Stress = runif(250, 0, 10))
CountData$Complaints <- rpois(250, lambda = exp(.2 + .15 * CountData$Stress))
PoissonModel <- glm(Complaints ~ Stress, data = CountData, family = poisson())
summary(PoissonModel)
```

```
Call:
glm(formula = Complaints ~ Stress, family = poisson(), data = CountData)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.11942    0.09195   1.299    0.194
```

```
Stress       0.16045    0.01332  12.044   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 410.48  on 249  degrees of freedom
Residual deviance: 257.08  on 248  degrees of freedom
AIC: 896.71

Number of Fisher Scoring iterations: 5
```

Exponentiated Poisson coefficients are multiplicative changes in the expected count. If $e^{\beta_1} = 1.20$, a one-unit increase in X multiplies the expected count by 1.20, or increases it by about 20%, holding other predictors constant.

25.6 Diagnostics Change with the Distribution

Binary and count residuals do not behave like Gaussian residuals. Check:

- Separation in logistic regression.
- Overdispersion in count models.
- Excess zeros.
- Nonlinearity on the link scale.
- Influential cases.
- Calibration and predictive performance.

> ❗ Choose the Family from the Outcome, Not the P-Value
>
> Do not try several distributions and keep whichever one produces the friendliest asterisk. Use the outcome's support, generating process, diagnostics, and scientific meaning to choose a model.

25.7 The Bridge to Generalized Mixed Models

When binary or count observations are repeated or clustered, we add random effects:

$$g(E[Y \mid b]) = X\beta + Zb.$$

That is a generalized linear mixed model. The graduate mixed-model section uses `glmmTMB` so the same framework can handle Gaussian, binomial, Poisson, negative-binomial, and zero-inflated outcomes.

25.8 Deviance and Model Comparison

GLMs are commonly fit by maximum likelihood. **Deviance** measures discrepancy between the fitted model and a saturated model that reproduces the observed data as closely as possible:

$$D = 2\left[\ell(\text{saturated}) - \ell(\text{fitted})\right].$$

For nested models, the difference in deviance is a likelihood-ratio statistic under regularity conditions. This lets us ask whether a block of predictors improves fit without pretending every coefficient must be judged separately by a Wald z-test.

25.9 Overdispersion and Count Outcomes

A Poisson model assumes

$$E(Y \mid X) = Var(Y \mid X) = \mu.$$

Psychological counts often have variance much larger than the mean because people differ in unmodeled ways or because zeros breed in dark corners of the dataset. This is **overdispersion**. It can make Poisson standard errors too small. A negative-binomial model adds a dispersion parameter; a zero-inflated model represents an additional process generating excess zeros. Do not choose among them solely by whichever one produces the most attractive p-value.

25.10 The Short Story

GLMs combine a distribution, a linear predictor, and a link function. Logistic coefficients live in log-odds; count-model coefficients often live on a log scale. Translate results back into probabilities or expected counts before asking ordinary humans to interpret them.

26 Advanced: Mediation and Indirect Effects

26.1 The Question Changes from Whether to How

Mediation asks whether the relationship between X and Y may operate through a third variable, M.

$$X \longrightarrow M \longrightarrow Y.$$

Suppose a study-skills workshop increases practice time, which then improves exam performance. The proposed mediator is practice time.

> **A Mediator Is a Causal Claim Wearing Three Variables**
>
> Regression can estimate paths consistent with mediation. It cannot prove temporal order, eliminate unmeasured confounding, or show that changing the mediator would change the outcome. Causal mediation requires design and assumptions, not merely three significant coefficients standing in a trench coat.

26.2 The Path Model

The mediator model is:

$$M = i_M + aX + e_M.$$

The outcome model is:

$$Y = i_Y + c'X + bM + e_Y.$$

The indirect effect is:

$$ab.$$

The total effect in a simple linear system is:

$$c = c' + ab.$$

26.3 Simulated Workshop Data

```r
set.seed(343)
n <- 220
MediationData <- data.frame(
  Workshop = rbinom(n, 1, .5)
)

MediationData$Practice <-
  3 + 2.2 * MediationData$Workshop + rnorm(n, 0, 1.6)

MediationData$Exam <-
  58 + 2.5 * MediationData$Workshop +
  3.2 * MediationData$Practice + rnorm(n, 0, 7)
```

```r
MediatorModel <- lm(Practice ~ Workshop, data = MediationData)
OutcomeModel <- lm(Exam ~ Workshop + Practice, data = MediationData)
TotalModel <- lm(Exam ~ Workshop, data = MediationData)

a <- coef(MediatorModel)["Workshop"]
b <- coef(OutcomeModel)["Practice"]
c_prime <- coef(OutcomeModel)["Workshop"]
c_total <- coef(TotalModel)["Workshop"]

c(a = a, b = b, indirect = a * b, direct = c_prime, total = c_total)
```

```
       a.Workshop        b.Practice indirect.Workshop   direct.Workshop
         2.302613          2.633241          6.063335          4.128035
   total.Workshop
        10.191370
```

26.4 Do Not Require Every Historical Step

The traditional Baron-and-Kenny procedure required a sequence of significant paths, including a significant total effect. Modern mediation analysis focuses on the indirect effect itself (Preacher and Hayes 2008). A total effect can be small when indirect pathways operate in opposite directions or when the direct and indirect effects cancel.

> **The Indirect Effect Is the Target**
>
> Do not infer mediation merely because a and b are individually significant. Test and report ab with an interval that reflects its skewed sampling distribution.

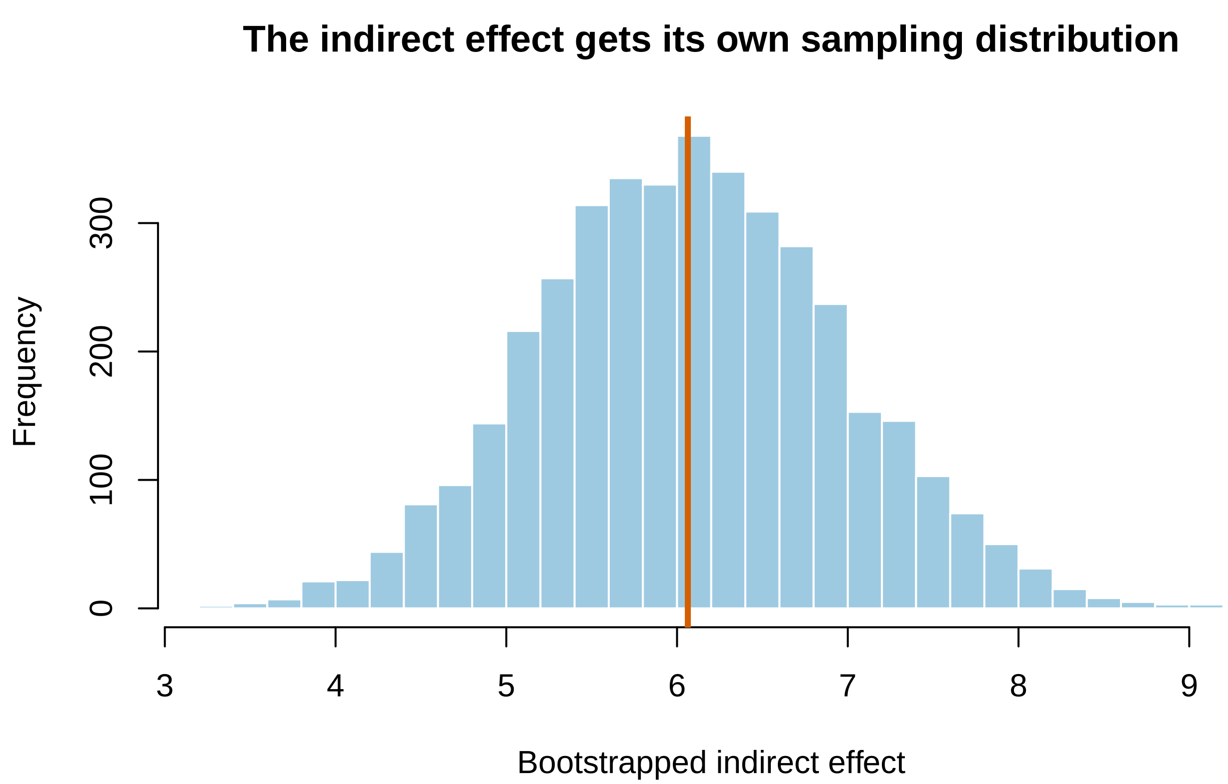
26.5 Bootstrap the Indirect Effect

The product ab is often asymmetric, especially in smaller samples. A bootstrap interval avoids forcing the product into a symmetric normal approximation.

```r
set.seed(344)
B <- 4000

IndirectBoot <- replicate(B, {
  rows <- sample(seq_len(nrow(MediationData)), replace = TRUE)
  d <- MediationData[rows, ]
  a_star <- coef(lm(Practice ~ Workshop, data = d))["Workshop"]
  b_star <- coef(lm(Exam ~ Workshop + Practice, data = d))["Practice"]
  a_star * b_star
})

quantile(IndirectBoot, c(.025, .975))
```

```
    2.5%    97.5%
4.400216 7.843485
```

```r
hist(
  IndirectBoot, breaks = 35,
  col = "#9ECAE1", border = "white",
  xlab = "Bootstrapped indirect effect",
  main = "The indirect effect gets its own sampling distribution"
)
abline(v = 0, lty = 2, lwd = 2)
abline(v = a * b, col = "#D55E00", lwd = 3)
```



26.6 What Must Be True for a Causal Interpretation?

At minimum, we need defensible assumptions about:

- Temporal order: X precedes M, which precedes Y.
- No unmeasured confounding of $X \to M$.
- No unmeasured confounding of $M \to Y$.
- No mediator–outcome confound created by X.
- Correct functional forms and measurement.
- No major post-treatment selection problems.

Randomizing X helps with paths beginning at X. It does not automatically randomize the mediator. Participants who practice more may differ in motivation, sleep, prior knowledge, terror of disappointing their mother, or several variables we did not measure.

26.7 Partial and Complete Mediation

Avoid treating "complete mediation" as a trophy awarded when c' becomes nonsignificant. Significance depends on sample size and uncertainty. Report the direct effect, indirect effect, total effect, and their intervals. Let the estimates describe the pattern.

26.8 Moderated Mediation

An indirect effect may depend on another variable W. For example, the workshop may increase practice more strongly among students with enough available time:

$$M = i_M + a_1 X + a_2 W + a_3 XW + e_M.$$

The conditional indirect effect becomes:

$$(a_1 + a_3 W)b.$$

This is not a separate magical procedure. It combines mediation with an interaction, then evaluates the indirect effect at meaningful values of the moderator.

> **! More Arrows Require More Theory**
>
> Every added mediator, moderator, and covariate creates additional assumptions and alternative explanations. A path diagram is not evidence merely because the arrows are straight and the font is attractive.

26.9 Identification: What the Three Regressions Cannot See

For the indirect effect to support a causal mediation story, we need strong assumptions about confounding of the $X \to M$, $X \to Y$, and $M \to Y$ relations. We also need no mediator-outcome confound that was itself caused by X. The last condition is especially rude because controlling for the post-treatment confound can create additional bias.

Randomizing X protects comparisons beginning at X. It does not randomize the mediator. If motivated students both practice more and perform better, the estimated b path can mix the mediator with motivation. A beautiful indirect-effect interval does not reveal the missing variable.

26.10 Competing Mediators and Temporal Order

With several mediators, the total indirect effect can be divided into specific indirect effects only under additional assumptions. Parallel mediators may influence one another; serial mediators require a defensible order. Measuring every variable in one survey and drawing arrows from left to right does not create time. Longitudinal measurement, experimental manipulation of mechanisms, and sensitivity analysis make the argument stronger.

26.11 The Short Story

Mediation estimates an indirect effect ab. Bootstrap its uncertainty, report direct and total effects, and separate "the data are consistent with this pathway" from "we proved this causal mechanism." The second claim requires much more than regression output.

27 P-Values and Multiple Comparisons

Advanced topic

This chapter assumes that you already understand null hypotheses, Type I and Type II errors, power, effect sizes, and confidence intervals. Undergraduates may continue, but should bring snacks and a responsible graduate student. Graduate students should bring two snacks because they are expected to pretend this is easy.

27.1 First, Let Us Put a Dead Fish in an MRI

Bennett and colleagues placed a mature Atlantic salmon in an fMRI scanner and showed it photographs of people in social situations. The salmon was asked to decide what emotion each person was experiencing. There was a slight methodological complication: **the salmon was dead** (Bennett et al. 2010).

The researchers ran tens of thousands of voxel-by-voxel tests. With an uncorrected threshold of $p < .001$, they found apparently active voxels in the salmon's brain cavity and spinal column. After proper familywise-error and false-discovery-rate corrections, the salmon's social-cognitive abilities vanished. Even relaxed corrected thresholds found nothing, which is generally what we hope to discover about the thoughts of a dead fish.

If Atlantic salmon become difficult to recruit, Alex can substitute for the fish because he is approximately as brain-dead before coffee. The major design problem is that Alex may twitch, complain, or ask whether the scan counts toward tenure.

The study was deliberately absurd, but the statistical problem was real. If you conduct enough tests, fickle chance receives enough opportunities to manufacture something exciting. A tiny p-value attached to one selected result does not tell us how many other tests were run before that result crawled out of the computer wearing a crown.

Before we control many p-values, however, we should establish what one p-value means. This seems obvious. It is not. Researchers have spent nearly a century proving that they can misunderstand a definition with remarkable consistency.

27.2 What a P-Value Actually Is

A p-value answers this conditional question:

> **Assuming the null hypothesis and the statistical model are correct, how probable are results at least as incompatible with the null hypothesis as the result we observed?**

In shorthand:

$$p = P(\text{data this extreme or more extreme} \mid H_0, \text{model})$$

Small p-values say that the observed result does not sit comfortably inside the null model. They do not identify which assumption is wrong. Perhaps the null hypothesis is a poor description. Perhaps independence failed, the model was misspecified, the measurement was terrible, or one participant completed the experiment while being chased by a goose. Statistics can tell you that the whole package looks strained. It cannot interrogate the goose.

Statistical significance is a decision rule. If the p-value is at or below a threshold chosen in advance, usually $\alpha = .05$, we call the result statistically significant. The threshold is a convention, not a cliff in nature. A result with $p = .049$ and one with $p = .051$ are not members of different species.

27.2.1 The Conditional-Probability Trap

The most famous mistake is reversing the condition:

$$P(D \mid H_0) \neq P(H_0 \mid D)$$

The probability of screaming **given that** a foam-chainsaw-wielding research assistant is chasing you is probably high. The probability that a foam-chainsaw-wielding research assistant is chasing you **given that** you are screaming is not necessarily high. You may have seen your tuition bill.

Cohen emphasized that a p-value gives $P(D \mid H_0)$, not the probability that the null is true after seeing the data (Cohen 1994). To calculate $P(H_0 \mid D)$, we would need additional information, including prior probabilities and an explicit alternative model. Quietly turning one conditional probability around does not make it Bayesian. It makes it wrong.

27.2.2 A P-Value Is Not Any of These Things

A p-value is **not**:

- The probability that the null hypothesis is true.
- The probability that the research hypothesis is false.
- The probability that the result happened "because of chance."
- The Type I error rate for this particular result.
- The probability that the study will replicate.
- The size or practical importance of the effect.
- A tiny certificate declaring the study free of confounds, bias, fraud, or nonsense.

Lane-Getaz documented thirteen recurring p-value misconceptions in statistics students and experienced researchers (Lane-Getaz 2005). Even graduate students who had survived multiple statistics courses often confused p-values with the probability of a hypothesis, replicability, effect size, or Type I error. Taking another statistics course helps, but apparently does not exorcise the demon automatically.

27.3 Cohen's Complaint: The Ritual Ate the Science

Cohen's title, *The Earth Is Round ($p < .05$)*, mocked the habit of testing a nearly certain or scientifically uninteresting null and then acting impressed when it was rejected (Cohen 1994). In much of psychology, an exact **nil hypothesis**—a population difference or correlation of precisely zero—is implausible. With a large enough sample, a trivial departure from zero can become statistically significant.

That produces two failures:

1. Researchers mistake "detectable" for "important."

2. Researchers treat rejection of one null model as confirmation of their favorite theory, even though confounds, measurement problems, alternative theories, and several Probability Gods remain available.

Cohen did not offer a new sacred button. He recommended exploratory data analysis, graphs, effect sizes, confidence intervals, better measurement, power planning, and replication. In other words: look at what happened, estimate how much happened, admit the uncertainty, and see whether it happens again. Annoyingly, the solution is scientific judgment rather than purchasing a newer ritual.

27.4 Significant Here, Not Significant There

Gelman and Stern identified another extremely common error: concluding that two effects differ because one is statistically significant and the other is not (Gelman and Stern 2006).

Suppose two independent studies estimate effects of:

$$25 \pm 10 \qquad \text{and} \qquad 10 \pm 10$$

The first estimate is statistically significant and the second is not. But their estimated **difference** is 15, with a standard error of:

$$SE_{difference} = \sqrt{10^2 + 10^2} = 14.14$$

The two effects are only about 1.06 standard errors apart. That difference is not statistically significant. "Treatment A worked but Treatment B did not" is not evidence that Treatment A worked **better** than Treatment B. Test the comparison you want to claim.

This matters for interactions, subgroups, replications, and breathless headlines. If an effect is significant for women but not men, younger adults but not older adults, or Tuesday but not Wednesday, you have not established a group difference until you test that difference directly.

27.5 One Test at a Time Creates a Family Problem

Set $\alpha = .05$ for one true null hypothesis. In repeated use, that test falsely rejects the null about 5% of the time. Now run m independent tests, all with true null hypotheses. The probability of **at least one** false positive is:

$$FWER = 1 - (1 - \alpha)^m$$

With 20 independent tests:

$$1 - .95^{20} = .642$$

There is about a 64% chance of at least one false positive. You did not change α for any individual test. You simply gave fickle chance twenty auditions and published whichever one could sing.

```r
set.seed(343)

alpha <- .05
m_grid <- 1:100
theoretical_fwer <- 1 - (1 - alpha)^m_grid

simulated_at <- c(1, 5, 10, 20, 50, 100)
simulated_fwer <- vapply(simulated_at, function(m) {
  p_values <- matrix(runif(20000 * m), ncol = m)
  mean(rowSums(p_values <= alpha) > 0)
}, numeric(1))

plot(
  m_grid, theoretical_fwer,
  type = "l", lwd = 3, col = "#1F4E79",
  ylim = c(0, 1),
  xlab = "Number of tests in the family",
  ylab = "Probability of one or more false positives"
)
points(simulated_at, simulated_fwer, pch = 19, col = "#D55E00")
abline(h = .05, lty = 2, col = "gray50")
legend(
  "bottomright",
  inset = c(.01, .10),
  legend = c("Theoretical FWER", "Simulation"),
  col = c("#1F4E79", "#D55E00"),
  lty = c(1, NA), pch = c(NA, 19), bty = "n"
)
text(78, .075, "5% reference", col = "gray40", cex = .82)
```

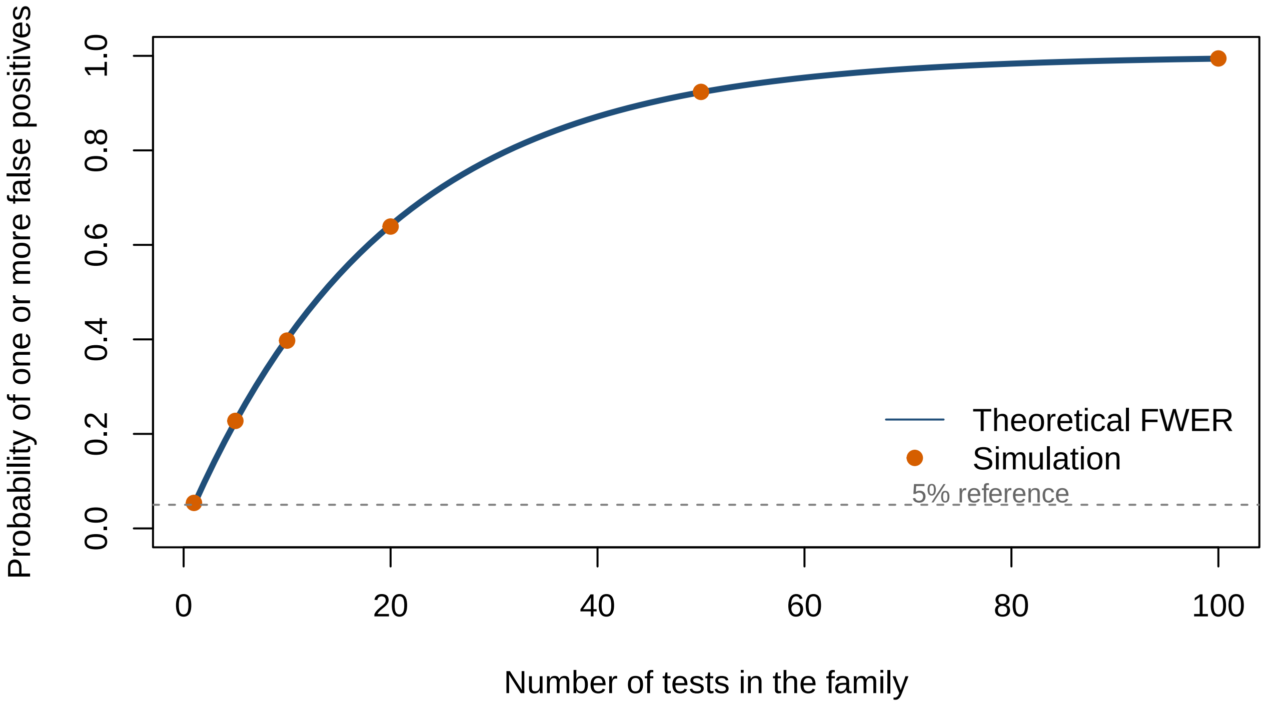


Figure 27.1: The probability of at least one false positive grows rapidly as more independent true null hypotheses are tested at alpha = .05. Simulated points sit on the theoretical curve.

The independence formula is not exact for correlated tests, such as neighboring fMRI voxels. Dependence changes the arithmetic but does not make the problem disappear. Specialized methods can exploit that dependence; pretending the other tests never happened cannot.

27.5.1 What Counts as a Family?

A **family** is the set of tests over which you want a joint error guarantee. Defining it requires scientific judgment.

- All pairwise comparisons following one omnibus test often form a family.
- All outcomes used to support one broad claim may form a family.
- Thousands of voxels in one brain contrast form a family.
- Every analysis ever conducted by everyone named Alex does not form one useful family, although it may form a cry for help.

The family should be defined by the claim and analysis plan, preferably before looking at results. Dividing twenty tests into twenty "families of one" after seeing the p-values is not a clever solution. It is p-hacking with office supplies.

27.6 Familywise Error Rate: Protect the Whole Family

The **familywise error rate** (FWER) is the probability of making **one or more Type I errors anywhere in the family**:

$$FWER = P(V \geq 1)$$

Here, V is the number of false rejections. FWER control is strict. It is appropriate when even one false positive would be costly or when a small, confirmatory set of claims is being tested.

27.6.1 Bonferroni

For m tests, Bonferroni compares each p-value with α/m, or multiplies each p-value by m and caps the result at 1:

$$p_{Bonferroni} = \min(mp, 1)$$

Bonferroni works under arbitrary dependence, which is useful, but it can sacrifice substantial power when there are many tests.

27.6.2 Holm's Sequential Method

Holm's procedure orders the p-values and tests them sequentially (Holm 1979). It controls FWER under the same broad conditions as Bonferroni and is never less powerful. If your entire justification for plain Bonferroni is "that is the one I remember," R remembers Holm for you.

```r
raw_p <- c(.001, .012, .021, .046, .210, .730)

data.frame(
  test = paste("Hypothesis", 1:length(raw_p)),
  raw = raw_p,
  bonferroni = p.adjust(raw_p, method = "bonferroni"),
  holm = p.adjust(raw_p, method = "holm")
)
```

```
          test   raw bonferroni  holm
1 Hypothesis 1 0.001      0.006 0.006
2 Hypothesis 2 0.012      0.072 0.060
3 Hypothesis 3 0.021      0.126 0.084
4 Hypothesis 4 0.046      0.276 0.138
5 Hypothesis 5 0.210      1.000 0.420
6 Hypothesis 6 0.730      1.000 0.730
```

27.7 False Discovery Rate: Permit Some Risk Among Discoveries

In large exploratory analyses, strict FWER control may erase too many real signals. The **false discovery rate** (FDR) instead controls the expected proportion of false positives among the rejected hypotheses:

$$FDR = E\left(\frac{V}{\max(R, 1)}\right)$$

V is the number of false rejections and R is the total number of rejections. The `max(R,1)` convention sets the proportion to zero when nothing is rejected.

Benjamini and Hochberg's procedure controls FDR under independence and certain forms of positive dependence (Benjamini and Hochberg 1995). It is commonly available in R as `method = "BH"`.

FDR of .05 does **not** guarantee that exactly 5% of one study's discoveries are false. Some studies will have none, some will have more, and some will reject nothing. The control applies to the expected false-discovery proportion over repeated uses of the procedure.

```r
data.frame(
  test = paste("Hypothesis", 1:length(raw_p)),
  raw = raw_p,
  holm_fwer = p.adjust(raw_p, method = "holm"),
  bh_fdr = p.adjust(raw_p, method = "BH")
)
```

```
          test   raw holm_fwer bh_fdr
1 Hypothesis 1 0.001     0.006  0.006
2 Hypothesis 2 0.012     0.060  0.036
3 Hypothesis 3 0.021     0.084  0.042
4 Hypothesis 4 0.046     0.138  0.069
5 Hypothesis 5 0.210     0.420  0.252
6 Hypothesis 6 0.730     0.730  0.730
```

27.8 What the Corrections Trade Away

No method creates information. Tight false-positive control usually reduces the ability to detect real effects. The useful question is not "Which correction is best?" It is "Which error promise matches this research goal?"

The simulation below contains 20 tests per study. Five hypotheses have real standardized signals and fifteen null hypotheses are true. We compare no adjustment, Bonferroni, Holm, and Benjamini–Hochberg over 5,000 simulated studies.

```r
set.seed(343)

n_families <- 5000
n_tests <- 20
n_real <- 5

z_values <- matrix(rnorm(n_families * n_tests), nrow = n_families)
z_values[, 1:n_real] <- z_values[, 1:n_real] + 2.5
p_matrix <- 2 * pnorm(-abs(z_values))

adjust_rows <- function(method) {
  t(apply(p_matrix, 1, function(p) p.adjust(p, method = method) <= .05))
}

rejections <- list(
  Unadjusted = p_matrix <= .05,
  Bonferroni = adjust_rows("bonferroni"),
```

```r
  Holm = adjust_rows("holm"),
  BH_FDR = adjust_rows("BH")
)

summarize_method <- function(reject) {
  true_discoveries <- rowSums(reject[, 1:n_real, drop = FALSE])
  false_discoveries <- rowSums(reject[, (n_real + 1):n_tests, drop = FALSE])
  total_discoveries <- true_discoveries + false_discoveries
  false_discovery_proportion <- ifelse(
    total_discoveries == 0,
    0,
    false_discoveries / total_discoveries
  )

  c(
    average_true_discoveries = mean(true_discoveries),
    probability_any_false_positive = mean(false_discoveries > 0),
    average_false_discovery_proportion = mean(false_discovery_proportion)
  )
}

round(t(vapply(rejections, summarize_method, numeric(3))), 3)
```

```
           average_true_discoveries probability_any_false_positive
Unadjusted                    3.501                          0.526
Bonferroni                    1.503                          0.037
Holm                          1.539                          0.039
BH_FDR                        2.076                          0.122
           average_false_discovery_proportion
Unadjusted                              0.152
Bonferroni                              0.018
Holm                                    0.019
BH_FDR                                  0.039
```

The unadjusted method finds more real signals but also produces at least one false positive far too often. Bonferroni and Holm protect strongly against any false positive, with Holm usually retaining a little more power. BH accepts a higher chance that a study contains some false positives in exchange for finding more real signals, while controlling their expected proportion among the discoveries.

Goal	Sensible default
A few planned, confirmatory tests where any false claim is costly	Holm FWER control
A simple method that remains valid under broad dependence	Bonferroni, accepting lower power
Many exploratory tests where some false leads can be tolerated and checked later	Benjamini–Hochberg FDR control
A small number of clearly preregistered primary tests	Test those primary hypotheses as planned; correct the separate exploratory family

27.9 Multiple Comparisons in Regression

A regression with many coefficients is also a collection of tests. The correct family depends on the claim.

If one coefficient was the preregistered primary hypothesis and the others were prespecified covariates, automatically correcting every coefficient together may not represent the design. If you added 84 predictors, 31 interactions, six subgroups, and the participant's astrological sign until something became significant, you have a multiple-testing problem plus a confession.

Store the p-values you intend to treat as one family and adjust them directly:

```
# Example after fitting a model called model:
# coefficient_p <- coef(summary(model))[, "Pr(>|t|)"]
# p.adjust(coefficient_p, method = "holm")
# p.adjust(coefficient_p, method = "BH")
```

For planned contrasts, use software that can adjust the contrast family explicitly. Always report the method, the family of tests, and whether the analyses were confirmatory or exploratory.

27.10 The Point Is Not to Fear P-Values

P-values are not evil. They are also not tiny judges who determine whether a theory is innocent or guilty. They describe how incompatible the data are with a specified null model, under a pile of assumptions that must remain attached.

A defensible statistical argument uses:

- The design and quality of the measurement.
- The estimated effect and its confidence interval.
- The p-value as one piece of evidence rather than a personality test for the hypothesis.
- A correction that matches the family and scientific goal.
- Transparency about how many analyses were attempted.
- Replication, because one clever correction cannot make a single study carry an entire theory on its back.

The dead salmon did not teach us that fMRI is fake. It taught us that if we give chance enough opportunities, eventually even dinner appears to think. Correct the tests, show the effects, admit the uncertainty, and please stop asking deceased seafood to perform social cognition without preregistration.

28 Advanced Measurement: Exploratory Factor Analysis

28.1 Why Factor Analysis Appears in a Regression Book

Psychologists frequently average several questionnaire items and call the result "anxiety," "belonging," or "willingness to volunteer for another suspicious experiment." Exploratory factor analysis asks whether the items behave as if fewer latent dimensions could explain their correlations (Fabrigar et al. 1999).

This is measurement work. A perfectly fitted regression cannot rescue a scale whose items measure three unrelated constructs and one participant's desire to finish quickly.

> ⚠ Factors Are Models, Not Hidden Organs
>
> A factor is a statistical representation of shared variation. It is not a psychological substance discovered inside the covariance matrix. Theory and validation determine whether the factor deserves a construct name.

28.2 The Common-Factor Model

For observed item vector $\mathbf{x}$,

$$\mathbf{x} = \Lambda \mathbf{f} + \epsilon,$$

where:

- Λ contains factor loadings.
- $\mathbf{f}$ contains latent factor scores.
- ϵ contains item-specific variation and measurement error.

For item j,

$$x_j = \lambda_{j1} f_1 + \cdots + \lambda_{jm} f_m + \epsilon_j.$$

A loading describes how strongly an item relates to a factor, conditional on the model and rotation.

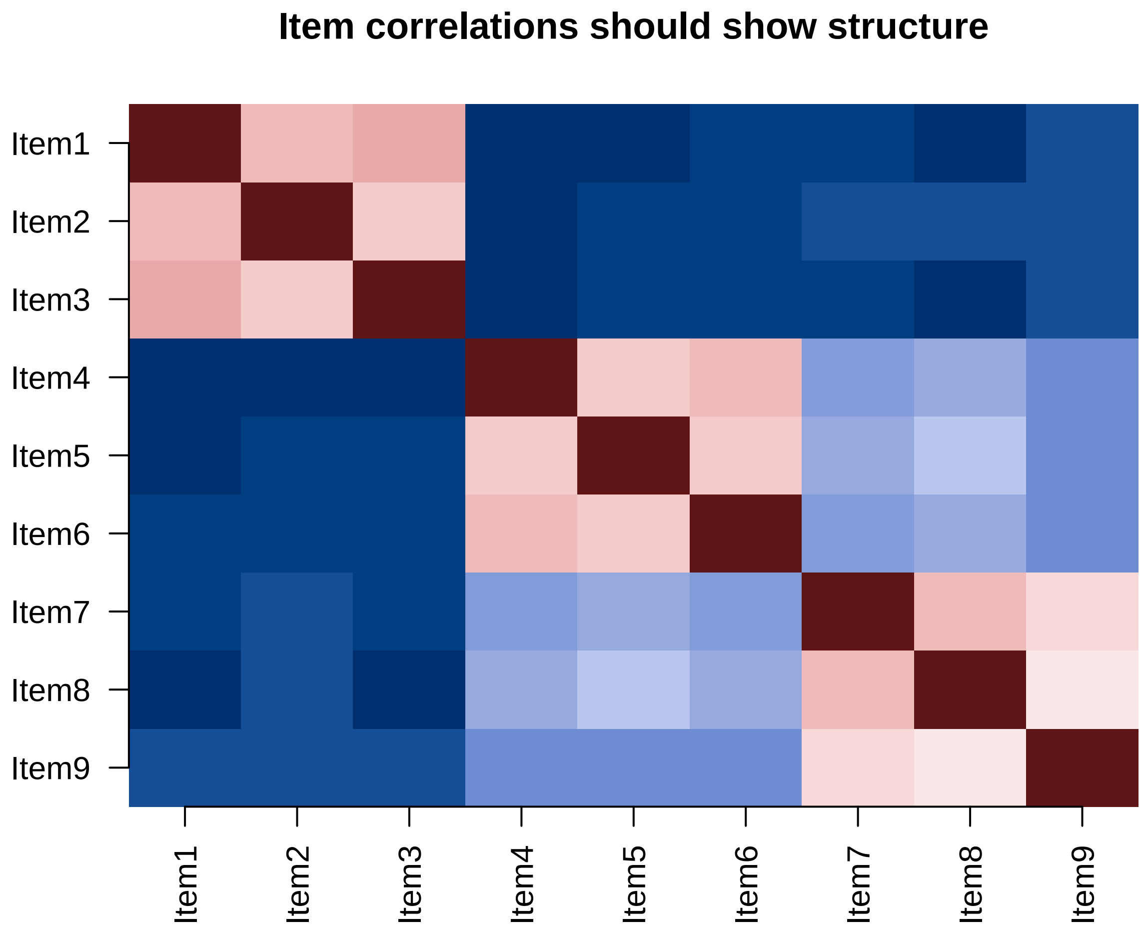
28.3 Simulating Nine Questionnaire Items

We will simulate three related dimensions: statistics anxiety, R confidence, and research enthusiasm. The final construct may be "graduate student emotional weather."

```r
set.seed(343)
n <- 450

FactorCor <- matrix(
  c(1, -.35, -.20,
    -.35, 1, .30,
    -.20, .30, 1),
  nrow = 3
)

Latent <- MASS::mvrnorm(n, mu = c(0, 0, 0), Sigma = FactorCor)

Loadings <- matrix(0, nrow = 9, ncol = 3)
Loadings[1:3, 1] <- c(.82, .76, .70)
Loadings[4:6, 2] <- c(.80, .74, .68)
Loadings[7:9, 3] <- c(.78, .72, .66)

Items <- Latent %*% t(Loadings) + matrix(rnorm(n * 9, 0, .65), nrow = n)
Items <- as.data.frame(Items)
names(Items) <- paste0("Item", 1:9)
```

28.4 First Inspect the Correlations

```r
ItemCor <- cor(Items)
image(
  1:9, 1:9, t(ItemCor[9:1, ]),
  axes = FALSE, col = hcl.colors(30, "Blue-Red 3"),
  xlab = "", ylab = "", main = "Item correlations should show structure"
)
axis(1, at = 1:9, labels = names(Items), las = 2)
axis(2, at = 1:9, labels = rev(names(Items)), las = 2)
```



Factor analysis needs correlations worth explaining. Items with almost no shared variation will not become a coherent scale because we rotated them enthusiastically.

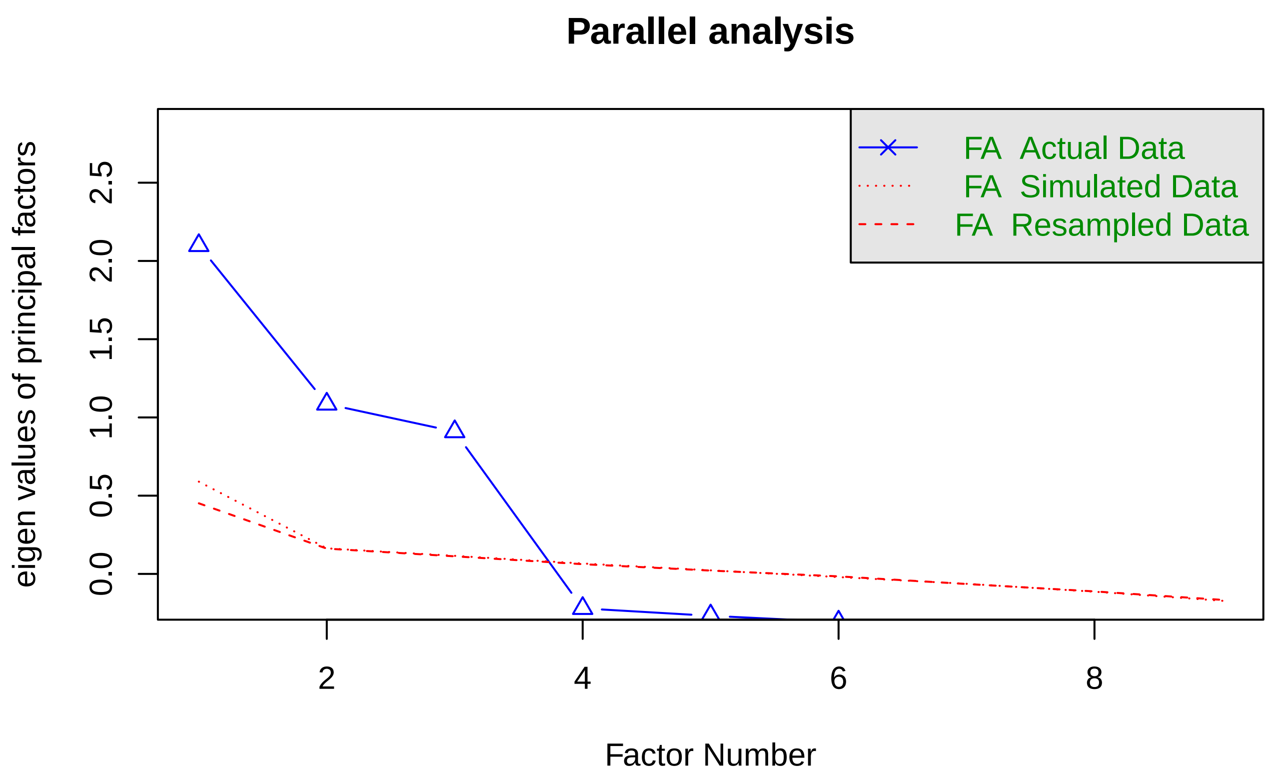
28.5 How Many Factors?

Do not use eigenvalues greater than one as the sole decision rule. Combine:

- Theory.
- Scree plots.
- Parallel analysis.
- Interpretability.
- Replication in new data.

Parallel analysis compares observed eigenvalues with eigenvalues from random data of the same dimensions. Retain factors whose observed eigenvalues exceed the random-data benchmarks.

```
psych::fa.parallel(Items, fa = "fa", fm = "minres", n.iter = 50,
                   main = "Parallel analysis")
```




```
Parallel analysis suggests that the number of factors =  3  and the number of components =  NA
```

The Computer Suggests; the Theory Must Still Speak

Parallel analysis is evidence about dimensionality, not a command. A factor solution should also make theoretical sense, contain adequately measured factors, and survive replication.

28.6 Extraction and Rotation

```
EFA.Model <- psych::fa(
  Items,
  nfactors = 3,
  fm = "minres",
  rotate = "oblimin"
)
print(EFA.Model$loadings, cutoff = .30)
```

```
Loadings:
      MR3    MR2    MR1
Item1  0.805
Item2  0.729
```

```
Item3  0.762
Item4         0.759
Item5         0.737
Item6         0.777
Item7                0.797
Item8                0.714
Item9                0.674

                 MR3   MR2   MR1
SS loadings    1.765 1.729 1.604
Proportion Var 0.196 0.192 0.178
Cumulative Var 0.196 0.388 0.567
```

Extraction estimates the factor model. Minimum residual (`minres`) and maximum likelihood are common choices.

Rotation finds a simpler orientation of the loading pattern:

- Orthogonal rotations force factors to be uncorrelated.
- Oblique rotations allow factors to correlate.

Psychological constructs commonly correlate, so oblique rotation is usually the more defensible starting point. Forcing anxiety and confidence to be independent because the graph looks tidier is an aesthetic decision masquerading as theory.

28.7 Pattern and Structure Matrices

With oblique rotation:

- The **pattern matrix** contains regression-like weights linking items to factors.
- The **structure matrix** contains item–factor correlations.

They differ when factors correlate. State which matrix you interpret and avoid selecting whichever one gives prettier loadings.

28.8 Communalities and Uniqueness

An item's communality is the proportion of its variance explained by the common factors:

$$h_j^2 = \sum_{k=1}^{m} \lambda_{jk}^2$$

for an orthogonal solution; correlated factors require the factor-correlation structure in the calculation.

Uniqueness is:

$$u_j^2 = 1 - h_j^2.$$

High uniqueness means the common factors explain little of that item. The item may be noisy, unusually specific, or measuring a construct that the other items neglected.

28.9 Factor Scores Are Estimates

Factor scores can be useful predictors or outcomes, but they are estimated with uncertainty and depend on the fitted solution. A simple average may sometimes be more transparent when items form a clear unidimensional scale. Complexity is not automatically psychometric sophistication.

> **!** EFA Explores; CFA Tests a Specified Structure
>
> EFA searches for a plausible loading structure. Confirmatory factor analysis tests a structure specified in advance. Ideally, explore in one sample and confirm in another. Confirming the structure in the same data used to discover it is statistical déjà vu, not independent evidence.

28.10 EFA Is Not PCA Wearing a Fake Mustache

Principal components analysis decomposes **total observed variance** into weighted components. Common-factor analysis models the covariance shared among items while separating unique and error variance:

$$\boxtimes = \boxtimes\boxtimes\boxtimes' + \boxtimes.$$

- $\boxtimes$ contains factor loadings.
- $\boxtimes$ contains factor correlations.
- $\boxtimes$ contains unique variances.

PCA can be useful for data reduction. EFA is generally better aligned with a latent-construct question. Similar-looking loading tables do not make the goals interchangeable.

28.11 Rotation Changes the View, Not the Reproduced Correlations

Rotation seeks a simpler interpretation of the same factor space. **Orthogonal** rotations force factor correlations to zero. **Oblique** rotations permit correlations and produce both a pattern matrix and a structure matrix. Psychological constructs are often correlated, so forcing orthogonality needs a theoretical reason stronger than "the table looked cleaner."

Factor retention should combine parallel analysis, scree plots, interpretability, theory, communalities, and replication. Kaiser's eigenvalue-greater-than-one rule is not a tiny oracle. A factor solution is a proposal about measurement that needs confirmation in new data.

28.12 The Short Story

EFA models item correlations using fewer latent factors. Decide factor count with theory plus empirical evidence, usually allow psychological factors to correlate, inspect loadings and communalities, and validate the solution in new data before naming a factor after a human emotion.

29 Reporting Models Without Making the Reader Solve a Puzzle

29.1 The Analysis Is Not Finished When R Stops Printing

A results section should let a reader reconstruct the statistical argument without opening your R session, reading your mind, or contacting the undergraduate who disappeared with the codebook.

Report:

- What was analyzed.
- How variables were coded.
- Which model was fitted and why.
- Estimates in interpretable units.
- Uncertainty.
- Effect size or model fit.
- Diagnostics and consequential deviations.
- What the result supports and what it does not.

! Lead with the Scientific Result

"A regression was significant" is about software output. "Each additional hour of sleep predicted about four more memory points" is about the research question. Start with the second sentence.

29.2 One Reproducible Example

```
set.seed(343)
n <- 180
ReportData <- data.frame(
  Sleep = runif(n, 3, 9),
  Caffeine = pmax(rnorm(n, 180, 85), 0),
  Condition = factor(sample(c("Control", "Training"), n, replace = TRUE))
)

ReportData$Memory <-
  38 + 3.8 * ReportData$Sleep -
  .018 * ReportData$Caffeine +
  5.5 * (ReportData$Condition == "Training") +
  rnorm(n, 0, 7)

ReportModel <- lm(Memory ~ Sleep + Caffeine + Condition, data = ReportData)
```

29.3 Descriptives Before Models

Describe the sample and variables needed to understand the model. Do not create a 47-column Table 1 containing every variable collected since the lab opened.

```r
Descriptives <- data.frame(
  Variable = c("Memory", "Sleep", "Caffeine"),
  Mean = c(mean(ReportData$Memory), mean(ReportData$Sleep), mean(ReportData$Caffeine)),
  SD = c(sd(ReportData$Memory), sd(ReportData$Sleep), sd(ReportData$Caffeine)),
  Minimum = c(min(ReportData$Memory), min(ReportData$Sleep), min(ReportData$Caffeine)),
  Maximum = c(max(ReportData$Memory), max(ReportData$Sleep), max(ReportData$Caffeine))
)

knitr::kable(Descriptives, digits = 2, caption = "Descriptive statistics")
```

Table 29.1: Descriptive statistics

Variable	Mean	SD	Minimum	Maximum
Memory	59.41	9.98	35.72	87.23
Sleep	5.81	1.71	3.03	8.89
Caffeine	187.00	77.65	0.00	385.28

29.4 Extract Results from the Model Object

Never type changing results manually into a table when R can extract them. Manual transcription creates opportunities to report the coefficient from Model 4, the p-value from Model 6, and the confidence interval from a model no one remembers fitting.

```r
CoefficientTable <- broom::tidy(ReportModel, conf.int = TRUE)
knitr::kable(
  CoefficientTable,
  digits = 3,
  caption = "Regression predicting memory performance"
)
```

Table 29.2: Regression predicting memory performance

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	38.130	2.487	15.330	0.000	33.221	43.038
Sleep	3.600	0.319	11.283	0.000	2.970	4.229
Caffeine	-0.013	0.007	-1.845	0.067	-0.027	0.001
ConditionTraining	5.064	1.092	4.637	0.000	2.909	7.219

For mixed models, use `broom.mixed::tidy()` because fixed effects, random-effect parameters, and model components require more supervision.

29.5 What Belongs in a Regression Table?

At minimum:

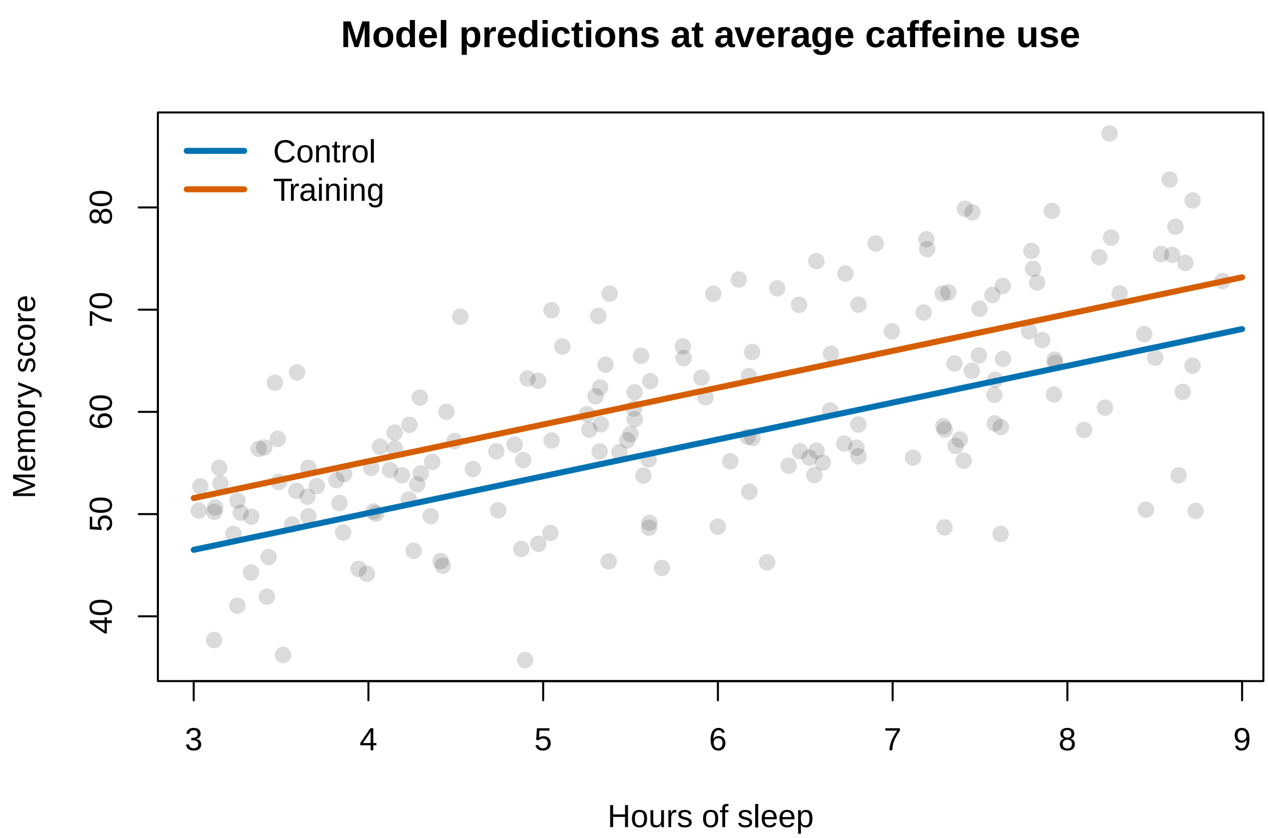
- Predictor names readers understand.
- Unstandardized estimates.
- Standard errors or confidence intervals.
- Test statistics and p-values when required.
- Sample size.
- Model R^2 and adjusted R^2 for ordinary regression.

Standardized coefficients may supplement, not automatically replace, coefficients in meaningful original units.

> ⚠ Asterisks Are Not a Results Section
>
> Stars can help scan a table. They do not describe magnitude, uncertainty, practical importance, or whether the model made sense. Never make the reader infer the entire conclusion from celestial punctuation.

29.6 Plot Model-Based Predictions

```
PredictionData <- expand.grid(
  Sleep = seq(3, 9, length.out = 100),
  Caffeine = mean(ReportData$Caffeine),
  Condition = levels(ReportData$Condition)
)

PredictionData$Predicted <- predict(ReportModel, newdata = PredictionData)

plot(
  ReportData$Sleep, ReportData$Memory,
  pch = 19, col = rgb(.2, .2, .2, .18),
  xlab = "Hours of sleep", ylab = "Memory score",
  main = "Model predictions at average caffeine use"
)

colors <- c("#0072B2", "#D55E00")
for (i in seq_along(levels(ReportData$Condition))) {
  group <- levels(ReportData$Condition)[i]
  d <- PredictionData[PredictionData$Condition == group, ]
  lines(d$Sleep, d$Predicted, lwd = 3, col = colors[i])
}
legend("topleft", legend = levels(ReportData$Condition),
       col = colors, lwd = 3, bty = "n")
```



Figures should show the comparison the model actually tested. Label axes in meaningful units, show uncertainty when it matters, and do not use a bar chart to hide a continuous distribution.

29.7 Example Write-Up

> A multiple regression examined whether sleep, caffeine consumption, and training condition predicted memory scores. Holding caffeine and condition constant, each additional hour of sleep predicted an estimated increase in memory performance. Training participants also had higher predicted scores than control participants at the same sleep and caffeine values. Caffeine showed a small negative adjusted association with memory. We report unstandardized coefficients and 95% confidence intervals in Table X. Diagnostic plots did not reveal a consequential violation of linearity or constant residual variance.

Replace that language with the actual numerical estimates from the model. The paragraph should interpret the most important coefficients rather than reciting every cell in the table aloud.

29.8 Reproducibility Checklist

- Use a script or Quarto document.
- Set random seeds for simulations.
- Record package versions when reproducibility is important.
- Keep raw data unchanged.

- Document exclusions and transformations.
- Generate tables and figures from fitted objects.
- Store enough code to reproduce every reported number.

> **Future You Is an Unreliable Collaborator**
>
> Future You will not remember why `Model.Final.2.REAL.useThisOne` excludes participants 14 and 88. Write the reason now. Future You is busy, confused, and probably blaming Past You with considerable justification.

29.9 The Short Story

Report the scientific result, estimate, uncertainty, effect size, model fit, and limitations. Generate output from model objects, use figures to clarify relationships, and leave a reproducible trail for readers and Future You.

Part III

Mixed Models: An Ultra Quick Tour of the Basics

30 Advanced Mixed Models I: Clustering, Levels, and the Equation

30.1 This Is the Quick Tour

This chapter begins a compact graduate-level overview of mixed-effects models. It will not replace a full mixed-models course. It will give you the conceptual structure, essential mathematics, a consistent `glmmTMB` workflow, and enough judgment to notice when your model has climbed onto the roof.

> ⚠ Advanced Means You Need the Earlier Regression Chapters
>
> Before starting, you should understand regression coefficients, interactions, residuals, likelihood-based model comparison, and categorical coding. Mixed models extend regression; they do not replace the need to understand it.

30.2 The Study That Nearly Ruined Math Education

Suppose you want to test a computer-based active-learning method in mathematics classes. You measure:

- each student's math score;
- the proportion of class time each student spends using the active-learning program; and
- the classroom each student belongs to.

You expect more active-learning time to predict higher math scores. Unfortunately, each classroom contains only 20 students, which feels too small to publish. You therefore combine the classrooms and pretend all 80 students came from one enormous classroom taught by a many-headed math teacher.

Let us collect the data using my favorite method: making them up in R.

```
set.seed(9)

n_per_class <- 20
ClassroomStory <- factor(rep(paste0("Classroom ", 1:4), each = n_per_class))

ActiveTimeStory <- c(
  runif(n_per_class, 0.00, 0.35),
  runif(n_per_class, 0.30, 0.55),
  runif(n_per_class, 0.50, 0.75),
  runif(n_per_class, 0.70, 1.00)
)

ClassroomBaseline <- rep(c(80, 70, 60, 50), each = n_per_class)
```

```r
ClassroomStoryData <- data.frame(
  Classroom = ClassroomStory,
  ActiveTime = ActiveTimeStory
)

ClassroomStoryData$MathScore <-
  ClassroomBaseline +
  10 * ave(
    ClassroomStoryData$ActiveTime,
    ClassroomStoryData$Classroom,
    FUN = function(x) x - mean(x)
  ) +
  rnorm(nrow(ClassroomStoryData), 0, 2.5)
```

The true within-classroom slope is positive: within any one class, students who use the program more tend to score higher. However, classrooms using the program more also happen to have lower baselines. Perhaps their teachers adopted the program because those classes were already struggling. The predictor is tangled up with classroom membership.

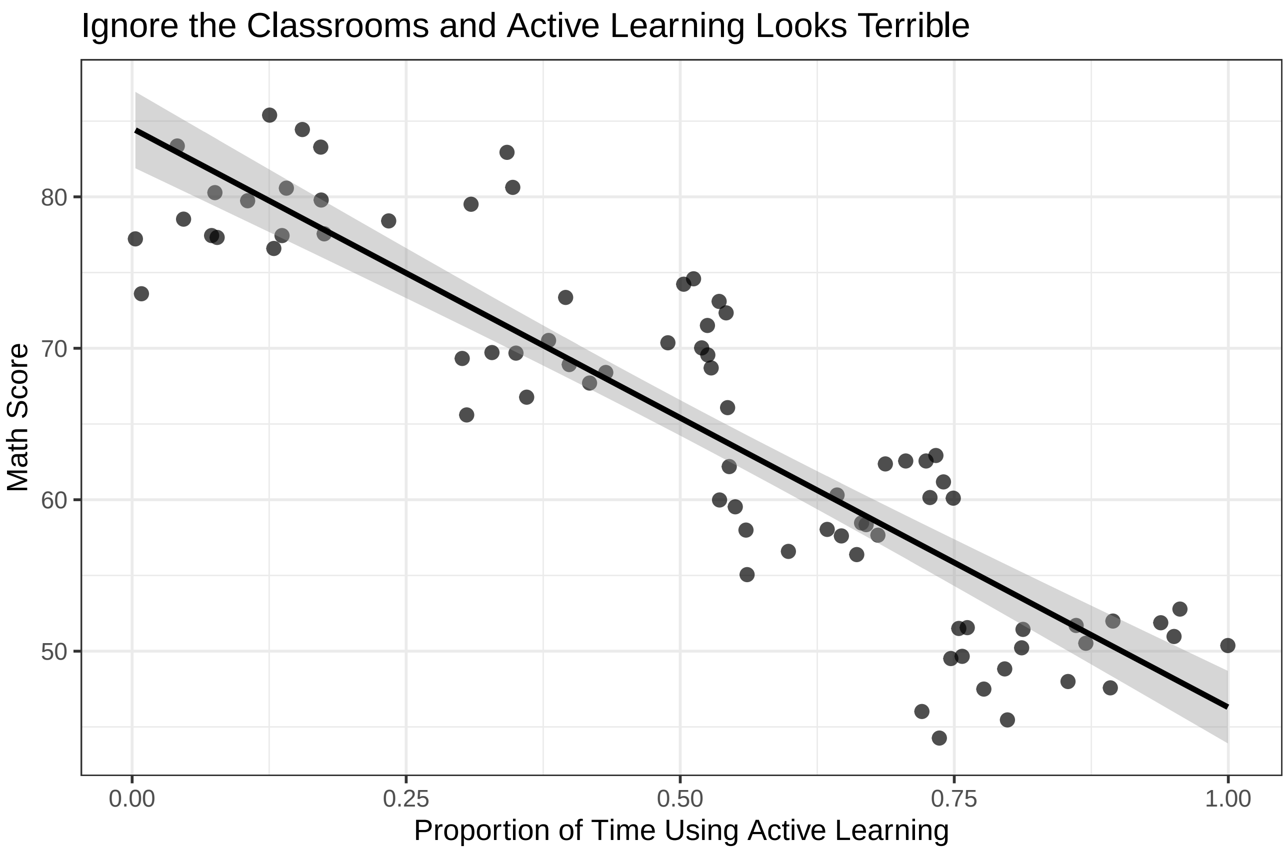
30.2.1 Ignore the Classrooms and Publish Disaster

```r
PooledModel <- lm(MathScore ~ ActiveTime, data = ClassroomStoryData)
summary(PooledModel)
```

```

Call:
lm(formula = MathScore ~ ActiveTime, data = ClassroomStoryData)

Residuals:
     Min       1Q   Median       3Q      Max 
-12.1043  -3.8995  -0.5892   4.2604  11.4905 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   84.522      1.274   66.35   <2e-16 ***
ActiveTime   -38.234      2.180  -17.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.263 on 78 degrees of freedom
Multiple R-squared:  0.7978,	Adjusted R-squared:  0.7952 
F-statistic: 307.7 on 1 and 78 DF,  p-value: < 2.2e-16
```

```r
ggplot(
  ClassroomStoryData,
  aes(x = ActiveTime, y = MathScore)
) +
  geom_point(size = 2, alpha = 0.7) +
```

```r
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  labs(
    title = "Ignore the Classrooms and Active Learning Looks Terrible",
    x = "Proportion of Time Using Active Learning",
    y = "Math Score"
  ) +
  theme_bw()
```



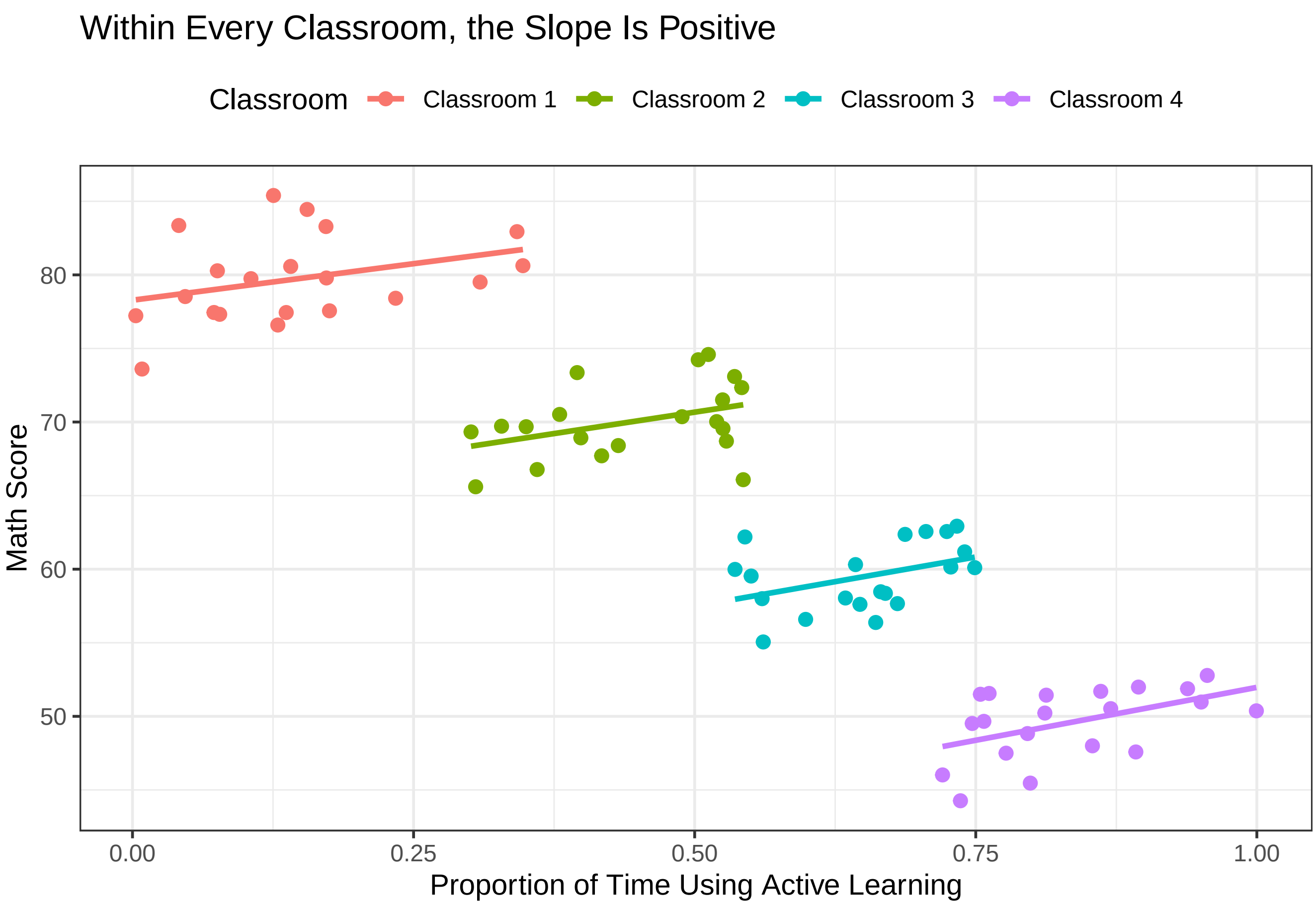
The pooled slope points in the wrong direction and looks impressively large. You rush to publish *Active Learning Is a Flop: Computers Have Betrayed the Children.* Math teachers across the country abandon the program. The next generation scores worse. You have ruined America's future. Good job!

> ⚠ A Huge R-Squared Can Describe the Wrong Relationship
>
> Statistical significance and impressive model fit cannot repair a model that misunderstood the design. The pooled model mostly compares **different classrooms**. Our scientific question concerned differences **among students within the same classroom**.

30.2.2 Put the Classrooms Back

```r
ggplot(
  ClassroomStoryData,
  aes(x = ActiveTime, y = MathScore, color = Classroom)
```

```r
) +
  geom_point(size = 2) +
  geom_smooth(aes(group = Classroom), method = "lm", se = FALSE) +
  labs(
    title = "Within Every Classroom, the Slope Is Positive",
    x = "Proportion of Time Using Active Learning",
    y = "Math Score"
  ) +
  theme_bw() +
  theme(legend.position = "top")
```



Now the story changes. Students are **nested** within classrooms. Students in the same classroom share a teacher, schedule, curriculum, room temperature, class clown, and whatever motivational speech occurred before the exam. They are not independent replications of classroom context. In one sense, the students are repeated observations of their classroom.

This reversal between the pooled and within-classroom patterns is related to **Simpson's paradox**. The point is not that mixed models perform an exorcism. The point is that relationships can differ across levels, and a pooled regression can answer a completely different question from the one you thought you asked.

30.3 The Two Old-School Solutions

Before mixed models became ordinary software, researchers had several creative ways to make this problem worse.

30.3.1 Old-School Approach 1: Analyze Every Classroom Separately

```r
SeparateSlopes <- lapply(
  split(ClassroomStoryData, ClassroomStoryData$Classroom),
  function(d) coef(lm(MathScore ~ ActiveTime, data = d))["ActiveTime"]
)

unlist(SeparateSlopes)
```

```
Classroom 1.ActiveTime Classroom 2.ActiveTime Classroom 3.ActiveTime
              9.922347              11.637781              13.420646
Classroom 4.ActiveTime
             14.382490
```

```r
mean(unlist(SeparateSlopes))
```

```
[1] 12.34082
```

The slopes are positive, but each regression uses only 20 students. Individual p-values will stagger around because chance is fickle. Averaging the slopes also treats a noisy estimate and a precise estimate as equally trustworthy unless we carry their uncertainty forward correctly. With 40 classrooms, this becomes 40 regressions and an excellent reason to begin screaming into a desk drawer.

30.3.2 Old-School Approach 2: Center Everything Within Classroom

```r
ClassroomStoryData$MathWithin <- with(
  ClassroomStoryData,
  MathScore - ave(MathScore, Classroom, FUN = mean)
)

ClassroomStoryData$ActiveWithin <- with(
  ClassroomStoryData,
  ActiveTime - ave(ActiveTime, Classroom, FUN = mean)
)

CenteredModel <- lm(MathWithin ~ ActiveWithin, data = ClassroomStoryData)
summary(CenteredModel)
```

```

Call:
lm(formula = MathWithin ~ ActiveWithin, data = ClassroomStoryData)

Residuals:
   Min     1Q Median     3Q    Max
-5.127 -1.827 -0.105  1.462  5.910
```

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -5.581e-17  2.613e-01   0.000 1.000000
ActiveWithin  1.199e+01  3.119e+00   3.845 0.000245 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.337 on 78 degrees of freedom
Multiple R-squared:  0.1593,    Adjusted R-squared:  0.1485
F-statistic: 14.78 on 1 and 78 DF,  p-value: 0.0002449
```

This recovers the positive within-classroom slope, but the outcome's classroom means have been forced to zero. We threw away the between-classroom intercept differences instead of modeling them. The ordinary regression also still pretends the denominator degrees of freedom came from 80 fully independent students.

The old approaches can work in narrow cases. They become awkward with unequal classroom sizes, multiple predictors, random slopes, three levels, missing observations, or basically any dataset that survived contact with reality.

30.4 Why Ordinary Regression Fails

Suppose students are nested within classrooms. Students in the same classroom share a teacher, schedule, building, curriculum, and possibly one malfunctioning radiator. Their outcomes tend to be more similar than outcomes from students in different classrooms.

Ordinary regression assumes independent residuals:

$$Cov(\epsilon_i, \epsilon_j) = 0 \quad \text{for } i \neq j.$$

Clustering violates that assumption. Treating 600 students from 30 classrooms as 600 completely independent pieces of information usually makes uncertainty too small because the classrooms are quietly repeating themselves.

Mixed models keep the student-level observations while explicitly representing how classrooms vary. We do not have to erase the classroom means or run a separate model for every room.

30.5 Two-Level Notation

Let student i belong to classroom j.

30.5.1 Level 1: Students

$$Y_{ij} = \beta_{0j} + \beta_{1j} X_{ij} + e_{ij}.$$

Both the intercept and slope carry the classroom subscript j. That notation leaves open the possibility that classrooms differ in their starting points, their slopes, or both.

30.5.2 Level 2: Classrooms

$$\beta_{0j} = \gamma_{00} + u_{0j}.$$

The classroom slope can also vary:

$$\beta_{1j} = \gamma_{10} + u_{1j}.$$

- γ_{00} is the population-average intercept.
- γ_{10} is the population-average slope.
- u_{0j} is classroom j's intercept deviation.
- u_{1j} is classroom j's slope deviation.

Substituting the level-2 equation into level 1 gives:

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + u_{0j} + u_{1j}X_{ij} + e_{ij}.$$

Every classroom can receive its own intercept adjustment u_{0j} and slope adjustment u_{1j}, drawn from a population of classroom effects. In this chapter we fit the simpler random-intercept model first. The next chapter lets the slopes escape too.

$$u_{0j} \sim N(0, \tau_{00}), \qquad e_{ij} \sim N(0, \sigma^2).$$

With random slopes, the additional variance and covariance components are:

$$Var(u_{1j}) = \tau_{11}, \qquad Cov(u_{0j}, u_{1j}) = \tau_{01}.$$

The covariance tells us whether classrooms with higher intercepts tend to have steeper or flatter slopes. Random effects are allowed to know one another.

30.6 The Matrix Version

Mixed models are often written:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{b} + \epsilon.$$

- $\mathbf{X}\beta$ contains fixed effects shared across the population.
- $\mathbf{Zb}$ contains cluster-specific random effects.
- ϵ contains observation-level residuals.

The random effects and residuals have covariance structures:

$$\mathbf{b} \sim N(\mathbf{0}, \mathbf{G}), \qquad \epsilon \sim N(\mathbf{0}, \mathbf{R}).$$

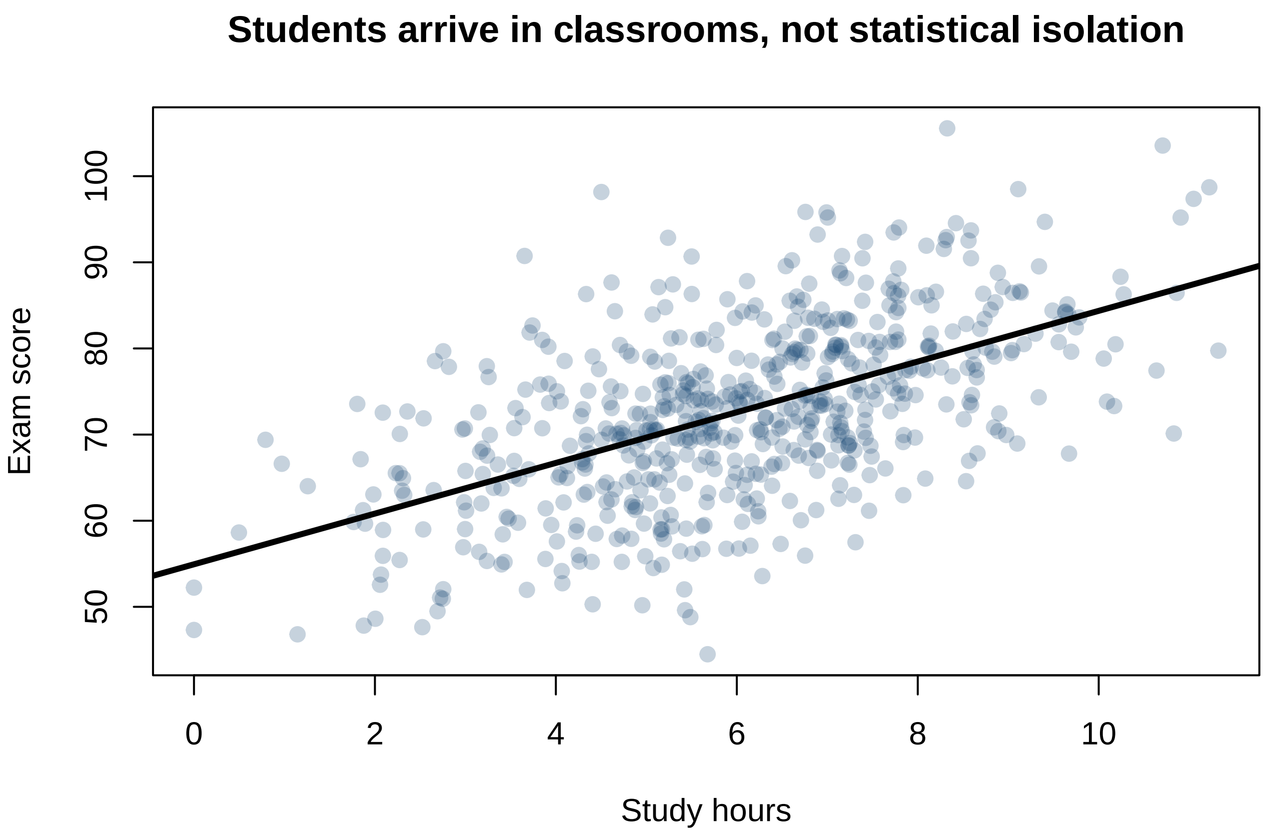
Random Does Not Mean Unimportant or Chaotic

A random effect represents variation across sampled levels of a grouping factor. We usually care about the variance of classroom effects, participant effects, or item effects rather than estimating a separate fixed coefficient for every possible classroom, person, or word stimulus in the population.

30.7 Simulating Students in Classrooms

```r
set.seed(343)
n_class <- 30
n_student <- 20

Classroom <- factor(rep(seq_len(n_class), each = n_student))
ClassIntercept <- rnorm(n_class, 0, 6)
StudyHours <- pmax(rnorm(n_class * n_student, 6, 2), 0)

MixedData <- data.frame(Classroom, StudyHours)
MixedData$Exam <-
  55 + 3.2 * MixedData$StudyHours +
  ClassIntercept[MixedData$Classroom] +
  rnorm(nrow(MixedData), 0, 7)
```

```r
plot(
  MixedData$StudyHours, MixedData$Exam,
  pch = 19,
  col = grDevices::adjustcolor("#1F4E79", alpha.f = .25),
  xlab = "Study hours", ylab = "Exam score",
  main = "Students arrive in classrooms, not statistical isolation"
)
abline(lm(Exam ~ StudyHours, data = MixedData), lwd = 3, col = "black")
```



30.8 Fit the Empty Model First

The empty or unconditional model separates classroom and student-level variation:

```r
library(glmmTMB)

EmptyModel <- glmmTMB(
  Exam ~ 1 + (1 | Classroom),
  data = MixedData,
  family = gaussian(),
  REML = TRUE
)

summary(EmptyModel)
```

```
 Family: gaussian  ( identity )
Formula:          Exam ~ 1 + (1 | Classroom)
Data: MixedData

      AIC       BIC    logLik -2*log(L)  df.resid
   4406.0    4419.2   -2200.0    4400.0       597
```

```
Random effects:

Conditional model:
 Groups    Name        Variance Std.Dev.
 Classroom (Intercept) 24.99    4.999
 Residual              81.61    9.034
Number of obs: 600, groups:  Classroom, 30

Dispersion estimate for gaussian family (sigma^2): 81.6

Conditional model:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  72.6554     0.9843   73.81   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The formula `(1 | Classroom)` says each classroom receives a random intercept.

30.9 The Intraclass Correlation

For a random-intercept Gaussian model:

$$ICC = \frac{\tau_{00}}{\tau_{00} + \sigma^2}.$$

The ICC is the expected correlation between two observations from the same cluster. It also describes the proportion of outcome variance attributable to between-cluster differences in this simple model.

```r
performance::icc(EmptyModel)
```

```
# Intraclass Correlation Coefficient

    Adjusted ICC: 0.234
  Unadjusted ICC: 0.234
```

An ICC near zero means classrooms add little clustering. A larger ICC means students within classrooms resemble one another. No universal ICC is “small enough to ignore”; consequences also depend on cluster sizes and the analysis.

> **!** Six Hundred Rows Are Not Always Six Hundred Independent Observations
>
> The effective information for a classroom-level predictor depends heavily on the number of classrooms, not merely the number of students. Adding 300 students to the same four classrooms does not magically create 304 classrooms.

30.10 Add a Fixed Effect

```r
StudyModel <- glmmTMB(
  Exam ~ StudyHours + (1 | Classroom),
  data = MixedData,
  family = gaussian(),
  REML = TRUE
)

broom.mixed::tidy(
  StudyModel,
  effects = "fixed",
  component = "cond",
  conf.int = TRUE
)
```

```
# A tibble: 2 x 9
  effect component term        estimate std.error statistic   p.value conf.low
  <chr>  <chr>     <chr>          <dbl>     <dbl>     <dbl>     <dbl>    <dbl>
1 fixed  cond      (Intercept)    54.3      1.32       41.3 0            51.7
2 fixed  cond      StudyHours      3.05     0.143      21.4 3.53e-101     2.77
# i 1 more variable: conf.high <dbl>
```

The fixed slope estimates the population-average relationship between study hours and exam scores while accounting for classroom-specific intercepts.

30.11 Conditional and Population-Level Predictions

- A **conditional** prediction includes the estimated random effect for a known classroom.
- A **population-level** prediction sets classroom random effects to zero.

```r
MixedData$ConditionalPrediction <- predict(StudyModel)
MixedData$PopulationPrediction <- predict(StudyModel, re.form = NA)
head(MixedData)
```

```
  Classroom StudyHours     Exam ConditionalPrediction PopulationPrediction
1         1   5.374921 56.45697              63.82018             70.75529
2         1   6.684974 67.55878              67.80988             74.61025
3         1   4.122426 64.95678              60.00577             67.06969
4         1   7.303397 74.94600              69.69325             76.43003
5         1   6.059187 59.87663              65.90408             72.76881
6         1   7.230732 78.76105              69.47195             76.21620
```

Use conditional predictions when describing known clusters. Use population-level predictions when describing an average new classroom from the modeled population.

30.12 Centering Is About the Question, Not Tidying the Numbers

The original active-learning disaster mixed two relationships together:

1. **Within classrooms:** Do students who study more than their classmates score higher?
2. **Between classrooms:** Do classrooms whose students study more on average have higher scores?

Those are not automatically the same relationship. We can separate them by decomposing Study Hours into a classroom mean and each student's deviation from that mean:

```
MixedData$ClassStudyMean <- with(
  MixedData,
  ave(StudyHours, Classroom, FUN = mean)
)

MixedData$StudyWithin <-
  MixedData$StudyHours - MixedData$ClassStudyMean

MixedData$ClassStudyMean.C <-
  MixedData$ClassStudyMean - mean(MixedData$ClassStudyMean)

WithinBetweenModel <- glmmTMB(
  Exam ~ StudyWithin + ClassStudyMean.C + (1 | Classroom),
  data = MixedData,
  family = gaussian(),
  REML = TRUE
)

broom.mixed::tidy(
  WithinBetweenModel,
  effects = "fixed",
  component = "cond",
  conf.int = TRUE
)
```

```
# A tibble: 3 x 9
  effect component term          estimate std.error statistic   p.value conf.low
  <chr>  <chr>     <chr>            <dbl>     <dbl>     <dbl>     <dbl>    <dbl>
1 fixed  cond      (Intercept)     72.7       0.999    72.7   0            70.7
2 fixed  cond      StudyWithin      3.05      0.143    21.4   2.22e-101     2.77
3 fixed  cond      ClassStudyMe~    0.873     2.23      0.391 6.96e-  1    -3.51
# i 1 more variable: conf.high <dbl>
```

`StudyWithin` answers the student-level question while holding the classroom average constant. `ClassStudyMean.C` answers the classroom-level question. **Grand-mean centering** merely moves zero and changes the intercept. **Group-mean centering** changes which source of variation a coefficient describes. Treating both procedures as cosmetic subtraction is how a very clean spreadsheet tells the wrong story.

! A Predictor Can Have Two Slopes Hiding Inside It

Whenever a predictor varies both within and between clusters, the raw mixed-model coefficient can blend those relationships. Separate them when theory requires separate questions. The next chapter makes this distinction considerably more mathematical and therefore, according to tradition, more frightening.

30.13 Mixed-Model R-Squared

Marginal and conditional R^2 separate fixed and total modeled variation:

- **Marginal R^2:** variation explained by fixed effects.
- **Conditional R^2:** variation explained by fixed plus random effects.

For a Gaussian random-intercept model, the basic logic is:

$$R^2_{marginal} = \frac{\sigma^2_{fixed}}{\sigma^2_{fixed} + \sigma^2_{random} + \sigma^2_{residual}},$$

$$R^2_{conditional} = \frac{\sigma^2_{fixed} + \sigma^2_{random}}{\sigma^2_{fixed} + \sigma^2_{random} + \sigma^2_{residual}}.$$

```
performance::r2_nakagawa(StudyModel)
```

```
# R2 for Mixed Models

  Conditional R2: 0.585
     Marginal R2: 0.332
```

i Random Effects Can Explain Variation Without Being the Hypothesis

A large difference between conditional and marginal R^2 means clusters account for substantial variation. It does not mean classroom identity is a causal treatment or that naming every classroom would explain why they differ.

Someone may look at the population-level line over the raw data and announce that your fit is terrible. The fixed effect is not trying to reproduce every classroom's baseline. Conditional predictions add those classroom deviations and will hug the observed clusters more closely. That can make conditional R^2 enormous. You have not become the greatest modeler alive. You told the model which students shared a classroom.

30.14 Why `glmmTMB`?

We use one engine throughout the advanced chapters. For Gaussian outcomes, `glmmTMB` fits linear mixed models. By changing `family`, the same framework handles binary, count, negative-binomial, beta, and zero-inflated outcomes (Brooks et al. 2017).

That consistency lets us focus on modeling rather than spending half the chapter translating between packages that all print the intercept in a slightly different location.

30.15 The Short Story

Mixed models account for dependence by combining fixed population effects with random cluster effects. Begin with the study's levels, fit an empty model to understand variance, calculate the ICC and mixed-model R^2 deliberately (Nakagawa et al. 2017), and remember that the number of clusters—not merely rows—limits cluster-level information.

31 Advanced Mixed Models II: Random Slopes, Centering, and Cross-Level Interactions

31.1 When One Slope Does Not Fit Every Cluster

A random-intercept model allows classrooms to start at different exam-score levels but forces the study-hours slope to be identical everywhere. That may be unreasonable. Study time could help more in classrooms with organized instruction and less in classrooms where the final exam covers material communicated through rumors.

31.2 Random-Slope Equations

Level 1:

$$Y_{ij} = \beta_{0j} + \beta_{1j} X_{ij} + e_{ij}.$$

Level 2:

$$\beta_{0j} = \gamma_{00} + u_{0j},$$

$$\beta_{1j} = \gamma_{10} + u_{1j}.$$

Combined:

$$Y_{ij} = \gamma_{00} + \gamma_{10} X_{ij} + u_{0j} + u_{1j} X_{ij} + e_{ij}.$$

The random intercept and slope have covariance matrix:

$$\mathbf{G} = \begin{bmatrix} \tau_{00} & \tau_{01} \\ \tau_{01} & \tau_{11} \end{bmatrix}.$$

τ_{01} describes whether clusters with higher intercepts tend to have steeper or flatter slopes. Centering choices determine which within- and between-cluster questions those coefficients answer (Enders and Tofighi 2007).

> **A Random Slope Requires Replication Within Clusters**
>
> To estimate how a slope varies across people or classrooms, the predictor must vary within those units. You cannot estimate a person's random treatment slope if each person experienced only one treatment level.

31.3 Simulating Classroom Slopes

```
set.seed(344)
n_class <- 36
n_student <- 18

ClassInfo <- data.frame(
  Classroom = factor(seq_len(n_class)),
  TeacherSupport = rnorm(n_class)
)

RandomEffects <- MASS::mvrnorm(
  n_class,
  mu = c(0, 0),
  Sigma = matrix(c(30, 2.8, 2.8, 1.2), 2, 2)
)

SlopeData <- do.call(rbind, lapply(seq_len(n_class), function(j) {
  Study <- rnorm(n_student, mean = runif(1, 4.5, 7.5), sd = 1.5)
  data.frame(
    Classroom = factor(j, levels = seq_len(n_class)),
    TeacherSupport = ClassInfo$TeacherSupport[j],
    StudyHours = Study,
    u0 = RandomEffects[j, 1],
    u1 = RandomEffects[j, 2]
  )
}))

SlopeData$StudyMean <- ave(SlopeData$StudyHours, SlopeData$Classroom)
SlopeData$StudyWithin <- SlopeData$StudyHours - SlopeData$StudyMean
SlopeData$StudyMean.C <- SlopeData$StudyMean - mean(SlopeData$StudyMean)

SlopeData$Exam <-
  68 +
  3.2 * SlopeData$StudyWithin +
  2.0 * SlopeData$StudyMean.C +
  2.5 * SlopeData$TeacherSupport +
  1.3 * SlopeData$StudyWithin * SlopeData$TeacherSupport +
  SlopeData$u0 +
  SlopeData$u1 * SlopeData$StudyWithin +
  rnorm(nrow(SlopeData), 0, 6)
```

31.4 Within- and Between-Cluster Effects

The raw predictor contains two relationships:

1. **Within classrooms:** Do students who study more than their classmates score higher?
2. **Between classrooms:** Do classrooms with higher average study time score higher?

Decompose the predictor:

$$X_{ij} = (X_{ij} - \bar{X}_j) + \bar{X}_j.$$

- $X_{ij} - \bar{X}_j$ is the cluster-centered within effect.
- $\bar{X}_j$ is the cluster mean and carries between-cluster information.

> ⚠ Group-Mean Centering Changes the Question
>
> Grand-mean centering changes the zero point. Group-mean centering separates within- and between-cluster variation. Those operations are not interchangeable merely because both subtract a mean.

31.5 Fit the Random-Slope Model

```
library(glmmTMB)

RandomInterceptModel <- glmmTMB(
  Exam ~ StudyWithin + StudyMean.C + TeacherSupport +
    StudyWithin:TeacherSupport +
    (1 | Classroom),
  data = SlopeData,
  family = gaussian(),
  REML = FALSE
)

RandomSlopeModel <- glmmTMB(
  Exam ~ StudyWithin + StudyMean.C + TeacherSupport +
    StudyWithin:TeacherSupport +
    (1 + StudyWithin | Classroom),
  data = SlopeData,
  family = gaussian(),
  REML = FALSE
)

summary(RandomSlopeModel)
```

```
 Family: gaussian  ( identity )
Formula:
Exam ~ StudyWithin + StudyMean.C + TeacherSupport + StudyWithin:TeacherSupport +
    (1 + StudyWithin | Classroom)
Data: SlopeData

      AIC       BIC    logLik -2*log(L)  df.resid
   4253.7    4294.0   -2117.9    4235.7       639

Random effects:
```

```
Conditional model:
 Groups    Name        Variance Std.Dev. Corr
 Classroom (Intercept) 25.312   5.031
           StudyWithin  1.018   1.009    0.71
 Residual              33.973   5.829
Number of obs: 648, groups:  Classroom, 36

Dispersion estimate for gaussian family (sigma^2):   34

Conditional model:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                 67.8781     0.8724   77.80  < 2e-16 ***
StudyWithin                  3.2521     0.2372   13.71  < 2e-16 ***
StudyMean.C                  2.2022     0.7800    2.82 0.004754 **
TeacherSupport               0.8281     0.7997    1.04 0.300433
StudyWithin:TeacherSupport   0.7952     0.2067    3.85 0.000119 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The fixed `StudyWithin` coefficient is the expected within-classroom study slope when teacher support is zero. The interaction says that slope changes as teacher support changes. The random-slope variance says classrooms still differ around the modeled slope.

31.6 Compare Random Structures Carefully

```r
anova(RandomInterceptModel, RandomSlopeModel)
```

```
Data: SlopeData
Models:
RandomInterceptModel: Exam ~ StudyWithin + StudyMean.C + TeacherSupport + StudyWithin:TeacherSupport + , zi=~
RandomInterceptModel:     (1 | Classroom), zi=~0, disp=~1
RandomSlopeModel: Exam ~ StudyWithin + StudyMean.C + TeacherSupport + StudyWithin:TeacherSupport + , zi=~0, d
RandomSlopeModel:     (1 + StudyWithin | Classroom), zi=~0, disp=~1
                     Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
RandomInterceptModel  7 4269.5 4300.8 -2127.8   4255.5
RandomSlopeModel      9 4253.7 4294.0 -2117.9   4235.7 19.763      2   5.11e-05

RandomInterceptModel
RandomSlopeModel     ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Likelihood-ratio tests for variance components require caution because zero variance lies on a boundary. Use the comparison as evidence alongside estimated variances, convergence, design, and theoretical need.

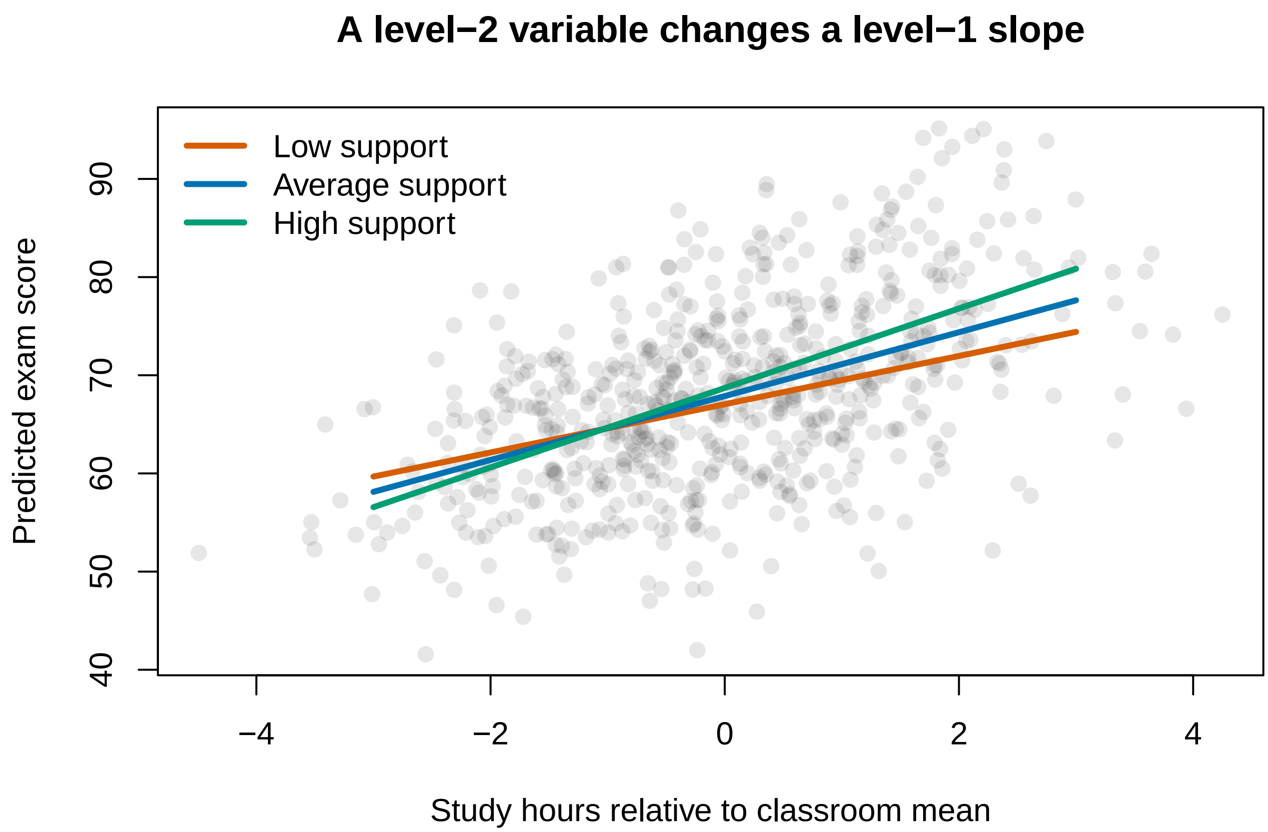
Do not automatically choose the largest possible random-effects structure or delete every small variance. A defensible model represents important sampling and repeated-measures structure while remaining estimable by the data.

❗ A Zero Variance Is Information

If a random-slope variance approaches zero, the data may not support meaningful slope differences, the design may contain too few clusters, or the model may be overparameterized. Simplification should follow investigation, not a panicked sequence of deleted parentheses.

31.7 Plot the Cross-Level Interaction

```r
support_values <- c(Low = -1, Average = 0, High = 1)
study_grid <- seq(-3, 3, length.out = 100)

plot(
  SlopeData$StudyWithin, SlopeData$Exam,
  pch = 19, col = rgb(.2, .2, .2, .12),
  xlab = "Study hours relative to classroom mean",
  ylab = "Predicted exam score",
  main = "A level-2 variable changes a level-1 slope"
)

line_colors <- c("#D55E00", "#0072B2", "#009E73")
for (i in seq_along(support_values)) {
  newdata <- data.frame(
    StudyWithin = study_grid,
    StudyMean.C = 0,
    TeacherSupport = support_values[i],
    Classroom = SlopeData$Classroom[1]
  )
  lines(
    study_grid,
    predict(RandomSlopeModel, newdata = newdata, re.form = NA),
    col = line_colors[i], lwd = 3
  )
}
legend("topleft", legend = paste(names(support_values), "support"),
       col = line_colors, lwd = 3, bty = "n")
```



31.8 Three Levels

Students may be nested in classrooms nested in schools. A random-intercept model becomes:

$$Y_{ijk} = \gamma_{000} + v_{00k} + u_{0jk} + e_{ijk},$$

where k indexes schools, j classrooms, and i students.

In `glmmTMB`, nested intercepts can be written:

```
ThreeLevelModel <- glmmTMB(
  Score ~ Predictor + (1 | School / Classroom),
  data = ThreeLevelData,
  family = gaussian()
)
```

This expands to school intercepts plus classroom-within-school intercepts. Additional random slopes require enough clusters and variation at the level where the slope is estimated.

31.9 Effect Size

```
performance::r2_nakagawa(RandomSlopeModel)
```

```
# R2 for Mixed Models

  Conditional R2: 0.630
     Marginal R2: 0.331
```

```
performance::icc(RandomSlopeModel)
```

```
# Intraclass Correlation Coefficient

    Adjusted ICC: 0.446
  Unadjusted ICC: 0.298
```

Marginal R^2 summarizes fixed-effect variation; conditional R^2 includes random effects. The ICC in a random-slope model can depend on predictor values because cluster similarity changes with X.

31.10 The Short Story

Random slopes allow predictor effects to vary across clusters. Separate within- and between-cluster information before interpreting level-1 predictors. Cross-level interactions explain some slope variation; random-slope variance describes what remains.

32 Advanced Mixed Models III: Repeated, Nested, Crossed, and Longitudinal Designs

32.1 Start with the Sampling Structure

Mixed-model formulas should follow how observations were generated.

- **Nested:** Every lower-level unit belongs to one higher-level unit. Students are nested in classrooms.
- **Crossed:** Units encounter multiple levels of another factor. Participants respond to many stimuli, and each stimulus is seen by many participants.
- **Longitudinal:** Repeated observations are ordered in time within people or other units.

> ⚠ The Spreadsheet Does Not Reveal the Design
>
> One row per observation does not make rows independent. You must know which person, item, classroom, therapist, or time series produced each row. The design lives in the data-generating process, not the rectangular shape of the file.

32.2 Crossed Participants and Items

In many psychology experiments, participants respond to stimuli sampled from a broader population: words, faces, scenarios, voices, or images of objects that researchers insist are emotionally neutral.

A crossed model can include participant and item effects:

$$Y_{si} = \beta_0 + \beta_1 X_{si} + u_{0s} + u_{1s} X_{si} + v_{0i} + v_{1i} X_{si} + e_{si}.$$

- u terms vary across subjects.
- v terms vary across items.

```
set.seed(345)
n_subject <- 48
n_item <- 24

CrossedData <- expand.grid(
  Subject = factor(seq_len(n_subject)),
  Item = factor(seq_len(n_item))
)

CrossedData$ConditionNum <- ifelse(
  (as.integer(CrossedData$Subject) + as.integer(CrossedData$Item)) %% 2 == 0,
```

```
  -.5, .5
)

SubjectRE <- MASS::mvrnorm(
  n_subject, c(0, 0), matrix(c(35, 2, 2, 2), 2, 2)
)
ItemRE <- MASS::mvrnorm(
  n_item, c(0, 0), matrix(c(18, 1, 1, 1.2), 2, 2)
)

s <- as.integer(CrossedData$Subject)
i <- as.integer(CrossedData$Item)

CrossedData$Attention <-
  70 + 5 * CrossedData$ConditionNum +
  SubjectRE[s, 1] + SubjectRE[s, 2] * CrossedData$ConditionNum +
  ItemRE[i, 1] + ItemRE[i, 2] * CrossedData$ConditionNum +
  rnorm(nrow(CrossedData), 0, 5)
```

```
library(glmmTMB)

CrossedModel <- glmmTMB(
  Attention ~ ConditionNum +
    (1 + ConditionNum | Subject) +
    (1 + ConditionNum | Item),
  data = CrossedData,
  family = gaussian(),
  REML = TRUE
)

broom.mixed::tidy(CrossedModel, effects = "fixed", conf.int = TRUE)
```

```
# A tibble: 2 x 9
  effect component term         estimate std.error statistic  p.value conf.low
  <chr>  <chr>     <chr>           <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
1 fixed  cond      (Intercept)     70.6      1.17       60.2 0            68.3
2 fixed  cond      ConditionNum     5.23     0.476      11.0 4.52e-28     4.30
# i 1 more variable: conf.high <dbl>
```

Ignoring item variation treats the particular stimuli as fixed and perfectly representative. That may exaggerate generalization to new stimuli. Forty participants responding to the same twelve items do not provide forty independent samples of the item population.

! Participants and Items Can Both Be Random Factors

If conclusions should generalize to new participants and new stimuli, model both sampling sources when the design supports it. “But the software output was shorter without items” is not a sampling argument.

32.3 Repeated Measures as a Mixed Model

Repeated observations are nested in people. A random intercept accounts for stable individual differences. A random treatment slope allows people to respond differently to the manipulation.

```
RepeatedModel <- glmmTMB(
  Outcome ~ Condition + (1 + Condition | Participant),
  data = RepeatedData,
  family = gaussian()
)
```

If stimuli are also sampled, participant and item random effects can appear together. This is one reason mixed models scale beyond classical repeated-measures ANOVA.

32.4 Longitudinal Growth Models

Longitudinal models treat time as a predictor. A random intercept allows different starting points; a random time slope allows different rates of change.

```
set.seed(346)
n_person <- 90
times <- 0:5

LongData <- expand.grid(
  Person = factor(seq_len(n_person)),
  Time = times
)
LongData$Treatment <- factor(
  rep(sample(c("Control", "Therapy"), n_person, replace = TRUE), each = length(times))
)

PersonRE <- MASS::mvrnorm(
  n_person, c(0, 0), matrix(c(40, -2, -2, 1.5), 2, 2)
)
p <- as.integer(LongData$Person)

LongData$Symptoms <-
  55 - 1.0 * LongData$Time -
  1.8 * LongData$Time * (LongData$Treatment == "Therapy") +
  PersonRE[p, 1] + PersonRE[p, 2] * LongData$Time +
  rnorm(nrow(LongData), 0, 4)

# Mimic ordinary dropout without deleting entire people.
LongData$Symptoms[LongData$Time > 3 & runif(nrow(LongData)) < .12] <- NA
```

```
GrowthModel <- glmmTMB(
  Symptoms ~ Time * Treatment + (1 + Time | Person),
  data = LongData,
```

```r
  family = gaussian(),
  REML = TRUE,
  na.action = na.omit
)

summary(GrowthModel)
```

```
 Family: gaussian  ( identity )
Formula:          Symptoms ~ Time * Treatment + (1 + Time | Person)
Data: LongData

      AIC       BIC    logLik -2*log(L)  df.resid
   3285.7    3319.8   -1634.9    3269.7       513

Random effects:

Conditional model:
 Groups   Name        Variance Std.Dev. Corr
 Person   (Intercept) 38.777   6.227
          Time         1.372   1.171    -0.08
 Residual             16.959   4.118
Number of obs: 521, groups:  Person, 90

Dispersion estimate for gaussian family (sigma^2):   17

Conditional model:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)            54.5944     0.8201   66.57  < 2e-16 ***
Time                   -1.0760     0.1969   -5.47 4.61e-08 ***
TreatmentTherapy       -0.7469     0.7360   -1.01     0.31
Time:TreatmentTherapy  -1.6397     0.2582   -6.35 2.16e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The time-by-treatment interaction asks whether the expected rate of change differs between groups. The random time slope allows individual trajectories to vary around their group's expected slope.

32.5 Center Time Where the Intercept Matters

If Time $= 0$ represents baseline, the intercept is baseline symptoms. If time is centered at the final session, the intercept becomes expected final symptoms. Predictions remain the same; the coefficient questions change.

> **Centering Time Is a Scientific Decision**
>
> Choose zero to represent a meaningful occasion: baseline, treatment completion, a developmental milestone, or another theoretically important point. Centering at the mean is convenient but not automatically the most informative choice.

32.6 Time as Continuous or Categorical

Treat time as continuous when a linear or specified nonlinear trajectory is meaningful. Treat it as categorical when change may be irregular and each occasion deserves its own estimate.

```
LongData$TimeFactor <- factor(LongData$Time)

OccasionModel <- glmmTMB(
  Symptoms ~ TimeFactor * Treatment + (1 | Person),
  data = LongData,
  family = gaussian()
)
```

The continuous model tells a compact growth story. The categorical model tells an occasion-by-occasion story and spends more degrees of freedom doing it.

32.7 Residual Correlation Across Time

Random intercepts and slopes may not absorb every temporal dependency. Nearby observations can remain more similar than distant ones. `glmmTMB` can represent structures such as autoregressive correlation:

```
LongData$TimeFactor <- factor(LongData$Time)

AR1Model <- glmmTMB(
  Symptoms ~ Time * Treatment +
    (1 + Time | Person) +
    ar1(TimeFactor + 0 | Person),
  data = LongData,
  family = gaussian()
)
```

Structured covariance models require correctly ordered occasions and enough information. Do not add `ar1()` merely because the formula looks graduate-level.

32.8 Missing Repeated Observations

Likelihood-based mixed models can retain people with incomplete outcomes under the model's missing-data assumptions. Classical repeated-measures ANOVA may discard an entire participant after one missed visit.

This advantage is not permission to ignore dropout. Report missingness, consider why observations are missing, and conduct sensitivity analyses when dropout may depend on unobserved outcomes.

32.9 The Short Story

Nested designs have one-way membership; crossed designs sample from multiple overlapping populations; longitudinal designs add ordered repeated observations. Build random effects from the sampling structure, model time deliberately, and do not mistake more rows for more independent units.

33 Advanced Mixed Models IV: Inference, Diagnostics, and Effect Sizes

33.1 The Model Fitted. Now What?

Mixed models contain fixed effects, random effects, variance components, and sometimes non-Gaussian distributions. There is no single p-value ritual that is best in every design.

This chapter uses `glmmTMB` as the default engine. If a field requires ANOVA-style denominator degrees of freedom for a Gaussian model, we deliberately refit with `lmerTest`. That is a specialized detour, not the main road.

33.2 A Model for the Examples

```
set.seed(347)
n_group <- 32
n_each <- 16

InferenceData <- data.frame(
  Clinic = factor(rep(seq_len(n_group), each = n_each)),
  Sessions = runif(n_group * n_each, 0, 10)
)

ClinicRE <- MASS::mvrnorm(
  n_group, c(0, 0), matrix(c(20, 1.5, 1.5, .8), 2, 2)
)
g <- as.integer(InferenceData$Clinic)

InferenceData$Improvement <-
  4 + 2.3 * InferenceData$Sessions +
  ClinicRE[g, 1] + ClinicRE[g, 2] * InferenceData$Sessions +
  rnorm(nrow(InferenceData), 0, 5)

library(glmmTMB)

InferenceModel <- glmmTMB(
  Improvement ~ Sessions + (1 + Sessions | Clinic),
  data = InferenceData,
  family = gaussian(),
  REML = FALSE
)
```

33.3 Wald z-Tests

`glmmTMB` reports a Wald statistic:

$$z = \frac{\hat{\beta}}{SE(\hat{\beta})}.$$

```
summary(InferenceModel)
```

```
 Family: gaussian  ( identity )
Formula:          Improvement ~ Sessions + (1 + Sessions | Clinic)
Data: InferenceData

      AIC       BIC    logLik -2*log(L)  df.resid
   3253.3    3278.7   -1620.6    3241.3       506

Random effects:

Conditional model:
 Groups   Name        Variance Std.Dev. Corr
 Clinic   (Intercept) 23.2344  4.8202
          Sessions     0.6411  0.8007   0.30
 Residual             25.0569  5.0057
Number of obs: 512, groups:  Clinic, 32

Dispersion estimate for gaussian family (sigma^2): 25.1

Conditional model:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   3.5671     0.9689   3.682 0.000232 ***
Sessions      2.5907     0.1635  15.842  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
broom.mixed::tidy(
  InferenceModel,
  effects = "fixed",
  component = "cond",
  conf.int = TRUE
)
```

```
# A tibble: 2 x 9
  effect component term        estimate std.error statistic  p.value conf.low
  <chr>  <chr>     <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
1 fixed  cond      (Intercept)     3.57     0.969      3.68 2.32e- 4     1.67
2 fixed  cond      Sessions        2.59     0.164     15.8  1.60e-56     2.27
# i 1 more variable: conf.high <dbl>
```

The p-value uses a large-sample standard-normal approximation. It does not estimate denominator degrees of freedom.

Wald inference is convenient and often reasonable when the model is stable and information is adequate. It may be unreliable with few clusters, weakly identified parameters, extreme coefficients, boundary estimates, separation, or an unhealthy Hessian.

> ⚠ A z-Test Does Not Create a Large Sample
>
> If you have four clinics, the software can print six decimal places but cannot manufacture a rich population of clinic effects. Asymptotic inference deserves more caution when the number of independent clusters is small.

33.4 Likelihood-Ratio Tests

Compare nested models fitted with maximum likelihood:

$$LR = 2\left(\ell_{Full} - \ell_{Reduced}\right).$$

```r
ReducedModel <- glmmTMB(
  Improvement ~ 1 + (1 + Sessions | Clinic),
  data = InferenceData,
  family = gaussian(),
  REML = FALSE
)

anova(ReducedModel, InferenceModel)
```

```
Data: InferenceData
Models:
ReducedModel: Improvement ~ 1 + (1 + Sessions | Clinic), zi=~0, disp=~1
InferenceModel: Improvement ~ Sessions + (1 + Sessions | Clinic), zi=~0, disp=~1
               Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
ReducedModel    5 3321.6 3342.7 -1655.8   3311.6
InferenceModel  6 3253.3 3278.7 -1620.6   3241.3 70.285      1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Use LRTs when a hypothesis concerns one or more terms and the models are nested, fitted to the same observations, and use compatible likelihoods.

Do not compare fixed-effect structures under REML. The restricted likelihood changes with the fixed-effect design matrix, making the likelihoods unsuitable for that comparison.

> ℹ Boundary Problems Affect Random-Effect Tests
>
> A variance cannot be less than zero. Testing whether a random-effect variance equals zero puts the null hypothesis on the boundary of the parameter space, so the ordinary chi-square reference can be conservative. Treat random-structure decisions as a combination of design, estimates, diagnostics, and model comparison.

33.5 Confidence Intervals

Wald intervals are fast:

```
confint(InferenceModel, method = "wald", parm = "beta_")
```

```
                2.5 %   97.5 % Estimate
(Intercept) 1.668086 5.466213 3.567149
Sessions    2.270148 2.911193 2.590671
```

Profile intervals follow the likelihood and can behave better when it is asymmetric, but they require more computation:

```
confint(InferenceModel, method = "profile", parm = "beta_")
```

Important results deserve intervals, not only stars. If Wald and profile conclusions disagree substantially, investigate the model rather than selecting the friendlier one.

33.6 If You Need ANOVA-Style Degrees of Freedom

For a **Gaussian linear mixed model**, `lmerTest` provides Satterthwaite and Kenward–Roger approximations (Kuznetsova et al. 2017).

```
LmerTestModel <- lmerTest::lmer(
  Improvement ~ Sessions + (1 + Sessions | Clinic),
  data = InferenceData,
  REML = TRUE
)

summary(LmerTestModel, ddf = "Satterthwaite")
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Improvement ~ Sessions + (1 + Sessions | Clinic)
   Data: InferenceData

REML criterion at convergence: 3241.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-3.15593 -0.62835  0.02849  0.64829  2.90545

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 Clinic   (Intercept) 24.2217  4.9216
          Sessions     0.6679  0.8172   0.29
 Residual             25.0574  5.0057
Number of obs: 512, groups:  Clinic, 32
```

```
Fixed effects:
            Estimate Std. Error      df t value Pr(>|t|)
(Intercept)   3.5626     0.9843 29.8923   3.619  0.00108 **
Sessions      2.5915     0.1660 31.4715  15.610 2.27e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
         (Intr)
Sessions 0.024
```

```r
anova(LmerTestModel, ddf = "Kenward-Roger")
```

```
Type III Analysis of Variance Table with Kenward-Roger's method
         Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
Sessions 6098.5  6098.5     1 30.886  243.38 3.401e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

These are estimated denominator degrees of freedom designed to provide t- and F-style inference. They are not available as a general solution for binomial, Poisson, negative-binomial, or zero-inflated `glmmTMB` models.

> **! Choose the Inferential Framework Before Reading the Result**
>
> Do not run Wald, LRT, Satterthwaite, Kenward–Roger, and bootstrap tests, then report whichever one crosses .05. Choose a defensible primary method from the design, model family, field expectations, and sample structure.

33.7 Diagnostics Before Inference

Check convergence and the Hessian before interpreting coefficients.

```r
InferenceModel$sdr$pdHess
```

```
[1] TRUE
```

```r
glmmTMB::diagnose(InferenceModel)
```

```
Unusually large Z-statistics (|x|>5):

                 Sessions          disp~(Intercept) theta_1+Sessions|Clinic.1
                15.841689                 48.259685                  9.587909

Large Z-statistics (estimate/std err) suggest a *possible* failure of
the Wald approximation - often also associated with parameters that are
at or near the edge of their range (e.g. random-effects standard
deviations approaching 0).  (Alternately, they may simply represent
```

```
very well-estimated parameters; intercepts of non-centered models may
fall in this category.) While the Wald p-values and standard errors
listed in summary() may be unreliable, profile confidence intervals
(see ?confint.glmmTMB) and likelihood ratio test p-values derived by
comparing models (e.g. ?drop1) are probably still OK.  (Note that the
LRT is conservative when the null value is on the boundary, e.g. a
variance or zero-inflation value of 0 (Self and Liang 1987; Stram and
Lee 1994; Goldman and Whelan 2000); in simple cases these p-values are
approximately twice as large as they should be.)
```

A non-positive-definite Hessian means the likelihood surface is flat or poorly behaved in at least one direction. Common causes include:

- An overcomplicated random-effects structure.
- Variances approaching zero.
- Correlations approaching ± 1.
- Poorly scaled predictors.
- Separation in binary models.
- An inappropriate outcome family.

Models with unresolved convergence or Hessian problems should not be treated as ordinary successful fits simply because `summary()` produced a table.

33.8 Maximal Versus Parsimonious Random Effects

Random effects should reflect the sampling and manipulation structure. A maximal structure can protect against treating varying effects as fixed (Barr et al. 2013), but data may not support every possible variance and correlation.

A practical sequence is:

1. Specify scientifically necessary grouping factors.
2. Include random slopes for within-unit manipulations when the design and data support them.
3. Inspect variances, correlations, convergence, and Hessian behavior.
4. Simplify unsupported correlations or slopes deliberately.
5. Document what changed and whether fixed-effect conclusions are sensitive.

`glmmTMB` can specify a diagonal random-effect covariance structure when correlations are unnecessary:

```r
DiagonalModel <- glmmTMB(
  Improvement ~ Sessions + diag(1 + Sessions | Clinic),
  data = InferenceData,
  family = gaussian()
)
```

33.9 Effect Sizes

33.9.1 Marginal and Conditional R-Squared

```r
performance::r2_nakagawa(InferenceModel)
```

```
# R2 for Mixed Models

  Conditional R2: 0.814
     Marginal R2: 0.402
```

- Marginal R^2 describes fixed-effect variation.
- Conditional R^2 describes fixed plus random-effect variation.

33.9.2 Intraclass Correlation

```r
performance::icc(InferenceModel)
```

```
# Intraclass Correlation Coefficient

    Adjusted ICC: 0.689
  Unadjusted ICC: 0.412
```

In random-slope models, similarity within clusters can depend on predictor values, so one ICC may be an approximation to a changing covariance structure.

33.9.3 Effect Sizes for Specific Contrasts

Report model-estimated differences and confidence intervals in meaningful units. Standardized mean differences can supplement those estimates when a defensible standardizer exists.

For generalized models:

- Logistic coefficients can be reported as odds ratios.
- Count-model coefficients can be reported as rate ratios.
- Predicted probabilities and expected counts are often most interpretable.

> ⚠ There Is No Universal Mixed-Model Effect Size
>
> Different effect sizes describe different targets: fixed-effect magnitude, variance explained, cluster similarity, standardized group differences, odds ratios, or predicted changes. Name the measure and explain what its denominator contains.

33.10 Same Engine, Non-Gaussian Outcome

Suppose improvement is coded as success or failure:

```r
set.seed(348)
InferenceData$Success <- rbinom(
  nrow(InferenceData), 1,
  plogis(-2 + .35 * InferenceData$Sessions +
           ClinicRE[g, 1] / 8)
)

BinaryMixedModel <- glmmTMB(
  Success ~ Sessions + (1 | Clinic),
  data = InferenceData,
  family = binomial()
)

broom.mixed::tidy(
  BinaryMixedModel,
  effects = "fixed",
  component = "cond",
  conf.int = TRUE,
  exponentiate = TRUE
)
```

```
# A tibble: 2 x 9
  effect component term        estimate std.error statistic  p.value conf.low
  <chr>  <chr>     <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
1 fixed  cond      (Intercept)    0.129    0.0329     -8.04 9.19e-16   0.0785
2 fixed  cond      Sessions       1.39     0.0554      8.25 1.62e-16   1.29
# i 1 more variable: conf.high <dbl>
```

The exponentiated slope is an odds ratio. The mixed-model structure remains; the outcome distribution and link have changed.

33.11 The Short Story

Use `glmmTMB` Wald z-tests and likelihood-based comparisons deliberately, examine confidence intervals, and diagnose the likelihood before trusting inference. Use `lmerTest` only when a Gaussian model requires ANOVA-style denominator degrees of freedom. Report effect sizes that match the scientific question and name them clearly.

Part IV

R Help and Review

34 R Help and Review: Tidyverse Without Tears

34.1 Why Is This at the End of the Book?

This is not a second statistics course hiding behind a fake mustache. It is a compact reference for the R operations you will repeatedly forget five minutes after learning them. That is normal. Professional R users also look things up constantly; they have merely developed more dignified facial expressions while doing it.

The **tidyverse** is a collection of R packages designed around consistent ways of storing, transforming, summarizing, and plotting data. We will focus on two:

- `dplyr` for manipulating data.
- `ggplot2` for making graphs.

Install them once on a computer:

```
install.packages(c("dplyr", "ggplot2"))
```

Load them again whenever you begin a new R session:

```
library(dplyr)
library(ggplot2)
```

Installing is moving the furniture into your apartment. Loading is taking the furniture out of the closet so you can sit on it. Do not reinstall the sofa every morning.

34.2 Our Deeply Serious Caffeine Study

The old lecture expected a file called `caffeine.csv` to materialize from a working directory last seen in 2019. We will simulate the data instead, which makes the example reproducible and removes the need to perform archaeological research inside Alex's Dropbox.

In this completely fictional study, 160 students consume different amounts of caffeine. We record sleep, anxiety, hatred of group projects, and whether the student begins hearing colors. No students actually consumed these doses. The IRB has stopped returning our calls for unrelated reasons.

```
set.seed(343)

n_students <- 160

lab_students <- data.frame(
  student_id = seq_len(n_students),
  caffeine_mg = sample(c(0, 100, 200, 400), n_students, replace = TRUE),
```

```r
  sleep_hours = round(pmin(10, pmax(2, rnorm(n_students, 6.2, 1.4))), 1),
  group_project_hatred = sample(1:10, n_students, replace = TRUE)
)

color_probability <- plogis(
  -4 + .010 * lab_students$caffeine_mg - .20 * (lab_students$sleep_hours - 6)
)

lab_students$hears_colors <- factor(
  ifelse(rbinom(n_students, 1, color_probability) == 1, "Yes", "No"),
  levels = c("No", "Yes")
)

lab_students$anxiety <- round(pmin(
  10,
  pmax(
    0,
    4 + .009 * lab_students$caffeine_mg -
      .30 * lab_students$sleep_hours +
      .12 * lab_students$group_project_hatred +
      rnorm(n_students, 0, 1.2)
  )
), 1)

head(lab_students)
```

```
  student_id caffeine_mg sleep_hours group_project_hatred hears_colors anxiety
1          1           0         4.8                    6           No     1.8
2          2         400         7.1                    9           No     6.2
3          3         200         3.4                    5           No     2.5
4          4         100         6.9                    6           No     1.9
5          5         400         7.0                    1          Yes     5.3
6          6         400         5.6                    5           No     4.7
```

Because `set.seed(343)` fixes the random-number sequence, everyone gets the same fictional students. The Probability Gods are still fickle, but for this example we temporarily confiscated their dice.

34.3 Data Frames

A data frame is a rectangular collection of variables:

- Each **row** is one observational unit, here one student.
- Each **column** is one variable.
- Each **cell** is one student's value on one variable.

This structure is often called **tidy data** when each variable has its own column, each observation has its own row, and each value has its own cell.

Useful first inspections include:

```r
dim(lab_students)
```

```
[1] 160   6
```

```r
names(lab_students)
```

```
[1] "student_id"           "caffeine_mg"          "sleep_hours"
[4] "group_project_hatred" "hears_colors"         "anxiety"
```

```r
str(lab_students)
```

```
'data.frame':	160 obs. of  6 variables:
 $ student_id          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ caffeine_mg         : num  0 400 200 100 400 400 400 100 100 200 ...
 $ sleep_hours         : num  4.8 7.1 3.4 6.9 7 5.6 8.4 5.6 3.8 6.5 ...
 $ group_project_hatred: int  6 9 5 6 1 5 1 8 10 6 ...
 $ hears_colors        : Factor w/ 2 levels "No","Yes": 1 1 1 1 2 1 1 1 1 1 ...
 $ anxiety             : num  1.8 6.2 2.5 1.9 5.3 4.7 6.5 5.4 4.5 5.8 ...
```

```r
summary(lab_students)
```

```
   student_id      caffeine_mg     sleep_hours    group_project_hatred
 Min.   :  1.00   Min.   :  0.0   Min.   :2.000   Min.   : 1.0
 1st Qu.: 40.75   1st Qu.:100.0   1st Qu.:5.600   1st Qu.: 3.0
 Median : 80.50   Median :150.0   Median :6.400   Median : 5.0
 Mean   : 80.50   Mean   :176.2   Mean   :6.379   Mean   : 5.5
 3rd Qu.:120.25   3rd Qu.:200.0   3rd Qu.:7.200   3rd Qu.: 8.0
 Max.   :160.00   Max.   :400.0   Max.   :9.600   Max.   :10.0
 hears_colors    anxiety
 No :144      Min.   :0.000
 Yes: 16      1st Qu.:2.875
              Median :4.200
              Mean   :4.247
              3rd Qu.:5.500
              Max.   :8.500
```

`str()` is especially useful. It reveals whether R thinks a variable is numeric, character, or a factor before you spend 45 minutes accusing the regression model of personal betrayal.

34.4 The Native Pipe

Modern R uses `|>` to pass the result of one step into the next step:

```r
lab_students |>
  head(3)
```

```
  student_id caffeine_mg sleep_hours group_project_hatred hears_colors anxiety
1          1           0         4.8                    6           No     1.8
2          2         400         7.1                    9           No     6.2
3          3         200         3.4                    5           No     2.5
```

Read it as **"and then."**

> Start with `lab_students`, **and then** keep highly caffeinated students, **and then** select the variables we want.

```
lab_students |>
  filter(caffeine_mg >= 200) |>
  select(student_id, caffeine_mg, anxiety, hears_colors) |>
  head()
```

```
  student_id caffeine_mg anxiety hears_colors
1          2         400     6.2           No
2          3         200     2.5           No
3          5         400     5.3          Yes
4          6         400     4.7           No
5          7         400     6.5           No
6         10         200     5.8           No
```

Older R code often uses `%>%`, the pipe from `magrittr` and `dplyr`. It is still valid and appears throughout older lectures, published code, and the ancient inscriptions found beneath BSB. We will use the native `|>` in new code and eventually update the rest of the book.

34.5 The Five Verbs You Will Use Constantly

34.5.1 `select()`: Keep Columns

```
small_data <- lab_students |>
  select(student_id, caffeine_mg, sleep_hours, anxiety)

head(small_data)
```

```
  student_id caffeine_mg sleep_hours anxiety
1          1           0         4.8     1.8
2          2         400         7.1     6.2
3          3         200         3.4     2.5
4          4         100         6.9     1.9
5          5         400         7.0     5.3
6          6         400         5.6     4.7
```

Remove columns with a minus sign:

```r
lab_students |>
  select(-student_id) |>
  head()
```

```
  caffeine_mg sleep_hours group_project_hatred hears_colors anxiety
1           0         4.8                    6           No     1.8
2         400         7.1                    9           No     6.2
3         200         3.4                    5           No     2.5
4         100         6.9                    6           No     1.9
5         400         7.0                    1          Yes     5.3
6         400         5.6                    5           No     4.7
```

34.5.2 `filter()`: Keep Rows

```r
chromatic_students <- lab_students |>
  filter(hears_colors == "Yes")

nrow(chromatic_students)
```

```
[1] 16
```

Combine conditions with `&` for **and** and `|` for **or**:

```r
lab_students |>
  filter(caffeine_mg >= 200 & sleep_hours < 6) |>
  head()
```

```
  student_id caffeine_mg sleep_hours group_project_hatred hears_colors anxiety
1          3         200         3.4                    5           No     2.5
2          6         400         5.6                    5           No     4.7
3         11         200         5.1                    9           No     5.8
4         14         400         5.9                    9           No     8.5
5         17         200         5.0                    1           No     3.8
6         23         200         4.3                    3           No     4.1
```

Use `==` to test equality. One `=` assigns an argument; two `==` ask whether values are equal. This tiny distinction has destroyed entire afternoons.

34.5.3 `mutate()`: Create or Change Columns

```r
lab_students <- lab_students |>
  mutate(
    caffeine_per_hour_sleep = caffeine_mg / sleep_hours,
    extremely_caffeinated = caffeine_mg >= 400
  )

head(lab_students)
```

```
  student_id caffeine_mg sleep_hours group_project_hatred hears_colors anxiety
1          1           0         4.8                    6           No     1.8
2          2         400         7.1                    9           No     6.2
3          3         200         3.4                    5           No     2.5
4          4         100         6.9                    6           No     1.9
5          5         400         7.0                    1          Yes     5.3
6          6         400         5.6                    5           No     4.7
  caffeine_per_hour_sleep extreme_caffeinated
1                 0.00000               FALSE
2                56.33803                TRUE
3                58.82353               FALSE
4                14.49275               FALSE
5                57.14286                TRUE
6                71.42857                TRUE
```

`mutate()` keeps the existing variables and adds or replaces columns. The expression is calculated once per row.

34.5.4 `arrange()`: Sort Rows

```
lab_students |>
  arrange(desc(anxiety)) |>
  select(student_id, caffeine_mg, sleep_hours, anxiety) |>
  head(10)
```

```
   student_id caffeine_mg sleep_hours anxiety
1          14         400         5.9     8.5
2         152         400         5.6     8.0
3          72         400         8.0     7.9
4         130         400         5.0     7.9
5         129         400         8.1     7.8
6         145         400         7.1     7.8
7         114         400         5.6     7.6
8          22         200         6.4     7.5
9          73         400         5.4     7.2
10        158         400         6.4     7.2
```

These are the students most likely to email at 2:13 a.m. with the subject line `QUICK QUESTION!!!!!`.

34.5.5 `summarise()`: Collapse Many Rows into Answers

```
lab_students |>
  summarise(
    n = n(),
    mean_anxiety = mean(anxiety),
    sd_anxiety = sd(anxiety),
    median_sleep = median(sleep_hours),
```

```
  proportion_hearing_colors = mean(hears_colors == "Yes")
)
```

```
    n mean_anxiety sd_anxiety median_sleep proportion_hearing_colors
1 160       4.2475   1.863315          6.4                       0.1
```

Unlike `mutate()`, `summarise()` reduces many rows to a smaller summary. `n()` counts rows. A logical statement such as `hears_colors == "Yes"` becomes `TRUE` or `FALSE`; its mean is therefore the proportion that is `TRUE`.

34.6 Group, Then Summarize

`group_by()` tells later verbs to operate separately within categories.

```
caffeine_summary <- lab_students |>
  group_by(caffeine_mg) |>
  summarise(
    n = n(),
    mean_anxiety = mean(anxiety),
    sd_anxiety = sd(anxiety),
    mean_sleep = mean(sleep_hours),
    proportion_hearing_colors = mean(hears_colors == "Yes"),
    .groups = "drop"
  )

caffeine_summary
```

```
# A tibble: 4 x 6
  caffeine_mg     n mean_anxiety sd_anxiety mean_sleep proportion_hearing_colors
        <dbl> <int>        <dbl>      <dbl>      <dbl>                     <dbl>
1           0    36         2.48       1.45       6.23                    0.0278
2         100    44         4.01       1.33       6.16                    0     
3         200    41         4.18       1.36       6.47                    0.0732
4         400    39         6.22       1.28       6.66                    0.308 
```

`.groups = "drop"` removes the grouping after the summary. This prevents the invisible ghost of a previous grouping decision from possessing later analyses.

For simple frequency tables, `count()` is faster:

```
lab_students |>
  count(caffeine_mg, hears_colors)
```

```
  caffeine_mg hears_colors  n
1           0           No 35
2           0          Yes  1
3         100           No 44
4         200           No 38
```

```
5          200          Yes  3
6          400           No 27
7          400          Yes 12
```

For the same summaries across several columns, use `across()`:

```
lab_students |>
  group_by(caffeine_mg) |>
  summarise(
    across(
      c(sleep_hours, anxiety, group_project_hatred),
      list(mean = mean, sd = sd),
      .names = "{.col}_{.fn}"
    ),
    .groups = "drop"
  )
```

```
# A tibble: 4 x 7
  caffeine_mg sleep_hours_mean sleep_hours_sd anxiety_mean anxiety_sd
        <dbl>            <dbl>          <dbl>        <dbl>      <dbl>
1           0             6.23           1.57         2.48       1.45
2         100             6.16           1.13         4.01       1.33
3         200             6.47           1.29         4.18       1.36
4         400             6.66           1.26         6.22       1.28
# i 2 more variables: group_project_hatred_mean <dbl>,
#   group_project_hatred_sd <dbl>
```

34.7 Reading a Real Data File

Quarto renders the book from the project directory. Use a project-relative path instead of `setwd()`:

```
caffeine_data <- read.csv("data/caffeine.csv")
```

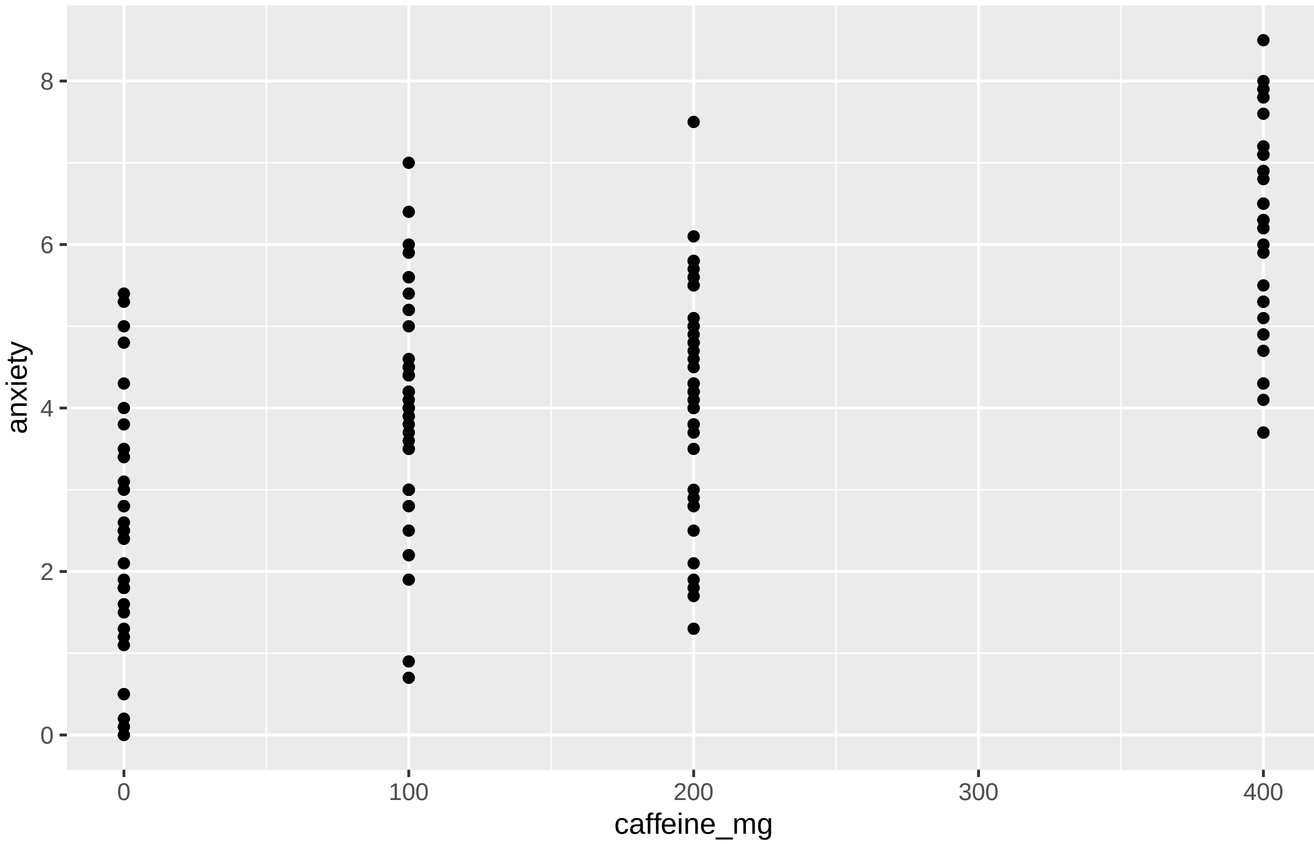
This says the file lives in a folder named `data` inside the project. Hard-coding a path to one person's Dropbox guarantees that the code will fail on another computer, next semester, or immediately after Alex reorganizes his folders during an avoidable burst of optimism.

34.8 `ggplot2`: Build a Graph in Layers

A `ggplot` usually has three core ingredients:

1. **Data**: the data frame.
2. **Aesthetics**: variables mapped to position, color, shape, or size with `aes()`.
3. **Geometries**: the marks placed on the graph, such as points, bars, or boxes.

```
ggplot(
  data = lab_students,
  mapping = aes(x = caffeine_mg, y = anxiety)
) +
  geom_point()
```



The **+** adds layers to a ggplot. It is not interchangeable with the pipe. The pipe moves data through operations; the plus sign adds components to a graph. R is very committed to making punctuation carry emotional weight.

34.8.1 Scatterplot with a Model Line

```
ggplot(lab_students, aes(x = caffeine_mg, y = anxiety, color = hears_colors)) +
  geom_jitter(width = 12, height = 0, alpha = .65, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  scale_color_manual(values = c("No" = "#1F4E79", "Yes" = "#D55E00")) +
  labs(
    x = "Caffeine (mg)",
    y = "Anxiety rating",
    color = "Hears colors",
    title = "At some point the espresso begins explaining the wallpaper"
  ) +
  theme_minimal(base_size = 12)
```

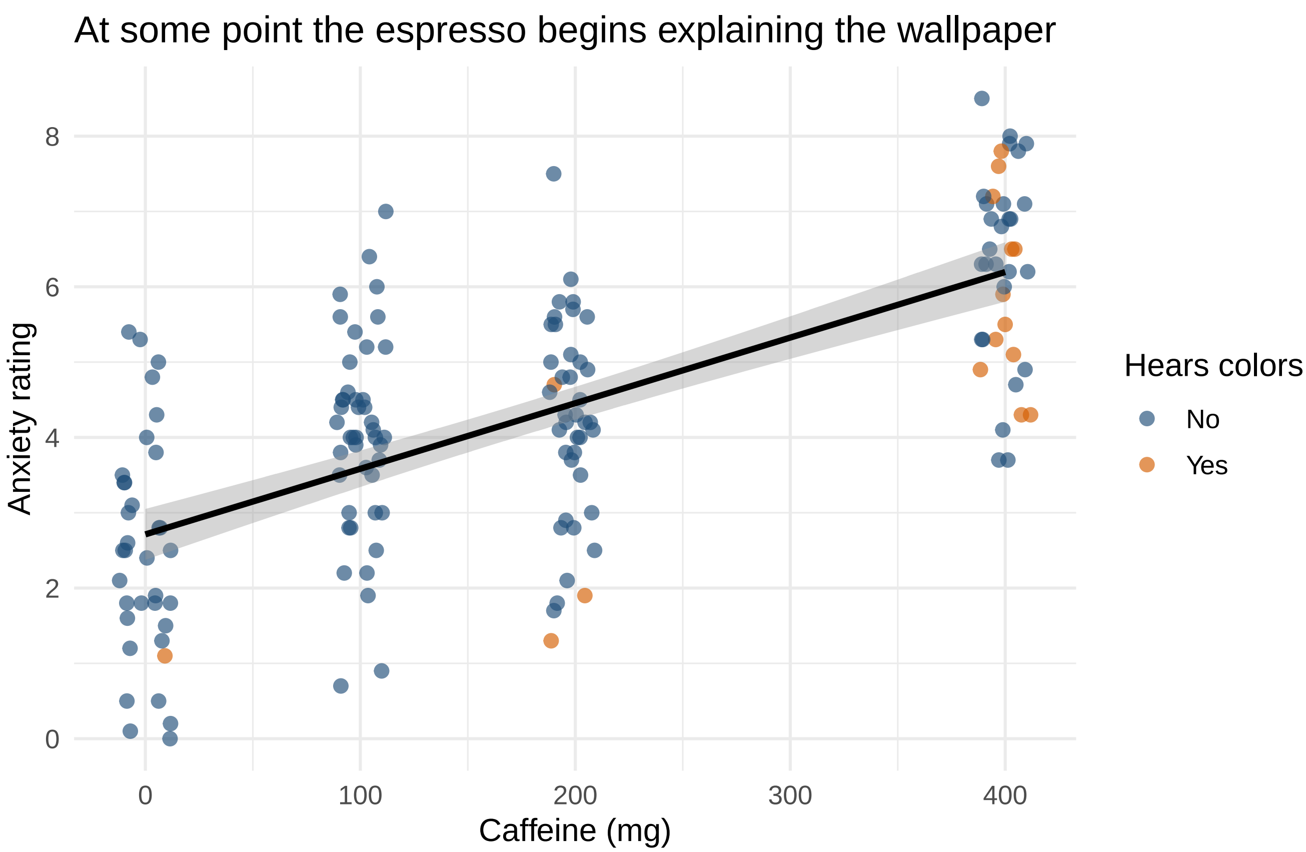


Figure 34.1: Anxiety across caffeine doses. Jitter separates students who otherwise occupy the same caffeine value.

`geom_jitter()` moves overlapping points by a tiny random amount horizontally so we can see them. It does not change the caffeine condition stored in the data.

34.8.2 Boxplots, Correctly Explained

```
ggplot(lab_students, aes(x = factor(caffeine_mg), y = anxiety)) +
  geom_boxplot(fill = "#9ECAE1", color = "#1F4E79", outlier.color = "#D55E00") +
  labs(
    x = "Caffeine (mg)",
    y = "Anxiety rating",
    title = "Boxes: useful, compact, and frequently lied about"
  ) +
  theme_minimal(base_size = 12)
```

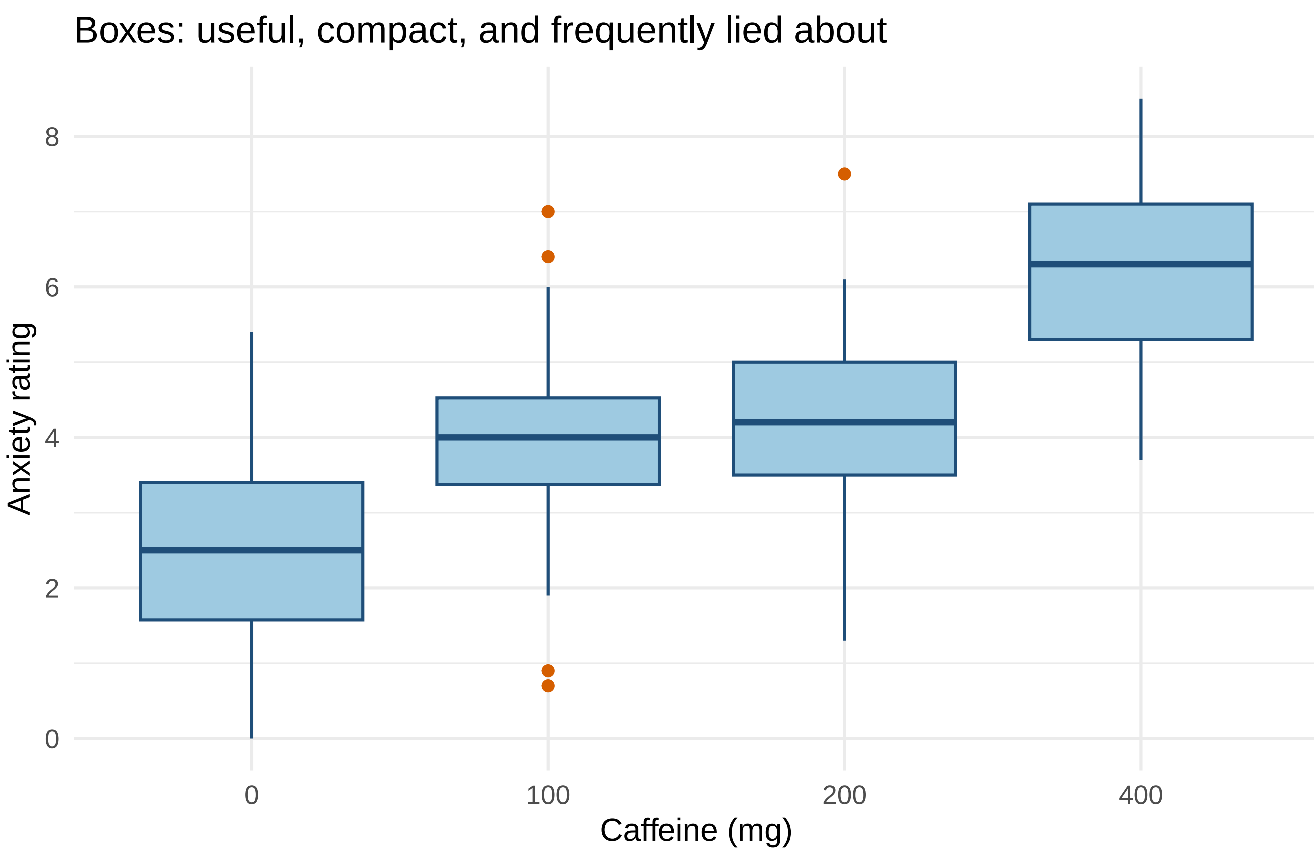


Figure 34.2: Anxiety distributions at each caffeine dose.

The line inside each box is the **median**. The bottom and top of the box are the first and third quartiles, so the box contains the middle **50%** of scores. Its height is the interquartile range (IQR). The whiskers ordinarily extend to the most extreme observed values within 1.5 IQRs of the box; points beyond them are plotted separately.

Those separate points are not automatically errors, criminals, or volunteers for deletion. They are observations that deserve inspection and an explanation.

34.8.3 Plotting a Summary

```
ggplot(caffeine_summary, aes(x = factor(caffeine_mg), y = proportion_hearing_colors)) +
  geom_col(fill = "#CC79A7", width = .70) +
  scale_y_continuous(limits = c(0, 1)) +
  labs(
    x = "Caffeine (mg)",
    y = "Proportion hearing colors",
    title = "Eventually teal starts whispering"
  ) +
  theme_minimal(base_size = 12)
```

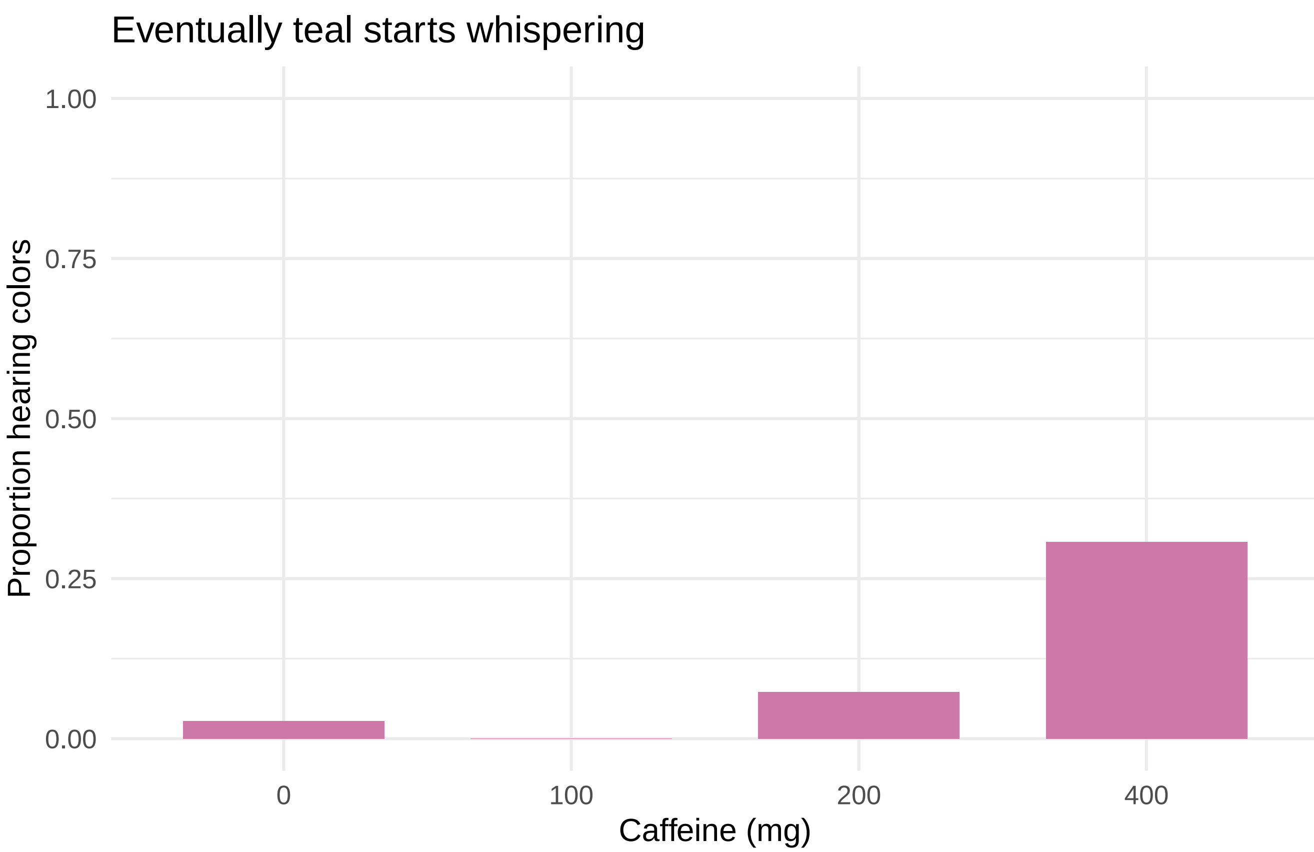


Figure 34.3: Simulated proportion of students reporting chromatic audition by caffeine dose.

`geom_col()` uses the y-values already calculated in `caffeine_summary`. `geom_bar()` would count raw rows instead. This difference is small, annoying, and worth remembering.

34.9 A Compact Workflow

Most routine work follows this sequence:

```
analysis_data <- lab_students |>
  filter(!is.na(anxiety)) |>
  mutate(caffeine_group = factor(caffeine_mg)) |>
  select(student_id, caffeine_group, sleep_hours, anxiety, hears_colors)

analysis_summary <- analysis_data |>
  group_by(caffeine_group) |>
  summarise(
    n = n(),
    mean = mean(anxiety),
    sd = sd(anxiety),
    .groups = "drop"
  )

analysis_summary
```

```
# A tibble: 4 x 4
  caffeine_group     n  mean    sd
  <fct>          <int> <dbl> <dbl>
1 0                 36  2.48  1.45
2 100               44  4.01  1.33
3 200               41  4.18  1.36
4 400               39  6.22  1.28
```

1. Start with the original data.
2. Filter only with a defensible rule.
3. Create or recode needed variables.
4. Select the variables used in the analysis.
5. Check and summarize the result.
6. Graph the data before asking a model to explain them.

Keep the original data unchanged and create a new analysis object. Overwriting the only copy of a variable is how `Final_Data_REAL_v7_USE_THIS_ONE.csv` enters the world.

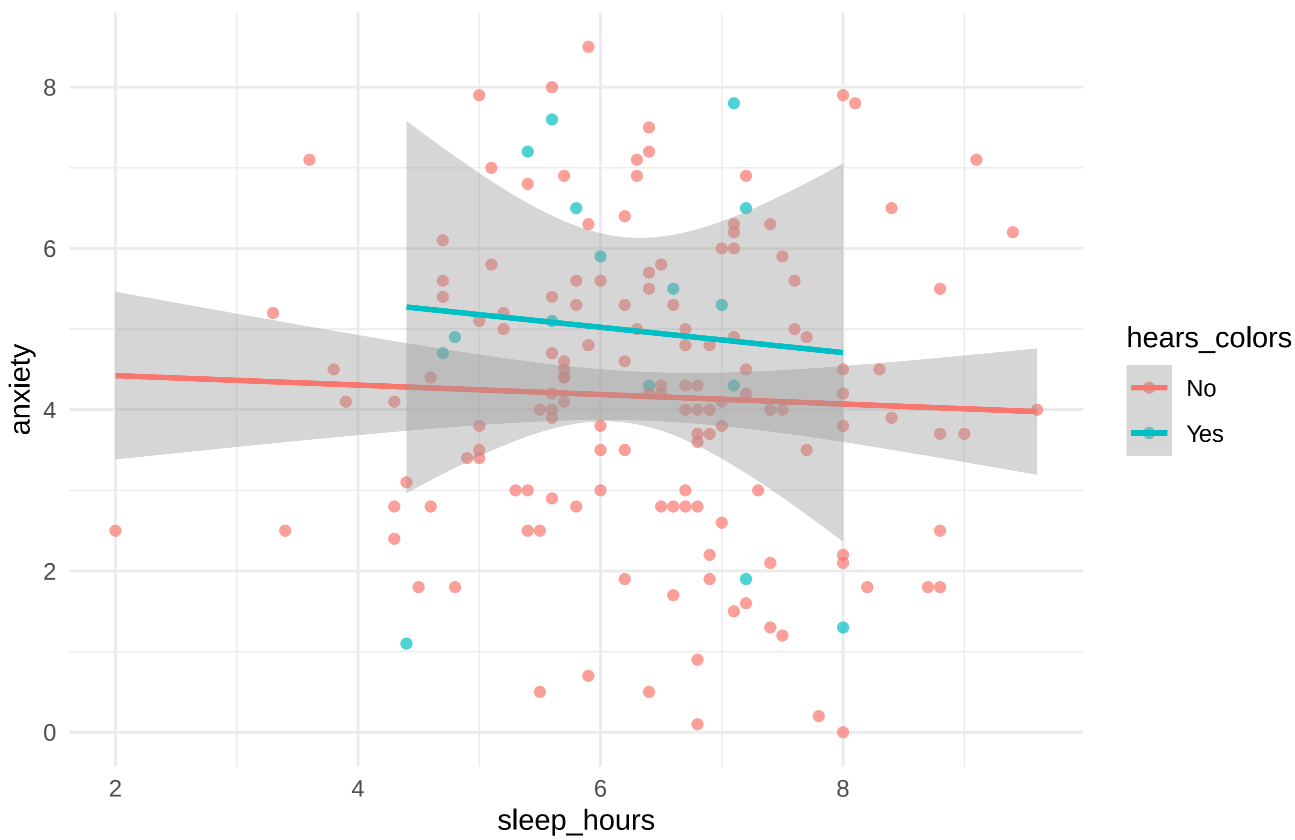
34.10 Practice: Make the Caffeine Data Confess

Try these before looking at the solutions:

1. Keep students who slept fewer than six hours and consumed at least 200 mg of caffeine.
2. For each caffeine dose, calculate sample size, median anxiety, IQR of anxiety, and the proportion hearing colors.
3. Create a scatterplot of sleep and anxiety, coloring points by whether the student hears colors.
4. Find the ten students with the highest caffeine-per-hour-of-sleep values.

💡 Solutions, because suffering has limits

```
sleep_deprived_and_caffeinated <- lab_students |>
  filter(sleep_hours < 6, caffeine_mg >= 200)

practice_summary <- lab_students |>
  group_by(caffeine_mg) |>
  summarise(
    n = n(),
    median_anxiety = median(anxiety),
    iqr_anxiety = IQR(anxiety),
    proportion_hearing_colors = mean(hears_colors == "Yes"),
    .groups = "drop"
  )

ggplot(lab_students, aes(x = sleep_hours, y = anxiety, color = hears_colors)) +
  geom_point(alpha = .70) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal()
```




```
most_concerning_students <- lab_students |>
  arrange(desc(caffeine_per_hour_sleep)) |>
  select(student_id, caffeine_mg, sleep_hours, caffeine_per_hour_sleep) |>
  head(10)

practice_summary
```

```
# A tibble: 4 x 5
  caffeine_mg     n median_anxiety iqr_anxiety proportion_hearing_colors
        <dbl> <int>          <dbl>       <dbl>                     <dbl>
```

34.11 Final Advice

Write code as a sequence of small, named decisions. Inspect the result after each important transformation. Make graphs before interpreting models. When something fails, read the error message from the bottom up and identify the first object or argument R says it cannot find.

Most importantly, do not confuse code that runs with an analysis that makes sense. R will cheerfully calculate the mean of participant ID numbers, regress anxiety on alphabetical order, and draw a confidence interval around almost any mistake you can encode. The software is obedient. You still have to be the adult in the room, which seems unfair but is apparently how science works.

References

Abelson, Robert P. 1995. *Statistics as Principled Argument*. Lawrence Erlbaum Associates.

Aiken, Leona S., and Stephen G. West. 1991. *Multiple Regression: Testing and Interpreting Interactions*. Sage.

Ang, L. H., and Masitah Shahrill. 2014. "Identifying Students' Specific Misconceptions in Learning Probability." *International Journal of Probability and Statistics* 3 (2): 23–29.

Barr, Dale J., Roger Levy, Christoph Scheepers, and Harry J. Tily. 2013. "Random Effects Structure for Confirmatory Hypothesis Testing: Keep It Maximal." *Journal of Memory and Language* 68 (3): 255–78. https://doi.org/10.1016/j.jml.2012.11.001.

Benjamini, Yoav, and Yosef Hochberg. 1995. "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society: Series B (Methodological)* 57 (1): 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

Bennett, Craig M., Abigail A. Baird, Michael B. Miller, and George L. Wolford. 2010. "Neural Correlates of Interspecies Perspective Taking in the Post-Mortem Atlantic Salmon: An Argument for Proper Multiple Comparisons Correction." *Journal of Serendipitous and Unexpected Results* 1 (1): 1–5.

Best, John B. 1979. "Item Difficulty and Answer Changing." *Teaching of Psychology* 6 (4): 228–30. https://doi.org/10.1207/s15328023top0604_13.

Brooks, Mollie E., Kasper Kristensen, Koen J. van Benthem, et al. 2017. "glmmTMB Balances Speed and Flexibility Among Packages for Zero-Inflated Generalized Linear Mixed Modeling." *The R Journal* 9 (2): 378–400. https://doi.org/10.32614/RJ-2017-066.

Button, Katherine S., John P. A. Ioannidis, Claire Mokrysz, et al. 2013. "Power Failure: Why Small Sample Size Undermines the Reliability of Neuroscience." *Nature Reviews Neuroscience* 14 (5): 365–76. https://doi.org/10.1038/nrn3475.

Buuren, Stef van, and Karin Groothuis-Oudshoorn. 2011. "mice: Multivariate Imputation by Chained Equations in R." *Journal of Statistical Software* 45 (3): 1–67. https://doi.org/10.18637/jss.v045.i03.

Cohen, Jacob. 1992. "A Power Primer." *Psychological Bulletin* 112 (1): 155–59. https://doi.org/10.1037/0033-2909.112.1.155.

Cohen, Jacob. 1994. "The Earth Is Round ($p < .05$)." *American Psychologist* 49 (12): 997–1003. https://doi.org/10.1037/0003-066X.49.12.997.

Cohen, Jacob, Patricia Cohen, Stephen G. West, and Leona S. Aiken. 2003. *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*. 3rd ed. Lawrence Erlbaum Associates.

Efron, Bradley, and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman; Hall/CRC.

Enders, Craig K., and Davood Tofighi. 2007. "Centering Predictor Variables in Cross-Sectional Multilevel Models: A New Look at an Old Issue." *Psychological Methods* 12 (2): 121–38. https://doi.org/10.1037/1082-989X.12.2.121.

Errington, Timothy M., Maya Mathur, Courtney K. Soderberg, et al. 2021. "Investigating the Replicability of Preclinical Cancer Biology." *eLife* 10: e71601. https://doi.org/10.7554/eLife.71601.

Fabrigar, Leandre R., Duane T. Wegener, Robert C. MacCallum, and Erin J. Strahan. 1999. "Evaluating the Use of Exploratory Factor Analysis in Psychological Research." *Psychological Methods* 4 (3): 272–99. https://doi.org/10.1037/1082-989X.4.3.272.

Gelman, Andrew, and Hal Stern. 2006. "The Difference Between 'Significant' and 'Not Significant' Is Not Itself Statistically Significant." *The American Statistician* 60 (4): 328–31. https://doi.org/10.1198/000313006X152649.

Hoenig, John M., and Dennis M. Heisey. 2001. "The Abuse of Power: The Pervasive Fallacy of Power Calculations for Data Analysis." *The American Statistician* 55 (1): 19–24. https://doi.org/10.1198/000313001300339897.

Holm, Sture. 1979. "A Simple Sequentially Rejective Multiple Test Procedure." *Scandinavian Journal of Statistics* 6 (2): 65–70.

Ioannidis, John P. A. 2005. "Why Most Published Research Findings Are False." *PLoS Medicine* 2 (8): e124. https://doi.org/10.1371/journal.pmed.0020124.

Ioannidis, John P. A. 2012. "Why Science Is Not Necessarily Self-Correcting." *Perspectives on Psychological Science* 7 (6): 645–54. https://doi.org/10.1177/1745691612464056.

Kuhn, Thomas S. 1962. *The Structure of Scientific Revolutions*. University of Chicago Press.

Kuznetsova, Alexandra, Per B. Brockhoff, and Rune H. B. Christensen. 2017. "lmerTest Package: Tests in Linear Mixed Effects Models." *Journal of Statistical Software* 82 (13): 1–26. https://doi.org/10.18637/jss.v082.i13.

Lane-Getaz, Sharon J. 2005. "Summary of p-Value Survey Research." *Proceedings of the 2005 United States Conference on Teaching Statistics*, 13–17.

Marcoci, Alexandru, David P. Wilkinson, Ans Vercammen, Bonnie C. Wintle, et al. 2025. "Predicting the Replicability of Social and Behavioural Science Claims in COVID-19 Preprints." *Nature Human Behaviour* 9: 287–304. https://doi.org/10.1038/s41562-024-01961-1.

Marek, Scott, Brenden Tervo-Clemmens, Finnegan J. Calabro, et al. 2022. "Reproducible Brain-Wide Association Studies Require Thousands of Individuals." *Nature* 603: 654–60. https://doi.org/10.1038/s41586-022-04492-9.

McCullagh, Peter, and John A. Nelder. 1989. *Generalized Linear Models*. 2nd ed. Chapman; Hall.

Meehl, Paul E. 1978. "Theoretical Risks and Tabular Asterisks: Sir Karl, Sir Ronald, and the Slow Progress of Soft Psychology." *Journal of Consulting and Clinical Psychology* 46 (4): 806–34. https://doi.org/10.1037/0022-006X.46.4.806.

Meehl, Paul E. 1990a. "Appraising and Amending Theories: The Strategy of Lakatosian Defense and Two Principles That Warrant It." *Psychological Inquiry* 1 (2): 108–41. https://doi.org/10.1207/s15327965pli0102_1.

Meehl, Paul E. 1990b. "Why Summaries of Research on Psychological Theories Are Often Uninterpretable." *Psychological Reports* 66 (1): 195–244. https://doi.org/10.2466/pr0.1990.66.1.195.

Michell, Joel. 1997. "Quantitative Science and the Definition of Measurement in Psychology." *British Journal of Psychology* 88 (3): 355–83. https://doi.org/10.1111/j.2044-8295.1997.tb02641.x.

Miller, Lulu. 2020. *Why Fish Don't Exist: A Story of Loss, Love, and the Hidden Order of Life*. Simon & Schuster.

Nakagawa, Shinichi, Paul C. D. Johnson, and Holger Schielzeth. 2017. "The Coefficient of Determination R^2 and Intra-Class Correlation Coefficient from Generalized Linear Mixed-Effects Models Revisited and Expanded." *Journal of the Royal Society Interface* 14 (134): 20170213. https://doi.org/10.1098/rsif.2017.0213.

Nature Neuroscience. 2024. "Reducing Publication Bias with Registered Reports." *Nature Neuroscience* 27: 1635. https://doi.org/10.1038/s41593-024-01762-9.

Open Science Collaboration. 2015. "Estimating the Reproducibility of Psychological Science." *Science* 349 (6251): aac4716. https://doi.org/10.1126/science.aac4716.

Peters, Michael, Bruno Laeng, K. Latham, M. Jackson, R. Zaiyouna, and C. Richardson. 1995. "A Redrawn Vandenberg and Kuse Mental Rotations Test: Different Versions and Factors That Affect Performance." *Brain and Cognition* 28 (1): 39–58. https://doi.org/10.1006/brcg.1995.1032.

Preacher, Kristopher J., and Andrew F. Hayes. 2008. "Asymptotic and Resampling Strategies for Assessing and Comparing Indirect Effects in Multiple Mediator Models." *Behavior Research Methods* 40 (3): 879–91. https://doi.org/10.3758/BRM.40.3.879.

Rubin, Donald B. 1987. *Multiple Imputation for Nonresponse in Surveys*. Wiley.

Stevens, S. S. 1946. "On the Theory of Scales of Measurement." *Science* 103 (2684): 677–80. https://doi.org/10.1126/science.103.2684.677.

U.S. Census Bureau. 2024. *Historical Income Tables: Households*. https://www.census.gov/data/tables/time-series/demo/income-poverty/cps-hinc/hinc-01.2024.html.

Yeager, David S., Paul Hanselman, Gregory M. Walton, Jared S. Murray, et al. 2019. "A National Experiment Reveals Where a Growth Mindset Improves Achievement." *Nature* 573: 364–69. https://doi.org/10.1038/s41586-019-1466-y.